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LIST OF ABBREVIATIONS

BEB	Battery Electric Bus
BEV	Battery Electric Vehicle
CPO	Charging Point Operator
DMS	Distribution Management System
DNO	Distribution Network Operator
DSO	Distribution System Operator
EV	Electric Vehicle
EVCI	Electric Vehicle Charging Infrastructure
EDCL	Energy Development Corporation Limited
ESD	Energy Storage Device
EUCL	Energy Utility Corporation Limited
E2&3W	Electric Two and Three Wheelers
GE	Grid Extension
GHG	Greenhouse Gas Emissions
GIS	Geographic Information System
HDT	Heavy-Duty Truck
HDV	Heavy-Duty Vehicle
HEV	Hybrid Electric Vehicle
LCPDP	Least Cost Power Development Plan
LCV	Light Commercial Vehicle
LDV	Light-Duty Vehicle
LZEV	Low and Zero-Emission Vehicles
MININFRA	Ministry of Infrastructure of the Government of Rwanda
MSP	Mobility Service Provider
NEP	National Electrification Plan
RDB	Rwanda Development Board
REG	Rwanda Energy Group

RES	Renewable Energy Source
RRA	Rwanda Revenue Authority
RURA	Rwanda Utilities Regulatory Agency
RTDA	Rwanda Transport Development Agency
SAS	Stand-Alone (Home)-Systems
TN	Transmission Network
TSO	Transmission System Operator
V2G	Vehicle-to-Grid
ZEV	Zero-Emission Vehicle

1 INTRODUCTION

1.1 The EV Charging Master Plan

As Rwanda embarks on an ambitious journey towards sustainable mobility, the **Electric Vehicle Charging Infrastructure (EVCI) Master Plan** emerges as a key blueprint guiding this transition. The shift towards electric mobility is not only a response to global environmental imperatives but also aligns with Rwanda's vision of a green and resilient economy as outlined in the National Strategy for Transformation (NST1). This Master Plan aims to strategically **develop a comprehensive network of EV charging stations**, ensuring seamless integration of electric vehicles (EVs) into the country's transport system.

The EVCI Master Plan addresses critical aspects necessary for fostering an efficient and sustainable EV ecosystem. This includes the deployment of public, semi-public, and private charging infrastructures, the implementation of policies that support the phased transition from fossil fuel vehicles to electric alternatives. It highlights the **roles of various stakeholders**, including government entities, private sector players, emphasizing their contributions to the success of this initiative. Informed by detailed geospatial analyses, stakeholder consultations, and best practices from leading EV markets, this Master Plan provides a roadmap for the deployment of EVCI in Rwanda.

Developing an effective EV charging master plan involves strategically planning various crucial steps to **integrate EVs into the transportation system smoothly**. This requires a multi-faceted approach that covers infrastructure development, regulatory frameworks, and collaboration with stakeholders. **Key prerequisites** include the presence of EVs in the market to stimulate demand, alongside well-structured plans for expanding the charging infrastructure. Additionally, ensuring grid connections and upgrades are responsive to demand, availability of designated land slots for charging sites, and streamlined permitting processes are required. Moreover, operational charging infrastructure and accessible data, coupled with robust business and finance models, play pivotal roles in supporting the transition to widespread EV adoption. Beyond these foundational elements, proactive management of broader social, economic, and environmental effects, as well as informed stakeholder collaboration, are integral for a holistic and sustainable EV ecosystem.

To effectively kickstart the transition towards widespread EV adoption, prioritizing **three key actions becomes imperative**. Firstly, setting **long-term targets and regulations** serves as a catalyst for driving zero-emission vehicle adoptions. These targets not only signal demand to manufacturers but also provide a clear pathway for compliance. Secondly, establishing a **robust governance structure** is essential, empowering stakeholders with clear responsibilities and timelines to accelerate deployment efforts. A well-defined governance framework fosters accountability facilitates coordination among relevant stakeholders and ensures timely action. Finally, identifying future electricity needs and **developing long-term plans** to fortify the grid's capacity play a pivotal role in guaranteeing its readiness to meet the burgeoning demands of an electrified transportation landscape. Through these prioritized actions, coupled with comprehensive strategic planning, the foundation is laid for a sustainable and efficient EV charging infrastructure ecosystem.

This report is structured as follows:

- *Chapter 1 Introduction:* This chapter provides an overview of the goals of the Electric Vehicle Charging Infrastructure Master Plan and its relevance. It discusses the main trends in e-mobility, global developments towards e-mobility and charging infrastructure, developments in Africa, and the current status in Rwanda.
- *Chapter 2 eMobility Principles:* This chapter focuses on the fundamental definitions related to charging infrastructure, which are essential for understanding the subsequent chapters. It serves as a guideline for the basics of charging infrastructure concepts, including the categorization and types of EVs, charging systems, types of connectors, deployment strategies for EVCI, key insights on installation and management of EVCI, typical roles in eMobility, interoperability, payment modalities, and circularity and waste disposal for EVCI.
- *Chapter 3 Current Situation in Rwanda:* This chapter illustrates the current situation in Rwanda that impacts e-mobility. It provides an overview of current transport numbers and pattern behaviors, current numbers of electric vehicles and charging stations in the country, policies adopted to promote e-mobility, the governance structure, and main stakeholders involved. It also gives insights into the current status of the energy sector, which is fundamental for developing EVCI.
- *Chapter 4 Future Scenarios*: This chapter focuses on future scenarios, giving an overview of the expected growth of transport vehicles in Rwanda, including both current transport and EVs. It presents an overview of the different categorizations of vehicles and pattern behaviors toward private and public transport as well as non-motorized transport. The future scenario also includes a section on energy, providing information regarding the growth and required work of the grid to support the widespread use of EVs and their charging infrastructure.
- *Chapter 5 Technical Roadmap for Charging Infrastructure*: This chapter provides a detailed roadmap for the widespread deployment of charging infrastructure. It outlines the necessary steps to expand the infrastructure and the technical details that need to be considered for development. It includes guidelines for location and site planning for EV charging infrastructure in the country and insights on how to handle access and demand for EVCI.
- *Chapter 6 Institutional Roadmap*: This chapter focuses on the policies, regulations, governance structure, and roles of stakeholders essential for implementing a successful charging infrastructure roadmap. It provides insights

into governance roles, various financing models, and cost-benefit analyses for these models. Additionally, the chapter highlights the necessary collaboration between public and private organizations and potential operational and business models to support the efficient implementation, management, and operation of the proposed charging solutions.

This comprehensive structure ensures a thorough understanding of the current state and future needs for electric vehicle charging infrastructure in Rwanda.

Finally, it is worth mentioning that, as per the results required by the Terms of Reference, these have been integrated throughout the entire document in a coherent storyline. However, they can also be found as follows. *The Technical note on different [Charging] solutions (on or off the grid) and their advantages/disadvantages* and the *Technical note on different types of EV charging stations (AC, DC, Low and High Power)* can be found on **section 2.2 Charging systems**. The *EV charging standards for interoperability, best on international best practice*, could be found on **section 2.3 type of connectors** and in **section 2.7 Interoperability of EVCI**. The *Technical note on the proper disposal of waste generated from EV charging services* can be found on **section 2.9 Circularity and waste disposal**, and the overview of the *technology and technical solutions with an assessment of the resiliency to minimize technology obsolescence, as well as a detailed analysis of barriers and bottlenecks to overcome* can be found on **section 2.10 Technical considerations to minimize technology obsolescence**.

The digitalised Map of both existing and potential locations to install EVs charging stations in Rwanda ready to be accessed online can be found on **section 3.2 Current E-Mobility Sector in Rwanda**, additionally, in the **Annexes** all the maps can be found. The *Geospatial analysis and mapping of the most relevant sites to place EV charging stations in the country (City of Kigali, secondary cities and satellite cities and main national roads) and assessing energy/ grid connectivity requirements for each identified site, as well as minimum number of parking bays based on the projected demand per charging station* can be found on **Section 3.2 Current E-Mobility Sector in Rwanda** and **in Section 5.11 Geospatial Analaysis.** The *Governance structure of the EV charging ecosystem by identifying the regulatory and executive government agencies involved in charging infrastructure implementation, and by defining the roles of charge point operators and e-mobility service providers* can be found on the section 2.6 **Typical Roles in eMobility Ecosystem**, also in **section 3.3 Public and Private Current Stakeholder Situation** and in section **6.8.1 Final Recommendations**. The *Technical note on analysis on the energy grid, for now and the future, and plan infrastructure upgrades accordingly*, can be found in section 3.4 **Current Energy System of Rwanda** and in section **4.2 Energy Future Scenario: EVs' Chargers Embedded Energy System of Rwanda.**

The specific analysis of the use of off-grid charging infrastructure more specifically in remote areas and cost comparison with usage of grid energy and *the Technical note on the assessment of grid-level impacts, and then recommendations of a suitable charging technology to minimize adverse impacts of EV charging loads on the grid* could be found in Section 5.12 **Grid and Energy System Planning** and in Section **5.13 Proposed Solutions for Mitigating EVCI.**

The *Geospational Analysis* and the *Technical note on the planning process with an overview of the access- and demand-based approaches for setting targets (for*

number of public chargers required), and methodology for assessing energy demand for public EV charging can be found in **Chapter 5 Section 5.2 Deployment of EV charging, 5.3 Factors to consider the demand on EVCI** and **5.4 Proactive Charging Infrastructure Placement Approaches**. *The Technical note on location and site planning aspects for EV charging, by framing the principles of location planning and demonstrating a methodology for spatial allocation of charging demand and identifying enabling processes and policies to integrate public charging in urban planning* can be found on **Section 2.5 Deployment and Management of EVCI** and in **Section 5.11 Geospatial analysis and Proposed Locations for EVCI**. The *Advice on the mechanism to install charging stations at existing petrol stations* is on section **2.5.1. Key Considerations to install EVCCI**. The result referred to *Propose a set of regulations that govern electricity supply for EV charging infrastructure* can be found on **section 5.12** and on **6.8.1 Final Recommendations**.

The *mechanism to regulate and promote private charging (workplace charging and home charging) and charging for commercial fleets and public transport* can be found in **Section 6.4. Public Investment and Governance Roles.** The *Technical note on the typical roles within an implementation model for EV charging infrastructure and recommendations on the best models for charging infrastructure implementation* and *the detailed report on different financing models, associated cost benefit analysis* can be found on section **2.6 Typical Roles in eMobility Ecosystem** in **Chapter 6 Institutional Road Map.** The *detailed report on different financing models, associated cost benefit analysis and grid analysis* can be found in **Section 6.1**. **Financing Models and Cost-Benefit Analysis**, the *Technical note on strategic actions (with their costing) for effective management of EVs charging stations in* Rwanda can be found in **Section 2.5. Management of EVCI** and in **Section** Erreur ! S ource du renvoi introuvable. **EV Charging Infrastructure Costs.**

1.2 Goals of the Electric Vehicle Charging Infrastructure Master Plan

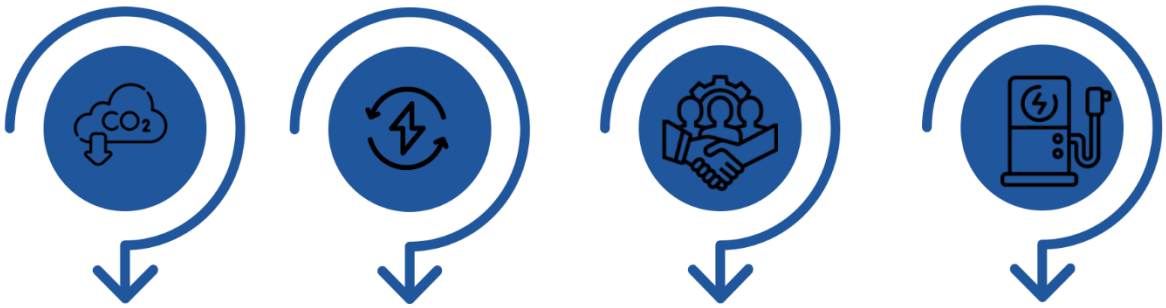
Rwanda aims to become a climate-resilient, carbon-neutral economy by 2050. To achieve the 2030 emissions reduction goal of 38%, Rwanda collaborates with national and international partners to attract sustainable green investments. Establishing an accessible and robust EV charging infrastructure is crucial for achieving these environmental objectives.

The primary objective of the EV infrastructure master plan for Rwanda is to accelerate the widespread adoption of EVs across various vehicle types, commuting segments, and usage patterns over the next decade. The plan envisions a comprehensive roadmap to anticipate EV growth and determine specific charging infrastructure requirements. It aims to optimize charging infrastructure locations, establish viable business models, assess grid impact, and anticipate energy demand. Additionally, the plan outlines roles for both public and private sectors, establishes interoperability standards, and adheres to international best practices. The goal is to incentivize and expedite EV adoption through comprehensive support, including incentives for EVs and their charging infrastructure, coupled with public awareness strategies highlighting the benefits of EV utilization.

The vision includes integrating electric mobility into Rwanda's national transportation framework, considering the country's unique landscape and settlement patterns.

This involves creating a resilient and flexible network of strategically positioned charging stations for various EV types, including two and three wheelers, passenger cars, light-duty vehicles and heavy-duty vehicles. The plan anticipates the adoption trajectory for EVs across different vehicle categories and commuting segments over the next decade.

Implementing the EV Charging Infrastructure Master Plan is crucial for reducing carbon emissions and fostering a cleaner environment. Additionally, it enhances energy efficiency and promotes economic growth through market-driven opportunities. This plan aims to achieve a sustainable mobility system by integrating renewable energy within charging stations, optimizing the power grid, and providing viable, zero-emission transportation alternatives across various modes. By doing so, it aligns with Rwanda's broader sustainability and climate resilience goals, contributing significantly to the country's environmental and social well-being.



Reducing CO2 Emissions

Promoting EVs reduces carbon emissions, aligns with climate resilience goals, and contributes to improved public health by decreasing air pollution from fossil fuel transport.

Energy Efficiency

EVs and their infrastructure significantly enhance energy efficiency and grid optimization, contributing to energy conservation and more reliable power grid management through smart integration. Incorporating renewable energy sources into charging stations further promotes energy efficiency and sustainability.

Economic Growth

The master plan is market-driven, promoting new business models that attract investments, create job opportunities, and drive economic growth while advancing technology in the eMobility sector.

Sustainable Mobility

The master plan underscores the crucial role of charging infrastructure in providing zero-emission transport across various modes. Aligning with sustainability principles, it ensures viable alternatives for citizens, enhancing environmental and social well-being.

1.3 General Trends on eMobility

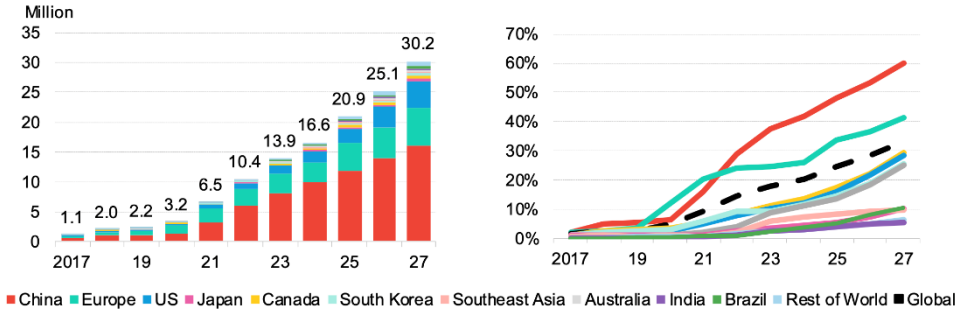
1.3.1 Global developments toward eMobility

EVs are increasingly becoming a key component of global transportation systems, driven by the urgent need to reduce carbon emissions and combat climate change. The adoption of EVs is rapidly growing across various vehicle categories, including passenger cars, two- and three-wheelers, light commercial vehicles, and heavy-duty vehicles. This widespread adoption is supported by advancements in battery technology, decreasing costs, and robust policy frameworks[1].

Passenger Cars

EV sales have been growing robustly as major markets advance and emerging economies adopt more EVs. In 2024, electric car sales are projected to reach around 17 million, making up over 20% of total global car sales. This trend is driven by increasing competition among manufacturers, falling prices for batteries and vehicles, and continued policy support. In the first quarter of 2024, electric car sales rose by 25% compared to the same period in 2023, reflecting consistent growth despite challenges like tight margins, volatile battery metal prices, high inflation, and the phase-out of purchase incentives in some regions[2].

Global passenger EV sales are expected to continue their upward trajectory in the coming years, though at a slower pace than previously observed. According to our Economic Transition Scenario, EV sales are projected to increase from 13.9 million in 2023 to over 30 million by 2027. This represents an average annual growth rate of 21% over the next four years, compared to a remarkable 61% average annual growth from 2020 to 2023. By 2027, the share of EVs in global new passenger vehicle sales is anticipated to rise to 33%, up from 17.8% in 2023. While China and Europe are expected to surpass the global average, some European markets will see even more rapid adoption, with the Nordics at 90% and Germany, the UK, and France all exceeding 40%[3].



Source: BloombergNEF. Note: Europe includes the EU, the UK and EFTA countries. EV includes BEVs and PHEVs.

Figure 1: Global near-term passenger EV sales and share of new passenger vehicle sales by market

[1] International Energy Agency. (2024). Global EV Outlook 2024. In International Energy Agency [Report]. https://iea.blob.core.windows.net/assets/a9e3544b-0b12-4e15-b407-65f5c8ce1b5f/GlobalEVOutlook2024.pdf

[2] International Energy Agency. (2024). Global EV Outlook 2024. In International Energy Agency [Report]. https://iea.blob.core.windows.net/assets/a9e3544b-0b12-4e15-b407-65f5c8ce1b5f/GlobalEVOutlook2024.pdf

[3] Bloomberg NEF. (2024). Electric Vehicle Outlook 2024. https://assets.bbhub.io/professional/sites/24/847354_BNEF_EVO2024_ExecutiveSummary.pdf

Two- and Three-Wheelers

Two- and three-wheelers (2/3Ws) play a critical role in daily passenger and commercial transportation in Latin America and Africa. Electrifying these vehicles presents significant opportunities for decarbonizing mobility and improving urban air quality. In 2023, electric 2/3Ws made up 13% of global sales, with 8% of the total stock being electric. China led the market with over 30% of 2/3W sales being electric, followed by India at 8% and ASEAN countries at 3%. Despite the smaller market sizes, these vehicles are essential for daily transportation needs in these regions.

Light Commercial Vehicles (LCVs)

The global market for electric LCVs continued to expand in 2023, with sales growing by more than 50% and the sales share increasing to just under 5%. Major markets such as China and Europe saw significant increases in sales, with China selling over 240,000 electric LCVs and Europe experiencing a 60% growth in this segment. In Korea, electric LCV penetration is moving faster than for electric passenger cars, with models like the Hyundai Porter and Kia Bongo leading the market.

Heavy-Duty Vehicles

Electric buses, including all medium- and large-sized buses, are leading the heavy-duty vehicle (HDV) segment in sales. In 2023, several European countries and China achieved sales shares above 50% for electric buses. Globally, nearly 50,000 electric buses were sold, representing 3% of total bus sales and bringing the global stock to approximately 635,000. However, the market penetration of electric buses remains low in many emerging markets and larger markets such as the United States and Korea[4].

Future Worldwide Perspectives to achieve a zero-emission fleet by 2050 without additional policy intervention

The three-wheeled vehicle segment is currently the only one poised to achieve a zero-emission fleet by 2050 without requiring additional policy measures. Sales of two- and three-wheeled vehicles continue to climb in emerging economies, with electric models projected to surpass 90% of global sales by 2040. In 2023, over 40% of two-wheelers and more than 80% of three-wheelers sold were electric. This positions three-wheelers as the only vehicle segment on a clear path to attaining a net-zero emission fleet by 2050

[4] Bloomberg NEF. (2024). Electric Vehicle Outlook 2024. https://assets.bbhub.io/professional/sites/24/847354_BNEF_EVO2024_ExecutiveSummary.pdf

Segment	Current share of road transport CO2 emissions	Current estimated global fleet size	Zero-emission vehicle (ZEV) fleet share in 2050 – Economic Transition Scenario	Level of policy intervention needed to hit Net Zero Scenario (100% ZEV share) by 2050
Three-wheeled vehicles	<1%	120 million	95%	On track
Two-wheeled vehicles	5%	1.1 billion	80%	Almost on track: minor additional measures needed
Municipal buses	1%	3.4 million	86%	Almost on track: minor additional measures needed
Light commercial vehicles	11%	170 million	77%	Positive trajectory: moderate additional measures needed
Passenger vehicles	53%	1.3 billion	69%	Positive trajectory: moderate additional measures needed
Medium + heavy commercial vehicles	30%	83 million	28%	Not on track: strong additional measures needed urgently

Source: BloombergNEF, various government sources. Note: Fleet size represents vehicles of all drivetrain types and are estimates based on various sources and BNEF data. Some values rounded. Current emissions and fleet size data are for 2023.

Figure 2: Road transport segment progress toward net zero

Charging Infrastructure

The expansion of public charging infrastructure is crucial to support the growing number of EVs. In 2023, the global number of installed public charging points increased by 40% compared to 2022, with fast chargers growing faster than slower ones. Adequate public charging infrastructure is essential for a mass-market switch to electric transport, enabling longer journeys even though most charging will likely continue to take place at homes and workplaces. To meet the anticipated EV deployment levels, public charging points need to increase sixfold by 2035.

In addition to charging infrastructure, policy support is significantly boosting industry investment, enhancing confidence in continued rapid electrification. Policies like the US Inflation Reduction Act, the EU Net Zero Industry Act, and China's 14th Five-Year Plan are fostering investments across the EV supply chain. These industrial incentives encourage value addition and job creation in the EV sector, contributing to the overall growth of e-mobility.

Home charging remains the most prevalent method for electric car owners, especially those with access to private parking spaces where they can charge overnight. This method not only offers convenience but also capitalizes on lower electricity rates during off-peak hours. However, the availability of home charging varies significantly across regions, influenced by factors such as urban density, housing types, and income levels. In densely populated cities, where many residents live in multi-unit buildings, access to home charging is more restricted, increasing reliance on public charging infrastructure.

For two-wheelers (2Ws), the charging landscape differs notably. With rising sales and stock in countries like India and ASEAN nations, there is a growing adoption of battery swapping technologies, particularly in India, to address the limitations of traditional charging setups.

Heavy-duty vehicles (HDVs) face unique challenges in charging. Although they can use the same points as light-duty vehicles (LDVs), their larger batteries and longer charging times can interfere with regular operations. This necessitates dedicated HDV charging equipment and facilities, which are still in the nascent stages of widespread development. Progress is being made globally towards establishing standards for megawatt-scale chargers to ensure maximum interoperability for electric HDVs.

1.3.2 Developments Toward eMobility in Africa

Emerging Market and Developing Economies EMDEs

Emerging market and developing economies (EMDEs) outside China currently represent only a small portion of the global electric car market. In these regions, road transport tends to be dominated by smaller mobility solutions within urban centers, such as electric two- and three-wheelers, which have achieved more significant electrification success. Shared mobility solutions are also more prevalent for regional commuting. Car purchasing behavior in EMDEs is distinct, characterized by lower rates of personal car ownership and a higher tendency to buy used vehicles. Looking ahead, while it is anticipated that both new and used electric car sales will rise in EMDEs, many countries are likely to continue to primarily depend on two- and three-wheelers for their transportation needs[5].

The lack of small and affordable electric car models significantly hinders broader market adoption. Most available car models are SUVs or large vehicles targeting high-end consumers, making them too expensive for the mass market, where personal car ownership is less common. Consequently, electrification efforts in many EMDEs focus on two- and three-wheelers (2/3Ws) and public transport. For example, Colombia prioritizes electrifying mass transportation systems, cargo vehicles, and taxis. Similarly, countries like Cambodia, Morocco, and the Dominican Republic have set targets for electric 2/3Ws.

Africa

The African Electric Vehicle Market, including all modes of transport, is projected to grow significantly, reaching $21.39 billion by 2027, up from $11.94 billion in 2021. The market for motorcycles in Africa is also expected to expand, growing to $5.07 billion by 2027 from $3.65 billion in 2021. Several trading partners in Africa have set 2035 as the deadline for banning the sales of internal combustion engine vehicles[6].

There is a significant market for motorcycles, particularly in West, East, and North Africa. Electric two-wheelers are poised to become a dominant force in sub-Saharan Africa's sustainable mobility transformation. Motorcycle taxis, commonly known as Boda Boda in regions such as Nigeria, Uganda, Kenya, Rwanda and other Eastern African countries, play a crucial role in boosting two-wheeler sales. These vehicles are preferred for their speed and maneuverability, especially in sub-Saharan Africa, where road conditions present significant opportunities for improvement. Additionally, motorcycles provide stable income opportunities.

To meet this increasing demand, numerous e-mobility startups have emerged across the continent. Notable examples include Rwanda’s Ampersand Electric Motorcycles, Shift EV in Egypt, BasiGo and Opibus in Kenya, eWAKA Mobility Limited, Zembo in Uganda, Mobility for Africa in Zimbabwe, and South Africa’s Forever powered and Magutsa Energy. According to Mordor Intelligence, the African electric car market is predominantly led by a few major companies, including Volkswagen AG, BMW AG, and Stellantis N.V. Additionally, firms as BYD are also entering the market in countries such as Rwanda, Kenya and South Africa[7]. The report highlights that as the demand for electric vehicles increases in the region, these companies are likely

[5] https://iea.blob.core.windows.net/assets/dacf14d2-eabc-498a-8263-9f97fd5dc327/GEVO2023.pdf
[6] https://theexchange.africa/investing/decarbonizing-transport-exploring-e-mobility-in-africa/
[7] https://africafinancetoday.beehiiv.com/p/chinese-byd-expands-to-africa

to form alliances, joint ventures, and acquisitions with significant industry competitors.

The African Development Bank (AfDB) has been actively promoting the adoption of e-mobility across the continent. Earlier this year, the bank's Sustainable Energy Fund for Africa (SEFA) announced a $1 million technical assistance grant to the Green Mobility Facility for Africa (GMFA). This fund aims to provide technical assistance and investment capital to accelerate and expand private sector investments in sustainable transport solutions in seven countries: Kenya, Morocco, Nigeria, Rwanda, Senegal, Sierra Leone, and South Africa[8].

References of Current Numbers and Growth

The EV market in Africa is growing, supported by increasing investments and supportive government policies. Various African nations are adopting e-mobility to combat climate change, reduce air pollution, and enhance energy security. This burgeoning sector encompasses a range of electric vehicles, from two- and three-wheelers to passenger cars and heavy-duty vehicles. However, the growth is uneven across the continent, with some regions advancing more rapidly than others, and significant challenges remain in scaling up infrastructure and overcoming market barriers.

Kenya: The electric vehicle market in Kenya is growing rapidly, with a significant increase in electric bicycles, motorbikes, tuk-tuks, passenger cars, buses, and specialized equipment. Newly registered EVs jumped from 65 units in 2018 to 4,047 units in 2023, making up over 2.4% of all new vehicle sales. [9]

Tanzania: With approximately 5,000 EVs, Tanzania is one of the leader in East Africa for EV adoption. The country has 11 e-mobility businesses.[10]

Malawi: The e-mobility sector is emerging, with companies like SGV Electric Mobility and Sky Energy Africa driving growth. SGV has converted or assembled around 150 electric motorcycles and installed three charging stations, focusing on B2B sales.[11]

Egypt: Egypt is advancing in transport electrification with pilot projects and initiatives to convert over 100,000 internal combustion vehicles to electric within five years, as approved by the Supreme Council for Vehicle Construction in June 2023.

1.3.3 Developments toward eMobility in Rwanda

Rwanda is one of the most densely populated countries in Africa, with a population density of 414 inhabitants per square kilometer. The population is projected to increase to 16.9 million by 2032. In Kigali, non-motorized trips account for around 52% of all trips, a rate even higher in secondary cities[12].

Over the past decade, Rwanda has made significant strides in economic development, with the transport sector playing a key role. The country is committed to creating an integrated, efficient, and environmentally friendly transport network, which is crucial for supporting economic growth and improving living standards. Key

[8] https://furtherafrica.com/2023/08/28/decarbonizing-transport-exploring-e-mobility-in-africa/
[9] Electric Mobility Association of Kenya. (2024, April). Electrifying Kenya's Transportation Sector. https://e-mobilitykenya.org/downloads/
[10] Africa E-Mobility Alliance (AfEMA). (2023). Barriers to E-Mobility in Tanzania. In *Barriers to E-Mobility in Tanzania*. https://www.africaema.org/resources/AfEMA_country_report_2023_Tanzania.pdf
[11] Africa eMobility Alliance. (2024). Malawi E-Mobility Brief. In *AfEma* (pp. 1–3) [Report]. https://africaema.org/resources/AfEMA_technical_brief_2024_Malawi.pdf
[12] https://sustmob.org/STR/NMTStrategy_Rwanda_200402.pdf

policies such as the National Transport Policy (NTP), the National Strategy for Transformation (NST 1), and the Sustainable Development Goals (SDGs) emphasize improving connectivity and ensuring that transport assets are accessible and beneficial to the population and businesses.

Rwanda's commitment to promoting e-mobility is reflected in several initiatives. The National Environment and Climate Change Policy (2019) aims to reduce air pollution from urban transport, while Rwanda's Intended Nationally Determined Contribution (INDC) under the Paris Climate Accord emphasizes transitioning to e-mobility to mitigate climate change impacts and enhance energy security. The National Transport Policy and Strategy for Rwanda (2021) outlines specific directions to support e-mobility, including developing necessary infrastructure, providing investment incentives, and integrating renewable energy sources into the charging network.

To achieve carbon neutrality and climate resilience by 2050, Rwanda prioritizes the transition to e-mobility for both public and private vehicles. Road transport contributes 13% of the country's emissions, with buses accounting for 40% of transport sector emissions. Rwanda aims to have 20% of all buses transitioned to electric by 2030, which would avoid 72,000 tonnes of carbon dioxide equivalents[13].

The National Sustainable Mobility Policy facilitates Rwanda's transition to e-mobility, promoting low-carbon economic growth through new transport innovations and encouraging public, electric, and non-motorized transport, such as walking and cycling. The strategic paper on electric mobility adaptation in Rwanda (2021) details the current e-mobility initiatives, assesses the main challenges, and introduces various incentives to promote e-mobility.

This transition is integral to Rwanda's national transport policy and strategy, aiming to reduce dependence on imported oil and promote the use of domestically generated electricity. The strategic paper on electric mobility adaptation outlines ongoing projects like "Exploring Enabling Energy Frameworks for Battery Electric Buses in Rwanda" by the World Bank and the "Kigali E-bus Charging Infrastructure Master Plan." These initiatives collectively support the goal of achieving a comprehensive e-mobility ecosystem within the country.

Significant advancements in Rwanda's energy sector over the past decade have supported the adoption of electric mobility. There is electricity access for 74.5% of the population by 2022 and it is expected to reach 100% by 2024The nation's power generation capacity has tripled from 76 MW in 2010 to 276 MW by mid-2022 and it is expected to reach 556 MW by 2024. The reliance on oil-fueled power has decreased, being replaced by hydropower (53%), lake methane-based power (20%), solar (6%), and peat-fueled power (17%).

Rwanda's National Strategy for Transformation (NST1) aims for universal access to electricity by 2024 through a mix of on-grid and off-grid solutions, reducing electricity supply costs, enhancing service quality and reliability, and promoting clean cooking alternatives. The Least Cost Power Development Plan (LCPDP) projects an increase in installed capacity to over 600 MW by 2030.

[13]https://www.environment.gov.rw/index.php?eID=dumpFile&t=f&f=55460&token=6003242e29667513f33c128466ffc760c62d81d8#:~:text=TAX%20BREAKS%20Electric%20vehicles%2C%20spare,on%20a%20rent%2Dfree%20basis

Tax breaks for EV buyers, support for local electric vehicle manufacturers, and plans to electrify public buses are part of Rwanda's efforts to reduce air pollution and greenhouse gas emissions. The government estimates the cost of transitioning to e-mobility at $900 million, with potential annual savings of $22 million from electric motorcycles alone. By 2030, Rwanda aims to electrify 20% of buses, 30% of motorcycles, and 8% of cars, though challenges such as high vehicle costs, limited charging infrastructure, and a skills gap remain[14].

[14] https://www.unep.org/news-and-stories/story/face-rising-air-pollution-rwanda-turns-electric-vehicles

2 EMOBILITY PRINCIPLES

This chapter introduces the fundamental principles of electric mobility, outlining key concepts vital for developing an EV charging ecosystem. It begins with an overview of vehicle categories and types, followed by a detailed description of charging systems, standards, and connectors. The chapter then explores deployment strategies for EVCI, delineates typical roles within the ecosystem, and highlights essential functionalities such as interoperability. It concludes with insights into circularity and waste disposal practices relevant to EVCI.

2.1 Electric Vehicles categorization and types

This master plan includes within its scope the categorization of vehicles, which encompasses two- and three-wheelers, passenger cars, light-duty vehicles, and heavy-duty vehicles, each described with their main characteristics as follows:



Two and Three wheelers (E2&3W)

Two-wheelers are lightweight vehicles primarily designed for individual or small passenger transport. Commonly used for personal mobility, they offer an efficient and agile solution for navigating urban environments, providing quick and convenient transportation over short distances.

Three-Wheelers: Three-wheelers, which are versatile in function, can be used for both passenger transport and goods delivery. These vehicles are commonly employed in urban shared mobility services and in rural areas for delivering goods.

Electric Bikes: Electric bikes are utilized for both private use and shared services. They offer an eco-friendly alternative for short-distance travel.



Passenger Cars

Passenger cars are vehicles primarily designed and constructed to transport people rather than goods. These vehicles come in various types, catering to different needs and uses:

Private Passenger Cars: Used for personal, family, and business transportation needs. These cars provide comfort and convenience for individual or family travel.

Commercial Passenger Cars: Utilized to transport other people, such as in taxis or transport services offered by companies. These vehicles are often part of a fleet and are designed to carry multiple passengers for hire.



Light-Duty Vehicles (LDVs)

LDVs, also referred to as **Light Commercial Vehicles** (LCVs), are primarily used for the transport of goods, while minibuses are focused on passenger transport. Both types of vehicles have a Gross Vehicle Weight (GVW) of up to 3.5 tons. Examples of LDVs include vans, small trucks, mini busses and pickup trucks. These vehicles are commonly utilized for local deliveries, small-scale logistics, service operations, and utility work. LDVs are ideal for last-mile delivery, urban goods transport, and small parcel delivery due to their maneuverability and efficiency in urban environments. Typically, the routes for these vehicles are predictable, allowing for efficient scheduling and operation



Heavy-Duty Vehicles (HDVs)

HDVs are designed to carry heavy loads or a large number of passengers. These vehicles are essential for both freight transport and public transportation systems, characterized by larger engines, robust structures, and the ability to withstand demanding use conditions. Some modalities are:

Buses: Large passenger vehicles designed to carry multiple passengers, often used for public transport. Typical uses include urban and intercity transport, school transport, and private hire for tours and events.

Medium-Duty Trucks: Vehicles with a Gross Vehicle Weight (GVW) of 3.5 – 15 tons. These trucks are versatile and serve as delivery trucks, utility vehicles, and for various other commercial purposes. Typical uses include local deliveries, utility services, municipal operations, last-mile delivery, urban goods delivery, regional goods distribution, refrigerated goods delivery, waste collection, and construction.

Heavy-Duty Trucks: Designed for the heaviest loads, typically having a GVW of 15 tons or more. These include long-haul trucks, tractor-trailers, and specialized vehicles for construction and heavy industries. Typical uses include long-haul freight transport, heavy construction, industrial logistics, and specialized transport such as large machinery.

Based on their level of electrification, EVs can be further classified as either battery electric vehicles (BEVs) or plug-in hybrid electric vehicles (PHEVs) or Hybrid Electric Vehicles (HEVs).

Battery Electric Vehicles (BEVs)

BEVs are vehicles powered solely by electricity. They do not rely on traditional internal combustion engines or liquid fuel. Instead, they utilize large battery packs to store and provide energy for propulsion. This technology enables BEVs to achieve significant driving ranges on a single charge, typically ranging from 160 to 500 kilometers.

Plug-In Hybrid Electric Vehicles (PHEVs)

PHEVs combine a conventional internal combustion engine with an electric motor that can be charged via an external electric source. By storing electricity from the grid, PHEVs can significantly reduce fuel consumption during regular driving. As it needs to contain two means of propulsion, it is generally considered a less efficient and more expensive approach but can be useful as part of the transition towards full electrification.

Hybrid Electric Vehicles (HEVs)

HEVs utilize both a conventional internal combustion engine and an electric motor for propulsion. Unlike PHEVs, HEVs cannot be directly plugged into the grid; instead, the battery that powers the electric motor is charged through the vehicle's combustion engine. In some modern models, batteries can also be charged through regenerative braking, converting kinetic energy into electrical energy. This is considered as an additional efficiency measure for combustion engines, but as it wholly relies on the combustion engine, is not considered to be a future solution for transport electrification.

The following examples illustrate the diverse battery capacities and ranges across different types of electric vehicles that are in the current market worldwide, reflecting their specific usage requirements and design characteristics. It is important to note that technology is rapidly evolving, with the market continually presenting new models with greater capacities.

Case Study Box No. 1 EV batteries and capacities

Two-Wheelers

Two-wheelers, such as electric scooters and motorcycles, typically have battery capacities ranging from 2 kWh to 5 kWh. For example, the Revolt RV 300, equipped with a 2.7 kWh battery, offers a range of around 180 km. The Simple One electric scooter, with a 5 kWh battery, provides an impressive range of up to 225 km[15][16].

Passenger Cars

Electric passenger cars usually have larger battery capacities, offering longer ranges. The Nissan Leaf, a well-known BEV, features a 62 kWh battery that allows it to travel up to 360 kilometers on a single charge. The Mitsubishi Outlander PHEV, a plug-in hybrid, comes with a 12 kWh battery, enabling it to cover approximately 50 km on electric power alone. For hybrid electric vehicles, the Toyota Prius hybrid model has a 1.3 kWh battery, providing an all-electric range of up to 25 km.[17]

Light-Duty Vehicles

[15] https://evreporter.com/rv400-and-rv300/
[16] https://www.bikedekho.com/simple-energy/simple-one#leadForm
[17] Sanguesa, J. A., Torres-Sanz, V., Garrido, P., Martínez, F. J., & Marquez-Barja, J. M. (2021).. *Smart Cities*, *4*(1), 372–404. https://doi.org/10.3390/smartcities4010022

They typically have battery capacities ranging from 30 kWh to 80 kWh. A typical example is the Ford Transit Electric, which offers a 67 kWh battery providing a range of up to 350 kilometers[18].

Heavy-Duty Vehicles

Heavy-duty vehicles, such as electric buses and trucks, require significantly larger battery capacities due to their size and weight. These vehicles often have battery capacities exceeding 200 kWh. For instance, electric buses like the Proterra ZX5 feature a 660 kWh battery, offering a range of up to 530 kilometers on a single charge[19].

2.2 Charging Systems

There are two primary systems regarding charging systems to EVs: conductive and inductive. Conductive charging, which involves the use of cables, is the established and commonly used approach compared to inductive charging, which is still in the research phase and has yet to achieve widespread adoption in electric transportation. Additionally, another emerging methodology is the Battery Swap Station (BSS), where users exchange depleted batteries for charged ones.

2.2.1 Conductive charging

Relies on direct contact with the vehicle to transfer power and is known for its simplicity and efficiency. It involves direct contact between the vehicle and power source (through physical cables), with variations like onboard and off-board charging. Conductive charging stands out for its economic viability, fast charging capabilities, straightforward operation, and high efficiency.

Inductive charging

Operates without physical contact, using electromagnetic fields to transfer power. While it offers advantages like enhanced electrical safety, it suffers from drawbacks such as power loss and lower efficiency. Inductive charging enables automatic charging through methods like static, dynamic, and quasi-dynamic charging, each offering unique benefits and applications, depending on whether the charging occurs while the EV is in motion or stationary.

Battery swap station (BSS)

Batteries are exchanged for uncharged ones, and the user receives a charged battery in return. Users rent batteries from the BSS owner on a monthly basis. This process, while slowing down charging, extends battery lifespan. BSS systems can integrate locally generated Renewable Energy (RE) like solar and wind power. Users can quickly swap batteries without leaving their vehicles. Moreover, this system supports vehicle-to-grid (V2G) initiatives. However, charging at BSS stations tends to be more expensive due to leasing fees. Additionally, multiple batteries and ample storage space are required, potentially raising costs in high-traffic areas. Compatibility issues may arise if the station uses a specific battery model while vehicles adhere to different standards.

[18] https://media.ford.com/content/dam/fordmedia/Europe/en/2020/11/E-Transit/E-TRANSIT_FACTSHEET.pdf

[19] https://www.sustainable-bus.com/electric-bus/proterra-zx5-electric-bus-new-transit/



Involves direct contact between the vehicle and power source (through physical cables)

Operates without physical contact, using electromagnetic fields to transfer power

Batteries are exchanged for uncharged ones, and the user receives a charged battery in return.

2.2.2 On-grid and Off grid charging solutions

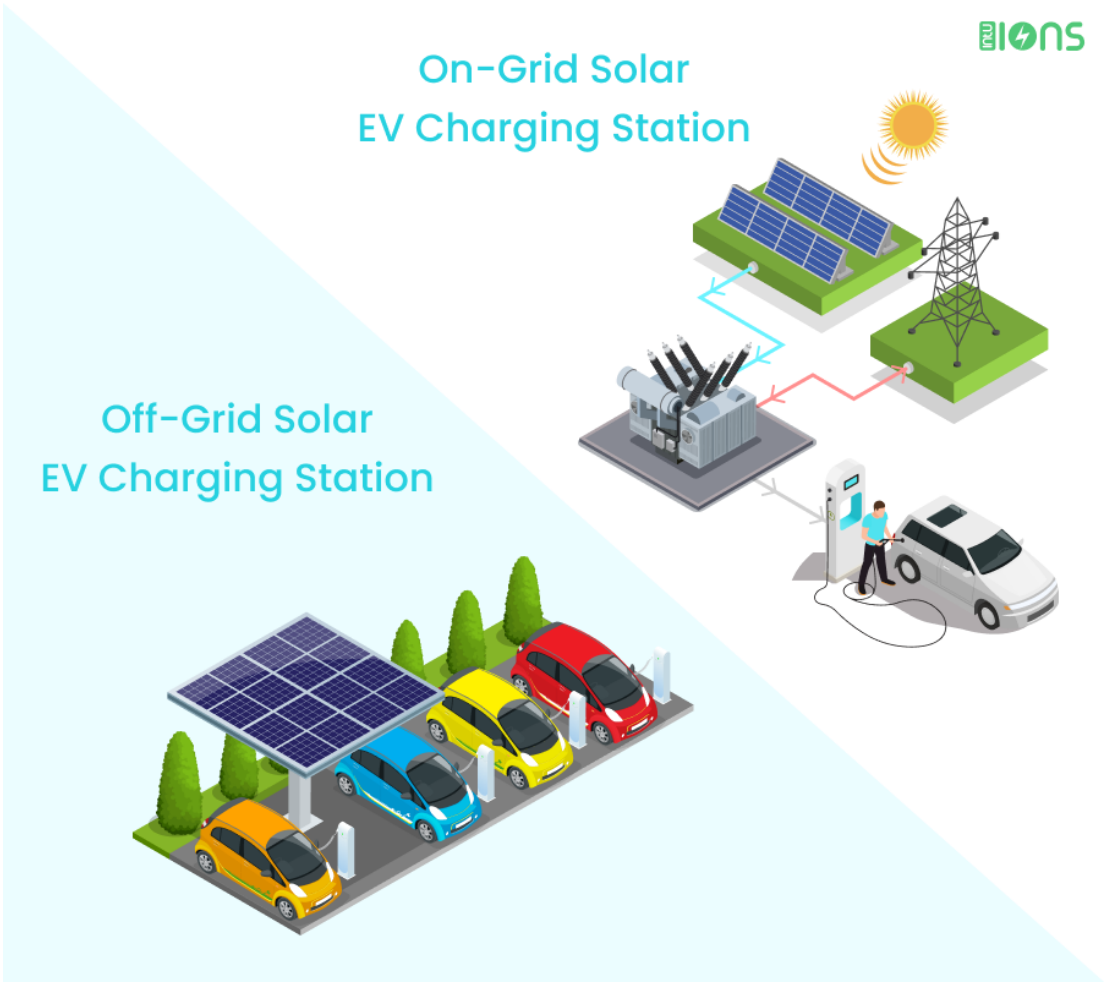


Figure 3: On-grid and off-grid solutions overview[20]

2.2.2.1 On-Grid Charging Solutions

On-grid charging solutions are connected to the main electrical grid, providing a stable and consistent source of electricity for EVs. These include regular charging stations, which are physical objects with one or more charging points and a common user identification interface. Charging stations may have features like RFID readers, buttons, displays, or LEDs. Some are 'Plug & Charge' stations that automatically identify the vehicle.

[20] https://intuions.com/blog/renewable-energy-ev-charging-station/

Charging plazas, on the other hand, consist of more than two charging stations that share a single grid connection rather than being individually connected. The choice between a charging plaza and a charging station depends on several factors such as: i) Spatial planning; ii) Traffic flow management; iii) Scalability; iv) User service and charging point reliability; v) Promotion of EV usage; vi) Financial considerations; vii) Integration with mobility hubs; vii) Charging process management[21].

Advantages:

- **Reliability:** On-grid systems offer a dependable power supply, ensuring that EVs can be charged consistently without interruptions.
- **Scalability:** These systems can be expanded easily to accommodate more charging stations as the number of EVs increases.
- **Integration with Renewable Energy:** On-grid charging can integrate with renewable energy sources such as solar and wind, enhancing sustainability.
- **Lower Initial Cost:** Installation and setup costs are typically lower than off-grid solutions because they utilize existing infrastructure.

Disadvantages:

- **Dependency on Grid Stability:** The efficiency of on-grid solutions is dependent on the stability of the local power grid. Power outages or grid failures can disrupt charging availability.
- **Grid Strain:** Increased demand for EV charging can strain the electrical grid, especially during peak hours, necessitating upgrades to the grid infrastructure.
- **Limited Remote Accessibility:** On-grid charging solutions are less feasible in remote areas without access to the main power grid.

2.2.2.2 Off-Grid Charging Solutions

Off-grid charging solutions operate independently of the main electrical grid, often using renewable energy sources combined with energy storage systems.

Advantages:

- **Independence from Grid:** Off-grid systems are not affected by grid instability or outages, providing a reliable charging option in remote or underserved areas.
- **Sustainability:** These systems often use renewable energy sources, such as solar panels, reducing the carbon footprint and reliance on fossil fuels.
- **Flexibility:** Off-grid solutions can be deployed in a variety of locations, including areas without existing grid infrastructure.

Disadvantages:

- **Higher Initial Costs:** The installation of off-grid systems, including renewable energy sources and energy storage, can be more expensive initially.
- **Energy Storage Limitations:** The efficiency and reliability of off-grid charging depend on the capacity and performance of energy storage systems, which can be limiting.

[21] The Netherlands Knowledge Platform for Charging Infrastructure. (2019). Guidelines for the realisation of charging plazas. In The Netherlands Knowledge Platform for Charging Infrastructure. https://nklnederland.nl/wp-content/uploads/2021/12/Guidelines-for-the-realization-of-charging-plazas.pdf

- **Maintenance:** Off-grid systems may require more maintenance and technical expertise to manage the energy production and storage components.

2.2.3 AC and DC charging

The hardware used to supply electrical energy to charge EV batteries is shown in

Figure 4: A Typical Electric Vehicle Charging Infrastructure. The EVCI receives Alternating Current (AC) from the grid. However, EVs charge their batteries with Direct Current (DC). Thus, two charging options are commercially available: AC or DC charging concepts. Firstly, for the AC charging concept, the EV receives AC power from the grid to be converted to DC power through an onboard charger to charge the EV battery. The onboard charger defines and limits the charging rate and time. AC charging rate can be either slow or fast based on the type of charging technology. Secondly, for the DC charging concept, the AC grid power is converted to DC power through a grid-connected charger, outside the vehicle. The EV receives converted DC power that bypasses the onboard charger directly to the EV battery.

Compared to AC chargers, DC chargers are much faster, however, also more complex in design and costly. For DC chargers, to deliver the high power required for fast charging, this technology requires higher capacity electrical systems and grid connections.

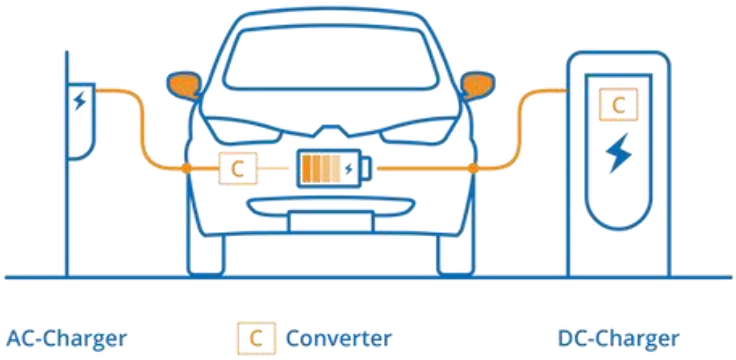


Figure 4: A Typical Electric Vehicle Charging Infrastructure[22]

2.2.4 Charging Levels

Charging EVs involves several levels of charging, each offering different power outputs and charging times:

Level of charging	*Description*
AC Level 1	These are standard electrical outlets providing a range from 1.9 kW to 7.40 Kw of power for slow charging. This might take from 8 to 12 hours to fully charge the EV.
AC Level 2	Standard outlets delivering a range from 19.2 kW to 44 Kw for accelerated charging. These charging stations are designed for private and public facilities, and it takes between 4 and 6 hours to fully charge the EV battery.

[22] https://go-e.com/en/magazine/ac-dc-charging

Level of charging	Description
DC Level 3	External chargers offering a range from 44 kW to 350 Kw of power, typically found in fast-charging stations. These stations significantly reduce charging time compared to AC charging. DC fast-charging station takes approximately 30 min to charge the battery in the vehicles.
DC Level 4	External chargers providing a range from 100 kW to more than 350 Kw of power, also found in fast-charging stations. They further accelerate charging times compared to lower levels.

Table 1: Charging Levels

Charging stations for EVs vary in size, cost, and charging times, with higher power levels corresponding to faster charging rates. Level 1 charging involves plugging the car to recharge using 120 to 230 volts and a power between 1.90 to 7.40 kW depending on, while Level 2 charging, the most common in EV stations, uses 240 volts and can recharge a vehicle in about 5 hours. Level 3, also known as DC Fast Charging, utilizes 480 volts, offering the quickest charging time of just 30 minutes.

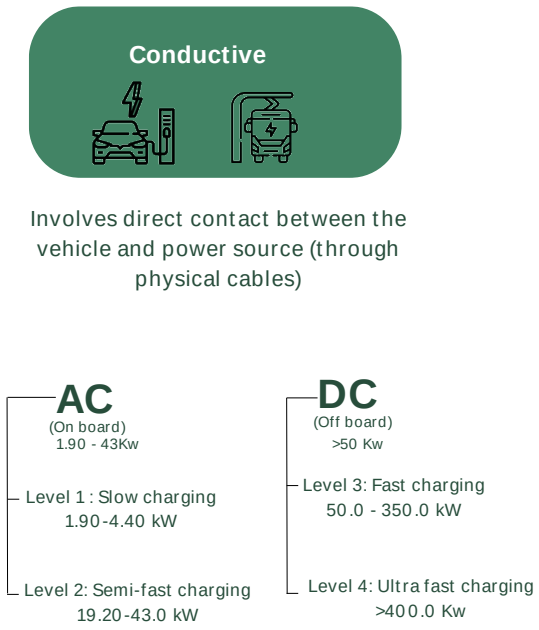


Figure 5: AC and DC charging levels

The following table outlines a typical scenario, including typical power outputs and durations for each charger type. However, the precise charging time will vary depending on the specific type of EV and its battery capacity.

Charging Type	Speed	Type of Location	Duration (hours)	Typical Power
Level 1 AC	Low	Households	4 to 36 hours	1.90 – 7.40 kW
Level 2 AC	Rapid	Households, workplaces, parking lots, retail stores, supermarkets, hotels, overnights charging.	2 to 13 hours	7,40 - 44 kW

Charging Type	*Speed*	*Type of Location*	*Duration (hours)*	*Typical Power*
Level 3 DC	Fast	Designated parking lots, highway stop areas	15 min to 2 hours	44 - 350 kW
Level 4 DC	Ultra Fast	Designated parking lots, highway stop areas	5 min to 2 hours	> 350Kw

Table 2: EV Charging Types

Note: The time to charge will depend on the exact power output of the station as well as to the EV battery size and capacity.

EV battery size vs Power Output to charge

The charging duration for an EV is contingent not only upon the power supplied by the charging station but also on the capacity of the EV's battery. Generally, smaller and lighter EVs are equipped with smaller batteries, while heavier-duty vehicles, such as trucks, boast larger ones. Consequently, the charging time for an EV varies depending on the power output of the charging station and the size of the battery being replenished. Essentially, it is similar to refueling a vehicle - the smaller the tank, the faster it fills. For example, a two-wheeler will recharge significantly quicker than a larger EV such as a passenger vehicle or a heavy-duty (for goods or passenger) vehicle using the same power output (kW/h) on a charging station.

As a reference, the next table shows the time to charge depending on factors such as the battery capacity, and the power output used in a charging station. The table outlines a typical scenario, including typical power outputs and durations for each charger type. However, the precise charging time will vary depending on the specific type of EV and its battery capacity.

	Battery Size kWh			
Power Output	25kWh	50 kWh	75 kWh	100 kWh
2.3 kW AC	10 hr 30 min	24 hr 30 min	32 hr 45 min	43 hr 30 min
7.4 kW AC	3 hr 45 min	7 ht 45 min	10 hr	13 hr 30 min
11 kW AC	2 hr	5hr 15 min	6 hr 35 min	9hr
22 kW AC	1 hr	3 hr	4 hr 30 min	6 hr
50 kW DC	36 min	53 min	1hr 20 min	1 hr 48 min
120 kW DC	11 min	22 min	33 min	44 min
240 kW DC	6 min	12 min	17 min	22 min
350 kW DC	4 min	8 min	12 min	17 min

Table 3: Time to Charge: Power output and Battery Size23

The State of Charge (SoC) of an EV battery is closely related to the charging power, as it influences the rate at which the battery can be charged. The values in this table do consider this relation. Typically, when the SoC is low, the battery can accept a higher charging power, allowing for faster charging rates. As the SoC increases, the charging power usually decreases to prevent overheating and to ensure the longevity of the battery. This behavior is managed by the battery management

[23] The results and methodology involve dividing the Battery Size by the Power Output to estimate the time required for charging.

system (BMS), which optimizes the charging process by reducing the power input as the battery approaches full charge. Consequently, the charging speed is highest at lower SoC levels and gradually slows down as the battery gets closer to 100% SoC.

The following figure illustrates the difference between AC and DC charging curves for EVs. The figure shows **AC Charging** is steady and slower, suitable for overnight charging or long-duration parking, while **DC Charging** is fast and efficient for quick recharges, especially when time is a constraint, but reduces power output after 80% SoC to protect the battery.

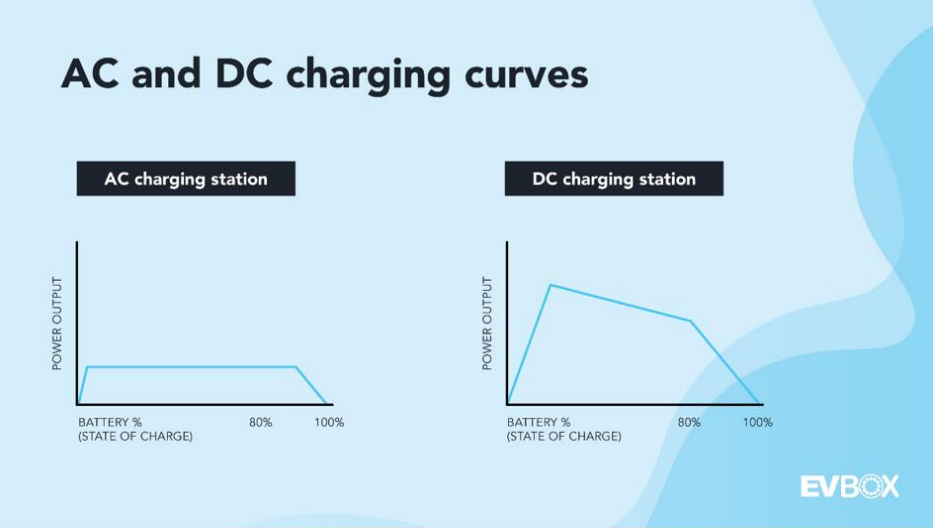


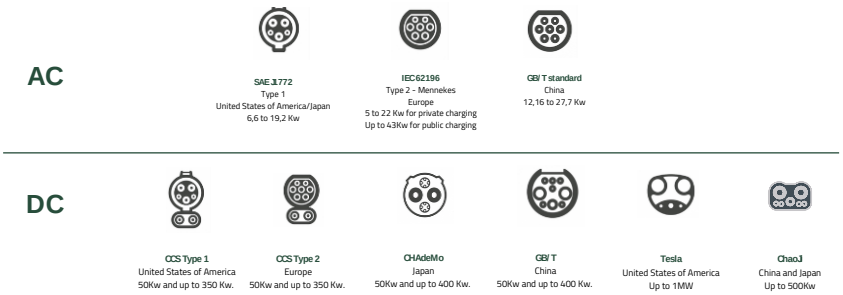
Figure 6: AC and DC charging curves[24]

2.3 Type of Connectors

The connector is the physical interface between the charging station and EVs. It allows the transfer of electrical power from the charging infrastructure to the vehicle's battery system. Connectors come in various types and standards, depending on factors such as geographical location, charging speed, and vehicle compatibility.

Achieving uniformity and standardization in EV connectors is an ongoing process due to variations in plug types, communication protocols, and voltage and current capacities across different regions. The connectors listed below are the primary standards for charging EVs, primarily used for four-wheeled cars. In certain instances, these connectors are also employed for charging heavy-duty vehicles or two- and three-wheelers, depending on the manufacturer and the region where the vehicle is sold.

[24] https://blog.evbox.com/difference-between-ac-and-dc



For a deep explanation, the Annex No. 1 Table with Type of Connectors shows the explanation and background of each of the connectors.

2.3.1 Connectors for two and three-wheelers:

Some connectors are compatible with two-wheelers. For instance, the Type 2 connector, the CHAdeMO connector, and the Type 2 connector (Mennekes) are implemented by some two- and three-wheelers. However, compatibility depends on the specific motorcycle and the region where it is commercialized.

In addition to the mentioned connectors, and regarding regional developments around the world the following are also used for two-wheelers

- *Type-6 Connector (TES-0D-01-01):* Commonly used in India and Taiwan for electric two-wheelers such as the Ola S1 Pro, Tork Kratos R, and Ultraviolette F77.
- *Bharat EV Charger Standards (DC-001):* Specifically developed for India's electric two- and three-wheelers.
- *GB/T 20234 Standard:* Widely adopted in China for electric two- and three-wheelers.
- *Regular home power outlets:* Additionally, electric motorcycles can also be connected and charged using regular power outlets available in homes, as some of them come equipped with appropriate connectors.

Connectors for Heavy Duty vehicles:

Heavy-duty vehicles (HDVs) require significantly more power than standard electric vehicles, necessitating the development of new connectors to meet these high-capacity needs. Type 1 and Type 2 connectors were primarily designed for light-duty vehicles and have limited applications for medium-duty vehicles. In contrast, the GB/T connector, widely used in China, supports heavy-duty vehicles and is designed to handle the higher power requirements of buses and trucks. The CHAdeMO connector, particularly its latest version known as ChaoJi, is also designed for both light and heavy-duty vehicles, offering higher power levels suitable for these applications. Additionally, the Megawatt Charging System (MCS) has emerged specifically to address the ultra-fast charging needs of heavy-duty vehicles, capable of delivering up to 3.75 megawatts.

2.3.2 Megawatt Charging

The Megawatt Charging System has been developed as a new standard designed specifically for high-power applications. The MCS is emerging as the preferred solution for heavy-duty vehicles, in-route charging, and other electric heavy modes of transportation.

The MCS initiative, coordinated by CharIN, provides a charging capacity of up to 4.5 MW for heavy-duty trucks. This system supports a voltage range between 500-1250V and has been tested with currents up to 3000A. This development is a crucial step towards the future of heavy-duty charging and other high-capacity applications, such as marine and aviation electric transportation.

CharIN's MCS concept extends beyond just a charging system and connector. It includes recommendations on various aspects like location considerations (e.g., 'drive-through charging') and software standards (such as OCPP and ISO 15118). These comprehensive guidelines ensure the efficient and effective implementation of the MCS for various heavy-duty and high-capacity electric transportation needs.

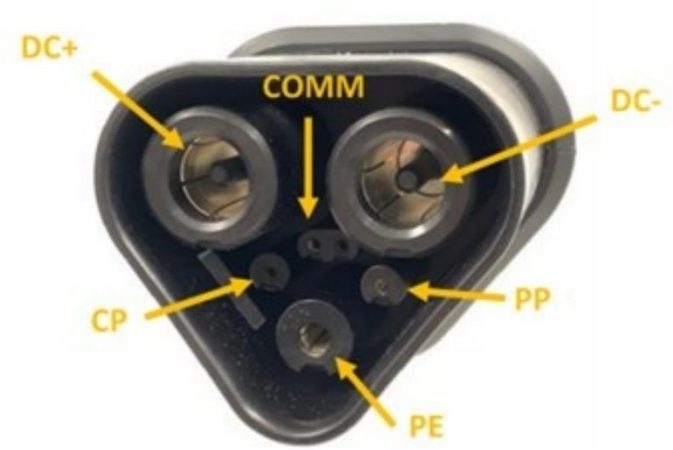


Figure 7: Megawatt Charging Plug Layout[25]

2.4 Distinguishing Charging Solutions

2.4.1 Public, semi-public, and private charging

When considering the function of charging stations, a distinction is commonly made between public and private charging. Public charging involves providing non-discriminatory access to charging, often with a corresponding business model, while private charging encompasses charging for personal use, fleet usage, or for known (guest) users. It's crucial to factor in the legal context, as charging stations can be situated on either public or private property, regardless of their function.

Based on these distinctions, the following categories emerge:

Public Charging:

Charging available for public use in the public domain, largely influenced by policy decisions of local and regional governments. This type of charging is subject to strict rules and procedures during planning and deployment as they are placed in the public domain, with unrestricted access for all EV drivers at any time.

Semi-Public Charging:

Charging intended for public use on private property, encompassing privately owned charging stations accessible to the public. Access may be restricted by factors such as location (e.g., within a gated parking garage), time (e.g., specific hours of

[25] Source: CharIN e. V. MCS Subgroup, 2022

operation), or user groups (e.g., different charging tariffs for customers and non-customers). The aim is to provide charging services to a broader audience within certain constraints.

Private Charging:

Charging designated for private use on private property, including home charging, fleet charging, and workplace charging. This category may also involve guest usage for known customers and may require authentication or charging fees. Unlike public or semi-public charging, there is no intention of providing charging services to the general public, and these stations are typically not displayed on public maps or navigation services.

2.4.1.1 Location-based categorization

Charging infrastructure can be categorized based on the location where the charging takes place. This classification encompasses a mix of public, private, and semi-private charging modalities: charging, depot charging, enroute charging and battery swapping. For every location, typical charging behavior can be expected, resulting in a specific choice for amongst others the charging power. Besides, an eventual charging implementation strategy contains the placement of a mix of charging solutions on various locations that result in an optimal charging experience and convenience for the EV users.

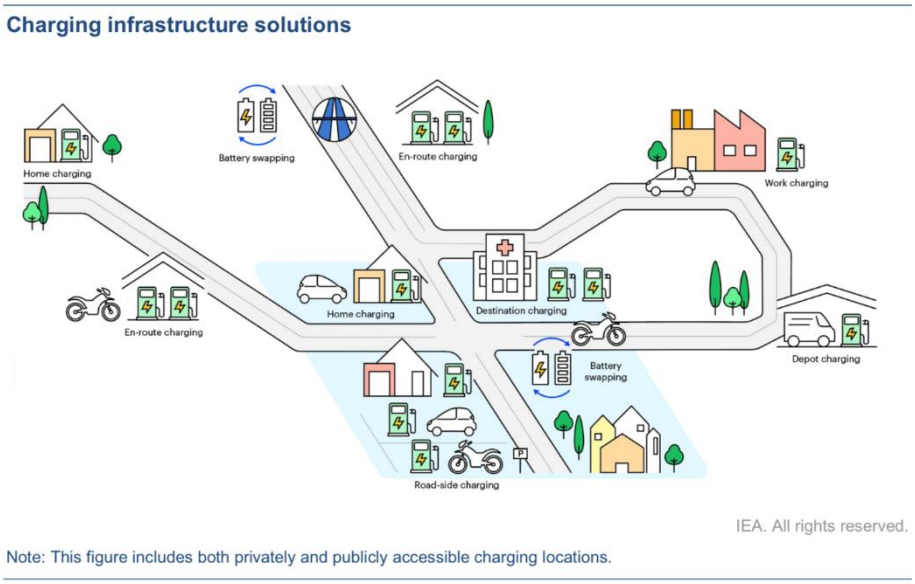


Figure 8: Charging Infrastructure Solutions[26]

The following table defines the different modalities of charging stations by location and highlights the key points regarding their locations, vehicle types, and advantages of each.

[26] International Energy Agency. (2021). Policy Brief on Public Charging Infrastructure: Promoting successful roll-out Strategies and Business models. In International Energy Agency [Policy Brief]. https://iea.blob.core.windows.net/assets/d560f6a6-8d40-4e63-aed2-4fc93139dd5c/PolicybriefonpubliccharginginfrastructurePromotingsuccessfulroll-outstrategiesandbusinessmodels.pdf

Name	Definition	Key Points
Home Charging	Charging an electric vehicle at the owner's residence.	Locations: Residential garages, streets or driveways. Vehicle Type: Light-duty vehicles, cars, motorcycles. Charging type: Typically AC (1.9 – 44Kw). Advantage: Convenient, usually slower AC charging, overnight charging.
Work Charging	Charging an electric vehicle at the driver's place of employment.	Locations: Office buildings, corporate parks. Vehicle Types: Light-duty vehicles, cars, motorcycles. Charging Type: Typically AC (1.9 – 44Kw). Advantages: Encourages EV adoption among employees, available during work hours.
Depot Charging	Charging commercial fleets, buses, or trucks at a centralized location with multiple charging points and restricted access, typically private stations for the operation of a specific fleet, allows efficient and streamlined energy management for large-scale electric vehicle operations.	Locations: Fleet depots, bus terminals, logistics centers. Vehicle Types: Heavy-duty vehicles. Charging Type: Typically DC (44Kw -350Kw) Advantages: Centralized, multiple charge points, managed access.
Hub Charging	A group of EV charging stations situated together allows multiple drivers to charge their electric vehicles simultaneously. These stations can be public, private, or semi-private.	Locations: commercial spots (supermarkets, parking lots, shopping centers) bus terminals, logistics centers. Vehicle Types: passenger cars, light and Heavy-duty vehicles. Charging Type: Both AC (1.9 – 44Kw) and DC (44Kw -350Kw). Advantages: Centralized, multiple charge points.
Road-side Charging	Charging at a public or private parking spot next to the road, typically found in urban areas.	Locations: Urban streets, public parking lots. Vehicle Types: Light-duty vehicles, passenger cars. Charging Type: Typically DC (44Kw -350Kw). Advantages: Publicly accessible, various charging speeds, supports urban EV usage.
Destination Charging	Charging at locations such as shopping malls, restaurants, or public institutions, which are the end destinations of journeys, aside from home or work.	Locations: Shopping centers, restaurants, public institutions, hospitals. Vehicle Types: Light-duty vehicles, passenger cars. Charging Type: Both AC (1.9 – 44Kw) and DC (44Kw -350Kw). Advantages: Provides charging while engaged in other activities.
Enroute Charging	Charging at stations located along the route to the final destination.	Locations: Highway rest stops, fuel stations. Vehicle Types: Light-duty vehicles, passenger cars. Charging Type: Typically DC (44Kw -350Kw). Advantages: Supports long-distance travel, fast charging options available.
Battery Swapping	Replacing the depleted battery of an electric vehicle with a fully charged one at specialized stations, allowing for quick	Locations: Designated battery swap stations. Vehicle Types: Mainly two and three wheelers. Charging Type: Specialized infrastructure. Advantages: Quick energy replenishment, reduces downtime for charging.

Name	Definition	Key Points
	energy replenishment without waiting for a charge.	

Table 4: Charging Infrastructure Characterized by Location

As a reference, in a semi-public charging scenario, the following table provides an estimation of the time required to fully charge an EV such as a BYD ATTO 3 model and a Tesla S using different charging power outputs across various charging scenarios.

Estimation to fully charge BYD ATTO 3 (hours):		45	10	4	1	0.333
Charging Category	**Duration (hours)**	**AC 1 kW**	**AC 7 kW**	**AC 19 kW**	**DC 50 kW**	**DC 350 kW**
Home Charging		1	7	19	50	350
Overnight (21:00-07:00)	10	16.67%	100.00%	100.00%	100.00%	100.00%
Destination Charging						
Hospital	2.5	4.17%	29.17%	79.17%	100.00%	100.00%
Supermarket	0.45	0.75%	5.25%	14.25%	37.50%	100.00%
Shopping Center	4	6.67%	46.67%	100.00%	100.00%	100.00%
Day activities (zoo, golf)	8	13.33%	93.33%	100.00%	100.00%	100.00%
Work Charging						
Daytime	9	15.00%	100.00%	100%	100%	100%
En-route charging (highway)						
coffee or lunch	0.5	0.83%	5.83%	15.83%	41.67%	100.00%
dinner	1	1.67%	11.67%	31.67%	100.00%	100.00%
Road-side charging						
Overnight (21:00-07:00)	10	16.67%	100%	100%	100%	100%
1 hour visit	1	1.67%	11.67%	31.67%	100%	100%
4 hour visit	4	6.67%	46.67%	100%	100%	100%

Table 5: Estimation to fully charge a BYD ATTO 3

Estimation to fully charge Tesla S (hours):		45	10	4	1	0.333
Charging Category	**Duration (hours)**	**AC 1 kW**	**AC 7 kW**	**AC 19 kW**	**DC 50 kW**	**DC 350 kW**
Home Charging		1	7	19	50	350
Overnight (21:00-07:00)	10	10.00%	70.00%	100.00%	100.00%	100.00%
Destination Charging						
Hospital	2.5	2.50%	17.50%	47.50%	100.00%	100.00%
Supermarket	0.45	0.45%	3.15%	8.55%	22.50%	100.00%
Shopping Center	4	4.00%	28.00%	76.00%	100.00%	100.00%
Day activities (zoo, golf)	8	8.00%	56.00%	100.00%	100.00%	100.00%
Work Charging						
Daytime	9	9.00%	63.00%	100.00%	100.00%	100.00%
En-route charging (highway)						
coffee or lunch	0.5	0.50%	3.50%	9.50%	25.00%	100.00%
dinner	1	1.00%	7.00%	19.00%	50.00%	100.00%
Road-side charging						
Overnight (21:00-07:00)	10	10.00%	70.00%	100.00%	100.00%	100.00%
1 hour visit	1	1.00%	7.00%	19.00%	50.00%	100.00%
4 hour visit	4	4.00%	28.00%	76.00%	100.00%	100.00%

Table 6: Estimation to fully charge a Tesla S

Case Study Box No. 2 Battery Swapping as a charging methodology and business model

The model of E2&3W battery swapping stations has seen notable success, particularly in countries such as Taiwan, where an estimated 100,000 battery swaps occur daily. Unlike full-sized EVs, E2&3W vehicles require less capital investment for swapping stations due to their smaller, standardized battery packs. This simplicity in battery replacement contrasts with the challenges posed by the varying sizes of battery packs in full-sized EVs, making infrastructure setup easier for E2&3W swapping stations. However, standardization remains an issue for E2&3W batteries.

Implementing a battery swapping station policy would ensure interoperability between E2&3W vehicles and their batteries, regardless of make or model, thereby making the stations accessible to the general public. However, the lack of standardization in both charging stations and vehicles hinders the development of the E2&3W market. Some E2&3W models are equipped only with level 1 charging capability, rendering them incompatible with level 2 charging stations.

In fact, the battery swapping modality is suitable for electric two-wheelers and three-wheelers (E2W and E3W); however, it is not considered practical for larger vehicles

2.5 Deployment and Management of EVCI

2.5.1 Key Considerations to Install EVCI

In order to install an EVCI, stakeholders must first understand the existing regulations, which typically include mandatory infrastructure requirements such as standardized connectors, minimum output power, and communication protocols. The following topics should be considered for successfully installing EIVCI and offering it as a service[27].

Connector Standardization: Ensure compliance with standardized socket outlets and connectors as required by regulations to guarantee compatibility and ease of use.

Site Selection: Choose strategic locations for charging stations based on traffic patterns, proximity to amenities, and demand forecasts to maximize usage and revenue.

Accessibility and User Convenience: Design stations to be user-friendly and easily accessible for all users, including those with disabilities.

Safety and Compliance: Comply with all relevant electrical safety standards and regulations to ensure the safety of users and the infrastructure.

Local Permissions for Installation: Obtain all necessary local and municipal permissions to comply with zoning and other regulatory requirements.

Customer Support and Maintenance: Establish robust support and maintenance systems to ensure the reliability and longevity of the charging infrastructure.

Payment and Pricing: Provide flexible, transparent, and user-friendly payment options, including various methods such as contactless payments.

Interoperability and Open Data: Implement systems that guarantee interoperability across different charging networks and vehicles and adopt open data protocols to

[27] https://iea.blob.core.windows.net/assets/d560f6a6-8d40-4e63-aed2-4fc93139dd5c/PolicybriefonpublicchargeinginfrastructurePromotingsuccessfulroll-outstrategiesandbusinessmodels.pdf.

facilitate the exchange of information between EVs, charging stations, and grid operators.

Agreements with Electricity Providers: Secure reliable power supply agreements to ensure continuous operation of the charging infrastructure.

Smart Charging: Equip stations with smart charging capabilities that optimize energy use and integrate with grid management systems.

Integration with Renewable Energy Sources: Where feasible, incorporate renewable energy sources to power the charging stations, promoting sustainability.

Cybersecurity Measures: Implement strong cybersecurity measures to protect the infrastructure and user data from cyber threats.

Case Study Box No. 3 A benchmark to determine the quality of service of a charging network

A benchmark to determine the quality of service of a charging network

A service benchmark has been developed to measure the user-friendliness of public charging. As a reference, The National Knowledge Platform for Charging Infrastructure in the Netherlands (NKL) has developed an overview of KPI's that together make up the quality of service for a charging network. The KPI's have been measured, both via a field test and desk research, and have delivered a benchmark on 'service' for public charging operators in the country.

The longlist of potential indicators to define user-friendliness has been drawn up, based on desk studies and via a questionnaire. The following KPI's have been considered most critical:

1. Charger reliability
2. Charger availability
3. Incorrectly parked fuel car
4. Availability in apps
5. Location in apps
6. Helpdesk availability
7. Helpdesk expertise
8. Contact information of the helpdesk
9. User-friendliness of the charger
10. Accessibility of the charger
11. Verifiability of the invoice
12. Maximum charging speed
13. User input for "smart charging"
14. Price transparency
15. Renewable electricity provided

2.5.2 Home EV Charging

When installing EVCI at home, it is essential to follow specific guidelines to ensure safety and compliance. These include assessing the home's electrical capacity, adhering to local regulations, and using a certified electrician to complete the installation. The next table gives an overview of the main requirements to follow:

Home EV Charging	
Optimal Location	Choose a convenient location that minimizes the need for extensive wiring or trenching. EV charging cables typically range between 18-25 feet, so the placement should avoid tripping hazards and obstacles. Map out the location to minimize costs and disruption. Minimize the complexity of the installation by positioning the charging unit close to the electric panel, reducing trenching costs
Electrical Capacity	The electrical panel's capacity determines how much electricity it can safely handle for the entire home. EV charging, depending on the level, can require anywhere from 16 to 80 amps. Assess whether the home's panel (typically 30-100 amps) can handle the 16-80 amps required for EV charging. Upgrading to a 200-amp panel may be necessary depending on the home electrical capacity.
Equipment Selection	Ensure the EV charger is compatible with the electric vehicle, paying attention to the specific connector required by the vehicle
Permits and Regulations	Obtain local permits if applies. When installing an EV charger, particularly in a multi-unit building or rented property, permission from the building management or property owner might be required. Additionally, in some cases permission from the municipality may be necessary, for registration purposes or if the EV charger exceeds a certain kW/h power limit. Permits and regulation varies according to the specific building, country or municipality.
Installation Requirements	Hire certified electricians to ensure the installation adheres to safety standards and complies with all relevant codes. The installation should also strictly follow the guidelines provided by the charger manufacturer.

Table 7: EV Home Charging Installation Guidelines

2.5.3 Commercial EV Charging

On the other hand, when installing EVCI in commercial settings, it is crucial to adhere to specific guidelines that address the increased power demands and user accessibility requirements. Additionally, unique considerations for commercial EV Charging must be taken into account[28][29], which differs from home charging installations[30]. The following table outlines the primary requirements for installing commercial EV chargers:

[28] https://www.transportationandclimate.org/sites/default/files/EV_Siting_and_Design_Guidelines.pdf
[29] https://www.boston.gov/sites/default/files/file/2020/06/How%20To%20Install%20an%20EVSE%20.pdf
[30] https://evchargingsummit.com/blog/considerations-for-site-hosts-installing-ev-charging-stations/

Commercial EV Charging[31]	
Site Assessment	Conduct an assessment considering employee or consumer needs, existing power sources, and accessibility. In commercial spaces, a hybrid approach using a mix of Level 1, 2 and 3 chargers can serve both regular and high-demand users. The EV charging stations should be placed close to existing electrical infrastructure to reduce installation costs. Consider areas with convenient access for users, avoiding interference with pedestrian pathways or parking access. Identify the complete layout of existing parking spaces and proposed location of EVSE parking space(s) with respect to existing building and structures
Equipment Selection	Two key factors should be considered. First, choose the appropriate charging level (Level 1, 2, 3, or 4) based on the energy output and the specific needs of EV drivers. Second, consider the connectors or adapters to be installed. These should comply with national regulations, if applicable, or align with market strategies that prioritize connectors or adapters most commonly used by the target consumers.
Electrical capacity, installation plan and Safety Considerations	**Electrical Capacity**: Ensure the electrical system can handle the increased load. For multi-vehicle charging stations, load management devices and subpanels may be needed. Facilities should consider an electrical system upgrade depending on the scale of the project and the charging power needed. Additionally, It is crucial to assess the type of charging units (Level 1, 2, or 3) that best match the charging duration requirements and consumers needs. **Installation Plan**: Create the installation plan, single-line diagrams showing the system, point of connection to the power supply and the EVSE should be made as planning action. **Protection**: Install protective barriers (e.g., steel bollards) around charging stations to prevent damage from vehicles. Ensure the stations are compliant with accessibility codes
Cost Considerations	Installation, operational and maintenance costs: Consider the costs related to installation, electrical upgrades, permits, and labor. Beyond installation, ongoing operational and maintenance expenses need to be accounted for. The Rwanda government offer financial incentives and tax rebates, to support the installation of EV charging stations. Take into consideration government financial incentives to reduce installation costs or provide ongoing benefits for operating EV charging stations.
Financial Considerations	Financial and income considerations: EV charging stations can provide both direct and indirect revenue streams. Direct income comes from charging fees, while indirect income may arise when charging stations are strategically located, leading to increased customer traffic and consumption of other goods and services available near the charger.
Energy Efficiency and Renewable Integration	For businesses interested in sustainability, integrating EV charging infrastructure with renewable energy sources such as solar panels are an option worth considering. This can reduce long-term energy costs and improve the environmental impact of the charging infrastructure. Commercial installations should also consider using energy-efficient chargers and load management systems to balance the

[31] For more details on EVCI design and installation consult: https://www.gov.ie/pdf/?file=https://assets.gov.ie/294327/df80a986-ea9b-4db7-8bc2-fc9f65c73c02.pdf#page=null

Commercial EV Charging[31]	
	building's total energy consumption and avoid peak demand charges.
Permitting and Compliance	Verify the specific permit requirements for the installation in the specific location. Municipal authorities often require permits for installing EVCI, and the regulations can vary significantly between municipalities. Additionally, permits may be needed when the installation exceeds certain energy thresholds, such as a specific kW capacity. It's essential to consult local building and electrical codes to ensure compliance and avoid delays. In some cases, environmental and zoning regulations might also apply, particularly for larger installations or public-facing charging stations.
Maintenance and Support	Regular maintenance of charging stations ensures long-term reliability and safety. Maintenance includes inspecting cables, software updates for smart chargers, and checking for potential damage or wear. For larger commercial setups, entering a maintenance contract with the equipment provider could be cost-effective and ensure timely repairs.
EVCI Identification	Electric vehicle charging station spaces may only be used for parking and charging electric or hybrid vehicles. Electric vehicle charging station spaces shall be identified with signage indicating the intended use of the parking space.
Technical Installation Requirements	When installing EV charging equipment, it's crucial to hire certified electricians to ensure the installation adheres to all relevant safety standards, electrical codes, the specific guidelines provided by the EVCI manufacturer, and to ensure long-term durability and compliance. Proper conductor sizing is necessary to handle the current load, while grounding must be done correctly, and electrical connections must meet manufacturer specifications. For chargers rated over 60 amperes or 150 volts, a disconnecting means should be installed for emergency and maintenance purposes. Panelboard compatibility should also be confirmed, ensuring circuit breakers are compatible with the existing panelboard or service equipment, ideally from the same manufacturer. Additionally, consider surge protection, weatherproofing for outdoor installations, and local safety regulations.
Operation	When installing EVCI in public or commercial spaces, it's important to collaborate with a Charge Point Operator (CPO) for smooth operations. A CPO can provide integrated payment systems, making it easy for users to pay for charging, and ensure your station appears on charging maps to increase visibility. They also handle maintenance, software updates, and customer support, improving the overall user experience. Additionally, CPOs help scale up infrastructure as demand grows, making the partnership essential for efficient and customer-friendly operations.
User Experience and Monitoring	To enhance the user experience, consider installing smart chargers that offer real-time monitoring, scheduling, and payment options through mobile applications. This not only simplifies usage for employees or customers but also allows building managers to track energy consumption and manage power usage during peak hours.
Scalability	With the expected growth in EV usage, ensure that the charging infrastructure is scalable. Planning for future expansion, such as adding more chargers or upgrading the electrical capacity, can save money and reduce disruption to operations in the long run. Installing extra conduit or choosing equipment that can handle

Commercial EV Charging[31]	
	increased demand will make future expansions easier and more cost-effective.
Public Communication:	Once the installation is complete, publicize the availability of the charging stations to the community and within the business. This can increase employee engagement and attract customers, especially if the chargers are made available for public use.

Table 8: Commercial EV Charging installation

2.5.4 EV Charging in Petrol Stations

When installing EVCI in petrol stations, it is essential to follow the same general guidelines as other commercial installations for EVCI, shared in Table 8: Commercial EV Charging installation. While there is a perception that service stations may pose a higher risk due to the proximity of electricity and gasoline, this scenario does not present any significant additional hazard as long as the EVCI installation adheres to all safety and accessibility measures[32].

Ensuring proper safety protocols, optimal placement, and accessibility is critical for the successful installation of EVCI. These safety measures include proper grounding, separating EV charging points from gasoline pumps, and using certified equipment and installers who follow national electrical codes and manufacturer installation guidelines.

Special considerations for the installation of EVCI in petrol stations should follow the general commercial installation guidelines already described, along with the following key considerations:

EV Charging in Petrol Stations ***Should follow guidelines of commercial EVCI installation, and the next additional considerations:***	
DC Fast Charging Priority	The most appropriate EVCI type for service stations is DC fast chargers. These chargers allow customers to quickly charge their vehicles while in transit, making speed and convenience essential in this context.
Dedicated EV Charging Space	Install EV charging hubs in dedicated areas, separate from gasoline refueling stations. This minimizes perceived safety risks and ensures a smooth, efficient charging experience for EV drivers. It also ensures accessibility by preventing non-electric vehicles from occupying those spots, helping to maintain the smooth flow of traffic and charging availability. Additionally, the EVSE location should not obstruct vehicles accessing gasoline pumps.
Signage and Markings	Clear signage such as "Electric Vehicles Only" and pavement markings must indicate charging bays, avoiding confusion. Warning signs should also be displayed to highlight potential hazards.
Weather Protection	Installing weather protection, such as canopies, for DC fast chargers at outdoor locations not only provides customer convenience but also serves as a critical safety measure for the electrical equipment.
Adequate Lighting	Ensure sufficient lighting in the charging area to improve visibility and safety for both users and operators, especially during nighttime.
Compliance with Electrical Codes	All installations must comply with local building and electrical standards, including proper grounding, overcurrent protection, and the use of disconnect switches for high-power chargers.

[32] https://www.transportationandclimate.org/sites/default/files/EV_Siting_and_Design_Guidelines.pdf

EV Charging in Petrol Stations ***Should follow guidelines of commercial EVCI installation, and the next additional considerations:***	
Energy Management Systems	To handle the high energy demands of fast chargers, it is recommended to implement energy management systems that optimize electricity usage, reduce costs, and manage peak load periods.
Partner with EVCI Networks or CPOs	Service stations should consider partnering with established EVCI networks or create their own charge-for-charge models to manage payments and ensure a seamless customer experience.
Customer Amenities	Offering customer amenities such as seating areas or cafes is crucial, as EV drivers may need to wait up to 30 minutes while their vehicle charges. Standard service stations may need to expand convenience stores to offer additional services for these longer stays.

Table 9: EV Charging in Petrol Stations

2.5.4.1 Business Potential of Installing EVCI in Petrol Stations:

Petrol stations are uniquely positioned to capitalize on the growing demand for EV charging, particularly with the rollout of fast DC chargers, which can charge vehicles quickly, making them ideal for long-distance travelers. By installing EVCI, service stations can:

- **Attract New Customers**: Tap into a growing market of EV drivers, many of whom prefer convenient, fast charging options at familiar roadside stations.
- **Increase Foot Traffic and Revenue**: EV drivers typically spend more time at charging stations than those refueling gas, providing opportunities for upselling goods and services like convenience stores, food outlets, and car washes.
- **Transform the Business Model**: Service stations can diversify their offerings, future-proof their business, and stay competitive by providing essential charging services to the rapidly growing number of EVs on the road.

Service station charging can emulate the gasoline refueling experience by focusing on speed and convenience. Stations are perfectly positioned along highways, where fast charging is essential. While proximity to gasoline may raise safety concerns, adherence to these guidelines ensures there is no added user hazard. By investing in EVCI, petrol stations can maintain relevance, drive customer engagement, and secure a competitive edge in the transition to electric mobility.

Case Study Box No. 4 International Trends in EVCI for Petrol Stations



Service stations have a unique advantage due to their strategic locations along high-traffic roads. As they evolve into multi-service hubs, they are well-positioned to attract EV drivers, who tend to stay 3 to 4 times longer than traditional internal combustion engine drivers. This increased dwell time opens opportunities for stations to boost revenue through additional services such as convenience stores and cafés.

Although EVs do not require gasoline, the location of gas stations makes them ideal for installing EV charging infrastructure, particularly for fast charging. Without adapting to the growing demand for EV services, studies suggest that up to

Fuel retailers can tap into the electric mobility shift by leveraging their strategic locations, reducing “range anxiety” for drivers, and connecting with a younger, tech-savvy customer base. By offering EV charging, stations can attract affluent customers, increase revenue, and safeguard against fluctuations in fuel prices. Additionally, most stations already have the space and facilities, such as parking lots and convenience services, which can enhance the charging experience.

Examples of Major Industry Investments[33]:

- Shell plans to install 500,000 EV charging points globally by 2025.
- BP aims to expand its EV charging network to 70,000 charging points by 2030.
- Total plans to grow its European EV charging network to 150,000 charging points by 2025, up from 18,000 today.

2.6 Typical Roles in eMobility Ecosystem

2.6.1.1 Charge Point Operator – (CPO)

The CPO oversees the installation, management, maintenance, and operation of charging stations, handling both technical and administrative aspects. The CPO's

[33] https://evbox.com/en/ev-charging-fuel-retail-opportunities

role may be divided into administrative responsibilities (e.g., access control, billing) and technical maintenance, which is often outsourced to the manufacturer. The CPO is usually also the registered entity of the grid connection, therefore maintains a close relationship with the distribution system operator and the meter operator

2.6.1.2 Charge Location Owner

The charge location owner possesses ownership of the charging location and often the charging point itself. Depending on whether the location is public or private, the energy may be procured by the charge location owner or the charge point operator.

2.6.1.3 Mobility Service Provider (MSP or eMSP)

The eMobility Service Provider is the entity with which the EV driver enters into a contract for services related to EV operation. Typically, the MSP performs services, and/or integrates services from other market actors, such as access to charging networks. Navigation and payment services, parking services and smart charging services. Car manufacturers or utilities may also fulfill this role. In order to provide these services, roaming contracts and operational (data) interfaces are required.

2.6.1.4 Roaming Platform and Aggregator

A roaming platform or aggregator serves as a central organization facilitating information exchange among multiple market players in order to allow roaming. It connects different market participants to establish a digital and cross-border charging network for electric vehicles. Different variants exist as to the amount of services provided.

2.6.1.5 EV Driver

An EV driver refers to the individual driving the electric vehicle, responsible for its current operation and the need to recharge it for continued use.

2.6.1.6 Energy Supplier

The energy supplier provides the electricity for electric vehicles via the grid connection of the (public or private) charging points. Various suppliers produce or purchase energy for this purpose. The CPO or location owner is usually considered the customer of the energy supplier, depending on whether the charging point is public or private, the market model, contractual agreement etc.

2.6.1.7 Distribution System Operator (DSO) or Regional Grid Operator

The DSO designs, operates, and maintains the public distribution medium- and low voltage grid, which supplies power to the grid connection. The charging locations are either directly connected to the grid connection, or are part of a private grid (e.g., home, building) connected to the DSO grid.

The primary focus of the DSO is to ensure availability of electricity at every moment, assure power quality and prevent issues such as grid congestion and disruptive events.

2.6.1.8 Transmission System Operator (TSO)

The TSO is responsible for maintaining stable power system operation, including physical balance, through a high-voltage transmission grid within a geographical area. Additionally, the TSO determines and manages cross-border capacity and exchanges, adjusting allocated capacity as needed to ensure operational stability.

2.7 Interoperability of EVCI

Interoperability, as a concept, pertains to the ability of systems or entities to collaborate effectively. Various definitions of interoperability exist. The International Organization for Standardization (ISO) outlines interoperability in its standard ISO/IEC 2382-01 (Information Technology Vocabulary, Fundamental Terms) as “the capability to communicate, execute programs, or transfer data among various functional units in a manner that requires the user to have little or no knowledge of the unique characteristics of those units”. The European Commission's Intelligent Transport Systems Directive (ITS 2010/40/EU) defines interoperability as “the capacity of systems and the underlying business processes to exchange data and to share information and knowledge”.

Interoperability has proven successful in sectors such as telecommunications and IT, enabling seamless sharing of pertinent information among service providers. This allows users to benefit from infrastructure and services without being constrained by specific hardware or subscriptions.

The advantages of interoperability include:

- Reduction of installation and integration costs by eliminating the need for conversion or translation services/components.
- Streamlined scaling of services through the reuse of interoperable components.
- Efficient development of new services with minimal reliance on third parties.
- Fostering a competitive environment by preventing technology lock-in, thereby ensuring an equal playing field and facilitating better comparison of offerings.
- Shifting competition towards factors like price and reliability, as enhanced transparency in pricing fosters the provision of more advanced offerings.

2.7.1 Layers of interoperability

In the realm of EV charging, an open and market-driven sector hinges on both interoperability and openness. Each country or region has the authority to delineate which aspects of the sector are competitive and which fall under the realm of pre- or non-competitive domains (such as data exchange through open standards or government-owned data reporting). Policymakers and competitive authorities must periodically evaluate forthcoming advancements to ensure the maintenance of a competitive marketplace. It proves beneficial to scrutinize all facets of the value chain and differentiate between the various 'layers' where market participants engage.

The Smart Grid Architecture Model, crafted by the European standardization body CEN-CENELEC, delineates distinct 'layers of interoperability' essential for a thriving market-driven sector, comprising:

- The **Hardware layer**, focusing on interoperability of connectors and plugs.
- The **Communication layer,** which entails seamless communication between hardware and software systems to facilitate data exchange, akin to an IP or 4G protocol.
- The **Information layer,** where standardized data models and information protocols are required for the recognition and interpretation of exchanged information.

- The **Service layer,** necessitating standardization and interoperability to enable predictable and measurable services such as payment, roaming, navigation, and reporting.
- The **Business layer,** calling for a non-discriminatory regulatory framework to outline the 'rules of the game' and standardized contractual arrangements among market participants to ensure predictability and sustainability concerning aspects such as settlement, liability, and dispute resolution.

The level of interoperability may vary across markets, but a certain threshold is imperative to realize mature user-centric service definitions (pertaining to themes like pricing, navigation, payments, roaming) and streamlined backend processes for the installation and operation of charging infrastructure.

An open and interoperable market model

When combining the above insights into a future-proof market model, the following picture arises.

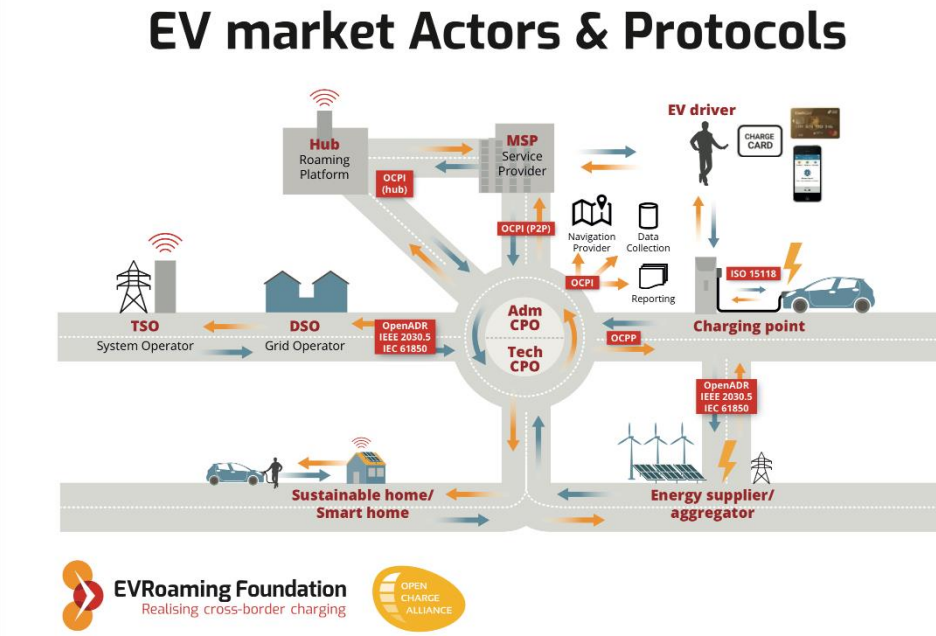


Figure 9: Ev Market and Protocols: EV market Actors & Protocols[34]

This market model makes use of the requirements for openness and interoperability for the hardware, communication and information layers. Interoperability requirements for the services layer and the business layer (to assure uniform user-centric propositions, and a market model which provides the ‘rules-of-the-game’, respectively) will need to be described via regulation.

[34] Source: EVRoaming Foundation

2.7.2 Charging Protocols

An overview of most widely used charging protocols and standards are described below[35]:

2.7.2.1 Charging Protocols in General

Open Charge Point Protocol (OCPP)/ IEC 63110

Operates between the charging station and the charging station management system: Open Charge Point Protocol (OCPP) has been developed to make it possible to connect different types/brands of charging stations to a single Charging Station Management System (CSMS) and vice versa, i.e. to connect a single type/brand of charging station to a range of charging station management systems. OCPP supports the management of charging stations and the handling of charging transactions, including the identification and authorization of the EV driver. In addition, the protocol can be used to control charging stations for smart charging. OCPP is used by the Charging Station Operator (CSO) to communicate with the charging stations it manages through its CSMS. OCPP has been developed into the international 'de facto' standard for managing charging stations and is used by many CPOs. OCPP is managed by the Open Charge Alliance[36].

IEC 63110 is the standardization effort by the IEC to develop a 'de jure' standard. Currently, the mandate has ended, and initiatives are ongoing to integrate the efforts of IEC63110 with OCPP.

Open ADR – Open Automated Demand Response Standard / Open ADR Alliance

The protocol's objective is to automate demand response communication, enabling systems and/or devices to adjust power consumption or production of demand-side resources. This adjustment can be triggered by various factors such as grid requirements, tariff adjustments, incentives, or emergency signals, all aimed at balancing demand with sustainable supply. [37]

IEC 61850

The document IEC 61850-90-8 does not constitute a standalone protocol but rather serves as a technical report delineating an object model for electric mobility. It conceptualizes Electric Vehicles within the framework of Distributed Energy Resources as outlined in the paradigms established by IEC 61850[38].

Charging protocols – roaming

EV Roaming empowers electric vehicle (EV) drivers to charge at any charging station, regardless of the Charge Point Operator (CPO) or station owner, while overseeing the billing process for the charging session. Implementing EV roaming necessitates an open charging infrastructure tailored for electric drivers, which mandates adherence to specific standard protocols for its functionality.

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[35] (Source "Position Paper on Open Markets & Open Protocols" (June, 2021), NAL Working Group Open Market & Open Protocols, Netherlands)

[36] (Source: Open Charge Point Protocol 2.0.1, Open Charge Alliance, [Online]. Available at: OCPP 2.0.1, Protocols, Home - Open Charge Alliance)

[37] The OpenADR protocol specification profiles A and B are accessible to the public at no charge via www.openadr.org.

[38] Access to the IEC 61850 specification is available to the public at a nominal cost through the IEC website: www.iec.ch/dyn/www/f?p=103:23:0::::FSP_ORG_ID:1255.

OCPI - Open Charge Point Interface protocol/ IEC 63119

OCPI stands as an autonomous roaming protocol designed for seamless data exchange. Operates between the charging station operator and the mobility service provider: OCPI protocol is used to exchange information between the Charging Station Operator (CSO) and the mobility service provider (MSP), but also with other market operators which require EV information. The OCPI protocol is used to set up a direct connection between two parties. It offers versatility, catering to both company-to-company interactions (peer-to-peer) and utilization through a roaming hub or platform, and supports the exchange of information on locations, tariffs, authorizations and charging transactions. It also supports smart charging through the management of charging profiles.

In addition to supporting roaming transactions, the OCPI protocol increasingly supports the reporting and data collection needs from e.g. governments, navigation service providers and location owners, to create a concise overview of all available charging stations, its performance, availability etc.

This protocol enjoys international support, providing EV drivers with visibility into charging point availability and costs. It is accessible at no charge through the EVRoaming Foundation, companies can become a contributor and thereby contribute to the development of the protocol [39].

IEC 63119 is the IEC standardization effort to develop a 'de jure' roaming protocol. Currently, efforts are ongoing to merge the initiatives of OCPI and IEC63119 in order to move to a single roaming protocol.

2.7.2.2 Proprietary Protocols

Proprietary protocols also exist, as developed by some roaming platforms for their own services, e.g. OCHP from e-clearing.net, OICP from Hubject and eMIP from Gireve. The general tendency is that all actors in the value chain move to a single protocol (OCPI) as the standard means of market communication.

ISO15118

Operates Between the car and the charging station: ISO15118-2 was developed with two important goals: providing a user-friendly mechanism for authentication, authorization, and payment at the charging station without further user interaction, known as Plug and Charge (PnC) and for Integration of the EV into the Smart Grid to enable flexible energy transfer (V2G) and thereby deliver added value for the grid without compromising the EV or its driver.

Its latest extension ISO15118-20 introduces significant enhancements such as support for wireless power transfer, improved data security with mandatory Transport Layer Security (TLS), and dynamic control modes for real-time power adjustments based on grid demands. It also includes multiplexed communication for efficient message processing and easier handling of multiple e-mobility contracts, allowing users to switch between different charging agreements effortlessly.

The following case study boxes showcase some of the best worldwide practices for adopting standards for EVs.

[39] More information: www.evroaming4.eu.

Case Study Box No. 5 Standards adopted by the European Union



In the European Union (EU), the standardization of charging infrastructure follows IEC 61851 for AC/DC charging and ISO 15118 for smart charging and vehicle-to-grid (V2G) communication. This ensures uniformity across all EU member states, enabling any EV to use public charging stations seamlessly. Furthermore, the EU mandates that all public stations must provide Type 2 connectors for AC charging and Combined Charging System Combo 2 (CCS2) connectors for DC fast charging. These practices guarantee high levels of interoperability between different EV brands, while roaming networks allow users to access multiple charging networks with a single payment system, simplifying EV charging for users.

Case Study Box No. 6 Standards adopted by the United States of America



In the United States, under the National Electric Vehicle Infrastructure (NEVI) program, federally funded charging stations must support open standards, ensuring compatibility across different EV brands. The U.S. predominantly uses the SAE J1772 (Type 1 connector) for AC charging and Combined Charging System Combo 1 (CCS1) for DC fast charging.

However, the North American Charging Standard (NACS), initially develop by Tesla for their EVs and now open for other manufacturers, which is now standardized as SAE J3400, is becoming the new standard. The Society of Automotive Engineers (SAE) recently announced that it will standardize Tesla's NACS connector[40], ensuring that any supplier or manufacturer can use, manufacture, or deploy the NACS port on EVs or charging stations across the continent.

This transition is expected to begin in 2025, with several major automakers—including Ford, GM, Toyota, Rivian, Volvo, Polestar, Nissan, Mercedes-Benz, Jaguar Land Rover, Fisker, Hyundai, Stellantis, Volkswagen, and BMW—planning to implement NACS as the standard for their EVs. This shift signals a significant move

[40]https://driveelectric.gov/charging-connector#:~:text=In%20June%202023%2C%20SAE%20announced,the%20openly%20available%20J3400%20connector.

towards unified charging systems in North America, making the charging infrastructure more accessible and streamlined for consumers and manufacturers alike.

In line with global trends, the U.S. also invests in V2G technology, which enables EVs to return electricity to the grid, supported by standardized communication protocols like ISO 15118 to ensure smooth integration.

Case Study Box No. 7 Standards adopted by China



In China, the government mandates the GB/T 20234.2 standard for AC charging and GB/T 20234.3 for DC charging, ensuring nationwide compatibility. China also has a centralized payment system, allowing users seamless access to charging stations across the country, further boosting EV adoption by simplifying the charging process.

2.8 Payment Modalities

The deployment of charging infrastructure has started with Charging Point Operators (CPOs) offering access to their charging networks through either a direct payment model or a subscription-based (post-payment) model. With the emergence of a market-driven operational model and multiple CPOs providing charging services, the role of an e-Mobility Service Provider (eMSP) has been defined to offer EV drivers mobility services across multiple CPO networks. These services include navigation, information, charging access, payment, invoicing, and more. It is worth noting that each CPO can also function as an eMSP, providing services that span all third-party charging networks.

2.8.1.1 Direct payment

Direct payment solutions utilize existing interoperable payment systems such as credit card terminals, QR codes, and in-app payments. Like payment at petrol stations, direct payments are an efficient and widely accepted payment method. However, depending on the specific direct payment solution and the physical interface needing to be installed (credit card reader, pinpad etc), it may result in higher cost for charging equipment. Moreover, there is minimal interaction between the EV driver and the CPO regarding the acquired charging service, limiting advanced customer services and smart charging propositions. [41] [42]

[41] https://www.avere.org/publications/item/2023/04/15/AVERE-position-paper-on-the-upcoming-proposal-for-a-new-Payment-Services-Directive-PSD-II

[42] https://www.chargeupeurope.eu/s/The-need-to-remove-PIN-pad-obligations-for-EV-charging-payments-2-March-1.pdf

2.8.1.2 Subscription model

Another well-established payment method is the subscription or post-payment model, commonly used for slow charging infrastructure. In this model, the eMSP grants access to a charging station through authentication via a token, RFID card, app, or Plug&charge feature in the vehicle. Once authentication between the eMSP and the respective operator is successful, the charging session commences. After the session, the eMSP provides an invoice and facilitates settlement between the eMSP and CPO.

For EV drivers, the subscription model offers several advantages:

- The eMSP can provide a customized customer experience by offering additional services to enhance the charging experience, such as navigation, price transparency, rebates, and aggregated monthly invoice.
- Functionalities like smart charging, vehicle-to-grid (V2G), and Plug&charge can only be successfully developed within a subscription model because knowledge about the user, vehicle and its requirements is crucial. Important factors include departure time, required kilowatts, and state-of-charge.

2.8.1.3 Roaming

While a direct payment solution serves as a minimum standard for accessing multiple charging networks, it does not constitute true roaming for EVs since it relies solely on existing payment systems without providing further information or intelligence during the transaction.

The concept of roaming originates from the telecom sector, with ISO 26927 defining it as “a service that enables users/terminals to use access networks and mobility services of a network operator different from the user’s home domain”. In the context of electric mobility, EV roaming refers to allowing an EV user to have a subscription with operator/service provider *A* and charge the electric vehicle at a charging station operated by operator *B*, with whom the EV driver does not have a direct contract.

Roaming in EV charging requires a contract-based model and necessitates layers of interoperability to ensure an optimal roaming service. Implementing a contract-based model with interoperable EV roaming offers several benefits:

- EV drivers can charge their vehicles at any charging network without the need for multiple access methods, both within a country and across countries.
- CPOs can expand their customer base, increasing charging sessions, and enhance their business case.
- CPOs can exchange information with user profiles and offer smart charging and other services, optimizing EVs as flexible storage assets and maximizing the use of renewable energy.
- Distribution System Operators (DSOs), through CPOs, gain better control over grid congestion and peak times, resulting in reduced grid investments.
- Public governments can efficiently invest in public charging networks and provide optimal charging experiences for citizens, thereby accelerating the deployment of zero-emission transport.

As stated by the UK government, “Implementing roaming across networks means consumers can access all public charge points with one membership card or

smartphone app"[43]. In France, decree n° 2017-26 mandates that public charging infrastructure owners make their charging stations open to eRoaming, either via direct links or roaming platforms. Public charge points must be accessible through direct payment or an MSP contract, at the EV driver's choice.

2.9 Circularity and waste disposal

As eMobility gains momentum, the number and variety of installed charging stations continue to rise presenting a challenge regarding their end-of-life management. The lifespan of a charging station is approximately 10 years, indicating that it will no longer function and provide its charging services after this duration. Since the supply chain of the charging stations is linear (Figure 10: Linear Supply Chain of EV Charging Stations), once they reach the end of their useful life, they undergo de-installation and disposal, resulting in the accumulation of substantial waste streams. This accumulation presents significant environmental and economic challenges.

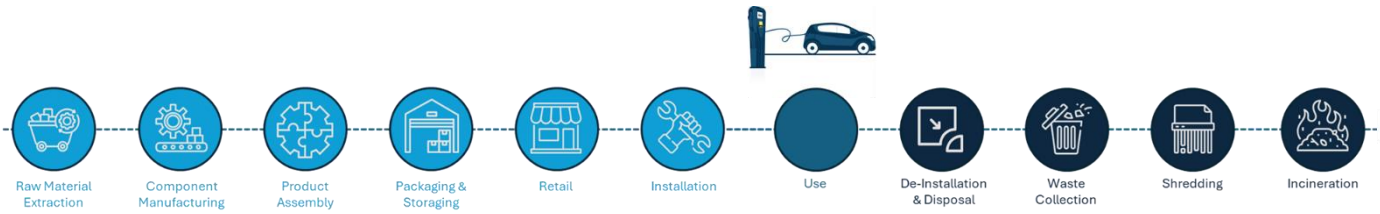


Figure 10: Linear Supply Chain of EV Charging Stations

To tackle the challenge of waste accumulation from charging stations, it is crucial to explore strategies that extend the lifespan and value of the infrastructure thereby promoting sustainability in the eMobility sector. Embracing the principles of the Circular Economy (CE) offers a promising solution. CE seeks to optimize resource utilization by minimizing waste and reducing reliance on primary resources. Instead of simply disposing of charging stations at the end of their useful life, methods such as refurbishing their foundation and casing and replacing components can effectively prolong their lifespan. The proactive approach not only reduces waste but also maximizes the utility of existing infrastructure, aligning with the goals of sustainability and resource efficiency.

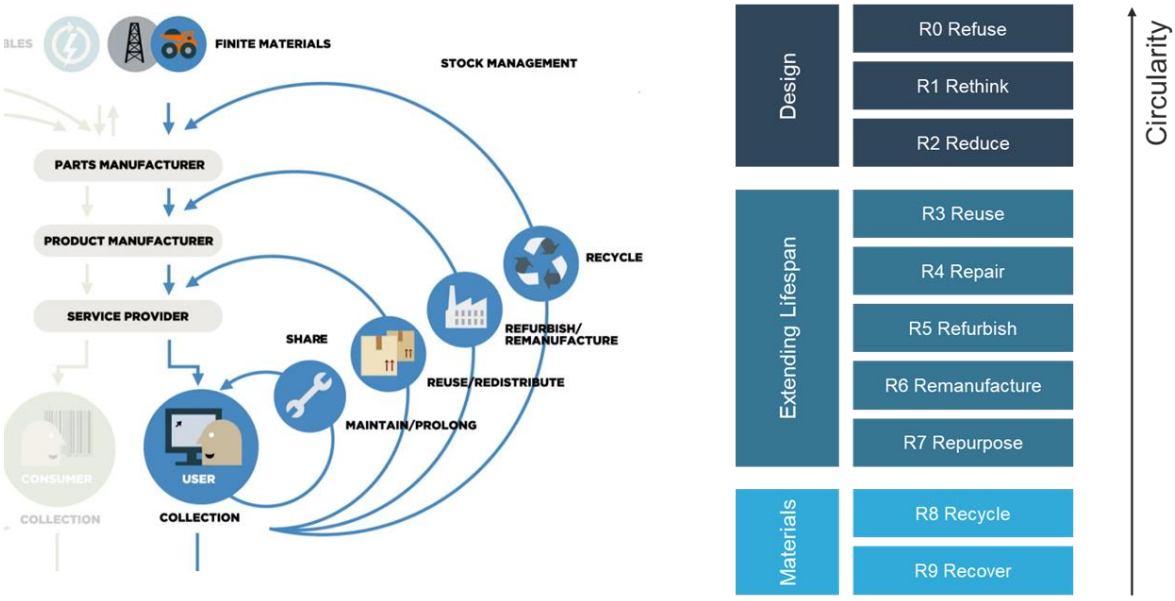
2.9.1 Circular Economy Frameworks

In the context of Circular Economy, two popular frameworks stand out: the 'Butter Diagram' of the Ellen MacArthur Foundation (EMF) and the 'R-strategies'. The Butterfly diagram illustrates a circular economy framework, emphasizing regenerative processes and minimizing waste through closed-loop systems. Figure 11: Popular CE Frameworks: The Butterfly Diagram & R-strategies, shows the technical cycle of the butterfly diagram which focuses on keeping items on the market as long as possible while maintaining their quality. Since the charging stations are (almost) completely made from technical nutrients, it makes sense to only consider the technical cycle of the Butterfly Diagram. The R-strategies encompass the technical strategies outlined in the butterfly diagram but also

[43] https://www.gov.uk/government/consultations/the-consumer-experience-at-public-electric-vehicle-chargepoints/the-consumer-experience-at-public-chargepoints

introduce additional approaches. These include Refuse, Rethink, and Reduce, which aim to optimize product design.

While the frameworks primarily emphasize extending the lifetime of components and products and reducing its waste, it is important to note that incorporating secondary materials as inputs for production also plays a significant role in advancing the Circular Economy. This is because its practice reduces dependance on primary resources and fosters sustainable resource management.



(a) Butterfly Diagram – Technical Cycle

(b) R-Strategies

Figure 11: Popular CE Frameworks: The Butterfly Diagram & R-strategies

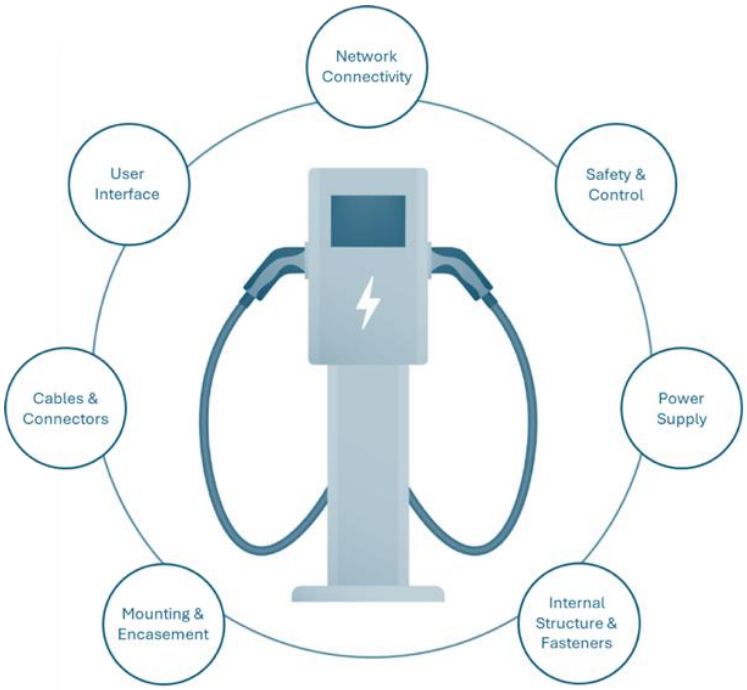
The next section delves into the most effective strategies and best practices for transitioning towards a circular charging station infrastructure in Rwanda.

Circular Economy Possibilities for Charging Stations

Charging stations vary in type and sized based on their intended purpose. Private chargers are typically installed at homes or private properties, while public chargers are found in places like parking lots or highways and require more robust infrastructure. Chargers are also tailored for different vehicles, with those for buses and trucks requiring heavier components due to higher energy demands. Despite these variations, charging stations share common components like power distribution units, charging connectors, and control systems (Figure 12: Typical components of a charging station). Adopting a CE approach entails designing these components to enhance the durability of the charging station and its parts and

facilitating material and component circulation. This promotes resource efficiency and minimizes environmental impact within electric vehicle charging infrastructure.

Figure 12: Typical components of a charging station



Cables and Connectors provide the conduit for a charge to be delivered.

Power Supply units supply the power to the EV's battery and are made up of wires, capacitors, transformers and other electronic parts.

User Interface may manifest as a touchscreen, button panel, or mobile application and includes empowers users to initiate and oversee charging process, status, and facilitate the payments.

Network Connectivity functions as the brain of the charging station, communicating with its network so data can be monitored and access can be managed.

Safety & Control is necessary for the deployment of several features, encompassing ground fault protection, meters, thermal management systems, cooling equipment, etc.

Mounting & Encasement of the charging station are essential for securing its position.

Internal Structure & Fasteners provide stability and support to the various components within the charging station.

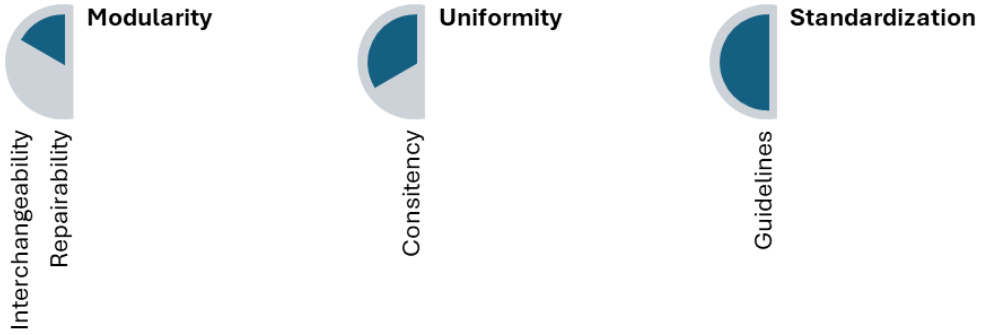
2.9.2 Circular Strategies for Charging Stations

An important takeaway to advance in the CE is to not solve the problem at the end, but seek solutions at the beginning of the chain.

This implies that strategies should be implemented during the manufacturing phase of the charging station production to facilitate the durability and recirculation components or product. For instance, if a component within the charging station malfunctions, ensuring its easy removal, repair, and reinstallation can prevent the

need to discard the entire product. Strategies to improve the CE early in the supply chain are driven by the integration of modularity, uniformity, and standardization (Figure 13: Early Stage Strategies to Improve CE).

Figure 13: Early Stage Strategies to Improve CE



By adopting modular design principles, manufacturers can create components that are easily interchangeable, repairable, and upgradable, reducing resource consumption and waste generation. Furthermore, uniformity ensures consistency and compatibility across products and standardization establishes guidelines and specifications for materials, facilitating the integration of circular practices. Together, these approaches optimize resource efficiency, minimize environmental impact, and promote circularity throughout the supply chain, ultimately leading to more sustainable and resilient CE ecosystems. It is, however, essential to consider the potential trade-offs of these strategies.

Modularity might lead to increased material and energy usage, uniform products may lack customization options to meet diverse needs or preferences and both uniformity and standardization may limit innovation and flexibility. It is essential to analyze the product's most critical components and carefully weigh the advantages and disadvantages of modularizing them. In this context, the critical components are those most prone to malfunctions, often including fuses, ground fault circuit interrupters, plugs, and smart meters. The ground fault circuit interrupter is a critical component that typically shows the issue of non-repairability since they are encased in plastic which makes it very hard to repair the component when it is defect. It is, therefore, important to expose the ground fault circuit interrupter so it is easily repairable. Evaluating and potentially modularizing these components can enhance the product's reliability, repairability, and overall performance.

In Rwanda, where the adoption of eMobility remains relatively niche, there exists a greater potential for influencing the manufacturing of components. The casing material for the charging station varies depending on their location and intended use. Indoor AC chargers are typically constructed from plastic, whereas outdoor charging stations may utilize heavier materials such as aluminum. For a Circular Economy perspective it would be interesting to explore the optimal material choice based on factors such as durability, recyclability, and reduced reliance on raw materials. Current waste streams and management, and policies in Rwanda influences the choice of material as well.

Best Practices for Circularity in Charging Stations

Despite the established production dominance of major suppliers, there are noticeable instances of modular practices emerging within the industry. Some examples of these best practices of modularity are:

Adjusting the foundation plates: If the charging stations are designed separately from its foundation, then the concrete blocks and anchors can remain in place, saving thousands of concrete blocks, anchors, and nuts.

Detachable cords: Customers often mishandle charger cords by driving on cords that have been left on the ground and thereby damaging them. With non-detachable cables, the entire charger would need to be replaced. Furthermore, detachable dabbles allow for regular disconnection, which reduces the wear and tear on the contact surfaces of both the charger socket and the cable plug, prolonging the lifespan of the charger.

Another emerging practice is the use of more sustainable materials in the charging stations. EVBox, a provider of flexible and scalable charging solutions for EVs, and Covestro, a pioneer in polymer solutions, are collaborating on a pilot initiative aimed at integrating more environmentally friendly plastics into the manufacturing of EV charging stations. This material, Makrolon® RE, comprises high-performance polycarbonates, sourced from renewable alternatives instead of solely relying on feedstock derived from fossil resources.

Drawing from the best practices researched in the world, along with an analysis of CE regulations and activities, as well as considering the level of maturity of eMobility in Rwanda, CE possibilities will be proposed to enhance circularity practices of charging stations.

Since the maturity of eMobility is still relatively small in Rwanda, the possibility of implementing CE practices becomes more feasible. However, this entails the need for the production of charging stations or components to take place within Rwanda. Basis systems should incorporate sustainable principles, such as modular designs, when production shifts to Rwanda. This shift would result in more local charging stations, fostering greater adherence to circular principles compared to large corporations that may prioritize market-wide product uniformity over customer preferences.

Rwanda's Circular Economy Action Plan and Strategies for Circular Charging Stations

The Ministry of Environment of Rwanda developed the National Circular Economy Action Plan to apply circular economy principles in key economic sectors, including the waste sector. The plan outlines several activities and policy interventions to be implemented by 2035, relevant to the transition towards circular charging stations.

Key activities and policy interventions include:

- Partnerships and Collaboration: Promoting partnerships across public and private sectors, academia, and civil society to enhance circular economy implementation.
- Circular Economy Accelerator: Launching programs to support entrepreneurs in scaling up circular ideas.
- Waste Collection and Transfer Centers: Establishing centers in each district to facilitate proper waste treatment and collection of useful materials.

- E-waste Collection Center: Investing in e-waste collection and recycling facilities and leveraging innovative businesses like Wastezon.
- Plastic Waste Reduction: Implementing activities to collect and repurpose plastic waste.
- Systemic Data Collection: Creating a waste data collection system to monitor and improve recycling rates.
- National Waste Regulation: Developing regulations for waste characterization and treatment to manage waste more effectively.
- E-waste Management Policy: Restricting the import of used electronics and offering incentives for e-waste collection.
- Plastic Ban: Banning single-use plastics to encourage alternative packaging solutions and mandatory waste management.

2.9.3 Strategy for Circular Charging Station Practices

To enhance circular charging station practices in Rwanda, two main strategies are proposed, inspired by European best practices and considering Rwanda's eMobility maturity level:

Modular Manufacturing:

Involves creating basic construction kits to facilitate the assembly and disassembly of components within Rwanda. This approach extends the lifespan of charging stations by allowing for the repair or replacement of damaged parts. It promotes local job creation and fosters independent production of charging stations. However, it requires substantial infrastructure for assembly and disassembly, thorough risk assessments for safety, and comprehensive employee training to ensure successful implementation.

Repurposing Charging Stations:

Entails refurbishing and testing charging stations from other parts of the world, to be used in Rwanda. This strategy is time and resource-efficient, providing a second life for these charging stations. However, it must be economically viable compared to purchasing new stations. Additionally, this approach must address potential waste accumulation in Rwanda and consider end-of-life management to prevent environmental impacts.

The first approach fosters local job creation and independent production by enhancing circular practices, but it necessitates sufficient infrastructure for assembly, collection, and disassembly. Conversely, the second approach is more time and resource-efficient but fails to address end-of-life management for the refurbished stations, potentially leading to waste accumulation in Rwanda instead of the country in origin. While it gives charging stations a second life in Rwanda, a third life might not be feasible.

The following table presents a detailed summary of these points, integrating existing practices and international best practices to guide future policy development.

Waste Disposal and EVCI in Rwanda

Waste collection in Rwanda is managed by private companies contracted by municipalities. After collection, there is no source separation, and typically all waste, including municipal solid waste (MSW), serviced, commercial, and industrial waste,

ends up in landfills. Manual sorting by authorized waste pickers occurs at these sites, but a significant amount of plastic waste is still mingled with organic waste, posing environmental and health risks. There is no advanced waste processing; waste management is limited to sorting after collection, waste spreading, soil coverage, and lacks leachate treatment or gas management.

The Government, in collaboration with the City of Kigali and the Global Green Growth Institute (GGGI), launched the project “Waste to Resources: Improving Municipal Solid Waste (MSW) and Hazardous Waste Management in Rwanda” in August 2021. This project aims to promote the recycling of organic and plastic waste, increase electronic waste collection, raise community awareness, build capacity, and improve the policy and regulatory environment for circular economy initiatives in the waste sector. The National Sanitation Policy and Implementation Strategy (2016) and a new Integrated Waste Management Strategy under development regulate waste management, including the recycling of various solid wastes under the Regulation of Solid Wastes Recycling.

Private sector initiatives also play a significant role. For example, Agruni Ltd. manages waste and recycling in Kigali, sorting waste into biodegradable, non-biodegradable, recyclable, and non-recyclable categories. COPED Rwanda, another key player, collects waste from residents and commercial institutions, sorting it into bio-organic, recyclable, non-recyclable, organic waste, and hazardous waste categories. COPED also has facilities to transform dry biomass waste into briquettes.

With the rising population, economic growth, and modernization, the demand for electrical and electronic equipment (EEE) increases, leading to more e-waste. A survey conducted in 2014-2015 estimated Rwanda's potential annual e-waste generation at 9,417 tons, with 81.52% from individuals, 12.14% from public institutions, and 6.43% from private institutions. Rwanda regulates e-waste through its national e-waste Management Policy, restricting the importation of used electronics to minimize adverse effects on health and the environment. Financial incentives are provided to individuals who collect e-waste and bring it to recycling facilities. In 2017, Rwanda’s Green Fund FONERWA invested in an e-waste collection and recycling facility run by Enviroserve in the Bugesera district. Additionally, innovative businesses like Wastezon use technology to connect homes with recycling facilities for e-waste management, complementing government efforts.

To manage waste from EVCI in Rwanda effectively, it is crucial to integrate comprehensive strategies aligned with the National Circular Economy Action Plan and ongoing initiatives for waste disposal. Key recommendations include developing specific regulations for EVCI waste disposal and enforcing Extended Producer Responsibility policies to ensure manufacturers manage end-of-life products. Establishing designated collection centers in each district and investing in advanced recycling facilities will facilitate the processing and recovery of valuable materials from EVCI waste.

Encouraging modular design and manufacturing practices can extend the lifespan of EVCI components and support refurbishment programs. Public-private partnerships and incentives for businesses innovating in EVCI waste management will further enhance these efforts. Additionally, education and awareness campaigns should promote responsible disposal and recycling, while integrating EVCI waste management with existing e-waste policies and facilities can optimize resource

recovery. Implementing these measures will help Rwanda develop a sustainable and efficient waste management system for its electric vehicle infrastructure, promoting environmental protection and resource efficiency.

2.10 Technical considerations to minimize technology obsolescence

EVCI relies on a variety of technologies to ensure efficient, reliable, and future-proof charging for electric vehicles. To minimize technology obsolescence and ensure long-term viability of the EVCI, it is essential to adopt the following strategies and take this into account at the moment to adopt the infrastructure.

2.10.1.1 Standards for Plugs and Connectors

Standardization is the process of developing and implementing rules, guidelines, or specifications to ensure consistency, compatibility, and uniformity. It is crucial for the circular economy and for reducing technology obsolescence for several reasons:

Standardization of plugs and connectors plays a crucial role in reducing technology obsolescence. By establishing and adhering to consistent guidelines, it ensures that new products and technologies remain compatible with existing systems and components, thereby extending their useful life. Interoperability between different manufacturers and products is facilitated through standardization, enabling seamless integration and reducing the need for frequent replacements due to incompatibility issues.

Quality assurance is another key benefit, as standardization ensures that recycled materials and components meet specified quality standards, thereby maintaining the performance and reliability of the products over time. This consistency builds market confidence, encouraging consumers and businesses to adopt standardized technologies with the assurance that they will remain functional and relevant.

Moreover, standardization helps streamline operations and reduce costs by eliminating redundancies and inefficiencies in resource use and manufacturing processes. This makes it economically viable to maintain and upgrade existing technologies rather than replace them entirely.

Overall, implementing universal standards for plugs, connectors and other components of the EVCI is essential. It simplifies user experience and ensures that as technology evolves, existing infrastructure and devices remain usable, thereby significantly reducing the risk of technology obsolescence.

2.10.1.2 Interoperability

Ensuring interoperability is key to future-proofing technology and reducing obsolescence. In the context of EV chargers, adhering to standards such as the Open Charge Point Interface (OCPI) is essential. OCPI facilitates seamless communication between EV charging stations and management systems. By ensuring that new platform versions are compatible with existing and future chargers, OCPI helps maintain the functionality and relevance of EV chargers over time. This ongoing compatibility prevents chargers from becoming obsolete quickly, thereby extending their useful life.

Adopting standards for each interoperability layer promotes an open market and reduces obsolescence by ensuring that various systems can work together seamlessly. Specifically:

- Standardize the type of connectors to be used to assure that EVCI can be used by everyone, and minimize hassle with adapters.
- Utilize standardized protocols and open data formats to facilitate communication between EVs, charging stations, and grid operators. This not only enhances compatibility but also fosters innovation and competition.
- Standardize services and ensure full overview of possibilities when it comes to services such as payment, navigation, roaming etc.
- Provide a long-term regulatory framework that is predictable for every investor and operator of EVCI

2.10.1.3 Backward Compatibility

Developing new technology with backward compatibility in mind is critical. This approach allows older devices and systems to function with new updates and innovations. For example, software updates for EV chargers should support older hardware models, ensuring that existing infrastructure remains useful and efficient.

2.10.1.4 Modular Design

Implement modular designs that allow for easy upgrades and replacement of outdated components. This ensures that the infrastructure can adapt to new technologies without requiring complete overhauls. Modular systems can extend the lifespan of charging stations and reduce the total cost of ownership by enabling incremental updates. Requirements of modularity should be part of the part of the specifications to be requested during procurement.

2.10.1.5 Regular Updates and Maintenance

Implementing a robust schedule for regular updates and maintenance is essential to keep technology current. Providing software updates, security patches, and performance enhancements helps address emerging issues and maintain system relevance. Ensuring that users can easily access and install these updates is equally important.

2.10.1.6 Future-Proof Infrastructure

Investing in future-proof infrastructure involves choosing technologies and components that are expected to remain relevant for an extended period. For instance, selecting high-capacity data cables or next-generation network protocols can prevent the need for frequent upgrades as technology advances.

2.10.1.7 Industry Collaboration

Collaborating with industry stakeholders, including manufacturers, regulatory bodies, and technology developers, helps establish and maintain standards. Industry-wide cooperation ensures that new developments are aligned with existing frameworks, reducing the risk of fragmentation and obsolescence.

2.10.1.8 Circularity

Requirements regarding the circularity of assets, such as the recycling of EV batteries and chargers, are expected to be further tightened in future global climate

agreements. It is therefore important to already take these requirements into account when designing a charging network, as well as to define processes and facilities to organize a circular approach

3 RWANDA CURRENT SITUATION: GOVERNANCE, TRANSPORTATION, ENERGY, BUSINESS MODELS AND STAKEHOLDERS

3.1 Current Transport System in Rwanda

Although Rwanda is geographically small, it is one of the most densely populated countries in Africa, with 414 inhabitants per square kilometer[44]. With a population of approximately 14 million and covering an area of 26,338 square kilometers, Rwanda is experiencing growth in its vehicle sector. The capital city, Kigali, houses the majority of the population and vehicles.

The transport system in Rwanda is primarily focused on road networks. The majority of urban residents walk, cycle, or use public transport to commute to work and access essential services. In Kigali, non-motorized trips account for around 52 percent of all trips, with even higher rates in secondary cities. However, car ownership is on the rise[45]. Additionally, two and three-wheelers are especially crucial in the transport system. In Rwanda alone, more than half of the vehicles on the roads at any given time are moto-taxis[46]. This is largely due to affordability and the convenience of using two and three-wheelers, making motorcycles the leading form of personal transport in the country[47].

Key statistics on the transport System in Rwanda include:

[44] https://www.unep.org/topics/transport/active-mobility/rwanda
[45] https://www.unep.org/topics/transport/active-mobility/rwanda
[46] https://www.weforum.org/agenda/2022/05/electric-motorbikes-rwanda-ampersand/
[47] World Economic Forum. (2020). Rwanda's e-Mobility Transition: Snapshot of the market opportunity [Report]. International Growth Centre (IGC). https://a.storyblok.com/f/109506/x/4a1d18e69c/rwanda-e_mobility-transition-smi-october-2021.pdf

- As of 3rd July 2020, the number of registered vehicles countrywide was 264,524 excluding security organs and government vehicles[48].
- The number of vehicles in Rwanda is increasing by 12% annually[49].
- Vehicle growth in Kigali is 6.2% per year, with the rest of the country experiencing higher growth rates.
- The total length of the road network is 2,750 km.

The following table provides an estimation of the number of vehicles in Rwanda according to different vehicle categories:

Category	*Vehicle Type*	*Numbers*
Two and Three Wheelers	Motorcycles/Moto Taxis	173,000 (primarily in urban areas)
Passenger Vehicles	Cars	Estimated 7 cars per 1,000 inhabitants
Light Duty Vehicles	Micro/Mini Buses	10,097
Heavy Duty Vehicles	Buses	2,621
	Trailers/Trucks	6,073

Table 10: Vehicle Numbers in Rwanda

Two and three-wheelers

According to Table 10: Vehicle Numbers in Rwanda, there are currently 173,000 motorcycles in Rwanda, most of which are used as moto-taxis in urban areas.

Case Study Box No. 8 E-bike adoption and expansion in Rwanda

The e-bike is rapidly gaining popularity worldwide, as in countries such as China, The Netherlands, Germany, and France.[50] [51] The basic principle remains the same: you have to pedal, but you get support from an electric motor when needed, thereby increasing the range.

The e-bike is also becoming popular in several other countries. Regulations regarding maximum speed may vary between countries. Manufacturers are expanding their market by targeting mountainous and hilly regions as well. Currently, a prototype of an e-bike with an extra powerful motor is being tested in Kigali and its surroundings. This should easily handle the slopes in and around Kinyinya Hill. It would be fantastic if this "bee-bike" were to be launched and produced in Rwanda and made available when houses in Kinyinya Hill are handed over.

[48] REPUBLIC OF RWANDA. (2021). Strategic Paper on Electric Mobility Adaptation in Rwanda. in *Ministry oflinfrastructure*. https://www.mininfra.gov.rw/fileadmin/user_upload/Mininfra/Publications/Laws_Orders_and_Instructions/Transport/16062021_Strategic_Paper_for_e-mobility_adaptation_in_Rwanda-Final.pdf

[49] REPUBLIC OF RWANDA. (2021). Strategic Paper on Electric Mobility Adaptation in Rwanda. in *Ministry oflinfrastructure*. https://www.mininfra.gov.rw/fileadmin/user_upload/Mininfra/Publications/Laws_Orders_and_Instructions/Transport/16062021_Strategic_Paper_for_e-mobility_adaptation_in_Rwanda-Final.pdf

[50] https://www.mordorintelligence.com/industry-reports/china-e-bike-market/market-trends

[51] https://www.linkedin.com/pulse/future-electric-e-bike-market-size-rise-alex-jackson/

Passenger Cars

Currently, there are approximately 7 passenger cars per 1,000 inhabitants in Rwanda. This low number indicates that personal car ownership is below 2% of the population.[52]

Case Study Box No. 9 proportion of cars per capita worldwide

Proportion of cars per capita worldwide

Country	No. cars per habitants
United States of America	766 cars per 1,000 inhabitants
Germany	575 cars per 1,000 inhabitants
France	570 cars per 1,000 inhabitants
Netherlands	500 cars per 1,000 inhabitants
Romania	260 cars per 1,000 inhabitants
Uganda	40 cars per 1,000 inhabitants
Rwanda	7 cars per 1,000 inhabitants

Light Duty Vehicles

There are 10,097 micro and mini busses in Rwanda.

Heavy-Duty Vehicles

There are 2,621 buses operating in Rwanda, all of which run on fossil fuels. Although the first electric buses have arrived in the country, most buses operate in urban areas and service the connections between towns and cities. Furthermore, Rwanda has 16,073 trailers/trucks. Some of these vehicles are old, slow, and polluting. Trailers/trucks are crucial for delivering goods across all areas, including Kigali, other cities and towns, and rural areas.

3.2 Current E-Mobility Sector in Rwanda

3.2.1 EV and Charging Stations numbers in Rwanda

By March 2024, Rwanda has nearly 9,093 electric and hybrid vehicles, with the majority being electric motorcycles (4,823) and hybrid vehicles with spark-ignition internal combustion engines (3,901), as next table shows:

Description	*Total*
Hybrid vehicles[53]	3.901
Electrical vehicles with only electric motor for propulsion 60	363
Electrical unassembled vehicles transporting goods with only electric motor for propulsion	2
Electrical vehicles transporting goods with only electric motor for special propulsion	4
Electric motorcycles for propulsion	4.823
Total	9.093

[52] World Economic Forum. (2020). Rwanda's e-Mobility Transition: Snapshot of the market opportunity [Report]. International Growth Centre (IGC). https://a.storyblok.com/f/109506/x/4a1d18e69c/rwanda-e_mobility-transition-smi-october-2021.pdf

[53] Includes Hybrid vehicles with both spark-ignition internal combustion reciprocating piston 1,300; Hybrid with both compression-ignition internal combustion piston engine (d 2000), and Hybrid vehicles with both spark-ignition internal combustion reciprocating piston 3,500

Table 11: Electric and hybrid vehicles in Rwanda

Regarding charging stations, there are 22 EV charging stations in Rwanda, 17 in Kigali with plans to install 28 more, and 5 charging stations in secondary cities. There are 28 private Battery swapping stations in Kigali.

Charging Stations

The following map, shows the distribution of existing and planned charging stations in Rwanda. The red dots in the figure represent planned charging stations, while the green dots indicate existing ones

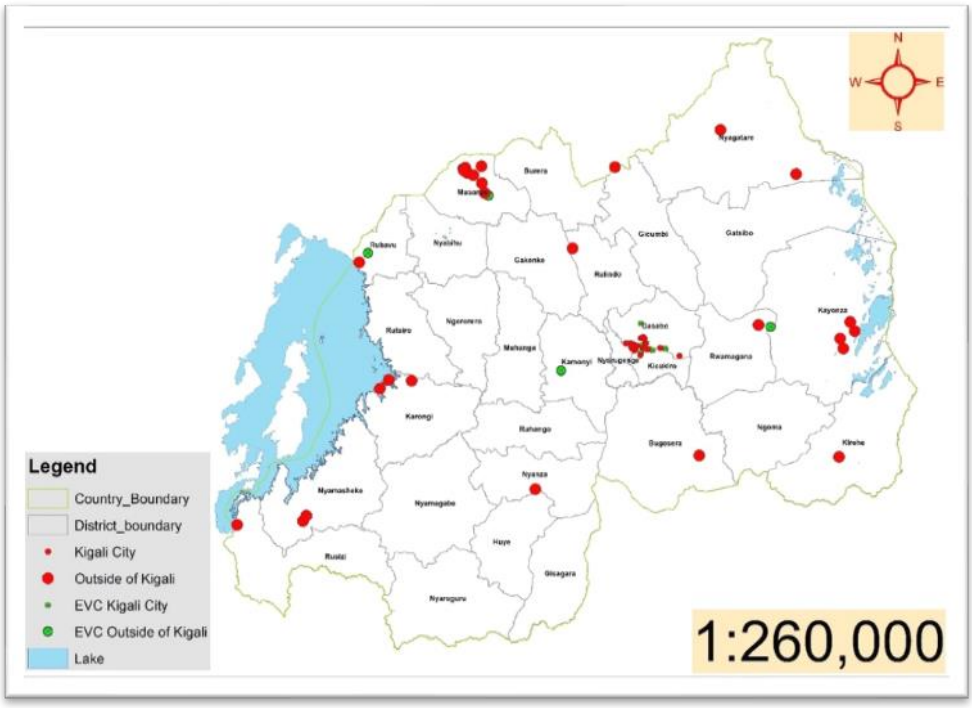


Figure 14: Distribution Map of Existing and Planned Charging stations in Rwanda

The distribution of EVCI (both planned and existing) is concentrated in Kigali, primarily along the trans-city road from Downtown to Kigali International Airport, as shown in Figure 14: Distribution Map of Existing and Planned Charging stations in Rwanda. The Southern Province has minimal EVCI. Most planned EVCI are to be installed in tourist areas such as coffee shops and hotels, as well as SP petrol stations.

There are two main types of charging solutions in Rwanda: battery swapping stations, serving primarily motorcycles, and conventional charging stations, designed mostly for passenger cars and in some cases e-buses but also serving some motorcycles. Throughout the data collection phase, existing charging facilities were visited. These stations are owned by key industry players, including Ampersand, EVP, Kabisa, Basigo, REM, and Spiro. Additionally, some of these companies plan to install more charging stations across Rwanda.

The following table summarizes the existing charging stations as of June 2024 in Kigali:

	Stations Operator	Nr of existing stations	Nr of planned stations
Charging stations	BasiGo	1	1
	EVP	6	-
	Volkswagen	2	-
	Kabisa	8	27
Total		17	28

Table 12: Charging Stations in Kigali[54]

Outside of Kigali, only Kabisa has started to install EVCI and plans to install more. These charging stations are primarily installed in tourist areas such as coffee shops and hotels. Each province has at least one EVCI already installed, with a few more planned. However, in the Southern Province, there is only one planned EVCI by Kabisa. The following table shows the number of both installed and planned charging stations as well as the district they were placed.

Province	Number of stations	Location (District or exact location)
Existing		
Eastern	1	Kayonza - Imigongo Art Cafe
Southern	1	Kamonyi – Stafford
Northern	2	Kinigi - Dian Fossy campus Musanze SP
Western	1	SP Rubavu
Potential Sites		
Eastern	9	Nyagatare, Kayonza, Bugesera, Kirehe, Rwamagana
Northern	9	Musanze, Gicumbi, Rulindo
Western	7	Karongi, Rusizi, Nyamasheke, Rubavu
Southern	1	Nyanza

Table 13: Table of EVCI Location and Station Count for Existing and Future Planned Stations outside Kigali

For more detailed information, Annex No. 2 Existing charging stations in Rwanda, includes the specifications of the EVCI, including the station name, location, owner, latitude and longitude, the number of parking bays, power (kW), charging level, and connector type, among other specific information.

Regarding battery swapping, there are currently 28 stations with plans to install more as the number of motorcycles increases. However, their installation is demand-based, so owners were unable to predict the future locations of the stations at this time.

[54] Some companies have future planned stations, but they were not shared on the data collection phase.

Charging methodology	*Stations*	*Number of existing stations*
Battery swapping stations	Ampersand	15
	Spiro	12
	REM	1
Total		28

Table 14: Table of Battery Swapping Ownership and Station Count for Existing Stations in Kigali

The following figure shows the distribution of battery swapping and charging stations in Kigali, strategically placed in residential zones and densely populated areas such as Kinyinya, Remera, Gikondo, Kimisagara, Kabeza, Busanza, and Kamonyi. These stations are located on national roads and major district roads, supporting the increasing number of electric motorcycles and promoting sustainable transportation in the city.

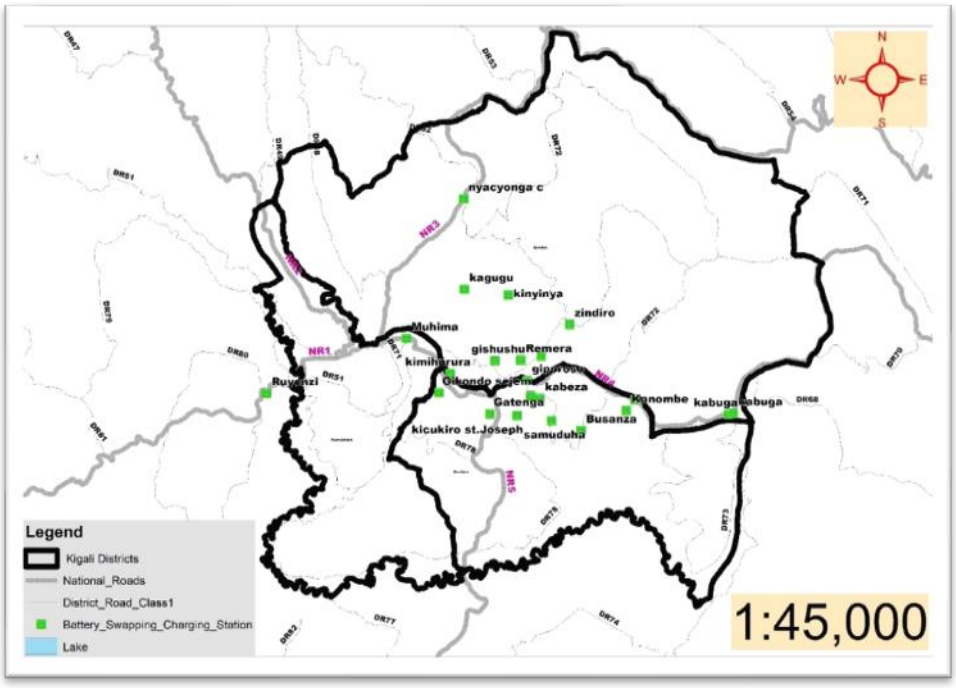


Figure 15: Distribution Map of Existing battery swapping stations in Rwanda.

Furthermore, regarding the visits, the following additional observations arose:

- Public Transportation and Parking: A notable lack of EV charging stations within public parking areas was observed, with the only forthcoming project being BasiGo's installation near the Nyanza taxi park. This highlights a significant opportunity for EVCI development in public parking spaces to support electric public transport and private EVs.
- Government Buildings: Buildings hosting offices of governmental entities, such as ministries, municipalities, and other agencies (e.g., RURA, WASAC, REG), were found to be viable locations for EVCI due to their strategic locations and high daily traffic, making them ideal for charging station installations.
- Healthcare and Market Areas: Hospitals and public markets with parking facilities were identified as potential sites for EVCI. However, it was noted that no charging

stations were installed in these areas during the visit, highlighting an untapped potential for infrastructure expansion.

The following maps provide an overview of the existing and planned EVCI in Kigali and the provinces. Additionally, it is worth noting that the annex contains a comprehensive set of maps, including current, planned, and potential sites in each of the provinces.

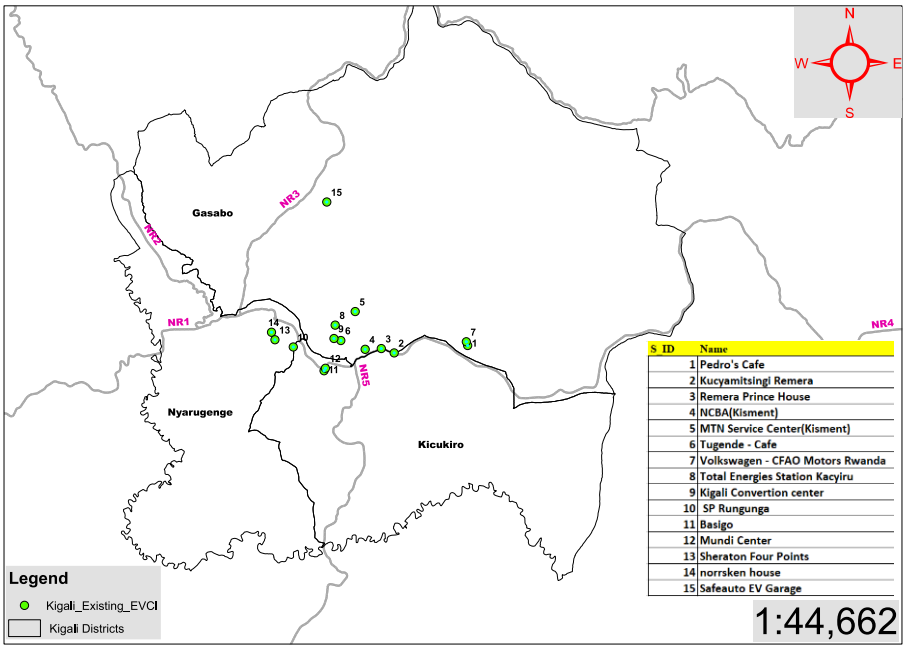


Figure 16 Kigali Existing EVCI

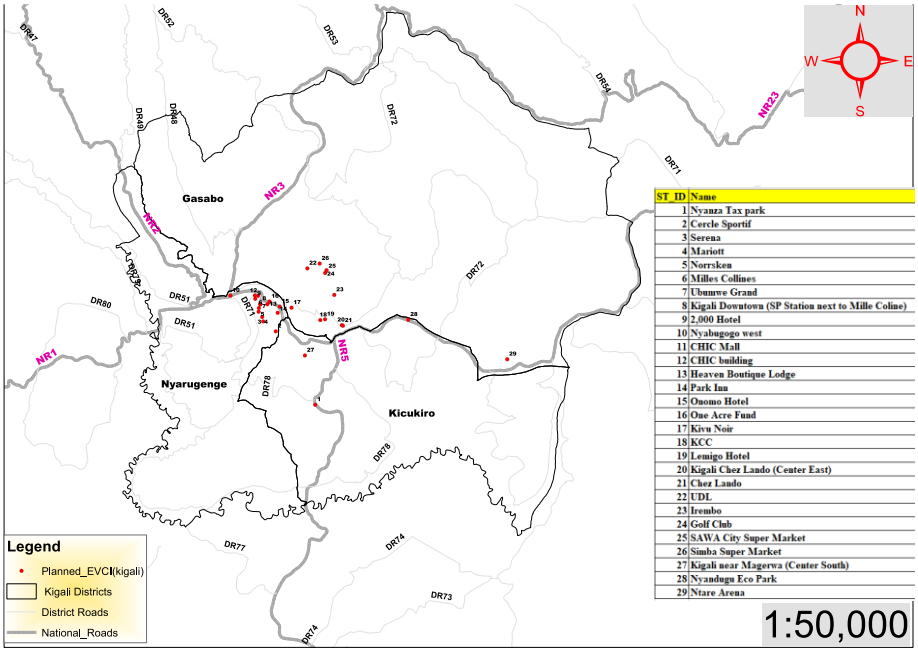


Figure 17 Kigali Planned EVCI

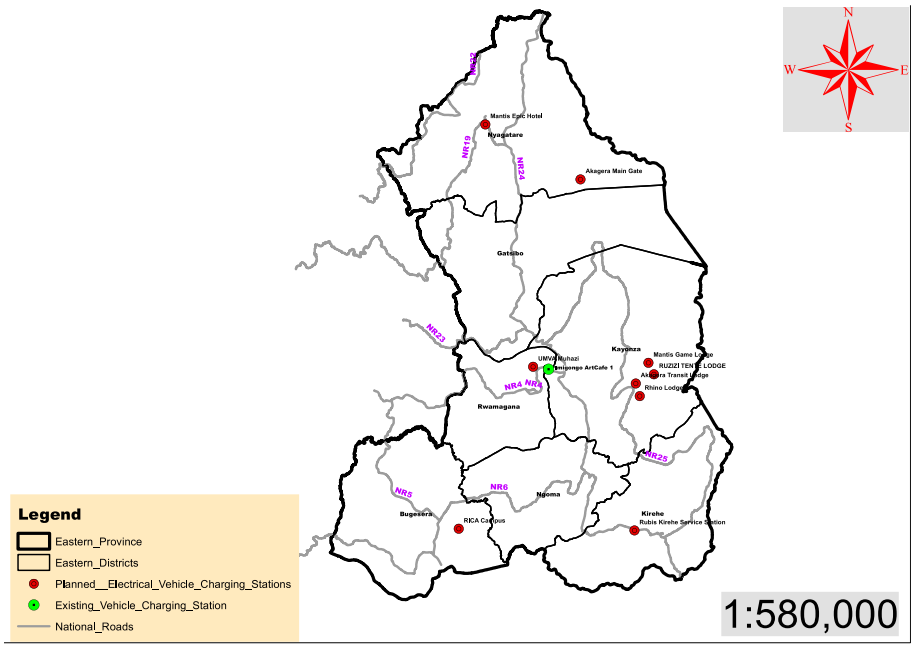


Figure 18 Eastern Province Planned & Existing EVCI

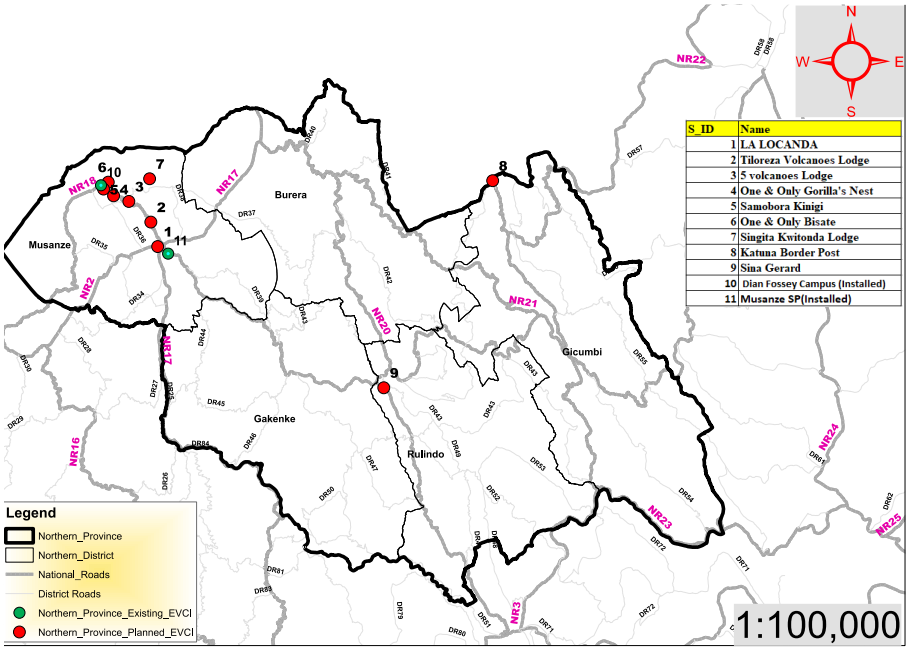


Figure 19 Northern District Planned & Existing EVCI

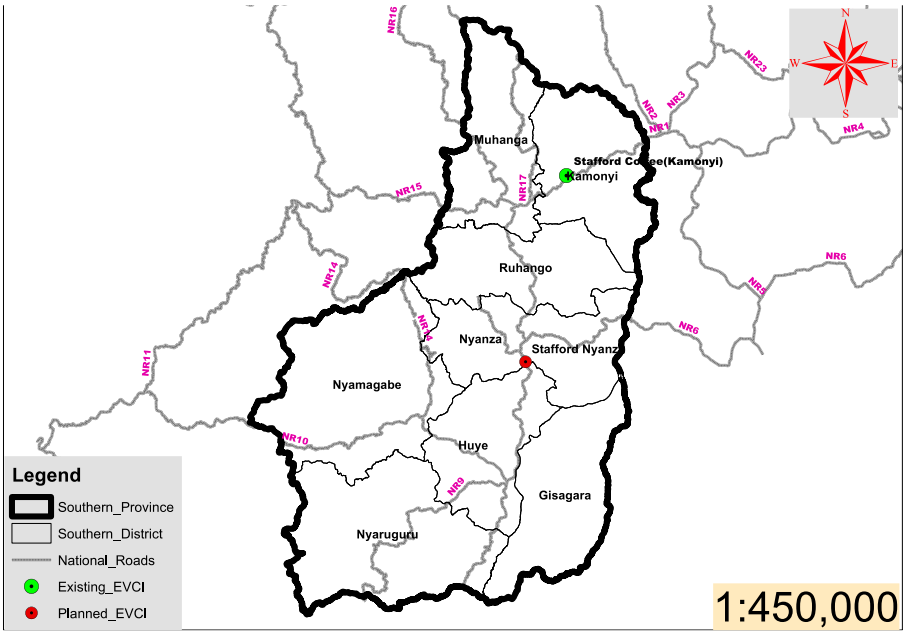


Figure 20 Southern Province Planned & Existing EVCI

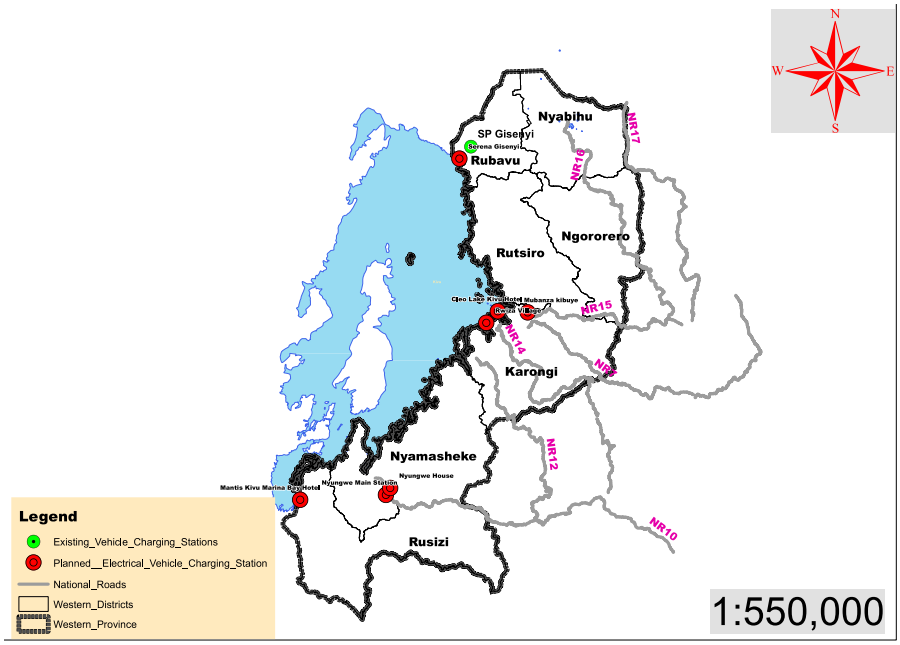


Figure 21: Western Province Planned & Existing EVCI

Mapping Overview

The maps of EVCI were created and are available as annexes. These maps show the locations of existing charging stations, planned stations by Kabisa, and proposed potential sites in underserved zones. Additionally, an online is shared for a clearer view. The maps attached in the report include:

- Existing EVCI locations
- Planned EVCI installations by Kabisa
- Proposed potential EVCI sites in underserved areas

The following Annexes are attached to the report:

- Annex No. 2 Existing charging stations in Rwanda (Attached)
- Annex No. 3 Current EVCI Landscape and Transport Infrastructure (Attached): This map provides a visualization of the existing EVCI in Kigali, juxtaposed with the city's transport infrastructure. It includes the layout of national and district roads, offering a visual representation that aid in understanding the accessibility and distribution of current charging stations in Kigali.
- Annex No. 4 Planned EVCI Developments in Kigali (Attached): This map illustrates planned EVCI facilities by the private operators for their future expansion. It highlights the ongoing efforts to expand the EVCI network. This map is instrumental in our proposing potential EV charging stations.
- Annex No. 5 Existing battery swapping stations (Attached): This map illustrates existing battery swapping stations and their access routes.
- Annex No. 6 Eastern Province Planned & Existing EVCI (Attached)This map combines already installed charging stations and Kabisa planned station in Eastern province.
- Annex No. 7 Northern Province Planned & Existing EVCI (Attached) This map combines already installed charging stations and Kabisa planned station in Northern province.
- Annex No. 8 Southern Province Planned & Existing EVCI (Attached) This map combines already installed charging stations and Kabisa planned station in Southern province.
- Annex No. 9 Western Province Planned & Existing EVCI (Attached)This map combines already installed charging stations and Kabisa planned station in Western province.
- Annex No. 10 Distribution map of potential charging station (Attached): This map shows existing, planned and proposed potential across the country.
- Annex No. 11 Potential EVCI for Kigali (Attached) This map shows proposed potential in Kigali.
- Annex No. 12 Potential EVCI for Nyarugenge (Attached): This map shows the proposed potential in Nyarugenge District, in Kigal. .
- Annex No. 13 Potential EVCI for Gasabo (Attached) This map shows the proposed potential in Gasabo District, in Kigali
- Annex No. 14 Potential EVCI for Kicukiro (Attached) This map shows the proposed potential in Kicukiro District, in Kigali.
- Annex No. 15 Potential EVCI for Southern Province (Attached) This map shows proposed potential in Southern Province.
- Annex No. 16 Potential EVCI for Eastern Province (Attached) This map shows proposed potential in Eastern Province
- Annex No. 17 Potential EVCI for Northern Province (Attached) This map shows proposed potential in Northern Province
- Annex No. 18 Potential EVCI for Western Province (Attached) This map shows proposed potential in Western Province.

3.2.2 Policies, Regulation Status and Current Practices toward eMobility on Rwanda

One of the key policy documents on e-mobility in Rwanda is the National Transport Policy and Strategy, dated April 2021, from the Ministry of Infrastructure. This document highlights several important aspects regarding the development and support of electric vehicles in Rwanda.

The National Transport Policy and Strategy emphasizes that the introduction of EVs should be paired with measures to encourage a shift from personal motor vehicles to walking, cycling, and public transport. There are significant opportunities for integrating EV technology into shared transport services, such as bike-sharing and public transport systems. The policy is driven by several national and international ambitions, including Rwanda Vision 2050, Transition Rwanda, the Paris Agreement on Climate Change, the United Nations 2030 Agenda for Sustainable Development, and the Sustainable Development Goals (SDGs).

The policy direction focuses on developing the necessary infrastructure to support electric mobility and implementing incentives to encourage investment in this sector. Additionally, the National Mobility Plan for Rwanda highlights EVs as a percentage of the total number of introduced cars, underscoring the country's commitment to sustainable transportation solutions.

The National Mobility Plan for Rwanda notes e-vehicles as a percentage of total introduced cars:

2020-2021	*2021-2022*	*2022-2023*	*2024-2025*	*2034-2035*
15%	20%	25%	30%	70%

Another key policy document issued by the government is the Strategic Paper for eMobility Adaptation, dated April 2021[55]. This document introduces a series of measures to promote the adoption of eMobility in Rwanda as follows:



Figure 22: Main Incentives on eMobility in Rwanda[56]

The following table provides a more specific overview of the policy measures described on the Strategic Paper for eMobility Adaptation:

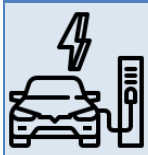
[55]https://www.mininfra.gov.rw/fileadmin/user_upload/Mininfra/Publications/Laws_Orders_and_Instructions/Transport/16062021_Strategic_Paper_for_e-mobility_adaptation_in_Rwanda-Final.pdf
[56] https://www.cliffedekkerhofmeyr.com/export/sites/cdh/sectors/downloads/eMobility-in-Africa-Guide.pdf

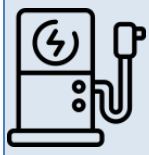
Strategic Paper for eMobility Adaptation Measures

Fiscal incentives	*Non-fiscal incentives*	*Administrative and other measures*
Electricity tariff for charging stations be capped at the industrial tariff level (large industry category)	Rent free land for charging stations (for land owned by Government)	Enforcement of existing emission standards to discourage the purchase of polluting vehicles
Electric vehicles to benefit from a reduced tariff during the off-peak time	Provisions of electric vehicle charging stations in the building code and City planning rules	Establish restricted zones for green transport
Electric vehicles, spare parts, batteries and charging station equipment be treated as VAT zero rated products	Green license plate to allow EVs getting preferential treatment in parking, free entry into congested zones that will be determined	Regulate importation of used vehicles by imposing age limit
Exemption of import and excise duties on electric vehicles, spare parts, batteries and charging station equipment	Free license and authorization for commercial EVs	Provide preference to electric vehicles for Government hired vehicles
Exemption of withholding tax of 5% at customs	Access to High Occupancy Vehicle lanes (Dedicated Bus Lanes).	Companies manufacturing and assembling EVs (battery electric vehicles, plug-in hybrid electric vehicles and hybrid electric vehicles) in Rwanda are given other incentives in the investment code such as 15% Corporate Income Tax (CIT) and tax holidays (irrespective of the investment value)
To introduce carbon tax to discourage polluting vehicles		

Table 15: Strategic Paper for eMobility Adaptation Measures

The development of eMobility practices in Rwanda has adopted some common practices from around the world, while other practices have not yet been undertaken. The following overview highlights the current situation of EVCI in Rwanda, showcasing existing practices and offering insight into how the country is managing various aspects of EVCI. Additionally, by referencing international best practices, the overview provides a comparative perspective to guide future policy development. The following table presents a detailed summary of these points.

Topic	Current Practice in Rwanda	International Best Practice
 **Electric Vehicle Charging and Licensing of CPOs**	There is no explicit policy stipulating the approach to electric vehicle charging in the country, though the existing practice is that electric vehicle charging is considered as a service to the transport sector. CPOs have established EV charging infrastructures without obtaining licenses from RURA. On the other hand, CPOs have to apply to RURA to avail the industrial tariff stipulated in the Strategic Paper on Electric Mobility Adaptation. On the other	In some Australian States, EV charging is considered as a retail sale of electricity and subjects it to regulations on electricity sales. In countries where EV charging is considered as retail sale of electricity, CPOs are required to obtain a license from the electricity sector regulatory authority. Most of the countries that advanced EV deployment, on the other hand, consider EV charging as a service to the transport sector, which is therefore not

Topic	Current Practice in Rwanda	International Best Practice
	hand, they are required to obtain business permits from Rwanda Revenue Authority (RRA).	governed by electricity regulations. For countries that consider EV charging as a transport service, EV charging is not subject to licensing but to general business permits, like any other commercial activity.
 **Regulations and Incentives for Private Charging**	There are no incentives related to the installation of EV chargers in domestic properties in Rwanda. The current fiscal incentives for EV purchase such as zero VAT, exemption of import and excise duties and exemption of withholding tax however covers charging equipment and indirectly benefits private charging of electric vehicles. The current policy caps the electricity tariff for EV charging stations at the industrial tariff level, which is typically lower than residential tariffs. This helps make public charging stations more affordable. Additionally, electric vehicles can take advantage of reduced tariffs during off-peak hours, further lowering the cost of charging and encouraging off-peak usage.	In many European countries, EV owners primarily charge their vehicles at home and work, using Type 2 AC chargers. Governments provide grants to cover the installation costs of EV chargers at homes, apartments, condominiums, and workplaces. Additionally, regulations mandate that a certain percentage of parking spaces in apartments, multi-story buildings, and workplaces be allocated for EVs. For existing buildings, the cost of installing EV chargers, including grid connection upgrades, is distributed according to association rules, with each EV owner paying for their own charging costs.
 **Regulations and Incentives for Public Charging**	No specific regulations for public EV charging stations. The fiscal and non-fiscal incentives, along with administrative measures outlined in the Strategic Paper on Electric Mobility Adaptation, are crucial for kick-starting the e-mobility market and stimulating investments in EV charging infrastructure	In the US and some European countries, governments rely on the private sector to build EV charging infrastructures. They have introduced regulations for deploying, managing, maintaining, and using these systems. Common regulations include mandatory requirements for commercial centers and parking areas to install EV chargers. Additionally, municipalities in several European countries issue competitive tenders for the installation and operation of public charging stations. Governments also provide various incentives, such as direct funding, grants, subsidies, tax benefits, and provision of city land, to attract stakeholders and stimulate private investments in public charging infrastructure.
 **Regulations and Incentives for Motorcycles**	The private sector has responded positively to current policies by introducing electric motorcycle financing for commercial users and establishing battery swapping stations to support the electric mobility ecosystem. Current financial and fiscal incentives, along with other government measures, support	Policies for electric motorcycles target those with engines above 500 W and speeds over 35 km/h, requiring registration. They are categorized into private (individual transport) and commercial (rent, ride-hailing, goods transport) users. Private users typically charge at home, while commercial users use home, destinations, and

Topic	Current Practice in Rwanda	International Best Practice
	both commercial and private usage of electric motorcycles.	swapping stations. Trips are about 10 km/day for private and over 50 km/day for commercial users. Battery swapping is ideal for commercial use. Taipei, China: Offered large subsidies for electric motorcycles and swapping stations (50% of construction cost, free land). Despite this, electric motorcycle sales did not exceed 15% of the market. People's Republic of China (PRC): Banned gasoline-powered motorcycles, increasing electric motorcycle sales. Battery standardization was attempted but withdrawn in Taipei and PRC due to limited success. It hampers competition and innovation.
 **Goals on Electrification of Transport**	No target for banning fossil fuel vehicles. By 2030, Rwanda aims to have 20% of its buses, 30% of its motorcycles, and 8% of its cars electrified.	Several countries have announced national bans on the sale of internal combustion engine (ICE) vehicles. Europe is leading the phase-out with the EU banning the sale of new petrol and diesel cars starting in 2035. Other national and local governments are mandating fleet and public transport electrification within the next decade. For example, California has mandated 90% fleet electrification for ride-hailing services by 2030, and Shenzhen has already transitioned to 100% electric buses.
 **Tariff Management Assessment**	CPOs benefit from industrial tariffs and off-peak rates. The current tariff for large industries is RWF 95 per kWh and the off-peak tariff amounts to RWF 94 per kWh. As a comparison, commercial tariff amounts to RWF 179 per kWh. CPOs set their own charging tariffs. Often, these tariffs, after considering their investment and O&M costs, are set at a rate competitive to the equivalent petrol or diesel price. Concrete Market Experience: In the case of BASIGO, the company estimated that the current equivalent price of diesel buses is RWF 1,500 per km. BASIGO offers a charging price of RWF 600 per km under its pay-as-you-drive financing model. KABISA's charging tariff is estimated to be around 30% higher than the industrial tariff to cover its capital and operational costs.	Market-Driven Prices: In most cases, EV charging prices are set freely by the market without specific regulations. Business-to-Consumer Pricing Schemes in Europe: Flat Rate Pricing: For example, €15 per month. Transaction-Based Pricing: For example, €0.30 per kWh. This approach is familiar to EV owners as it resembles conventional electricity billing. Time-Based Pricing: For example, €1.50 per hour of charging. CPOs may prefer this model to encourage turnover and maximize station usage. Price Caps in Developing Countries: Some developing countries introduce lower and upper price caps for charging to protect consumers and stabilize the market.

Topic	Current Practice in Rwanda	International Best Practice
Regulatory Framework for EV Charging	The current regulations for electrical connections in Rwanda are specified in the Regulations Governing Electrical Installations (2023). This regulation stipulates the procedures and requirements for electrical installations and conditions, inspection, monitoring and compliance. Electrical installations are classified for various voltage levels and for specific installation works. Fees are also based according to these classifications.	For charging stations, requirements and fees are similar to those for other consumption points. In some Latin American countries, fees are based on the cost of providing the new connection and are not differentiated for EV charging. In Germany, charging stations with a capacity of more than 12 kilovolt amperes (kVA) require approval from the network operator. Charging devices with outputs of less than 12 kVA shall be registered with the distribution network operator. If a single EV charging point is connected to the low voltage grid, the connection is treated like any other connection in the low voltage grid. If the installation of a public charging station is done via direct connection to the medium voltage grid, the same regulation applies as with other devices connected to the medium voltage grid. Also, to accelerate the installation of charging stations at common locations (supermarkets, hotels, restaurants, etc), the German Federal Ministry of Transport and Digital Infrastructure issues calls for tenders for the construction and operation of charging sites and charging points.
Technical Standards	The Rwanda Standards Board (RSB) is the main authority responsible for issuing technical standards, quality testing, metrology, and certification. Current standards in Rwanda are benchmarked against international standards, including those adopted by other African countries. No specific standards for EVs and charging infrastructures have been issued in Rwanda.	The EU has adopted the connector Type 2 for EV charging, which is widely used across many European countries. The US has adopted the Type 1 connector for EV charging. In Latin America, some countries like Colombia have adopted the Type 1 connector, while others, such as Chile and Uruguay, have adopted the Type 2 connector. Although governments outside China have not officially adopted the GB/T connector, these have gained market penetration in various regions.

Table 16 Current Practices and development on eMobility in Rwanda

3.3 Stakeholder Situation

3.3.1 Public Stakeholders Involved on eMobility

According to the rationale of the National Transport Policy, MININFRA and its agencies are mandated to plan, develop, manage, and maintain an efficient and

integrated national transport infrastructure network, including roads, bridges, airports, railways, and water transport. This effort aims to facilitate economic development, social equity, environmental improvement, and regional integration. MININFRA will establish dedicated Transport Authorities for urban centers. These TAs will develop infrastructure and engage private contractors to operate various elements of the transport system. Each TA should be a full-time professional body.

The following table lists the main entities that are either related to, could eventually play a role in, or are interested in the deployment of charging infrastructure due to their respective responsibilities:

Authority/Government agencies	***Main Roles and Responsibilities related with EVCI***
Ministry of infrastructure (MININFRA)	The Ministry of Infrastructure's primary role is to develop policies, regulations, and technical standards, and to evaluate project performance to monitor progress toward national transport system goals. It supports the implementation, staffing, and capacity building of local Transport Authorities and supervises activities related to the implementation of national policies in areas such as habitat, urbanism, transport, energy, water, and sanitation. The Ministry also supports infrastructure development programs under decentralized structures as per district development programs. Additionally, it is responsible for developing and monitoring policies and strategies for road public and freight transport, establishing a national information system for road transport, and coordinating institutions involved in public and freight transport. The Ministry of Infrastructure is improving public transport infrastructure by expanding, rehabilitating, upgrading and maintaining roads to enhance the capacity of the country's public transport network while also contributing to sustainable economic development
Rwanda Utilities Regulatory Authority (RURA)	The Rwanda Utilities Regulatory Authority (RURA) sets guidelines for implementing and enforcing laws, ensuring fair and transparent compliance by public utilities. It maintains continuous service delivery, protects public interest, and promotes effective competition. RURA ensures the availability, accessibility, and affordability of services for all consumers, including disadvantaged groups, and encourages sector development in line with government policies. It also handles energy tariffs, manages tax exemptions for vehicles, land use, and permitting, and regulates public utilities, including energy, industrial gases, pipelines, storage facilities, and transport. Additionally, RURA and the City of Kigali approve permits for operating charging points on specific parcels.
Rwanda Energy Group (REG)	Rwanda Energy Group (REG) oversees the strategic direction and overall management of the country's energy sector. REG is responsible for implementing national policies related to energy production, distribution, and regulation. It ensures alignment with Rwanda's Vision 2050 and other international commitments, focusing on expanding access to electricity, promoting renewable energy sources, and improving grid infrastructure.
The Energy Utility Corporation Limited (EUCL)	Energy Utility Corporation Limited (EUCL), a subsidiary of REG, manages the day-to-day operations of electricity distribution and supply across Rwanda. EUCL's responsibilities include maintaining and upgrading the national grid, addressing power outages, and ensuring efficient energy distribution. The corporation also implements customer service initiatives, such as installing smart meters

Authority/Government agencies	*Main Roles and Responsibilities related with EVCI*
	and managing billing systems to enhance service delivery and operational efficiency.
The Energy Development Corporation Limited EDCL	Energy Development Corporation Limited (EDCL), another subsidiary of REG, focuses on the development of energy infrastructure and generation projects. EDCL is tasked with expanding Rwanda's energy capacity by constructing new power plants, promoting renewable energy projects, and managing large-scale energy investments. The corporation plays a crucial role in achieving national targets for energy access and sustainability by overseeing projects that increase both grid and off-grid energy solutions.
Ministry of Information Communication Technology and Innovation	The Ministry of Information Communication Technology and Innovation plays a crucial role in facilitating the integration and application of ICT in the Transport Sector, enhancing the efficiency and effectiveness of transport services through technological innovations and solutions.
Rwanda Environment Management Authority (REMA)	REMA is tasked with ensuring that environmental issues and climate change are fully integrated into Rwanda's development strategy. Its mandate includes mainstreaming environmental concerns and climate change across all sectors to enhance performance and productivity, reducing the country's vulnerability to climate change, and controlling pollution while enforcing environmental regulations to promote sustainable development.
Rwanda Inspectorate, Competition and Consumer Protection Authority (RICA)	RICA is responsible for protecting and promoting fundamental consumer rights while fostering healthy competition among businesses. The unit works alongside other departments to ensure that products and services in the market are both safe and competitively priced. Additionally, it conducts annual inspections, awareness campaigns, and training programs to enhance compliance with relevant laws.
Rwanda Housing Authority (RHA)	RHA is tasked to improve the quality of life of Rwandans through planning, organizing and spearheading rural settlement, urban settlement, public building construction, affordable housing; management of public office space and Government Assets; and regulation of the construction industry. RHA is responsible for the development of the country's building codes and standards.
Rwanda Development Board (RDB)	The Rwanda Development Board (RDB) is responsible for facilitating the involvement of the private sector in the Transport Sector, encouraging private investment and participation to enhance the development and efficiency of transport infrastructure and services.
Republic of Rwanda Road Maintenance Fund (RMF)	The Republic of Rwanda Road Maintenance Fund (RMF) is a legally established institution responsible for collecting and funding maintenance services for Rwanda's road networks. It manages and disburses funds for maintaining Class 1 national, district, and Kigali roads. RMF collaborates with relevant bodies to prepare road maintenance programs, ensures legal compliance of proposals and tender documents, oversees work according to plans and contracts, and has the authority to expropriate and compensate for damaged works as per legal provisions.
Rwanda Transport Development Agency (RTDA)	The Rwanda Transport Development Agency (RTDA) plans public and freight transport, develops mobility and accessibility requirements, and establishes networks, routes, and schedules. It manages finances and service contracts, monitors demand, promotes sustainable transport, oversees

Authority/Government agencies	*Main Roles and Responsibilities related with EVCI*
	local administrative activities, and provides technical support. The RTDA also collects and manages transport data, advises the government and local authorities, and offers technical assistance to the City of Kigali and secondary cities.
City of Kigali and Secondary Cities	The municipal government of Kigali, overseeing Nyarugenge, Kicukiro, and Gasabo districts, along with Secondary Cities, is responsible for planning, designing, and managing public transport and taxi services. They prepare City Transport Plans or District Transport Plans based on their jurisdiction. They also implement and maintain roads and other urban transport facilities and develop an inventory of roads, bridges, terminals, depots, and other transport facilities to ensure comprehensive transport infrastructure management.
Rwanda National Police (RNP)	The Rwanda National Police (RNP), under its regulatory portfolio, is tasked with safeguarding transport safety and security, ensuring the protection of passengers, drivers, and transport infrastructure across the nation.
Rwanda Standards Board (RSB	The Rwanda Standards Board (RSB) is responsible for establishing design standards for all types of land and water transport infrastructure in Rwanda, ensuring uniformity, safety, and quality across the nation's transport systems.

3.3.2 Private Stakeholders Involved on eMobility

The electric mobility sector in Rwanda is rapidly evolving, with a growing number of companies actively promoting EVs. This burgeoning market encompasses two-wheelers, passenger cars, and light and heavy-duty vehicles, contributing to a diverse and expanding EV landscape in the country. Several companies are leading initiatives to enhance EV adoption, focusing not only on vehicle sales and leasing but also on the essential charging infrastructure. These initiatives include the implementation of battery swap stations for two-wheelers and the establishment of charging stations tailored to passenger cars and other companies pioneering charging solutions specifically for e-buses, ensuring a comprehensive approach to EV infrastructure development in Rwanda.

The following table provides an overview of a number of key stakeholders and their respective initiatives towards e-mobility in Rwanda.

Company	*Business*	*Description*
Ampersand	Two electric wheeler sales and lease, Battery Swap Stations	Ampersand offers a pay-as-you-drive electric motorcycle (e-moto) service. Drivers either purchase or lease these e-motos and use Ampersand's Driver App to monitor battery levels. When the battery is low, they visit a battery swap station. In less than 2 minutes, the depleted battery is swapped for a new one, and the driver pays for the energy used. This system allows drivers to continue their day seamlessly, with the convenience of quick battery swaps as needed.
Spiro	Offer affordable EVs (Two wheelers) and Battery Swap Stations	Spiro is an electric two-wheelers manufacturing company operating in Rwanda. The company accelerates access to affordable, renewable energy technology offering cost-effective and eco-friendly electric bikes and scooters models. Spiro is also establishing a battery swapping network. Their

Company	Business	Description
		multiple Battery Swapping Stations allow riders or drivers of electric vehicles to trade their discharged batteries for fully charged ones, ensuring uninterrupted mobility.
Safi Universal Links Ltd – (SUL)	Offer affordable EVs (Two wheelers, passenger cars, vans, busses), sales of spare parts and capacity building.	SUL Mobility provides quality and affordable electric vehicles, including e-scooters, e-motorcycles, e-cars, e-vans, and e-buses, along with spare parts, catering to the growing market of private users and taxi operators in Rwanda and Africa. The company's mission is to offer high-quality, convenient, and comprehensive electric vehicle sales/leasing, spare parts sales, after-sales services, and capacity-building training. SUL Mobility has initiated several noble programs to support communities, focusing on capacity building and employment creation to reduce the high rates of unemployment among the youth.
Rwanda Electric Motorcycle Company (REM)	Electric Two wheelers, Manufacturing e-bikes and offering Battery swap services.	REM manufactures fully electric e-bikes with dual batteries that are rented and swapped when they are ready to be recharged and offers battery swap services. Developed with the Rwandan Motorcycle Federation, these bikes use a carbon-neutral battery exchange system. REM's batteries are charged at a central depot and distributed via a fully electric truck developed by REM. Partnering with Energicotel, REM ensures a reliable and renewable energy source. Services include battery swap stations in Kicukiro, Remera, and Nyabugogo, emergency battery swaps for REM riders, and retrofitting services.
KABISA	Sells and leases EVs, provides charging and maintenance services.	KABISA is a company dedicated to building out the (EV ecosystem in East Africa. Their services include selling and leasing electric vehicles, operating a public charging network, and providing electric vehicle maintenance services. Each vehicle sold by KABISA comes with an installed charger, and they also offer repair services in Kigali. The KABISA Charging Network is open to all EV and hybrid electric cars and operates a network of EV chargers across Rwanda. The locations of these charging stations are marked on the Waze app and the PlugSurfing App, making them easily accessible to users.
Basigo	Provides electric buses and charging solutions	BasiGo provides sustainable electric mobility solutions for African cities by integrating expertise in electric vehicles, public transport, the power sector, asset finance, software, and data science. In December 2023, they introduced Kigali's first 100% Electric Bus, the R10 Rumuri, powered by renewable energy from Rwanda. BasiGo's Pay-As-You-Drive lease model helps bus operators expand their fleets profitably. To support Kigali bus operators, BasiGo is deploying Rwanda's first DC Fast Charge network for electric buses. These high-power chargers, located minutes from bus routes, can recharge buses in less than 2 hours, ensuring

Company	Business	Description
		convenient and reliable charging 24/7 through dedicated infrastructure from REG. This allows BasiGo customers to operate electric buses with ease and no disruption to their daily operations
Victoria Motors Rwanda Ltd	Distributors of Mitsubishi vehicles	Victoria Motors Rwanda Ltd are the distributors of Mitsubishi Vehicles in Rwanda. They have launched a Plug-In Hybrid Electric Vehicle in Rwanda, committing to the distribution and promotion of EVs in the country.
Volkswagen Mobility Solutions Rwanda	Distributors of Volkswagen vehicles, partnering to launch charging stations.	Volkswagen Mobility Solutions Rwanda is a distributor of Volkswagen vehicles in Rwanda. The company began integrating electric cars into the Rwandan market in 2019, with a current fleet of 20 e-Golfs. These vehicles are primarily used by public institutions or operate as cabs. The e-Golfs are supported by two charging stations operated by Siemens, with plans for additional charging stations in the pipeline to support the growing number of electric vehicles.
IZI Rwanda	IZI offers e-mobility-as-a-service, including fleet management and maintenance for ebuses.	IZI is a company that has introduced an e-mobility-as-a-service model tailored for the African transportation market. This model simplifies the transition of bus fleets to electric power by offering a full-service leasing package at a low per-kilometer fee. It includes the IZI Connect fleet management system, which manages over 9,000 vehicles in China, monitoring battery health and supporting predictive maintenance. A Battery Laboratory in Kigali facilitates on-site battery cell repair and replacement. This setup aims to create specialized jobs and boost EV adoption across Rwanda and East Africa. The service eliminates upfront investment, fuel costs, and maintenance expenses, significantly enhancing fleet operator efficiency and customer satisfaction.
CFAO	Distributes global car brands, offering EVs and services regarding charging stations and maintenance.	CFAO Mobility is the exclusive distributor for multiple global car brands in Rwanda, including Volkswagen. They recently began distributing BYD electric vehicles, offering a comprehensive e-mobility package that includes top-class vehicles, charging stations, and energy storage solutions. CFAO Mobility supports Rwanda's ecosystem with services such as installing recharging stations, supplying solar power generation solutions, vehicle maintenance, assistance, and spare parts.

3.4 Current Energy System of Rwanda

Rwanda’s energy systems are managed by various entities including the Ministry of Infrastructure (MININFRA), the Ministry of Natural Resources (MINIRENA), the Ministry of Local Government (MINALOC), the Ministry of Finance and Economic Planning (MINECOFIN), the Ministry of Trade and Industry (MINICOM), Rwanda Energy Group (REG), Rwanda Utilities Regulatory Agency (RURA), and Independent Power Producers (IPPs). These parties collaborate to implement the

Energy Sector Strategic Plan (2017-2024), with MININFRA overseeing off-grid energy development and national energy policies. REG, a government-owned private holding, coordinates energy and investment plans through two subsidiaries: the Energy Utility Corporation Limited (EUCL), which manages utility services and maintenance of generation plants, and the Energy Development Corporation Limited (EDCL), responsible for the implementation of development projects and electrification initiatives.

The following table shows the key data regarding Rwanda's electricity sector status in numbers as of 2024:

Parameter	*Details*
Conventional installed capacity	By 2024: 20% Methane, 17% Peat, and 5% Diesel.
Renewables installed capacity	By 2024: 53% Hydroelectric, 6% Utility Scale On-Grid PV, and 2% Small Scale Off-Grid PV.
Total installed capacity	276 MW in 2022, to reach 556 MW by 2024
Electricity access	74.5% of the population in 2022, to reach 100% by 2024 (52% On-Grid Connections and 48% Off-Grid Connections, mainly Stand-Alone Solar Systems (SAS) and Microgrids)
Electricity networks' voltage level	High Voltage: 70, 110 (main TNs), and 220 kV (for larger generation power plants & interconnection lines) Medium Voltage: 6.5, 15 (Kigali and Rwamagana), 17.32, and 30 kV (most DNs) Low Voltage: 0.4 kV

Table 17: Rwanda's electricity sector in numbers

3.4.1 On-grid Rwanda's electricity status

At the beginning of the 21st century, Rwanda's electricity sector began undergoing rapid changes. The following sections outline the on-grid and off-grid electricity status in Rwanda. Rwanda has developed a Least Cost Power Development Plan (LCPDP) for 2023-2050 to address generation expansion and meet forecasted demand. This plan is divided into two stages: the entry of committed projects (2023-2028) and the long-term phase (2028-2050).

Rwanda's Generation System:

As of 2023, Rwanda's total installed capacity is 332.6 MW, with an available capacity of 213 MW. The energy mix includes 51% from thermal power plants, 43.9% from hydropower plants, and 4.2% from solar photovoltaic (PV) sources. The share of installed capacity by generation source is illustrated the next figure:

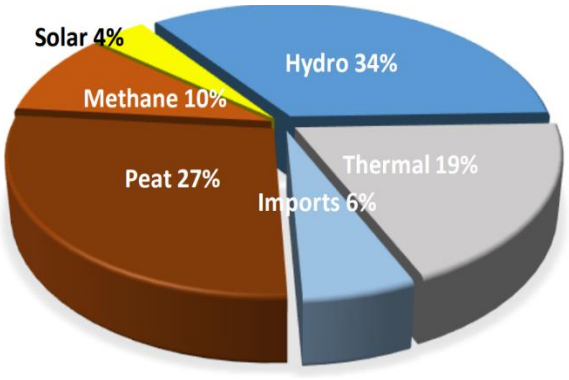


Figure 23: Share of installed capacity by generation source[57]

Rwanda's electricity sector faces several challenges across generation, transmission, and distribution networks. In generation, the key issues include limited resources, especially during dry seasons when hydro generation is insufficient, the need for generation expansion to match demand growth, the development of peaking power plants and energy storage devices for efficient load balancing, and overall system stability. Solutions proposed in the Long-Term Comprehensive Development Plan (LCPDP) include a load forecast study extending to 2050, a generation expansion plan in two stages (2023-2028 and 2028-2050), and an increased emphasis on renewable energy in the later stage.

Rwanda's Transmission System:

Rwanda's Transmission Development Plan (2023-2030) includes an updated model of the entire national transmission system. The next figure illustrates Rwanda's electricity networks.

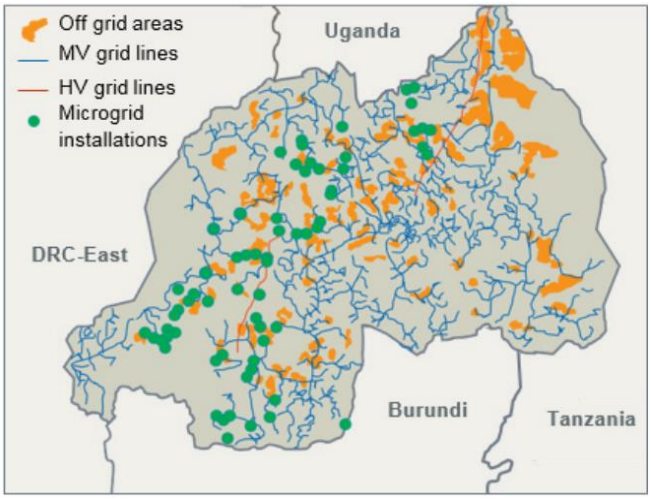


Figure 24: Rwanda's electricity networks

Transmission network challenges identified in the Rwanda's Transmission Master Plan include an isolated national grid with weak links to neighboring countries, numerous substations with single supply sources or lack of n-1 connections, single evacuation lines for major power plants leading to supply loss during line failures,

[57] Rwanda_LCPDP_Plan_2023-2050_-_June 2023: https://www.reg.rw/fileadmin/user_upload/Updated_Rwanda_LCPDP_Plan_2023-2050_-_June_2023.pdf

and frequent voltage drops due to overloaded high-voltage conductors and long distances. The plan proposes adhering to transmission guidelines for efficient distribution, installing 27 transmission projects by 2024 with an additional 21 projects from 2025 to 2029, and upgrading and installing transformers as detailed in the relevant tables.

It must be noted that almost all interviewed stakeholders for the construction of this master plan expressed concerns regarding the status of transformer funding policies, which have recently ended. The current limitations of distribution transformers are causing continuity of supply issues in distribution networks (DNs). In relation to e-mobility and the critical role of the energy system, upgrading and installing new transmission lines and transformers, as illustrated above, will significantly reduce the negative impacts caused by the installation of EV chargers.

Rwanda's Distribution System:

The Distribution Master Plan[58] is followed by DSOs. The Rwandan Distribution Network (DN) comprises 33 sub-stations (HV to MV) and is modeled using DigSilent Power Factory simulation software. DN's MV feeders' losses are 5.75% and 6.45% for peak and off-peak average scenarios, respectively. The overall aggregated load forecast for Rwanda predicts an average growth of 6.4% through 2030.

DNs are divided into five hubs aligned with the country's administrative boundaries:

- **Kigali Hub**: Nyarugenge, Kanombe, Kacyiru, Kicukiro, Rulindo, Bugesera, Gicumbi, and Jabana.
- **Northern Hub**: Gakenke, Musanze, Rubavu, Burera, Nyabihu, and Ngororero.
- **Southern Hub**: Muhanga, Nyanza, Huye, Gisagara, Nyamagabe, and Nyaruguru.
- **Eastern Hub**: Rwamagana, Kayonza, Ngoma, Kirehe, Gatsibo, and Nyagatare.
- **Western Hub**: Karongi, Rutsiro, Nyamasheke, and Rusizi.

Focusing on Kigali, the city's population is projected to reach 3.8 million by 2050 due to migration trends. Kigali is expected to grow between 6.5% and 10% over the next three decades, with electricity access at 96% in 2023, anticipated to reach 100% by 2024.

The distribution network also faces issues such as overstretched low-voltage networks causing voltage drops, weak medium-voltage feeders unable to handle load transfers, old cables causing power cuts and delays in fault identification, a weak single-phase network, the absence of accurate distribution network maps limiting expansion planning, and long distances from high-voltage substations to load centers. Solutions outlined in the Electricity Distribution Master Plan include upgrading substations, implementing a Distribution Management System (DMS) and smart meters, extending and upgrading medium and low-voltage lines, renovating existing electro-mechanical cabins, reducing feeder lengths, and transitioning from overhead to underground cables in developed areas of Kigali.

3.4.2 Off-grid Rwanda's electricity status

Rwanda's Rural Electrification Strategy was approved in June 2016. It aimed at providing households access to cost effective electricity and increasing private

[58] https://www.reg.rw/fileadmin/user_upload/Rwanda_Electricity_Distribution_Master_Plan_-_updated_June_2022.pdf

sector participation. EDCL signed MoUs with 27 private companies (i.e. suppliers) to increase SAS (14% of Rwandan's households).

3.5 Existing Business Models in Rwanda

E-mobility is an emerging sector in Rwanda and a handful e-mobility players have responded to government policies on e-mobility and have established their businesses providing e-mobility services including investments in EV charging infrastructures.

Electric mobility in the country at present is mainly driven by the private sector. A survey and interview with representative and dominant service providers, KABISA for passenger transport, AMPERSAND for motorcycles, and BasiGO for buses, were carried out to understand their business models.

The existing EV charging business models can be summarized as follows:

- The CPO and EMSP functions are vertically integrated by each company.
- Investments are sourced from debt and equity financing. Some companies received grants from either international or national innovation funds.
- These companies procure their sites through land rentals and made their own arrangements with respect to grid connection.
- Also, these companies are availed on the incentives provided by the government through the e-mobility policy.

	Passenger and Utility Vehicles (KABISA)	***Motorcycles (AMPERSAND)***	***Buses (BasiGO)***
Electric Mobility Business Model	Financing of electric vehicles. Public charging infrastructures were established as part of its own electricity mobility ecosystem.	Financing of Motorcycles for Commercial Use. Battery Charging at Swapping Stations	Pay-as-you-drive business model. Operators pay BasiGo based on kilometers driven each day. Pay-As-You-Drive includes charging and maintenance provided free-of-charge by BasiGo.
Funding Received by Service Providers and Incentive Mechanisms			
Funding	*Equity funds/grants* Not available	*Equity funds/grants* Beyond Capital Ventures AlphaMundi Group (AMG) StartupBootcamp, FactorE Ventures, Ecosystem Integrity Fund, Total Energies Venture arm, Rwanda Green Fund, USAID's Development Innovation Ventures, Shell Foundation	*Equity funds/grants* Grant funding from USAID's Development Innovation Ventures (DIV).

	Passenger and Utility Vehicles (KABISA)	***Motorcycles (AMPERSAND)***	***Buses (BasiGO)***
		the UK FCDO's Frontier Technology Livestreaming fund New Zealand Government	
	Loan Not available	*Loan* Blue Haven Initiative Catalytic Fund	*Loan* Not available
Incentives	Charging station equipment is exempted from import and excise duties (zero-rated VAT products and exempt from withholding tax). Electricity tariffs for charging stations are capped at the industrial tariff. Electric vehicles to benefit from reduced tariff during off-peak times. Rent free land for charging stations (for government-owned land)		
Charge Point Operator (CPO) Functions			
Real Estate	Private Rental by KABISA	Private Rental by AMPERSAND	Private Rental by BasiGO
EPC	KABISA	AMPERSAND	BasiGO
EVSE Supply			
O&M			
Power Supply	*Energy Utility Corporation Limited (EUCL)* *(state-owned utility)*	*Energy Utility Corporation Limited (EUCL)* *(state-owned utility)*	*Energy Utility Corporation Limited (EUCL)* *(state-owned utility)*
Electric Mobility Service Provider (EMSP) Functions			
Customer Service	KABISA	AMPERSAND	BasiGO
Payment Collection			
Interoperability	No	No	No

Table 18: Existing EV Charging Infrastructure Business Models in Rwanda

4 POTENTIAL FUTURE SCENARIOS

4.1 Transport Future Scenario

Rwanda has the potential to leapfrog the traditional fossil fuel-based infrastructure for traffic and transport due to its currently low motorization rate. Unlike Western countries, which have seen an unforeseen surge in private car ownership necessitating the development of the e-mobility market, Rwanda and other African nations are uniquely positioned to enhance alternative modes of transport such as public transportation and e-bikes, which can effectively compete with the car system.

Firm national policies, like those implemented by Rwanda, play a crucial role in fostering these developments. These policies can drive significant improvements compared to less progressive approaches. To anticipate future developments, we considered two scenarios, each subdivided into three regions:

- Kigali (the capital city) with (1,517,168 inhabitants)
- The ten major cities following Kigali (3,154,759 inhabitants)
- Rural areas, including transport between towns and cities (8,5 million inhabitants)

The projection scenarios are based on demographic data and the observation that the number of vehicles in Rwanda is increasing by 12% annually. In 2020, there were 264,524 registered vehicles in Rwanda, with 70% of these registered in Kigali. While vehicle growth in Kigali is 6.2% per year, the rest of the country is experiencing even higher growth rates. The National Mobility Plan for Rwanda indicates that by 2034-2035, 70% of newly introduced cars will be electric vehicles, reflecting the country's commitment to sustainable transportation solutions.

4.1.1 First scenario: Doubling Current Car Ownership Rates

Scenario 1 projects that by 2035, Rwanda will have a total of 200,000 private cars, equating to approximately 14 cars per 1,000 inhabitants. This scenario is based on doubling the current car ownership rate, which stands at 7 cars per 1,000 inhabitants, indicating there are currently around 100,000 private cars in Rwanda.

This scenario outlines the expected distribution and type of vehicles, along with the improvements in public transportation and other modes of transport. By 2035, Kigali's transportation landscape is projected to feature significant advancements in electric and public transport. The city is expected to have 100,000 e-bikes and 20,000 e-motor bikes, replacing traditional moto-taxis and playing a crucial role in urban mobility. Kigali will also have 140,000 private cars, with 70% (100,000) being electric, and 75% of these electric cars (75,000) will be used daily. The public transport system will be enhanced with 75 articulated e-buses on BRT lines and 300 e-buses on extended bus lines. Additionally, around 8,000 trailers and trucks will be based in Kigali, with 50% electrified, contributing to a cleaner and more efficient transport system.

In the ten major cities following Kigali, the transportation system will see a similar shift, with slightly more emphasis on non-motorized transport. These cities are projected to have 40,000 private cars, 28,000 of which will be electric, along with 50,000 e-bikes and 5,000 e-motor bikes[59]. Public transport will include 200 e-buses within towns and 100 e-buses between towns, with 50% of trailers and trucks being electrified. In rural areas, private car density will remain lower, with 20,000 cars (14,000 electric), complemented by 100 inter-city e-buses and 100 intra-rural e-buses, with the potential for e-bike adoption contingent on improved road conditions.

The following table summarizes those projections:

	Kigali	After Kigali: The Ten Major Cities	Rural Areas
Two wheelers	It is projected to have 100,000 e-bikes by 2035, playing a significant role in urban transport. It is projected to have 20,000 e-motor bikes replacing traditional moto taxis.	It is projected to have 50,000 e-bikes and to have 5,000 e-motor bikes.	E-bikes usage is contingent on improved road conditions.
Private and passenger Cars	It is projected to have a total of 200,000 private cars in Rwanda, with 70% of them (140,000) located in Kigali. Of these, 70% (100,000) will be electric cars, and 75% of these electric cars (75,000) will be used daily.	It is projected to have a total of 40,000 private cars, and 70% of these (28,000) will be electric cars.	It is projected to have a total of 20,000 private cars, and 70% of these (14,000) will be electric cars.
Public Transport	It is projected to have 75 articulated electric buses operating on Bus Rapid Transit (BRT) lines and an additional 300 electric buses running on extended bus lines.	It is projected to have 200 e-buses within towns. It is projected to have 100 e-buses between towns.	It is projected to have 100 inter-city e-buses. It is projected to have 100 intra-rural e-buses.
LDV and HDV	It is expected to be around 8,000 trailers and trucks based in Kigali, 50% will be electrified, contributing to a cleaner and more efficient transport system.	It is projected that trailers and trucks in these cities will be 50% electrified.	It is projected that trailers and trucks in rural areas will follow the electrification trend as infrastructure improves.

Table 19: Projected Transport Growth in Rwanda by 2035 (Scenario 1)

Modal Split in Kigali on the first scenario

By 2035, the modal split in Kigali is projected to be as follows: 52% of all trips will be made by cars and motorcycles, indicating a significant reliance on private vehicle usage. Non-Motorized Transport (NMT), which includes walking and cycling (and e-bikes), will account for 28% of the trips, highlighting the importance of sustainable and active transportation modes. Public Transport (PT), including buses and other forms of mass transit, will make up 20% of the total trips, reflecting the growing investment in and reliance on public transportation infrastructure to meet the city's mobility needs. The next figure shows the modal transportation split:

[59] An e-bike is primarily a bicycle propelled by the rider's pedaling. It typically includes an electric motor that assists by providing additional power, but this assistance only activates when the rider is pedaling. E-bikes are generally charged at home using a standard electrical outlet. In contrast, e-motorbikes are fully electric motorcycles where the motor provides 100% of the propulsion, without any need for human effort. They are designed for higher speeds and are typically much heavier, and can be charged at home or via battery swapping stations.

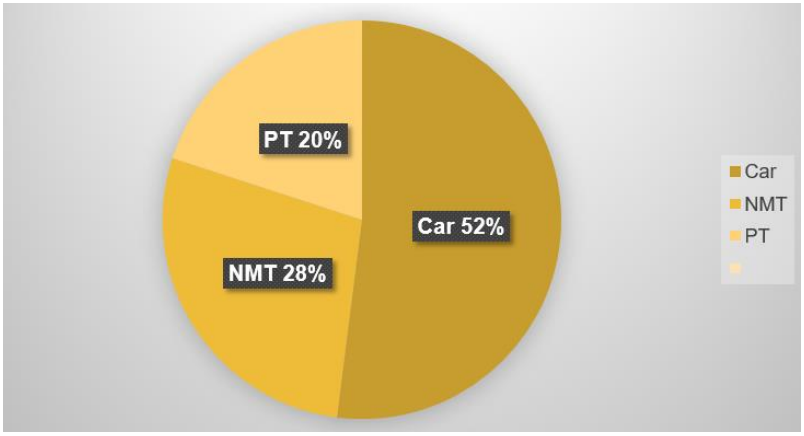


Figure 25: Scenario 1: Expected Growth on Modal Transportation by 2035

4.1.2 Second scenario: Tripling Current Car Ownership Rates

Scenario 2, which is based on a growth of the economic and demographic projections, states that by 2035, Rwanda will have a total of 300,000 private cars, equating to approximately 21 cars per 1,000 inhabitants. This scenario is based on tripling the current car ownership rate, which stands at 7 cars per 1,000 inhabitants, indicating there are currently around 100,000 private cars in Rwanda. This projection reflects a substantial increase in car ownership, highlighting the need for significant enhancements in infrastructure and sustainable transport solutions to manage the growing number of vehicles effectively.

This scenario projects that the city of Kigali will house 60% of these 300,000 cars; there will be 180,000 private cars, of which 125,000 will be electric. It is projected that 75% of these electric cars (93,750) will be used daily. Kigali will enhance its public transport infrastructure with 75 e-buses for BRT lines and an additional 300 e-buses for extended bus lines. The city will also see significant adoption of e-bikes (90,000) and e-motor bikes (25,000). In the ten major cities, there will be 50,000 private cars, with 35,000 being electric. These cities will also benefit from 200 intra-city and 100 inter-city e-buses, along with 50,000 e-bikes and 5,000 e-motor bikes. Rural areas will maintain lower car density, with 20,000 private cars, half of which will be electric, and improved public transport featuring 200 e-buses.

The following table summarizes those projections:

	Kigali	After Kigali: The Ten Major Cities	Rural Areas
Two Wheelers	It is projected to have 90,000 e-bikes by 2035. It is projected to have 25,000 e-motor bikes replacing traditional moto taxis.	It is projected to have 50,000 e-bikes. It is projected to have 5,000 e-motor bikes.	E-bikes usage is contingent on improved road conditions.
Private and Passenger Cars	Kigali is projected to have 180,000 private cars, and 70% of these (125,000) will be electric cars. It is projected that 75% (93,750) of these electric cars will be used daily.	It is projected to have a total of 50,000 private cars, and 70% of these (35,000) will be electric cars.	It is projected to have a total of 20,000 private cars, and 50% of these (10,000) will be electric cars.
Public Transport	BRT lines are projected to have 75 articulated e-buses. The bus lines are projected to have 300 e-buses running on extended bus lines.	It is projected to have 200 e-buses within towns. It is projected to have around 100 e-buses between towns.	It is projected to have 100 inter-city e-buses. It is projected to have 100 intra-rural e-buses.
LDV and HDV		It is projected that trailers and trucks will be electrified for 50%.	Electrification of trailers and trucks will follow as infrastructure improves.

Table 20: Projected Transport Growth in Rwanda by 2035 (Scenario 2)

Modal Split in Kigali on the second scenario

By 2035, it is assumed that the modal split for transportation in Rwanda will be comprised of 55% cars and motorcycles, 25% non-motorized transport (NMT), including e-bikes, and 20% public transport (PT). This projection indicates that more than half of all trips will be made by private cars, reflecting the increased car ownership anticipated in Scenario 2. Non-motorized transport, which includes walking and cycling, will account for a significant portion of trips at 25%, highlighting the importance of sustainable and active transportation modes. Public transport will make up 20% of the modal split.

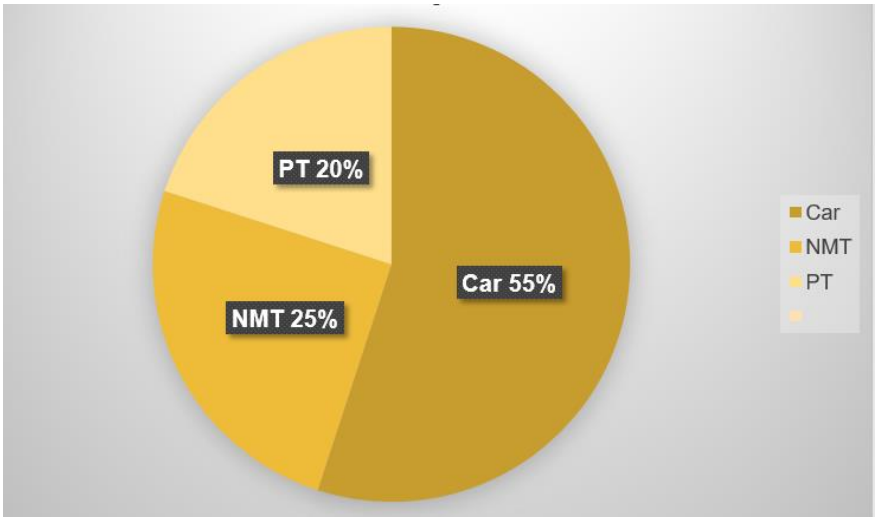


Figure 26: Scenario 1: Expected Growth on Modal Transportation by 2035

4.2 Energy Future Scenario: EVs'Chargers-Embedded-Energy System of Rwanda

Rwanda has developed a Least Cost Power Development Plan (LCPDP) for 2023-2050 to address generation expansion and meet forecasted demand. This plan is divided into two stages: the entry of committed projects (2023-2028) and the long-term phase (2028-2050).

Least Cost Power Development Plan (LCPDP) (2023-2050) concluded a forecasted total energy and peak demands as shown in Figure 27: LCPDP forecasted total energy demand and total peak demand growth. This demand forecast will be satisfied by a total installed capacity and peak demand energy mix resources shown in Figure 28: Total Installed Capacity and Peak Demand Figures.

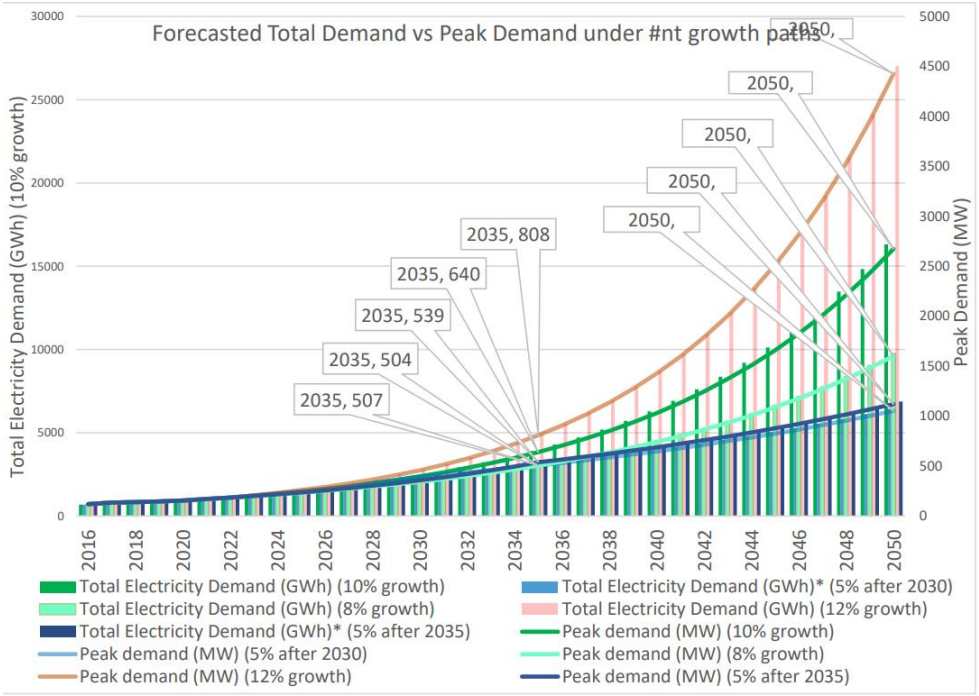


Figure 27: LCPDP forecasted total energy demand and total peak demand growth

In the last decades, raising awareness of global economic and environmental issues related to pollution of conventional fossil fuel-based vehicles promoted fast transition to electrify transportation. EVs' industry spread worldwide; however, this industry introduces various challenges that will impact electricity networks in two aspects; infrastructure and operation.

During dry seasons, when there is a water shortage, diesel generation units are rented at a high cost to fulfill peak demand and compensate for the loss of hydro generation. To address this issue and diversify Rwanda's energy mix, the Least Cost Power Development Plan (LCPDP) has considered incorporating other renewable resources. One of the main objectives of the LCPDP is to maximize the use of renewable energy within the country's energy mix.

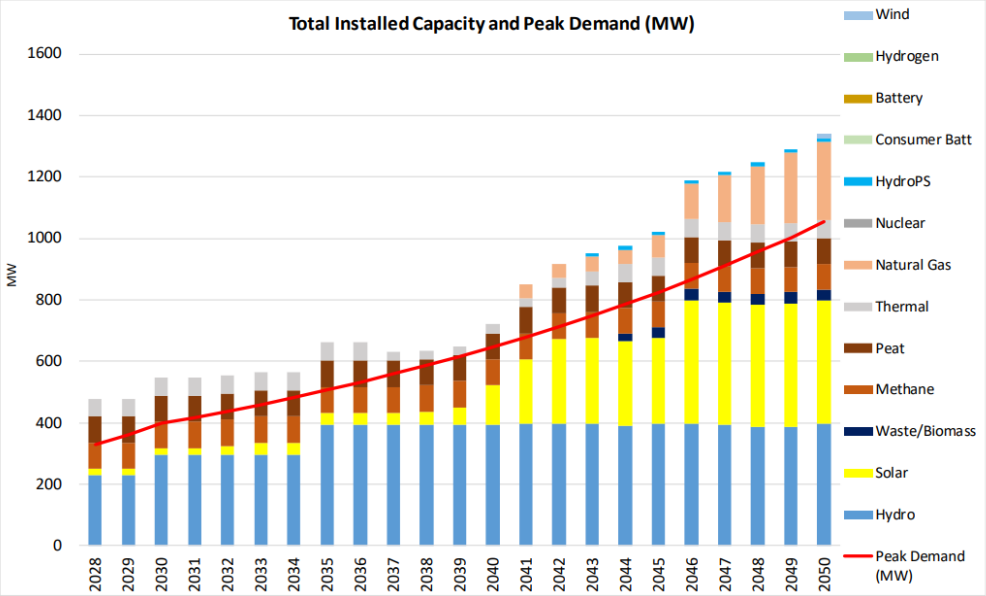


Figure 28: Total Installed Capacity and Peak Demand

4.2.1 Future grid infrastructure plans

From an electric utility infrastructure perspective, EVs need additional energy in Distribution Networks (DNs) to be charged. This need will increase the amount of energy required to be supplied from Transmission Networks (TNs) to DNs, if these networks have the proper capacity to transmit this additional amount of energy needed to charge EVs. Otherwise, if the existing networks' capacities are not sufficient, an upgrading (i.e. investment) is required along with associated equipment in both TNs and DNs; substations, transformers, control stations/networks (SCADA)/devices, circuit breakers and fuses, protective devices, etc.

On the other hand, another option to prove this additional energy can be obtained by converting DNs from passive to active networks; i.e. promoting the installation of RE-Based Distributed Generation (DGs) on DNs.

Having EVs as a mean of transportation requires additional infrastructure which was not part of electric networks earlier, which are:

- EV charging stations (technical equipment). Several technologies are used, AC & DC chargers each have their own requirements.
- Means of integration of such EVs charging stations with existing networks (i.e. On-Grid charging stations such as cables, safety equipment).

EV charging station allocations are obtained based on various factors, such as: access to the electric network while satisfying the necessary electrical and geographical constraints obtained from GIS software, drivers' behavior and routes, candidate electric-transportation type (2&3&4 wheels vehicles, buses).

EVs charging station sizing are obtained based on many factors, such as: access to an electric network while satisfying the necessary electrical constraints obtained from GIS software, drivers' behavior, numbers, routes, and EVs' battery sizes, and charging sessions durations.

These infrastructure selections must ensure proper operating conditions such as selecting candidate model of EV charging stations and evaluation of impacts of such

charging stations on electricity networks. Figure 29: Studies necessary for introducing EVs chargers into electric networks shows a summary of studies necessary for introducing EVs chargers into electric networks.

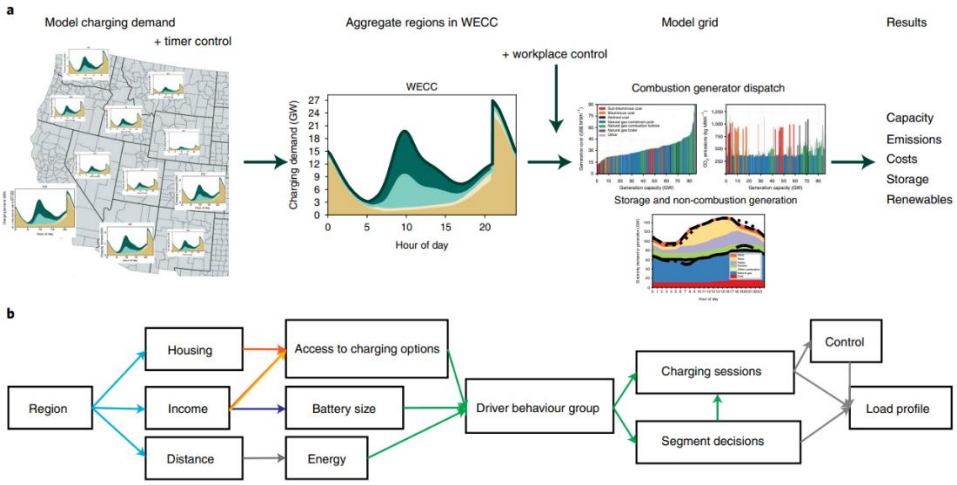


Figure 29: Studies necessary for introducing EVs chargers into electric networks

4.2.2 Renewable electricity generation

Most countries have RE targets following their electricity strategies. Rwanda has RE potential in biomass, solar, peat, geothermal, and hydropower technologies.

- Biomass (mainly wood fuel and animal waste) technology is the most used. Biogas is promising for thermal energy need for farms and small institutions.
- Solar irradiation satellite data shows that Rwanda has a daily global radiation (annual average) is between 4-6 kWh/m²/ day and a sunshine period of 8 hrs/day. Figure 30: *Rwanda's* NEP, 2023 4.2 shows the National Electrification Plan (NEP) with SAS.

In general, Figure 30: *Rwanda's* NEP, 2023 shows a significant contribution of solar energy in fulfilling the country's electrical demand along its territories. However, Figure 31: Rwanda's NEP, Kigali City, 2023 shows the NEP for Kigali city which relies mainly on GE not SAS.

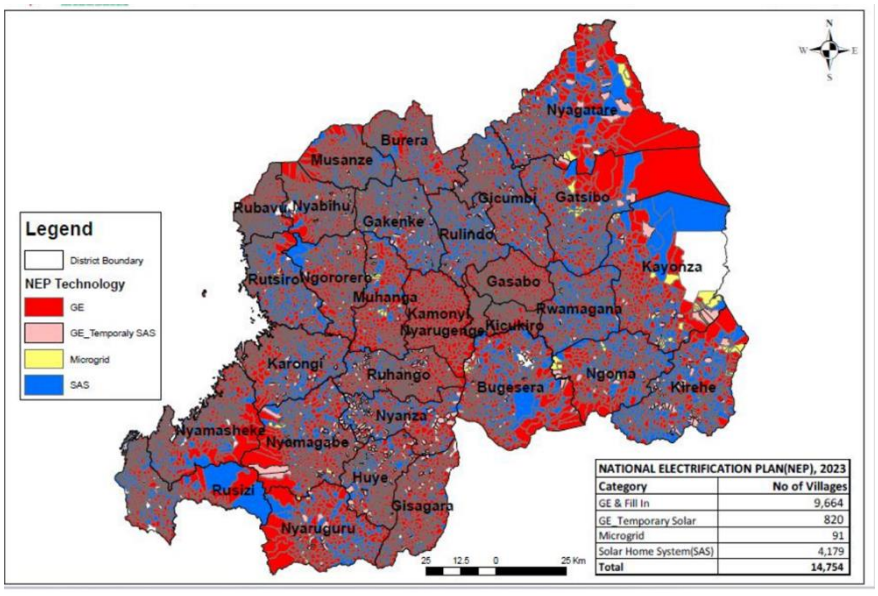


Figure 30: Rwanda's NEP, 2023

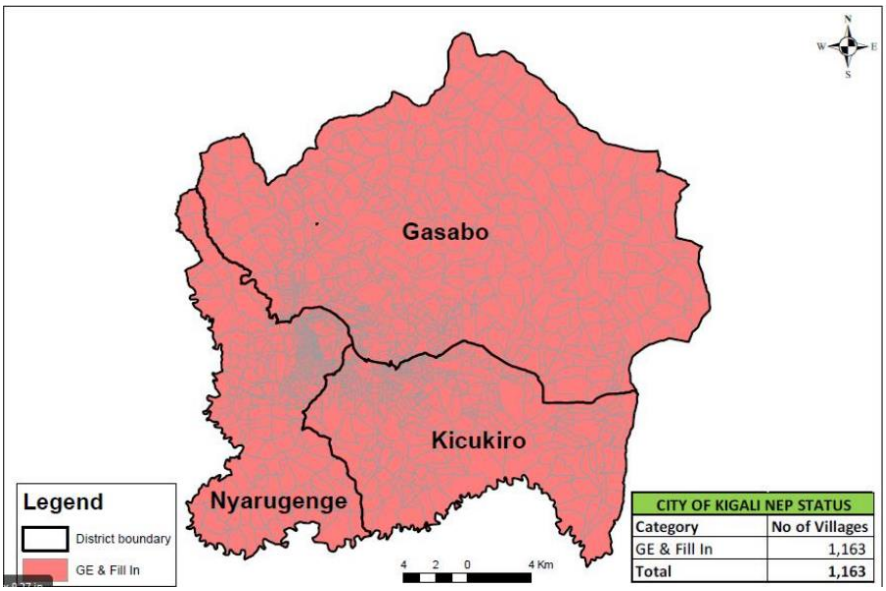


Figure 31: Rwanda's NEP, Kigali City, 2023

Geothermal technology exploration in Rwanda is still at the geoscientific surface exploration stage.

Hydro-generation technology is dominant (50% of Rwanda's energy use). The Hydropower Atlas has identified 70 hydro sites in the country. The hydropower plants are publicly owned and operated, leased to private companies, or privately owned i.e. IPP. While micro-hydropower technology supplies for off-grid customers constitute a significant potential for the rural power supply.

Promoting RE projects to provide the needed additional energy required for charging EV in electricity networks can be obtained through:

- Offering the installation of PV mega-projects integrated to TNs. However, this option requires pre-set regulations, proper technical codes (Grid Code and PV Code), agreed-upon contractual mechanisms/agreements (ex. Feed-in-Tariff (FiT)) with TNs as an off-taker, public/private/Public-Private-Partnership (PPP) investments plans, etc.
- Installing rooftop PV on-grid systems are considered a valuable option in Rwanda to provide the needed energy to EVs. This scenario will convert passive DNs to active ones. Like the above discussion, this option requires pre-set regulations, proper technical codes (Distribution Code and Small-Scale PV Code), agreed-upon contractual mechanisms/agreements (ex. Net-Metering and/or FiT) with DNs as an off-taker, investments, etc... The investment in this option can be for utility-owned and/or customer-owned projects.
- Installing PV off-grid systems along with batteries in remote areas which is the source of energy for stand-alone EVs charging stations. This investment must start as a utility investment case to promote this type of technology. After the maturity of this scheme, private investment can be implemented.

4.2.3 Assessment of Grid-Level Impacts: Grid impact and recommendations of suitable charging technology to minimize adverse impacts of EV charging loads on the grid

In general, EVs as Virtual Power Plant (VPP) can positively impact electricity networks' frequency and voltage regulations. Ancillary service contracts allow Distribution Network Operators (DNOs) to control contracted EVs to discharge certain amounts of energy, at certain locations in a network, and during certain periods to support electricity network. However, EVs during charging may negatively impact electric networks in various aspects. Mainly two aspects are considered; a) energy-related impacts which reflects infrastructure and operation issues and b) power quality impacts as shown in the next table

Network Energy Impact	*Network Power Quality Impact*
Local power shortage (lack of generation and/or transmission lines congestion) Transformers overloading Load profile non-homogeneity (unflatten) Changing networks' power factor	Voltage Imbalance Harmonics Stag/Swell Flickers Notches

Table 21: EVs charging impacts on networks' parameters

Impact of EVs charging on electric networks:

Network energy impact

Looking at EVs as heavy energy consumable loads, they will momentarily change DNs' power-balance and load profiles (unflatten) compared to a regular profile, as seen by DNOs in the next figure[8].

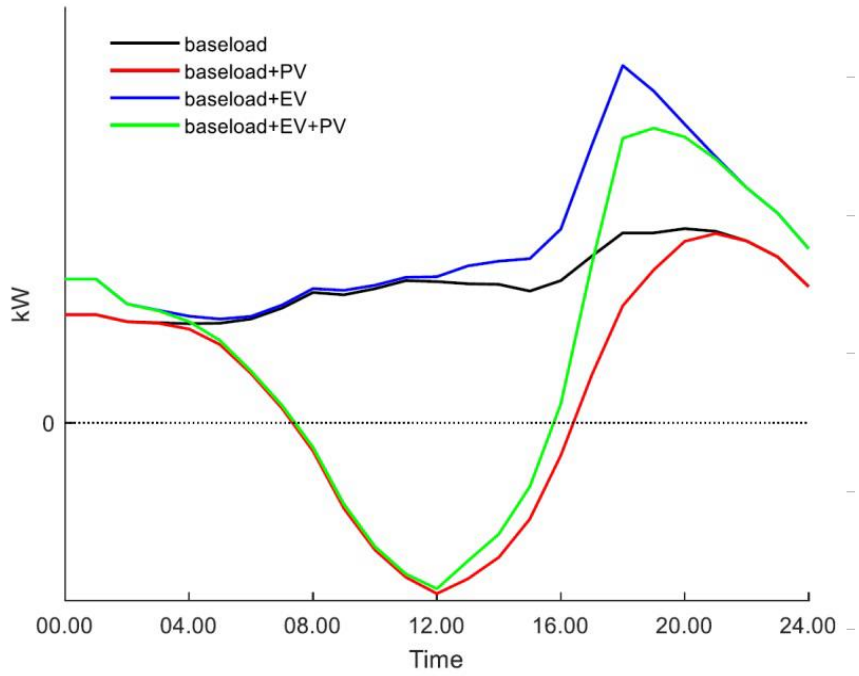


Figure 32: A residential load curves of different scenarios

Heavy EVs charging will increase the burden of estimating the amount of energy required to be supplied from TNs to DNs, both by the timing of EV charging as well as the peak load. Besides local equipment overloading, a change in the networks' power factor might occur, specifically in heavily inhabitant areas where EVs charging patterns are based on EVs owners' behavior. However, as shown in Figure 32: A residential load curves of different scenarios, combining the use of high PV penetration while having a high EV charging in DNs will reduce the resultant load peak compared to having EVs charging alone in DNs. Therefore an integral approach towards RE and EV is preferrable.

Network power quality impact

Power quality issues that are created by EVs charging stations are somehow near to those obtained during operating variable RE technologies (ex. PV and wind) and charging/discharging Energy Storage Devices (ESDs). EVs charging stations have power electronics devices that rely on switching semiconductor-based elements, thus, produce harmonics injection, and voltages' imbalance, stag/swell, flickers, and notches.

5 TECHNICAL ROAD MAP: CHARGING INFRASTRUCTURE DEPLOYMENT

5.1 Technical Road Map Chapters Overview

A charging implementation strategy should contain a mix of charging solutions at various locations that result in an optimal charging experience and convenience for electric vehicle users. An overview of different charging solutions was presented in chapter 2. Ideally, different charging options complement each other and consider the future charging needs of different users and various transport modalities. In this regard, a charging implementation strategy is partially reactive to insights gathered from forecasted demand and should be sufficient to fulfill the specified future charging needs. On the other hand, a charging implementation strategy is an instrument that shapes the transport system, as future electric vehicle users and other actors in the system will choose their mode of transport and shift towards electrified modes of transport based on the availability of charging infrastructure and the user experience of the charging infrastructure.

In Chapter 5, the focus is on the technical elements, their characteristics, and their technical interaction with other system elements. This chapter will contain recommendations on where to strategically place chargers for the different modes of transport considering expected transport system developments and the limitations of the electricity grid.

A charging implementation strategy should contain a mix of charging solutions at various locations that result in an optimal charging experience and convenience for electric vehicle users. An overview of different charging solutions was presented in chapter 2. Ideally, different charging options complement each other and take into account the future charging needs of different users and various transport modalities. In this regard, a charging implementation strategy is partially reactive to insights gathered from forecasted demand and should be sufficient to fulfill the specified future charging needs. On the other hand, a charging implementation strategy is an instrument that shapes the transport system, as future electric vehicle users and other actors in the system will choose their mode of transport and shift

towards electrified modes of transport based on the availability of charging infrastructure and the user experience of the charging infrastructure.

5.2 Deployment of EV charging

A typical long-term strategy for passenger vehicle charging infrastructure in any region involves a series of intertwined steps tailored to the specific local context. These steps, which evolve with the sector, include implementing a DC fast charging network to alleviate range anxiety, establishing an extensive AC charging network both privately and publicly for widespread charging during extended parking periods (such as overnight or at work), and introducing opportunity charging options that provide quick top-up charging services during short parking stays at locations like retail chains, public service locations, and guest parking areas. Privately, AC charging can be facilitated at residential homes, apartment complexes, and workplace parking facilities, including corporate fleets. Publicly, AC charging can be implemented along public streets.

5.2.1 Principles of Location Planning

- **Accessibility:** Charging stations must be easily accessible to users. This involves placing stations in high-traffic areas, near major roads, and within residential and commercial zones. Ensuring accessibility promotes convenience and encourages EV adoption.
- **Coverage:** A well-distributed network of charging stations is essential. Both urban and rural areas should have adequate coverage to support long-distance travel and reduce range anxiety.
- **Grid Capacity:** Locations should be chosen based on the existing grid infrastructure and capacity. Sites with robust electrical supply can support high-power charging stations with minimal additional investment.
- For areas without connectivity to the national grid or with insufficient energy demand, **off-grid solutions** such as solar panels and **battery storage** should be considered. This is particularly relevant for rural petrol stations and remote locations.
- **Land Use Compatibility:** Charging stations should be integrated into existing land use plans, ensuring they do not disrupt other urban functions. Preferred locations include parking lots, shopping centers, public transit hubs, and service stations.
- **Future Expansion:** Planning should consider future EV adoption rates and the potential need for additional infrastructure. Sites should have the flexibility to accommodate more charging points as demand grows.

5.3 Factors to consider the demand on EVCI

Determining the optimal placement and timing for installing additional EV charging stations requires careful consideration of several key elements. Knowing when to install more EV charging stations requires a holistic approach that considers factors such as both current and projected demand for EV, current and projected charging demand, parking space utilization and data-driven insights.

First, this involves evaluating factors such as the **number of EVs in use, projected EV sales, and the charging habits of drivers**. By understanding the charging

demand distribution, including the types of vehicles and their charging behaviors, stakeholders can better anticipate where and when additional charging infrastructure may be needed.

Furthermore, the availability and utilization of parking spaces play a crucial role in estimating charging demand. **Analyzing parking space utilization rates** provides insights into how many EVs are likely to be parked at any given time, indicating potential demand for charging services in those areas. Additionally, **considering the location and accessibility of existing charging stations helps identify gaps in coverage** and areas with limited charging infrastructure, signaling opportunities for expansion.

Data-driven approaches, such as utilizing traffic flow data and location-based models, can provide valuable insights into where to install additional charging stations. Algorithms for demand estimation and station planning can help optimize the placement of chargers based on factors like **traffic density, population density, and points of interest**.

Various methodologies are employed to estimate charging demand, including point-based and flow-based methods. Point-based methods utilize data such as parking information from trips and employ regression models incorporating variables like population density to estimate parking demand. These methods typically assume a proportional relationship between parking demand and charging demand. On the other hand, flow-based methods utilize traffic flow data to estimate charging demand, assuming that areas with heavier traffic may experience higher charging demand. These methodologies, along with others such as Origin-Destination (OD) trip analysis, provide valuable insights into potential locations for charging demand and aid in the strategic planning of charging infrastructure deployment.

By carefully analyzing these elements (parking space utilization rates, numbers of EVs in use, projected EV sales, current and projected charging demand, data-driven insights), stakeholders can make informed decisions to effectively expand EV charging infrastructure and meet the growing demand for electric vehicle charging services.

5.4 Proactive Charging Infrastructure Placement Approaches

Proactive placement means that a public or private organization takes the initiative to install charging infrastructure without a specific applicant or user for each charging location. Known approaches for proactive placement include[60]:

Strategic Placement: Organizations may install charging points at strategic locations that are important for specific target groups, such as business parks, sports clubs, and park-and-ride locations, serving tourists, commuters, or businesses. This approach helps to raise awareness and visibility of electric vehicles, contributing to increased public consciousness and adoption.

Placement Based on Expected Demand: An organization can proactively install charging infrastructure based on projected demand. This involves placing charging

[60] https://nklnederland.nl/wp-content/uploads/2020/01/Handreiking_visie_en_beleid_okt_2019.pdf

stations based on calculations of future needs. This forward-thinking strategy ensures that the infrastructure will be ready as demand grows.

Placement Based on Usage Data: As the charging network expands, the amount of usage data grows. Organizations can use this data to guide further expansion and densification of the charging infrastructure. Installing new charging stations based on usage data addresses insufficient coverage. If existing stations consistently exceed a certain threshold of kWh transactions or occupancy rates, the infrastructure can be expanded to meet the demand.

This proactive approach ensures a well-planned and strategically placed charging network that meets the needs of the community, enhances the visibility of electric vehicles, and adapts to growing demand. Both the Strategic Placement and Placement Based on Expected Demand approaches could indicate that charging points are needed at locations where a business case for a charging point is not (yet) positive.

5.5 Functions in Charging Infrastructure Implementation

A charging infrastructure implementation model involves multiple functions, which may be taken up by one stakeholder or delivered through partnerships between different stakeholders. Apart from the set-up of charging infrastructure, other roles include land provision, electricity supply, EVCI supply, charging software solutions, and customer services.

- **Procurement of charging infrastructure:** The stakeholder procuring the charging infrastructure is the driving partner for the implementation. Procurement can be undertaken by the primary user of the charging infrastructure, by the charging service provider, or by the governing authority responsible for providing charging infrastructure. The procurer may or not own the EV chargers.
- **Land provision:** The space required for EV charging may be owned by the procuring stakeholder or may be acquired on lease or alternative arrangements (revenue sharing, for example). Generally, private and semi-public charging facilities are installed on private land, while public ones may be installed on public or private land.
- **Energy supply:** Energy supply for all EV charging installations is provided by distribution utilities responsible for power distribution in the region where the charging facility is located.
- **EVSE supply, installation, and maintenance:** EV chargers may be supplied by an EVSE manufacturer or retailer. For semi-public or public charging, CPOs are usually responsible for the selection and installation of the required arrangement of chargers.
- **Charging software solutions:** System management software is used by CPOs to manage their network of charging points, to track and control charging sessions, run diagnostics on the charger equipment, and for other back-end services to manage customer subscriptions, pricing structures, etc. Charging solutions may be offered as white-label solutions from third-party vendors or may be developed in-house by CPOs.
- **Customer services:** CPOs and EMSPs are mainly responsible for customer services at public and semi-public charging facilities. Other responsibilities of

CPOs include planning and permissions, installation and commissioning, and operation and billing.

5.6 Standards to be adopted in Rwanda

5.6.1 AC and DC Charging Standards for Rwanda

As described in Chapter 2, AC charging is typically slower than DC charging and depends on the vehicle's onboard charger capacity. While AC charging ranges from slow to moderately fast, DC charging provides significantly faster charging times but requires higher grid capacity and comes with higher installation costs.

The levels of charging vary in power output, as presented in Table 2 of this report. AC charging includes Level 1 and Level 2, offering power ranges between 1.9 kW and 44 kW, with charging times ranging from 6 to 40 hours, depending on the battery size and vehicle type. On the other hand, DC charging, corresponding to Level 3 and Level 4, offers 44 kW to over 350 kW, reducing charging times to 5 to 45 minutes.

Given these characteristics, AC Level 1 charging is most suitable for household use, where EV drivers can charge their vehicles overnight. AC Level 2, offering faster charging speeds, is ideal for homes, workplaces, and public charging spots. DC Level 3 and 4 chargers are most appropriate for public and commercial locations such as highways, rest stops, and fuel stations, where fast charging is necessary for longer-distance travel, as described in deep on chapter 2 of this document.

Below are the recommended guidelines for implementing EVCI in Rwanda:

1. AC Charging Level 1:

- Purpose: For home charging installations, typically done by the EV user.
- Locations: Residential garages, streets, or driveways.
- Power Output: 1.9 kW to 7.4 kW.
- Vehicle Types: Light-duty vehicles such as passenger vehicles and motorcycles.
- Charging Time: 8 to 40 hours, depending on the vehicle and battery size.
- Advantage: Convenient, slow charging suitable for overnight use.

2. AC Charging Level 2:

- Purpose: Faster home charging, workplaces, and public facilities.
- Locations: Homes, office buildings, corporate parks, parking lots, and retail stores.
- Power Output: 7.4 kW to 44 kW.
- Vehicle Types: Light-duty vehicles such as cars and motorcycles.
- Charging Time: 4 to 6 hours to fully charge most EVs.
- Advantage: Accelerated charging, increasing EV adoption at work and public locations.

3. DC Charging Level 3:

- Purpose: Fast charging for public areas and commercial stations.
- Locations: Highway stop areas, designated public and commercial parking lots.
- Power Output: 44 kW to 150 kW.
- Vehicle Types: Light-duty passenger cars and heavy-duty vehicles.

- Charging Time: 30 minutes to 2 hours, depending on vehicle and battery size.
- Advantage: Reduced charging time, ideal for EV users during long-distance travel.

4. DC Charging Level 4:

- Purpose: Ultra-fast charging for high-demand locations.
- Locations: Highway rest areas, fuel stations, public charging hubs.
- Power Output: 150 kW to over 350 kW.
- Vehicle Types: Light-duty cars, heavy-duty trucks, and buses.
- Charging Time: 5 to 45 minutes, depending on the vehicle and battery size.
- Advantage: Extremely fast charging, designed for heavy-duty applications and high-demand scenarios.

Charging Infrastructure Categories

To complement these standards, the following charging infrastructure categories are recommended for different environments in Rwanda:

- **Home Charging**: EV owners charge their vehicles at residential locations. Ideal for slow overnight charging with AC Level 1 or 2.
- **Work Charging**: Charging at workplaces using AC Level 2 chargers, providing convenient charging for employees.
- **Depot Charging**: Centralized DC charging stations for commercial fleets such as buses or trucks. These are managed-access locations with multiple charging points.
- **Hub Charging**: Multiple charging stations grouped together at commercial locations such as supermarkets, shopping centers, or bus terminals. These use both AC and DC charging to accommodate various vehicle types.
- **Road-Side Charging**: Public parking lots or urban streets using DC Level 3 or 4 chargers. Designed for quick urban charging.
- **Destination Charging**: Stations at public institutions, shopping malls, or restaurants using both AC and DC charging.
- **Enroute Charging**: Located along highways or rest stops, offering ultra-fast DC charging for long-distance travelers.
- **Battery Swapping**: Stations that replace depleted EV batteries with fully charged ones, often used by two-wheelers and small electric vehicles.

Recommendations for Rwanda's EVCI:

For Rwanda, a hybrid infrastructure combining both AC and DC chargers is essential to accommodate the diverse range of EVs and their varying power requirements. This strategy will ensure efficient charging options are available at different types of locations, from urban centers to highways, as just described.

5.6.2 Interoperability Protocols to be Adopted in Rwanda: Software Layer

As stated in Section 2, the charging protocols allow openness and interoperability on the hardware, communication, and information layers. International standards on charging protocols, described in Chapter 2, reference the most widely used international protocols, such as the Open Charge Point Protocol (OCPP) / IEC

63110, Open ADR – Open Automated Demand Response Standard / Open ADR Alliance, IEC 61850, charging protocols for roaming, OCPI - Open Charge Point Interface protocol / IEC 63119, ISO 15118, and its latest extension, ISO 15118-20.

The **Open Charge Point Interface (OCPI)** Protocol is a standardized communication protocol designed to facilitate the exchange of information between various EV charging market stakeholders. It primarily operates between Charge Point Operators (CPOs) and Mobility Service Providers (MSPs), as well as with other market operators, allowing seamless interoperability across different charging networks. Key Features of OCPI:

- Roaming Capabilities: OCPI supports roaming by allowing EV drivers to access various charging stations without needing separate memberships or payment methods for each provider. This feature ensures that users can locate, authenticate, and charge their EVs at multiple networks with ease.
- Information Exchange: The protocol enables sharing information on charging locations, tariffs, availability, charging status, and session data. This promotes transparency for users and ensures that they have access to real-time information.
- Smart Charging: OCPI facilitates smart charging by supporting dynamic pricing and managing charging profiles based on peak demand or grid stability needs.
- Interoperability: It enables easy integration with navigation service providers, governments, and fleet managers, providing a holistic view of charging station performance, availability, and costs.

The **Open Charge Point Protocol (OCPP)**, is an open-source protocol used to manage communication between charging stations and Charge Point Management Systems (CPMS). It serves as the “language” that allows different brands and models of charging stations to communicate with the backend system effectively. Key Features of OCPP:

- Open Standard: As an open-source protocol, OCPP is widely adopted and supported by charging equipment manufacturers worldwide. This open standard reduces the risk of vendor lock-in and enhances system flexibility for future upgrades or replacements.
- Smart Charging Management: OCPP enables monitoring and controlling of charging stations, including starting or stopping charging sessions remotely, implementing load balancing, or scheduling charging based on electricity tariffs.
- Compatibility: OCPP ensures compatibility between the charging stations and the back-office software, making it easier for network operators to integrate different brands or types of charging points into a single management system.
- Security: OCPP includes security features such as encryption and authentication to secure communication between the charging station and the management system.

ISO15118-2 defines the communication protocol between battery electric vehicles (BEV) or plug-in hybrid electric vehicles (PHEV) and the EVCI. The application layer message set outlined in this standard facilitates energy transfer from the EVCI to the EV. Additionally, ISO 15118-1 introduces use cases for bidirectional energy transfer. ISO15118-2 enables the integration of the EV into the Smart Grid to allow flexible energy transfer (V2G). This delivers added value for the grid without compromising the EV or its driver.

This standard provides support for wireless power transfer, enhanced data security through mandatory Transport Layer Security (TLS), and dynamic control modes for real-time power adjustments in response to grid demands. It also features multiplexed communication, enabling efficient message processing and streamlined management of multiple e-mobility contracts, allowing users to effortlessly switch between different charging agreements. ISO15118-2 and its latest extension, ISO15118-20, are standards used in the European Union and the United States of America, and it is recommended that they also be implemented in Rwanda for V2G and smart charging for the EVCI. Key Features of ISO15118-2 include:

- Plug & Charge (PnC): ISO 15118-20 enables vehicles to automatically authenticate and begin charging without manual intervention by the user, streamlining the charging process and enhancing user experience.
- Vehicle-to-Grid (V2G) Integration: The protocol supports bi-directional energy transfer, allowing EVs to not only draw power from the grid but also feed excess power back into it. This capability is crucial for smart grids and renewable energy integration.
- Data Security: It provides robust security features, including Transport Layer Security (TLS), to protect data exchanges during charging and billing transactions.
- Energy Flexibility: By integrating EVs into the grid as flexible energy storage units, ISO 15118-20 supports grid balancing and peak shaving.

IEC 61851 is an international standard that specifies the requirements for electric vehicle conductive charging systems. It covers various aspects related to the equipment used for charging EVs through a conductive connection, including safety, compatibility, and communication between the vehicle and the charging station. Key aspects of IEC 61851 include:

- Charging Modes: Defines different charging modes (Mode 1, Mode 2, Mode 3, Mode 4), which dictate the power levels and communication requirements between the EV and the charging equipment.
    - Mode 1: Direct connection to a standard AC outlet with no dedicated charging interface.
    - Mode 2: Direct connection with an in-cable control and protection device.
    - Mode 3: Dedicated Electric Vehicle Supply Equipment with control and communication features.
    - Mode 4: DC fast charging with advanced communication protocols for safety and control.
- Connector Types: Establishes the standards for connectors and interfaces used for EV charging, ensuring compatibility between different types of EVs and charging stations.
- Safety and Protection: Outlines safety measures such as earthing, fault detection, and protection against overcurrent and overvoltage, ensuring a safe charging process for both the vehicle and users.
- Communication: Supports basic communication between the EV and the charging station to control the charging process and ensure proper connection and safety checks.

<u>Recommendations for standard protocols to be adopted in Rwanda's EVCI:</u>

Rwanda should adopt OCPP, OCPI, ISO 15118-20, and IEC 61851 to ensure the seamless integration, scalability, and future-proofing of its electric vehicle charging

infrastructure (EVCI). OCPP (Open Charge Point Protocol) provides an open, standardized method of communication between EV chargers and management systems, promoting vendor interoperability and flexibility. OCPI (Open Charge Point Interface) ensures smooth communication between charge point operators and e-mobility service providers, enabling users to locate and access different charging networks with ease. Implementing these protocols will allow Rwanda to build an interoperable, efficient, and user-friendly EVCI, crucial for the development of a robust EV market.

ISO 15118-20 supports advanced vehicle-to-grid (V2G) technologies and enhances user experience by enabling Plug & Charge capabilities, simplifying payment and authentication at charging stations. specifies the charging modes and systems for EVS, defining safety requirements for conductive charging and communication between electric vehicles and charging infrastructure.

These protocols are widely implemented in the USA and the EU, which are the most mature EV markets globally. By adopting these globally recognized best practices, Rwanda can align its EVCI standards with world-leading markets, ensuring a seamless and efficient EV charging experience for users while supporting effective energy and infrastructure management.

The recommended standards—such as OCPP, OCPI, ISO 15118, and IEC 61851—are applicable to passenger vehicles, light-duty, and heavy-duty vehicles. They do not directly address electric motorcycles, or electric buses.

For motorcycles, it is still early to set rigid standards, particularly in East Africa where hardware developments are still ongoing. However, on the software side, Rwanda could explore OCPP-light, a simplified version being proposed in India for electric motorcycles.

For electric buses, the choice of standards will largely depend on the manufacturer and the business model used for buses. In some cases, buses and depots operate as a closed system, making OCPP sufficient for communication. In scenarios where depots and chargers are shared among multiple brands, OCPI may not be necessary, as roaming across networks is less common.

5.6.3 Interoperability Hardware Layer: Standard Connectors to be adopted in Rwanda

For Rwanda to develop a scalable and user-friendly EV charging network, it is essential to ensure interoperability across charging stations on the hardware layer, being key to adopt a national standards connector for EVCI. This means EVs from different manufacturers must be able to charge using standardized connectors and communication protocols at all public and private stations.

There is multiple EV connectors used worldwide, and the choice of connector varies by region and market preferences, as described in deep in Chapter 2 and Annex No. 1. As EV adoption grows globally, different countries have established various connector standards based on their local markets and vehicle imports. In Rwanda, it is essential to adopt a flexible approach that supports a broad range of EVs while ensuring alignment with global standards.

The **Type 2 connector** (for AC charging and CCS for DC charging) is the official standard in Europe and other parts of the world, offering compatibility with multiple European automakers and covering a wide range of countries. Type 2 is not only

used in Europe but also adopted by countries in South America, such as Chile and Paraguay.

In this context, adopting the Type 2 connector makes it an ideal choice for Rwanda, as it aligns with international best practices, supports a wide range of EV models, and welcomes multiple EV brands to the Rwandan market. It provides flexibility, reliability, and high-quality standards, allowing all interoperability features to be fully utilized.

GBT connector, used predominantly in China, and the Chinese EVs, is another viable option for Rwanda. Given the growing influx of Chinese-manufactured EVs in the world and Rwanda, adopting GBT as an optional standard ensures that Rwanda can accommodate these vehicles and remain responsive to market needs.

Ideally, a single universal connector would simplify the EVCI infrastructure and make it easier for users to charge their vehicles anywhere. However, due to the diversity in the global EV market and the various vehicle models entering Rwanda, a hybrid solution that includes both Type 2 and GBT connectors offers a more practical approach. This solution accommodates a wider range of EV users by supporting vehicles from Europe and China, which are prominent markets for EVs. By adopting this dual-standard approach, Rwanda can ensure greater flexibility and convenience for current and future EV users.

However, it is important to recognize that charging standards are evolving worldwide. For example, and as referenced in Chapter 2, in the U.S., the NACS connector has now been adopted as the new standard in the US starting on 2025, replacing the type 1 connector currently used there as a standard. This demonstrates that the global EV landscape is continuously shifting and evolving, and Rwanda should remain agile, monitoring these developments.

While it is essential to start now with well-established standards like Type 2 and GBT to support the current market, future adjustments may be necessary as the industry evolves. Rwanda can begin with these standards to ensure high quality and broad compatibility while staying alert to emerging global trends for potential future adoption. To address these ongoing developments, it is recommended in the final chapter to establish an EVCI working committee. This committee will coordinate efforts among Rwanda's organizations and institutions as well as to track global advancements, ensuring Rwanda's e-mobility infrastructure remains adaptable and future-proof.

Recommendation standards per type of Vehicle

The recommendation for the standard connectors just mentioned applies primarily to public EVCI used by passenger vehicles. In the case of two-wheelers, Rwanda has seen the growth of battery swapping stations, which are expected to continue evolving as a viable business model. For light-duty, heavy-duty vehicles, and electric buses, logistics and operations typically follow a different methodology. These vehicles often rely on depots in closed systems, where dedicated charging infrastructure supports fleet operations, which as well depends on the EV manufacturer. On this model, EV fleets usually have their own infrastructure in place to ensure logistics and operational autonomy.

However, it is crucial to acknowledge the need for public charging infrastructure for heavy-duty vehicles in the long term. While the market for heavy-duty EVs is still in its early stages, Rwanda should adopt a flexible and future-proof approach.

To ensure readiness for future heavy-duty vehicle charging needs, the following steps are recommended:

- Pilot Projects for Heavy-Duty Charging: Pilot projects, such as implementing Megawatt Charging Systems (MCS) in strategic locations like logistics hubs or highways, should be initiated. This would allow Rwanda to explore the feasibility of public charging infrastructure for heavy-duty vehicles in a controlled and scalable manner.
- Collaboration with Private Sector: Engage private fleet operators early on to understand their charging infrastructure needs and preferences. This collaboration will help shape a public-private partnership model, ensuring that both public and private heavy-duty charging infrastructure can coexist efficiently.
- Ongoing Monitoring and Future-Proofing: While initially focusing on passenger vehicles, it is important for the recommended working committee to continuously monitor global standards for heavy-duty vehicles. This proactive approach will enable Rwanda to introduce relevant standards as the market matures, ensuring alignment with global best practices when necessary.

In conclusion, while it may not be feasible to establish a standard connector for heavy-duty vehicles in public infrastructure at this stage, starting with pilot projects and collaborating with stakeholders will position Rwanda to adapt as the heavy-duty EV market grows.

5.7 Technical Charging Infrastructure Implementation Roadmap for Vehicles

5.7.1 Current Situation

In Rwanda, the current charging infrastructure for EVs consists of a limited number of AC chargers primarily located at semi-public sites such as office and commercial buildings. Additionally, several DC fast chargers are concentrated mainly in the capital city of Kigali. Outside Kigali, access to charging infrastructure, particularly DC charging, is scarce. Some EV owners have access to home charging solutions. However, grid capacity limits and a resulting lowered voltage level sometimes result in limited home-charging options. Similarly, current operators of DC charging infrastructure, are limited in expanding their DC charging network due to insufficient grid capacity. More details regarding the current EVCI in Rwanda could be found in Chapter 3.

5.7.2 Intermediate State

In the coming years, a growing charging network for EVs in Kigali, secondary cities, and more rural areas is required to facilitate the adoption of a growing number of EVs. With range anxiety as an important consideration for vehicle owners to move from an ICE vehicle to an EV, sufficient charging options should be available to most EV drivers.

The charging infrastructure should be developed in such a way that it respects the limitations on the speed of executing grid upgrades. The recommended initial steps for further development of charging infrastructure for EVs therefore focus on growing the charging infrastructure as quickly as possible while limiting the number of locations where grid upgrades would be needed. This is built on the assumption that

smaller grid upgrades at various locations will require more time and resources from the grid operator than more significant upgrades on relatively fewer locations.

Taking this into consideration, the following is recommended to:

- Expand the number of strategically placed and publicly available DC chargers. A well-developed network of DC chargers addresses range anxiety and provides a reliable alternative to home charging. This expansion should at least include various locations in secondary cities to enable long-distance travel for those traveling from and to Kigali.
- Expand the semi-public charging network. As EV adaptation will be driven by, amongst others, fleet electrification, policies, employees should be provided the opportunity to charge their vehicles at work. This form of charging infrastructure expansion incentivizes charging during the daytime and limits charging peaks directly after office hours when electricity demand and the load on the grid is peaking. Besides, this form of charging could utilize solar energy generated on roofs of office and commercial buildings. Hereby, this charging solution could not only provide convenient charging options but also leverage renewable energy sources for sustainable transportation solutions.
- Similarly, expanding the number of semi-public charging stations for destination charging should be a high priority. Key locations such as supermarkets, hospitals, and other high-traffic areas will enhance the availability of charging solutions and accessibility to charging infrastructure. The latter will encourage EV adoption among a broader demographic. As described in the recommendation above, destination charging during the day provides opportunities to integrate distributed renewable energy generation into the charging solution and provides a convenient alternative for charging during off-peak hours.

5.7.3 Goal State

In the envisioned goal situation, the charging infrastructure for vehicles in Rwanda has undergone significant enhancement and expansion, facilitating widespread adoption of EVs and promoting sustainable transportation solutions. The expansion is mainly facilitated by strategic upgrades of the distribution grid. The upgrades make it possible to accommodate public charging solutions at the neighborhood level of the electricity grid and private home charging solutions.

To move from the intermediate state to the goal state, it is recommended to:

- Prioritize the further enhancement of the public and semi-public charging network with higher charging levels. With strategically located high-power (AC and DC) charging stations across urban centers, highways, and key destinations a large base of drivers can be provided with charging services.
- Facilitate charging at the neighborhood and household level, by ensuring sufficient grid capacity is available to charge at 7 to 22 kW charging power levels. This requires strategic grid planning from the grid operators and dedicated funding for the enforcement of the distribution grid to charge.
- Smart charging solutions should be adopted at this stage of the development of the charging infrastructure for EVs, allowing EV owners to optimize charging schedules based on energy prices, grid demand, and individual preferences. These solutions enable efficient use of charging infrastructure and promote grid stability and renewable energy use. After a roll-out of semi-public and public

charging infrastructure, as described in the intermediate state, smart charging pilots should create familiarity with the technologies required to embed smart charging in the charging infrastructure. Smart charging pilots should be carried out by a collective of private charging point operators, grid operators, and mobility service providers.

5.8 Technical Charging Infrastructure Implementation Roadmap for Motorcycles

5.8.1 Current Situation

The charging infrastructure for electric Motorcycles in Rwanda is expanding rapidly, particularly in the city center of Kigali, with an increasing number of battery-swapping stations. The number of swapping stations outside of Kigali is limited. These stations allow drivers to exchange batteries for fully charged ones. The further expansion of the number of swapping stations and their size is constrained by the current grid capacity. Swapping stations are operated by commercial companies that supply the bikes, own the batteries, and operate the swapping stations. The electric motorcycles drivers have displays that show the battery percentage and every battery is monitored by the operator. Operators incentivize drivers to return batteries at a decent state of charge to minimize battery degradation. The integration of distributed renewable energy sources into the energy supply of the swapping stations is currently minimal.

5.8.2 Intermediate situation

As electric motorcycles drivers demand highly reliable charging solutions, a growing number of electric motorcycles can only be reached having a charging infrastructure that has significant coverage and scale. The following is recommended to support the development of a growing charging infrastructure for electric motorcycles:

- Current grid limitations should be removed as a barrier for further expansion of the number of swapping stations by strategic grid planning processes and dedicated for this development.
- The current swapping station operators should be incentivized and supported in innovating their battery management and swapping operations to reduce the peak consumption of swapping stations. In this manner, current swapping stations can expand the number of electric motorcycles drivers they can provide charging services without increasing the need for grid reinforcement.
- Smaller-scale battery swapping stations should be rolled out in secondary cities and rural areas to increase the reach of the electric motorcycles charging services. These smaller-scale battery swapping stations should be part of multi-modal charging hubs.

5.8.3 Goal Situation

In the envisioned goal situation, the charging infrastructure for electric motorcycles in Rwanda has undergone significant enhancement and expansion. The expansion is mainly facilitated by a focus on efficient resource usage by aggregating charging facilities for different modes of transport at charging hubs in secondary cities and rural areas and by improving the accessibility of swapping stations for electric

motorcycles drivers by implementing interoperable technologies and standards that guarantee vendor-independent swapping station accessibility.

To move from the intermediate state to the goal state, it is recommended to:

- Promote interoperability and encourage the market to adopt a common battery standard. This will enable a network of swapping stations where batteries from different manufacturers can be exchanged, benefiting the entire market. For example, standardized connectors will facilitate seamless integration between batteries and stations.
- Support and incentivize swapping station facilities and business models that include renewable energy as main source of energy for charging batteries at the swapping stations.
- Further expand the network of multi-modal charging hubs in secondary cities and rural areas.

5.9 Technical Charging Infrastructure Implementation Roadmap for Buses

5.9.1 Current Situation

Rwanda is currently starting pilot programs for electric buses. These buses will primarily be charged at DC charging stations located at the outskirts of the city of Kigali. These buses utilize DC chargers for quick and efficient charging overnight. While the existing grid capacity can support these pilot programs, significant expansion of the number of DC charging locations would require electricity infrastructure upgrades.

5.9.2 Intermediate Situation

In order to support a growing number of electric buses that provide transport services within Kigali and from Kigali to secondary cities, the DC charging infrastructure should be expanded. A growing number of charging depots located at the outskirts of Kigali is assumed to be sufficient to expand the number of buses within Kigali, to secondary cities, and rural areas. The electric buses will be able to charge between driving their routes at the outskirts of the city center of Kigali.

To move to this charging infrastructure configuration for busses, it is recommended to:

- Scale up the number of bus depot charging facilities at the outskirts of the city center of Kigali.
- Promote smart charging and route planning in pilots.
- Remove current grid limitations as a barrier for expansion by implementing strategic grid planning processes.

5.9.3 Goal Situation

Because electric busses will become an important part of the transport services within Kigali and between Kigali and secondary cities, the goal is to develop a facilitating charging infrastructure for this transport system. To move from an intermediate situation with a growing number of electric buses to a completely electrified bus fleet, it is recommended to:

- Develop top-up charging facilities, utilizing pantograph-style chargers, within Kigali and various strategic locations to support additional charging convenience for electric buses. These chargers will enable quick and efficient charging during short stops, reducing downtime and ensuring continuous operation.
- Further expand the number of charging depots at the outskirts of Kigali.
- Promote interoperability by developing regulation and policies. By doing so, bus charging providers can support various bus companies by developing a reliable network of bus depots and innovative charging services.

5.10 Technical Charging Infrastructure Implementation Roadmap for Mini Buses

5.10.1 Current Situation

Mini-buses, vital for transport between secondary cities and in rural areas, are currently not electrified. There are no electric mini-buses in operation today, and thus, no charging infrastructure dedicated to them exists.

5.10.2 Intermediate Situation

The role of mini-buses will change as compared to their role today and they will have a specific role in the future electrified transport system. This mode of transport will be primarily used to connect passengers to larger transport hubs in secondary cities and rural areas. Most importantly, mini-buses will bring passengers to transport hubs where these passengers will be able to make use of electric buses that travel to Kigali and other transport hubs. Mini-buses should be able to top-up charge at these hubs, using fast AC and DC chargers and be able to charge at charging depots located at strategically placed locations in the areas in and around the secondary cities of Rwanda.

5.10.3 Goal Situation

In the envisioned future, Rwanda will have a comprehensive electrification strategy for mini-buses, contributing to sustainable transportation solutions. The goal situation includes the following key elements:

- Multi-modal Transport Hubs: Fully developed multi-modal transport hubs in secondary cities will integrate mini-buses into the transportation network, enhancing connectivity for passengers.
- DC Charging Infrastructure: Multi-modal transport hubs will be equipped with DC charging options tailored for mini-buses, ensuring seamless charging experiences for operators.
- Open Charging Network: An open charging network will be established for mini-bus operators, providing accessible and convenient charging services. This network will offer real-time information on charger locations, availability, and pricing, facilitating efficient planning and utilization of charging infrastructure.
- Grid Capacity Optimization: Grid capacity will be optimized to accommodate the electrification of mini-buses and the establishment of charging infrastructure, ensuring reliable and sustainable power supply.
- Stakeholder Collaboration: Collaboration between stakeholders, including mini-bus operators, local authorities, energy providers, and service providers, will be

fostered to ensure the successful deployment and operation of the charging infrastructure.

- Renewable Energy Integration: Renewable energy sources, such as solar power, will be integrated into the charging infrastructure wherever feasible, reducing reliance on the grid and enhancing sustainability.

5.11 Geospatial analysis and Proposed Locations for EVCI in Kigali

To develop a comprehensive EV Charging Infrastructure master plan for Rwanda, a detailed geospatial analysis was conducted to identify potential sites for EV charging stations. Initially, private owners such as Ampersand, EVP, Kabisa, Basigo, REM, and Spiro were interviewed to gather data on their existing and planned EVCI installations. This information was mapped to create an initial overview of the current and future charging network.

For identifying suitable locations for future expansion beyond the existing private operator zones, secondary data from various sources was utilized. The data included administrative boundaries to highlight major cities and towns, population density to maximize infrastructure usage, and residential and commercial zones from city master plans to reduce costs and implementation time. Additionally, road networks were considered to ensure accessibility, electric networks to ensure a source of power, and various types of infrastructure were categorized as follows:

- Government buildings such as city halls or RSSB building – Pension Plaza
- Commercial buildings with large parking areas such as shopping malls and office buildings
- Entertainment buildings such as arenas, stadiums, museums, and conference places
- Existing petrol stations
- Other public places such as airports, taxi parks, markets, hospitals, and large universities

Data sources included building locations from the City Masterplan, road network data from the Rwanda Transport Development Agency (RTDA), electric infrastructure information from the Rwanda Energy Group (REG), and population density data from the National Institute of Statistics of Rwanda (NISR). Petrol station locations were provided by the Rwanda Utilities Regulatory Authority (RURA). Additionally, field visits were conducted to ensure data accuracy and completeness.

During the geospatial analysis, all collected data layers were integrated to identify optimal locations for new EVCI. Key considerations included ensuring accessibility for EV users, making sure no car travels more than 50 km on national roads in upcountry areas without encountering a charging station, focusing on high population density and traffic areas, and leveraging existing petrol stations and commercial buildings. The selected potential EVCI sites are already connected to the electric grid.

Furthermore, the site selection strategy assumed that buildings with large parking areas and government buildings would accommodate users with sufficient time for charging, while petrol stations would cater to fast-charging needs for users requiring

quick turnaround times. This approach ensures both long-term and short-term charging requirements are addressed, optimizing convenience for all users.

An assessment of the current and potential EV charging infrastructure across various provinces in Rwanda reveals significant opportunities for development and expansion.

Kigali City, as the economic hub with high population density and significant traffic, hosts the highest concentration of EV charging stations with 17 existing, 28 planned, and 64 potential sites, totaling 109. This infrastructure is critical to support the city's growing demand for electric vehicle charging.

In the **Eastern Province**, 42 sites have been identified, including 1 existing, 9 planned, and 32 potential locations, indicating a strong potential for future expansion to meet anticipated demand. **The Northern Province**, with a total of 26 sites (2 existing, 9 planned, and 15 potential), shows a balanced distribution that suggests a moderate current demand with room for growth.

The **Southern Province** presents a considerable opportunity for infrastructure development, especially in currently underserved areas, with 20 sites identified (1 existing, 1 planned, and 18 potential). Similarly, the Western Province aims to enhance regional connectivity with 27 sites, including 1 existing, 7 planned, and 19 potential.

It is important to refer to the Annex, which includes maps with the current and potential locations of the proposed charging stations.

This comprehensive assessment highlights the strategic focus on expanding EV charging infrastructure to support the adoption of electric vehicles across Rwanda, ensuring accessibility, and promoting regional connectivity.

Regarding the intermediate and goal situations described earlier, the following table provides an overview of proposed strategic locations for EVCI in Kigali, which takes into account the Transport Plan Kigali Master Plan 2050. These strategic locations are crucial for activities of society integrating eMobility, ensuring the convenience and accessibility of charging infrastructure.

The geospatial analysis for the EV Charging Infrastructure Masterplan identified a total of 224 sites across Rwanda, categorized into existing, planned, and potential locations. The analysis focused on key regions: Eastern, Kigali City, Northern, Southern, and Western provinces. The table below summarizes the distribution of EV charging stations:

Row Labels	*Existing*	*Planned*	*Potential*	*Grand Total*
Eastern	1	9	32	42
Kigali City	17	28	64	109
Northern	2	9	15	26
Southern	1	1	18	20
Western	1	7	19	27
Grand Total	**22**	**54**	**148**	**224**

Table 22: Existing, planned and potential location of EVCI in Rwanda

Annex No. 19 Table of detail EVCI combined (Existing, Planned and Potential), shows the detailed specification of existing, planned, and potential sites for EVCI with the latitude and longitude information, the status of the EVCI, the name, province, district, and sector. Furthermore, the following guidelines were used to the assessment of the potential locations:

Public Transportation and Parking: It is recommended that EVCI be introduced in public parking spaces to support electric public transport and private EVs, addressing the observed shortage of EV charging stations.

Government Buildings: It is suggested that EVCI be installed in buildings hosting offices of governmental entities, such as ministries, municipalities, and other agencies, to leverage their strategic locations and high daily traffic for effective charging station installations.

Healthcare and Market Areas: To tap into the potential for infrastructure expansion, it is recommended that EVCI be deployed in hospitals and public markets with parking facilities, where no charging stations currently exist.

National Roads: The installation of DC chargers along national roads connecting Kigali to secondary and satellite cities is proposed. This strategy is designed to provide rapid charging capabilities and enable efficient travel to more remote areas. Sites with existing infrastructure and amenities, such as petrol stations or notable landmarks, should be selected to ensure high visibility and alignment with user distribution and demand.

Motorcycle Mobility: Given the rising use of electric motorcycles, Battery Swapping Stations (BSS) are recommended due to their minimal space requirements and ability to serve multiple users simultaneously. These facilities should be strategically placed along national and district roads, particularly in highly populated areas, to support transportation between Kigali and satellite cities and enhance internal city movements.

5.12 Grid and Energy System Planning

Grid and energy planning are essential for further expansion of charging infrastructure. Especially when considering the expansion of transformer capacity in the distribution grid and renewable electricity generation needed to support a reliable charging infrastructure for various modes of transport. Some guiding principles for successful grid and energy system planning are:

Grid planning should be holistic planning

Grid and energy system planning is a process that involves holistic thinking, including urban and transport planning. It therefore requires input from not only energy system experts, but also transport and urban planning experts. Processes and tools should be in place to facilitate information and data exchange between authorities responsible for making prognoses and designs related to the transport and urban systems of the future.

Grid and Energy System Plans result in resource allocation

Grid and energy system planning should determine the resources allocated by a grid operator for expanding and upgrading the electricity grid. As grid operators worldwide face limited resources in terms of personnel and equipment due to the

growing pressure of the energy transition, developed plans need to be assessed carefully for feasibility and continuously aligned with the resources available to the grid operator. Additionally, a careful assessment of the importance of eMobility-related resource allocation compared to other goals of the grid operator is needed to avoid overwhelming the grid operator.

Data-driven, in-house planning is a critical success factor for system transformation.

Grid and energy system planning is an immense effort, especially when it comes to data gathering and evaluation. Embedding data management and utilizing advanced data management and grid planning software will significantly reduce the effort required for planning. This reduction in effort will allow planning teams to execute planning cycles more frequently, making the planning of future energy, transport, and urban systems more agile. Ideally, planning should be executed by a dedicated team of experts within the grid operator, as it is a core activity of a grid operator. This team should proactively conduct studies using innovative grid planning tools and data-driven approaches, rather than relying on external reporting. Training and capability building to enhance in-house capabilities, processes, and tools is an investment in efficient resource allocation and system transformation.

5.12.1 Remuneration of Transformer Costs

Power grids, being natural monopolies, are heavily influenced by regulatory and policy frameworks that vary worldwide. Management structures range from state-owned national utilities to privatized industries with separate transmission and distribution entities. The primary oversight goal is to ensure service availability, cost efficiency, affordability, and technological innovation[61].

National regulatory authorities, such as the Rwanda Utilities Regulatory Authority (RURA), manage remuneration via energy tariffs and ensure operational efficiency of the grid operator. Remuneration mechanisms have evolved from "cost-of-service" models, which led to overinvestment, to price or revenue cap mechanisms aimed at cost reduction and innovation. Performance-based schemes and yardstick competition have been introduced to balance cost efficiency with service quality. Performance-based regulation, increasingly used, targets specific outcomes to improve service quality and innovation, providing a balanced approach to CAPEX and OPEX.

As funding of transformers is perceived to be a bottleneck for current charging infrastructure expansion, it is critical to assess and potentially redesign current regulatory frameworks for the distribution grid operator(s) to encourage and enable them to invest in resources that are required to expand charging infrastructure. eMobility-related goals could in this context become part of a performance-based regulatory framework and incentivize strategic grid reinforcements needed to facilitate charging infrastructure expansion.

[61] IEA Electricity Grids and the Energy Transition

Regulatory framework for distribution and transmission operators

Regulatory framework	Grid operator					Customers		Both
	Cost recovery	Cost minimisation	Planning optimisation for total costs	Performance and operational efficiency	Innovation and best practice deployment	Affordability – cost minimisation	Quality of service improvement	Dynamically aligned cost and price
Cost of service/Cost-plus Rules are set to cover costs, with allowed earnings defined *a priori* (rate of return) starting from the determination of total cost of service	🟢	🔴	🔴	🟡	🔴	🔴	🟡	🟡
Price cap/Revenue cap Remuneration is based on a yearly cap that the grid operator can charge for each specific service (price cap) or on the amount that can be earned for each cluster of service (revenue cap)	🟡	🟢	🟡	🔴	🟡	🟢	🔴	🟡
Yardstick competition The operator's performance is compared with the other operators' performance of the sector, with penalties/awards consequently defined	🟡	🟢	🟡	🟡	🔴	🟢	🟡	🔴
Output/Performance-based Remuneration is based on performance monitoring for each service (e.g. quality, losses) to incentivise continuous improvement	🟡	🟡	🟢	🟢	🟢	🟡	🟢	🟢
Impact on indicator			🔴 Poor	🟡 Moderate	🟢 Beneficial			

Note: "Dynamically aligned cost and price" refers to the frequency of updating the information used to set the operators' remuneration based on the actual costs they sustain. Very infrequent updates, for example until the end of a five-year regulatory period, can result in customers paying higher prices or grid companies suffering financial losses for an extended period until the imbalance is corrected in the next cycle.

Figure 33: Regulatory framework for distribution and transmission operators

5.13 Proposed Solutions for Mitigating EVCI Negative Impacts on Rwanda's Electricity Networks

Introducing EVs as a mode of transportation impacts electricity networks in two aspects: infrastructure and operation. To identify associated impacts and their severity, extensive efforts and studies must be conducted, as shown in Figure 29: Studies necessary for introducing EVs chargers into electric networks. These studies are necessary for integrating EV chargers into electric networks and require a significant amount of time. In the meantime, detailed electricity network parameters, data on network conditions and loading history, electricity load profiles, historical weather data, and consumer behavior data must be collected, reviewed, analyzed, and evaluated. Unfortunately, most of the above-mentioned data is not readily available or is confidential in many countries. Due to the lack of such information, proposed solutions for mitigating the negative impacts of EV charging

on Rwanda's electricity networks are reported through various scenarios, as presented in Table 24: Proposed Modular EV Charger Scenarios.

Available solutions to mitigate negative impacts of EVs charging stations on electric networks

Expanding the use of EVs increases the electricity demand on DNs, which might lead to an energy availability crisis. This issue can be addressed by either installing new infrastructure and/or adding distributed generation and energy storage devices (ESDs). Introducing specific regulations can be a last resort to solve EV charging problems on DNs. The next table reports solutions to mitigate the negative impacts of EV charging on electric networks.

New infrastructures	*Energy Availability*	*Regulations*
▪ Upgrading TNs and DNs ▪ New EVs charging stations	▪ Install DGs: PV, Fuel-Cell. ▪ Installing ESDs: batteries, ultra-capacitor.	▪ Smart Charging: Dynamic pricing. ▪ Charging offers. ▪ Software-based methods.

Table 23: Solutions to mitigate negative impacts of EVs charging on electric networks

New infrastructure installations

To overcome the need for excess electricity at DNs due to electric network congestion, while there is no lack of electricity generation, two solutions based on infrastructure upgrading can be implemented. The first solution is to add new passive equipment such as transmission lines, distribution cables, transformers, etc. The aim of this solution is to evacuate bulk-generated energy from centralized power stations to all DN territories to charge EVs. The second solution is to increase the number of EV charging stations, ensuring they follow international technical standards and are optimally located to allow proper access for EVs to be charged

Energy availability solutions

If the EV energy charging needs are not fulfilled due to a lack of energy availability in distribution networks (DNs), two solutions can be implemented. The first solution is to install a PV-based distributed generation (DG) in DNs. The second solution is to install energy storage devices (ESDs), such as batteries, that can be charged during low tariff periods and discharged into EVs during high tariff periods. Furthermore, ESDs can be used if PV energy generation does not coincide with the EV charging periods. Thus, PV-generated energy can be stored in ESDs temporarily and used at desired times to charge EVs.

5.14 Recommendations and scenario development regarding the grid and EVCI

For uncontrolled charging, EV owners charge their vehicles based on their needs whenever and wherever they wish. There is no oversight or insight into the actual charging behavior, other than generic trend patterns based on historical developments. Thus, utilities must install new infrastructure (i.e., charging capacity) and increase the number of charging stations to facilitate drivers' charging needs. This solution is not economically optimal and will not flatten load profiles.

Therefore, it makes more sense to introduce regulation to direct charging behavior. Multiple methods can influence charging behavior. Smart charging methods can be applied to fulfill technical and financial objectives. Technical objectives can include balancing load power, realizing V2G schemes, regulating voltage and frequency, and reducing losses. Financial objectives can include reducing energy costs and increasing profits for charging providers. Under smart charging, EV charging is controlled by linking EVs to distribution networks (DNs), while EVs are also considered mobile energy storage devices (ESDs) that can reduce load fluctuations. This link can be established through developing software-based methods to direct drivers to the most optimal charging times and locations based on EV owners' locations/zones.

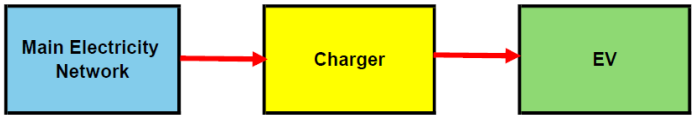
The following section focuses on presenting the impacts of EV charging on existing distribution networks to help select the appropriate technology for Rwanda. The proposed bundle of solutions is based on the following assumptions:

- EVs charging terminals/stations are connected to the DNs
- Installing Local Modular Slow & Fast AC Charging units and Modular Fast DC Charging units: i) alone; ii) with Local Modular PV Power Stations; iii) with Local Modular PV Power Stations and Local Modular Battery Banks.
- Local Modular AC Chargers, PV Power Stations, and/or Battery Banks can be multiplied to increase the charging capacity aiming at charging higher number of EVs on one charging terminal/station.

Furthermore, is worth mentioning that:

- Selecting a candidate solution or combinations of the below proposed ones is impacted by the main electricity network condition and resilience, and whether EVs charging stations are located ON-Grid or OFF-Grid.
- Selecting international standards AC Charge capacities (Level 1 (slow) and Level 2 (fast) chargers' capacities are up to 2.4 kW and 22 kW respectively) and DC Charging capacity (Level 3 (fast) charging capacity up to 50 kW) as shown in Table 24: Proposed Modular EV Charger Scenarios and in Table 25: Levels of EVs charging based on charging power and charging speed
- Selecting Modular PV Power Stations of 30.8 kW capacity is explained in the following scenario analyses based on the actual Rwanda's solar irradiance data in various locations.

The following table gives an overview of the Modular EV Charger Scenarios:

Scenario	*Figure*
Scenario-I: ON-Grid EV Charger Base case: The grid is strong and thus electric networks can deliver the required energy to charge EV.	 Figure 1 Scenario-I: ON-Grid EV Charger

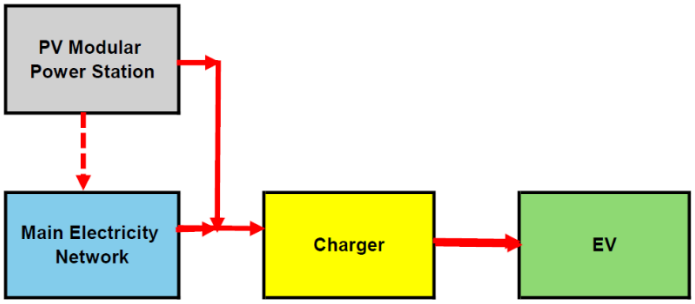
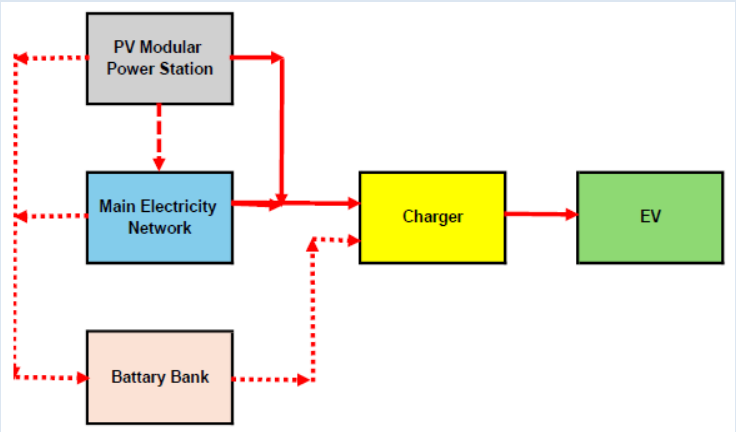
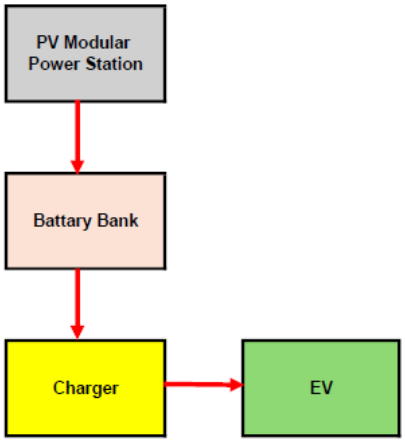
Scenario	*Figure*
Scenario-II: PV Power Station & ON-Grid EV Charger The grid is weak during daylight period and the EVs charging activities during same period, thus electric networks cannot deliver the required energy to charge EV. Therefore, PV Modular Power Station is proposed to support the grid to charge EVs. Case 1: PV Station, North-East Rwanda Case 2: PV Station, South-West Rwanda Case 3: PV Station, Center Rwanda	 Figure 2 Scenario-II: PV Power Station & ON-Grid EV Charger
Scenario-III: PV Power Station with Battery Banks & ON-Grid EV Charger The grid is weak all day or the main EVs charging activities in the nighttime, thus electric networks cannot deliver the required energy to charge EV. Therefore, PV Modular Power Station along as battery bank is proposed to support the grid to charge EVs all day long.	 Figure 3 Scenario-III: PV Power Station with Battery Banks & ON-Grid EV Charger
Scenario-IV: PV Power Station with Battery Banks & OFF-Grid EV Charger OFF-grid EVs charging that is supplied from PV system with batteries.	 Figure 4 Scenario-IV: PV Power Station with -Battery Banks & OFF-Grid EV Charger

Table 24: **Proposed Modular EV Charger Scenarios**

Level	*Form*	*Description*	*Charging Power*	*Charging Speed (additional range per hour)*
Level 1 ▪ Simplest, cheapest, slowest technology ▪ Can be used to top up daily use ▪ Requires a specialized cable usually supplied with the vehicle but no specialized installation equipment.	Single Phase	Produces a "trickle" charge through a standard power point	2.4kW	Will not fully charge an EV overnight
Level 2 – destination chargers ▪ Most common home and public charging level ▪ Connected directly to the electrical network via specific socket and plug and a dedicated circuit ▪ Can fully charge an EV overnight, usually within 5 to 8 hrs.	Single Phase	AC power supplied using dedicated EVCI	3.6kW–9.6kW	Up to 40 km/hr
	Three Phase		3.6kW – 22kW	Up to 120 km/hr
Level 3 ▪ Fast charging units typically found in commercial premises or en-route highway locations like fuel stations ▪ Requires significant panel and service upgrades and consequently is the most expensive to deploy.	DC	DC fast charging power dedicated EVCI and electrical infrastructure	25kW–350kW	Typically, 70 km/10min or 420 km/hr

Table 25: Levels of EVs charging based on charging power and charging speed

Proposed Solutions' Analyses:

Scenario-I: ON-Grid EV Charger

This scenario is proposed as the Base Case where a grid is strong enough and can deliver the required energy to charge EVs. The selection of the charger type will be based on:

- DN feeders' capacity that is connecting DN transformer/biller to EVs chargers taking into consideration DN feeders' voltage profile and losses.
- Voltage profile value at the connection point where EVs chargers to be connected to DNs.
- EVs batteries capacity and the required number of EVs to be charged at the same DN feeder and at same DN transformer and their charging time.

The proposed methodology to prevent negative impact from occurring due to adopting EV charging into DN will be based on the approval of DSO. There must be published regulations and steps to be followed for EVs owner to install an EV charger based on the calculated currents shown in Table 25: Levels of EVs charging based on charging power and charging speed.

Levels 1 and 2 (AC single phase): No installation permission is required from DSOs as the feeder's phase current is like current drawn by a home appliance such as an air-conditioner or even a single household load. In the meantime, slow charging will not cause demand peaks on the feeder and thus will not add a heavy burden on DNs.

Levels 2 and 3 (AC three-phase and DC up to 30kW): An installation permission is required from DSOs. EVs owners or investors will submit a request to install fast AC charger. DSOs will perform load flow analysis at candidate EVs chargers' locations to check assigned feeders' performance. DSOs will provide a reply on the installation request to EVs owners or investors. Even though fast charging causes stress on DNs, such EVs chargers will not cause demand peaks on the feeder as the withdrawn current is not high.

Level 3 (DC more than 30kW): An installation permission is required from both DSOs and TSOs and further network investigations are required.

Given the current status of Rwanda's DNs and the challenges stated in Chapter 3, the DNs are inadequate until the recommended upgrades by the Electricity Distribution Master Plan take place. To address the issue of current limits on DN transformers, there should be an additional supply to relieve the burden on DN equipment. Therefore, Scenario-II proposes installing PV power stations to support DNs in charging EVs.

Scenario-II: PV Power Station & ON-Grid EV Charger

This scenario is proposed if the grid is considered weak during the daylight period and EV charging activities occur during the same period. This means that the electric networks cannot deliver the required energy to charge EVs due to the lack of proper capacity of DN feeders connecting the DN transformer/biller to EVs or due to operational constraints such as DN feeders' voltage profile and losses.

Therefore, a PV Modular Power Station is proposed to support the main grid in charging EVs. This can be achieved by generating electric power from PV and injecting it into the grid at the Point of Common Coupling with the EV Charger. This penetration will:

- Delay any required investment in TNs and DNs needed to upgrade network cables' capacity;
- Relieve the overloaded DNs equipment; and
- Improve DN voltage profile and reduce losses.

PV Power Station Calculations:

Three cases are carried out to design the proposed Modular PV Power Station and estimate the output energy based on real solar irradiance in various locations in Rwanda. The three cases scenarios can be found the Annex, in the following order:

- Annex No. 20 Case 1 (North-East Rwanda:RE) - Modular pV Power Station: Project Description
- Annex No. 21 Case 1 (North-East Rwanda:RE) - Modular pV Power Station : Monthly Production and Sources of System Loss
- Annex No. 22 Case 1 (North-East Rwanda: RE) - Modular pV Power Station: Annual production and condition set
- Annex No. 23 Case 1 (North-East Rwanda:RE) - Modular pV Power Station: Components, Wiring Zones, Field Segments

- Annex No. 24 Case 1 (North-East Rwanda:RE) - Modular pV Power Station: Detailed Layout
- Annex No. 25 Case 2 (South-West Rwanda) - pV Power Station: Project Description
- Annex No. 26 Case 2 (South-West Rwanda) - Modular pV Power Station: Monthly Production and Sources of System Loss
- Annex No. 27 Case 2 (South-West Rwanda) - Modular pV Power Station : Annual Production and Condition set
- Annex No. 28 Case 2 (South-West Rwanda) - Modular pV Power Station : Components, Wiring Zones, Field Segments
- Annex No. 29 Case 3: PV Power Station, Center Rwanda - pV Power Station: Project Description
- Annex No. 30 Case 3: PV Power Station, Center Rwanda- pV Power Station: Monthly Production and Sources of System Loss
- Annex No. 31 Case 3: PV Power Station, Center Rwanda- pV Power Station: Annual Production and Condition set
- Annex No. 32 Case 3: PV Power Station, Center Rwanda - pV Power Station: Components, Wiring Zones, Field Segments

Scenario-III: PV Power Station with Battery Banks & ON-Grid EV Charger

This scenario is proposed if the grid is weak all day or if the main EV charging activities occur at night, making it difficult for the electric networks to deliver the required energy to charge EVs. Therefore, a Modular PV Power Station along with a battery bank is proposed to support the grid in charging EVs all day long

Scenario-IV: PV Power Station with Battery Banks & OFF-Grid EV Charger

This scenario is proposed for Off-grid EVs charging supplied from PV Power Station with battery banks to deliver steady charging for EVs.

The main solutions stated by the Electricity Distribution Master Plan directly related to the EV charging project for Kigali city (similar solutions are listed for other territories) are illustrated below:

- Some HV substations feeding DNs and MV feeders will be overloaded by 2025-2028, necessitating upgrades.
- While the majority of MV feeders are not expected to be overloaded by 2028, some feeders will need upgrading or reconfiguration in the event of faults or maintenance.
- MV line extensions, new transformer installations, and LV network upgrades are required to improve power quality and address voltage drop issues.
- Upgrading single-phase networks to three-phase networks.
- Focusing on accelerating the On-Grid Access Program Countrywide (RUEAP).
- Transitioning from overhead to underground cables in developed areas of Kigali city.

6 INSTITUTIONAL ROAD MAP: BUSINESS MODELS AND REGULATORY FRAMEWORK

This chapter focuses on the institutional layer of the charging system in the intermediate state and end state of this charging infrastructure. In general, from the current situation to the end state, the role of private commercial organizations will grow as the market for charging services becomes more mature and the return on investments for providing services and owning assets in the charging ecosystem becomes more predictable. Before reaching ecosystem maturity, public organizations that finance, manage, and in some cases operate parts of the charging system, will have a role. Public entrepreneurship and public-private partnerships in this manner can play an essential role in maturing the charging ecosystem and derisking investment opportunities to enable private entities to compete and innovate in the ecosystem.

6.1 Functions in Charging Infrastructure Business Ecosystem

A charging infrastructure ecosystem involves multiple functions, which may be taken up by one stakeholder or delivered through partnerships between different stakeholders. Apart from the set-up of charging infrastructure, other roles include land provision, electricity supply, EVSE supply, charging software solutions, and customer services.

These key functions, developed in section *5.5 Functions in Charging Infrastructure Implementation*, are as follows:

- **Procurement of charging infrastructure**
- **Land provision**
- **Energy supply**
- **EVSE supply, installation, and maintenance**
- **Charging software solutions**
- **Customer services**

These functions can be taken by the public sector, the private sector, or both through public-private partnership arrangements. The most direct participants in the EV recharging ecosystem are Charge Point Operators (CPOs) and eMobility Service Providers (eMSPs). The recharging infrastructure they provide varies depending on the use case (e.g., on-route, on-street, or destination recharging points) and the technology used (e.g., slow, fast, or ultra-fast rechargers).

Depending on the business model and regulatory framework, the roles of CPO, EMSP and other stakeholders can be separated or combined under one entity.

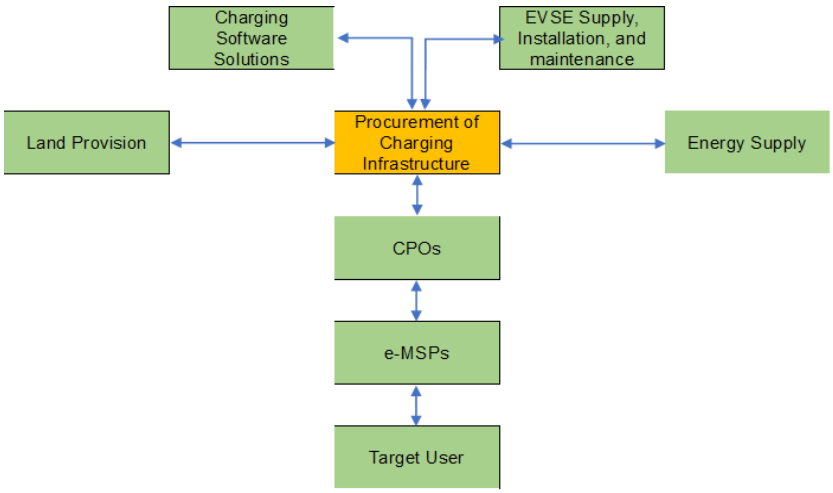


Figure 34: Key functions in EVCI business ecosystem

6.2 Business Models to Upscale EVCI Deployment

Though the private sector has made a significant stride in initiating the deployment of EV charging infrastructures in Rwanda, their current initiatives alone are not sufficient to scale up investments in EV charging infrastructures and to meet government e-mobility targets. The government, however, could leverage private sector investments by introducing various forms of partnerships with the private sector or by introducing frameworks that stimulate and sustain private sector investments.

This section presents 4 business models for EV charging that could be simultaneously introduced and developed in Rwanda. These are government driven public-private partnership model, private sector consumer-driven model and service provider model, and community-based deployment model.

The government of Rwanda, alongside the private sector, can adopt one of these implementation models to effectively deploy EV charging infrastructure. The financial models vary according to the approach implemented, offering flexibility in funding and operation, and allow different options for market engagement. Also, different modes of transport may benefit from different approaches, depending on the typical market situation, appetite and maturity phase.

Additionally, it is important to mention that these models are not mutually exclusive. Depending on the circumstances, needs, and actors involved, all of them can be implemented in combination to effectively deploy EV charging infrastructure. However, it could happen that depending on the market evolution and leading actors, some of these approaches may dominate the deployment strategy of EVCI. The following figure provides an overview of 3 of these approaches and their associated financial models.

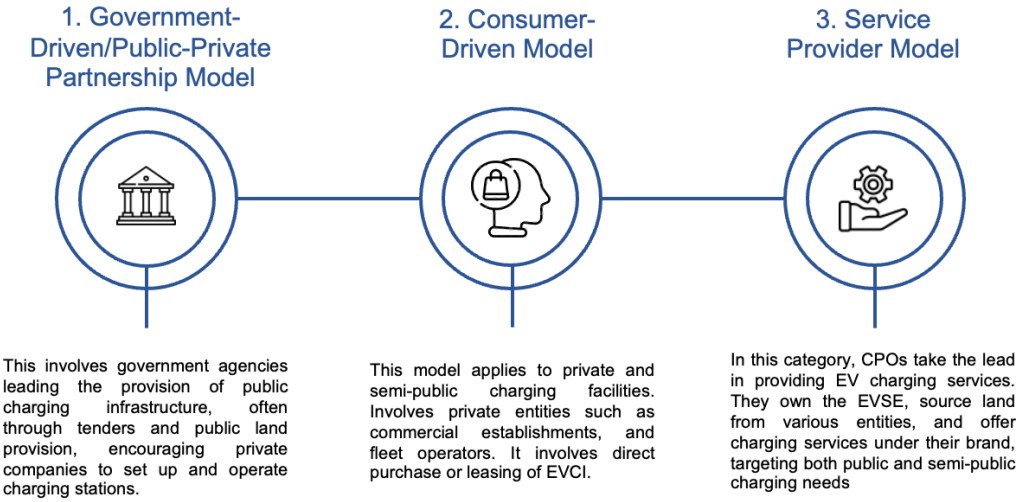


Figure 35: Approaches for the deployment of EVCI

These approaches, as well as the community-based deployment model, are explained in detail as follows:

6.2.1 Public-private partnership (PPP) model

This is the main model used by various cities of countries such as The Netherlands, China, Germany, achieving higher success rates in deploying EV charging infrastructure. It is particularly dominant in the early years of EV ecosystem development to establish a basic network of public charging facilities. This model applies to both public and private EV charging infrastructure.

In this model, government agencies—such as municipal governments, municipal corporations, urban development authorities, or national agencies—lead the provision of public charging infrastructure. These agencies issue tenders for the installation and operation of public charging infrastructures. Public land could be provided for free or under a rental arrangement for setting up charging facilities. The charging equipment may be owned by the government or by a CPO contracted to own and operate the charging services.

For self-owned EV charging facilities, public sector agencies can procure the EV charging infrastructure through an EPC (Engineering, Procurement, and Construction) contract. Charging services may be self-managed or outsourced to a CPO. Alternatively, government authorities may enter a public-private partnership (PPP) contract, inviting CPOs to install and operate EV charging facilities for public use. In this PPP model, governments often offer financial subsidies, concessional land provision, and/or energy supply to incentivize CPOs to reduce capital costs of implementation.

Application in Rwanda

- To apply the government-driven model for EV charging infrastructure in Rwanda, several steps can be envisioned. The local government, such as the City of Kigali, can issue a tender inviting private companies to establish and operate public EV charging stations throughout the city via a concession agreement. This tender will be awarded to companies that propose the lowest service fees.

- The selected concessionaire will be responsible for supplying, installing, testing, commissioning, maintaining, and operating the public EV charging stations at their own cost during the designated lease period. They will recover these costs through service fees.
- Land for the charging stations can be provided by the government either for free or at a fixed rental fee for the contract duration, which could be at least five years. The Energy Utility Corporation Limited (EUCL) will be tasked with providing electricity connections up to a specified capacity for these public charging facilities. The concessionaires will need to obtain EV metered connections and will be responsible for the energy charges for the electricity used.
- The City of Kigali or another mandated body will offer regulatory support for applications, permissions, quality checks, site feasibility, and approvals. They will also assist the concessionaire in processing the electricity connection with EUCL. At the end of the lease period, the concessionaire may opt to take ownership of the charging infrastructure assets, pending clearance by the land-owning agency and EUCL. This model ensures that the initial costs and risks are shared between the government and private sector, promoting the development of EV infrastructure in Kigali and potentially other parts of Rwanda.

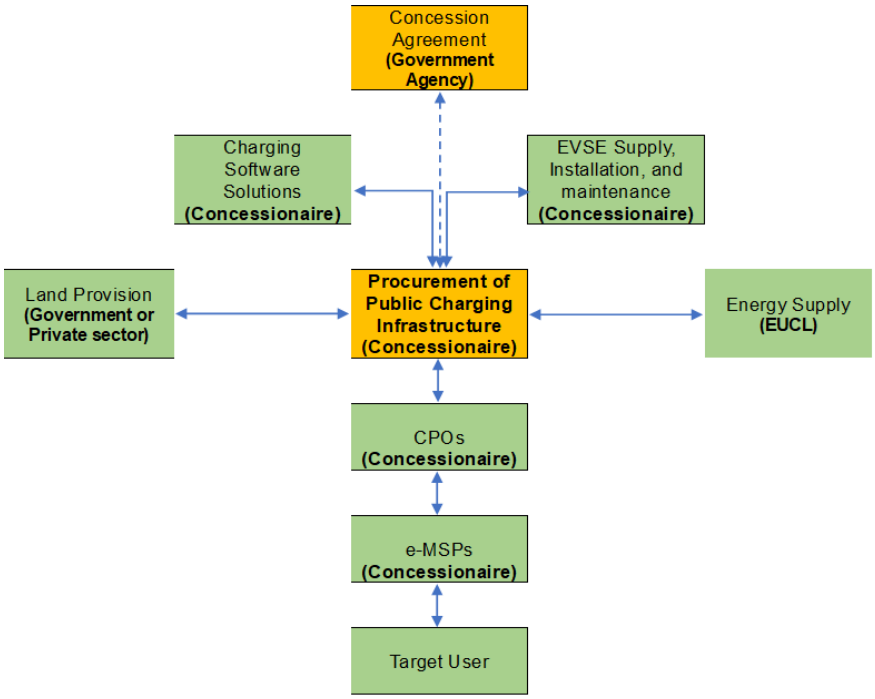


Figure 36: PPP Model

Benefits of Public-Private Partnerships

Financial Resources. One of the key advantages of PPPs is the ability to leverage both public and private financial resources. Governments can provide funding and grants, while private entities contribute investments and operational expertise. This collaboration allows for more substantial capital availability, reducing the financial burden on a single entity and enabling faster infrastructure development.

Expertise and Innovation. PPPs enable the exchange of knowledge and expertise between public and private entities. Governments bring regulatory insights and a

broader understanding of the public's needs, while private companies contribute technical know-how and market insights. This collaboration fosters innovation, driving the development of advanced charging technologies, efficient management systems, and user-friendly solutions.

Shared Risks and Responsibilities. By partnering through PPPs, risks and responsibilities are shared between public and private entities. This distribution of risk helps mitigate financial uncertainties, as costs and benefits are allocated appropriately. It also promotes a sense of accountability, ensuring that all parties are actively involved in the success of the charging infrastructure project.

Challenges and Limitations of Public-Private Partnerships

While PPPs offer numerous benefits, they also face certain challenges and limitations:

Balancing Interests. Aligning the diverse interests of public and private entities can be challenging. Governments aim to prioritize public welfare and equitable access, while private companies seek profitability. Striking a balance that addresses both objectives requires careful negotiation and compromise.

Regulatory Frameworks. Developing appropriate regulatory frameworks is vital to facilitate PPPs in the charging infrastructure sector. Clear guidelines and standards ensure fair competition, protect consumer rights, and address any potential monopolistic practices.

Ensuring Accessibility and Equity. Expanding charging infrastructure should prioritize accessibility and equity. It is crucial to consider the needs of underserved communities and rural areas, ensuring that the benefits of EV adoption and charging infrastructure reach all segments of society.

6.2.2 Consumer-driven model

The existing private sector-driven models could be further supported to maximize investments in the country. The current private sector led models can be categorized into consumer-driven model and service provider model.

6.2.2.1 Consumer-driven model

This model applies to private and semi-public charging facilities and to private EV owners and fleet operators.

The consumer-driven model applies to private and semi-public charging facilities, targeting private EV owners and fleet operators. In this model, primary stakeholders such as malls, commercial or institutional establishments, retail shops, and restaurants that have parking available on their premises partner with CPOs. These CPOs handle the supply, installation, maintenance, and management of EVCI. Typically, EVCI procurement by the private entity is through direct purchase, although internationally, leasing models from suppliers or CPOs are also emerging.

For private EV owners and fleet operators, the implementation is straightforward. EV owners can procure chargers from their automobile manufacturers, EVCI retailers, CPOs, or distribution utilities. Depending on the charger type and power connection, software services may be available to control charging sessions and take advantage of time-of-use or time-of-day tariffs. Fleet operators provide land for their EV fleets' charging facilities, owning or leasing it. They handle EVCI supply,

installation, and maintenance through direct contracts with suppliers or CPOs, managing charging services in-house or contracting them out.

Application in Rwanda

This model is already being piloted in Rwanda for commercial establishments, retail shops, and EV bus fleet operations. While there is no requirement yet for commercial establishments and retail shops to install EV chargers, some have begun installing EV charging infrastructure as an amenity for occupants and visitors. Access to charging infrastructure in office establishments is crucial to enable higher rates of EV adoption. For example, KABISA installed three chargers at the Norrsken Kigali Innovation Campus, offering free charging to EV users. This model is expected to scale up with policies requiring establishments with parking areas to install multiple EV charging stations.

Bus fleet operators in Kigali have also piloted electric buses. Instead of purchasing their own electric buses and installing EV charging infrastructure, these operators have entered a pay-as-you-drive lease agreement with BasiGo, including free electric bus charging and maintenance.

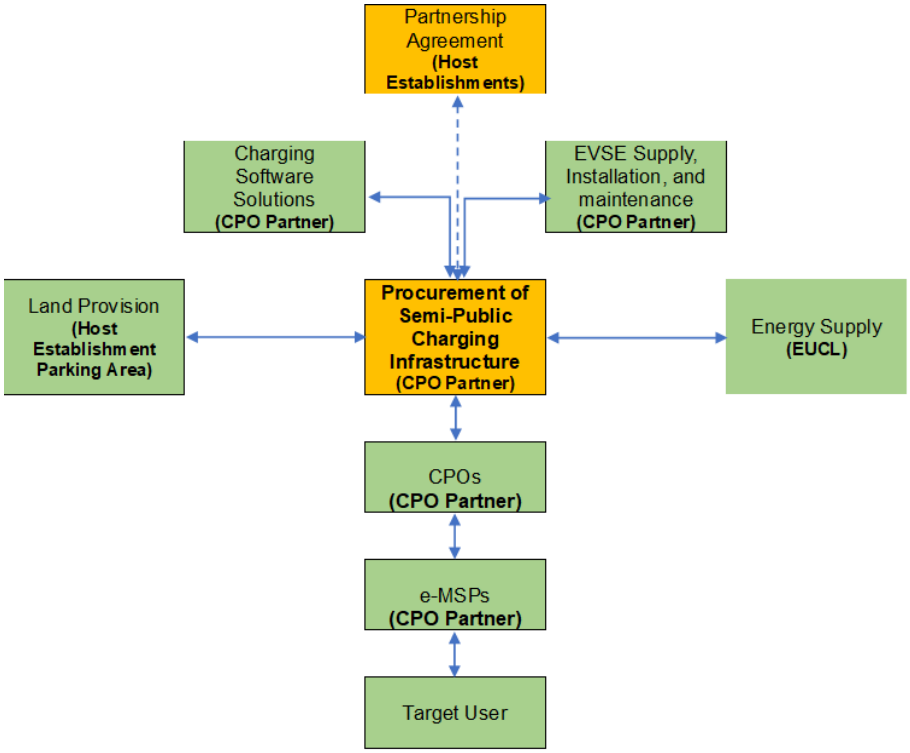


Figure 37: Consumer Driven Model

6.2.2.2 Service provider model

In this service provider model, it is the CPOs that drive EV charging provision for public and semi-public charging. The key distinguishing features of the service provider model are the following:

- EVCI equipment is usually owned by the CPO

- Land is sourced from a variety of owners, including public and private entities (this is especially true for private CPOs), and
- Charging services are offered under the brand of the CPO.

Private CPOs aim to establish a network of charging facilities in strategic locations with high potential charging demand. They source land parcels in selected locations from public or private entities, install EVSE equipment supplied by manufacturing partners, and operate paid EV charging services for public or semi-public use. CPOs may enter revenue-sharing arrangements with host establishments or other landowners for the use of land.

In some countries, distribution utilities are also entering the charging infrastructure market as CPOs. These utilities typically use their own land to set up public EV charging facilities and operate them as paid services. Distribution may also provide bundled charging services for private EV owners and recover the capital and operating costs through electricity tariffs. Other stakeholders driving the service provider model of EV charging implementation include industrial companies that are moving into charging infrastructure, and EV manufacturers that are setting up charging infrastructure networks as allied services.

Application in Rwanda

The private sector in Rwanda has been proactive in response to the government's electric mobility policy, with several companies already investing in EV charging infrastructure using the service provider model. Notably, KABISA and EVP have initiated the development of public EV charging stations targeting both private and commercial transport. As of now, KABISA operates six charging stations and plans to add 35 more in the future, while EVP has established five stations. Similarly, Ampersand, Spiro, and REM have created a network of electric motorcycle battery swapping stations in Kigali. Currently, Ampersand has installed 15 charging stations, Spiro has 12, and REM has one station.

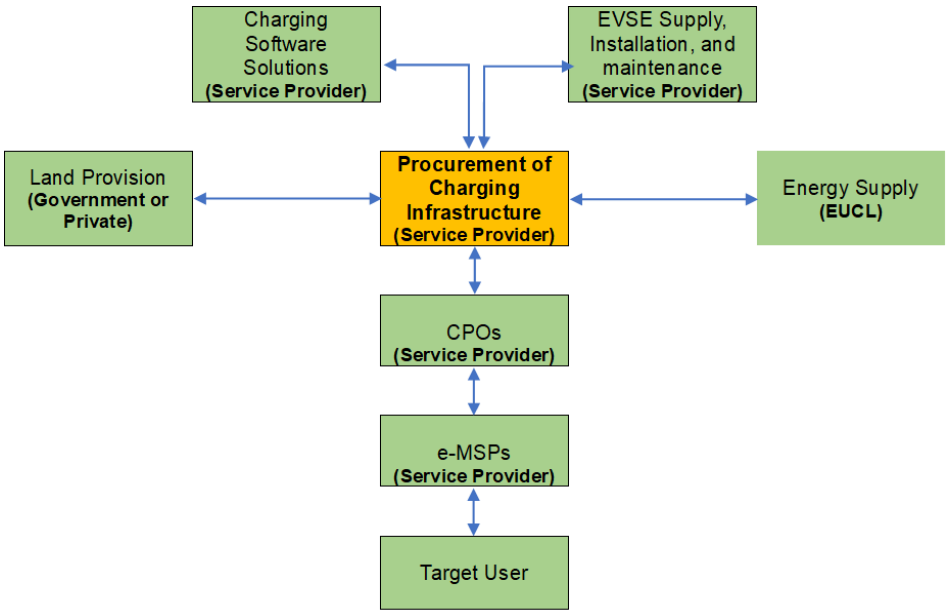


Figure 38: Service Provider Model

6.2.3 Community-based Model

Community-based funding involves mobilizing resources from local communities to support the development of EV infrastructure. It encourages grassroots participation and engagement, leading to a stronger commitment to maintaining, utilizing, and promoting the charging infrastructure. Community funding can complement government and private sector efforts, especially in areas where commercial investments may be limited.

This community investment model could be adapted to fund public charging infrastructure. Members who contribute towards the capital investment costs of charge points could have sole or preferential use of these charge points at certain times of pay a discounted tariff. Non-members could use the infrastructure but typically pay more.

One benefit of this investment model is that a greater proportion of charge point revenue will be retained within the community and could be reinvested to install further charge points as demand grows. Additionally, the charge points could integrate with renewable energy generation, recharging when there is surplus generation, which may otherwise be wasted.

The main barrier is likely to be a lack of demand in the short term as these groups invest based on the requirements of their members. Difficulties might be encountered in securing locations for the infrastructure. In addition, the community investment model may face limitations in generating significant capital compared to government or private sector investments as it relies heavily on voluntary contributions and community participation. Scaling up this approach to meet the national demand for charging infrastructure could pose challenges. Nevertheless, it can effectively complement other approaches and fill the gaps in specific regions in Rwanda.

6.3 EV Charging Infrastructure Costs

As previously discussed, there are several contracting models for infrastructure deployment. Some models require more public investment, while others rely more heavily on private sector efforts.

Below is a reference table outlining the typical costs involved in installing an EV charging station. These include: i) charger costs, ii) electricity connection costs, iii) civil works, iv) software and integration costs, and v) annual operation and maintenance costs.

The table presents typical EV charging infrastructure (EVCI) investment costs from developing countries, which can serve as a benchmark for assessing Rwandan EVCI investment costs. The following EV Charging Station Project Investment Costs per charger are presented in US dollars.

Cost Items	***Charger Type***			
	AC 2 wheeler	**AC 4 wheeler**	**30 kW**	**CCS-60 kW**
Charger costs	180	900	7,200	12,600
Electricity connection costs			3,000	6,000
Civil works	60	60	600	600

Software and integration costs	50	50	480	480
Operation and maintenance costs (yearly)	45	45	435	435

Source: Authors elaboration based on international survey

6.4 Public Investment and Governance Roles: Contractual models for rolling out DC charging and AC charging networks

When rolling out DC and AC charging networks, governments can utilize various contractual models[62] to effectively manage and deploy the necessary infrastructure, these approaches are mainly for a government-driven approach. Here are some key contracting models described, which vary according to the level of risk taken by public and private roles and the level of involvement the government desires to adopt.



Figure 39: Contractual models overview

6.4.1 Public Contract Model for EV Charging Infrastructure

The public authority maintains full control over the specification, installation, operation, and use of the EV charging infrastructure, including network connections. Known as the 'own and operate' model, this approach allows the local authority to directly manage or outsource tasks to third-party operators. The authority bears all upfront costs, including capital, operational, and maintenance expenditures, while retaining all generated revenues and assuming the full commercial risk. This model is typically applied to small-scale or low-cost projects that benefit from grant funding to minimize financial risks. Often, local authorities set up a special purpose vehicle to facilitate commercial operations.

[62] https://www.eib.org/attachments/publications/electric_vehicle_charging_concessions_en.pdf

Key Considerations

- **Full Control:** The local authority has complete autonomy to decide on the locations, types, and payment functions of the charging points, allowing for strategic installations in areas that may not be commercially viable for the private sector.
- **Revenue Retention:** The authority collects and retains 100% of the revenue generated from users.
- **Outsourcing Flexibility:** Authorities can leverage private sector expertise by outsourcing design, build, and operation tasks while using familiar procurement processes and available frameworks.
- **Scalability:** This model allows for testing the success of public EV charge points on a small scale before expanding to larger projects.
- **High Capital Costs:** The local authority is responsible for all upfront capital costs, and available grant funding may be limited.
- **Operational Responsibility:** The authority must manage the design, build, and operation directly or oversee outsourced contractors.
- **Technical Expertise:** Authorities need to understand and adapt to the technical requirements of charging infrastructure and market demands.
- **Reputational Risk:** Any network failures or reliability issues fall on the local authority, potentially affecting public perception and trust.

6.4.2 Joint Venture/Strategic Partnership Model for EV Charging Infrastructure

This model involves a collaborative approach where the public authority and a private partner share control of the EV charging infrastructure through a jointly created venture. Both parties share the risks and revenues according to their stakes in the venture. Financing can come from either or both parties or a separate third party. This model can encompass various commercial arrangements between public and private sector participants, and it is particularly suitable for larger-scale rollouts requiring private sector capital and expertise.

Key Considerations

- **Shared Risks and Resources:** Both parties contribute to and share the risks and resources, providing a balanced approach to project management.
- **Flexible Financing:** Funding arrangements can be tailored to the specific resources, objectives, and constraints of each party. Risk and benefits are distributed according to each party's stake in the venture.
- **Resource and Expertise Sharing:** Local authorities benefit from the private sector's resources and expertise as needed.
- **Cross-Authority Collaboration:** Ideal for projects that span multiple jurisdictions, maximizing shared resources and growth potential
- **Complex to Establish and Manage:** Setting up and managing a joint venture can be complex and requires a significant amount of effort and coordination.
- **Requires Alignment of Objectives:** Success depends on the ability of both parties to align their objectives and work together towards a unified goal.
- **Location Control:** Local authorities may have limited control over the location of charge points, potentially neglecting areas that are not commercially viable.

6.4.3 Concession Model

Under the concession model, the public authority retains some control over the EV charging infrastructure but transfers most operational and financial risks, including user-demand risk, to a private partner. This is achieved through a competitively tendered concession contract. The private partner finances the project, manages the installation and operation, and retains user revenues. There may be potential revenue-sharing arrangements with the public authority.

The operator may fully fund or match-fund the capital costs and take on the operational costs. In a match-funding scenario, the public authority can supplement grant funding to overcome capital constraints, while in a fully funded model, the private partner gains ownership and control of the assets, covering ongoing costs and liabilities.

Key Considerations

- **Risk Transfer:** Significant risks are transferred to the private sector.
- **Private Financing:** Utilizes private financing to support public infrastructure.
- **Cost Offset:** Leverages private funding to offset upfront costs for the local authority, with match funding allowing some retention of control.
- **Profit Sharing:** Potential for profit or revenue sharing, enhancing opportunities through competitive procurement.
- **Increased Funding:** Private funding extends public resources, enabling more extensive infrastructure development.
- **High Standards:** Operators are incentivized to maintain high operational and maintenance standards and future-proof technology due to profitability impacts.
- **Control Loss:** The public authority might have a potential loss of control over certain project aspects.
- **Market Understanding:** Needs a strong understanding of market offerings for successful, competitive procurement. The process may involve complex contract negotiations and require significant internal resources to manage the process.
- **Asset Ownership:** Ownership of assets at the end of the concession period is subject to negotiation.

This model is suitable for public authorities looking to leverage private sector investment and expertise while managing risks and optimizing resources through strategic partnerships.

6.4.4 Availability-Based Contract Model

Similar to the concession model, but the public authority pays the private partner based on a fixed fee for the infrastructure's availability for use rather than user demand. While most risks are transferred to the private partner, the user-demand risk remains with the public authority.

Key Considerations:

- **Risk Transfer:** Significant risks are transferred to the private sector, except for the user demand risk, which is assumed by the public authority.
- **Operational and Maintenance Risk:** Transfers operational and maintenance risks to the private partner.

- **Fixed Fee:** If well planned, there is more certainty on the fee that will be paid to the private partner. However, the risk of user demand remains with the public authority.
- **Infrastructure Availability:** Ensures infrastructure availability, providing a reliable service to the public.
- **Ongoing Public Payments:** Requires ongoing payments from the public authority, which can be a financial burden.
- **Limited Control:** The public authority has limited control over the use and management of the infrastructure.

6.4.5 License Model and Leasing Model

In the license model, the private partner controls the infrastructure and assumes most project risks from installation to operation. The private partner finances the capital and maintenance expenditures and collects and retains user revenues. This model typically includes conditions and limitations on the private partner's actions but allows more freedom compared to other partnering models. One version of this model is the leasing arrangement, in which the local authority leases available council land to commercial operators. The private sector operator undertakes the development and implementation of public EV charge points, bearing the risk and cost of operations.

Key Considerations:

- Risk Transfer: Most project risks, including installation, capital, and maintenance expenditures, are transferred to the private partner.
- Revenue Retention: The private partner collects and retains user revenues, providing an incentive for efficient operation and maintenance.
- Operational Freedom: The model typically offers the private partner more freedom in decision-making, leading to potentially more innovative solutions.
- Public Authority is paid based on the leasing for the installation of infrastructure on public domain.
- Control Limitations: The local authority may have less control over the specifics of operation, as the private partner has significant autonomy.
- Dependence on Private Partner: Success is heavily dependent on the private partner's performance and compliance with the licensing terms.

Indicative Allocation of Main Risks under the Five Contracting Models

The following table provides a comparative overview of different partnering models for EV charging infrastructure, the allocation of key risks.

Risk	*Public Contract*	*Joint Venture*	*Concession*	*Availability-Based Contract*	*Licence*
Site Identification and Provision	Public	Shared	Public or Private	Public	Private
Permitting	Public	Shared	Private or Shared	Private or Shared	Private
Installation	Public	Shared	Private	Private	Private
Operating	Public	Shared	Private	Private	Private
Demand	Public	Shared	Private	Public	Private

Risk	*Public Contract*	*Joint Venture*	*Concession*	*Availability-Based Contract*	*Licence*
Technology	Public	Shared	Private	Public	Private
Change in Law	Public	Shared	Shared	Shared	Private
Financing	Public	Shared	Private	Private	Private

Table 26: Risk allocation on Contracting Models for EVCI deployment[63]

6.5 Goals and Intermediate Scenario per mode of Transport

As referred for the cost of EVCI, the Annex No. 33 gives and estimation of the cost of equipment and installing EVCI. Institutional Charging Infrastructure Implementation Roadmaps per Transport Mode

Per transport mode, different choices need to be made from an institutional perspective on how to support the deployment of charging infrastructure, assuming uptake as mentioned in Chapter 4.

For each transport, a distinction will be made regarding the intermediate scenario and the final scenario. The intermediate scenario is useful when a mature environment is not yet in place.

6.5.1 Institutional roadmap for Electric Vehicles – Passenger Cars

Goals Scenario

For passenger cars, best practices demonstrate that mature market-driven operational and financial models are possible, with market actors providing charging infrastructure in urban areas and on highway routes. Public charging needs will be limited to cases where no positive business case can be achieved.

Charging type	*Charging use case*	*Financial drivers*
Private charging (home, office)	AC charging sales and operation via service model for private and business customers.	Consumer model, integral business case with EV sales
Semi-public charging (public service on private location)	Dominant service model for Kigali, AC charging for long stay operated by private locations DC charging (urban and en-route) operated by competing CPOs	Service model in Kigali for AC and DC charging. Access through full roaming is a pre-condition for the business case
Public charging	DC charging not present AC charging: limited use of public locations, mainly in secondary cities without a strong business case.	PPP models for AC charging outside Kigali, operations by selected commercial actors.

Table 27: EVCI for passenger vehicles - Goal scenario

[63] European Investment Bank. (2022). Electric vehicle charging concessions: A contract guide for public authorities. In Electric vehicle charging concessions: A contract guide for public authorities (pp. 1–5). https://doi.org/10.2867/418865

Intermediate Scenario

It is assumed that an intermediary scenario is needed for the first period to support uptake when more mature business models are not yet in place. Causes for this might be:

- Lack of scale for a certain business model, where additional (government) support is needed.
- An immature, not yet fully functioning market where government needs to provide sufficient guidance to assure proper development
- Not all required regulation and/or incentives are in place

The Semi-public charging network and destination charging network will need adequate support to develop a customer-driven model; EV targets and incentives should drive sales; commercial buildings should want to provide service; incentives for fleets; building regulations; thus indirectly forcing the EVCI for passenger vehicles - Intermediate scenario development there;

Charging type	*Charging use case*	*Financial drivers*
Private charging (home, office)	AC charging sales and operation via service model for private and business customers.	Consumer model, integral business case with EV sales
Semi-public charging (public service on private location)	DC charging Kigali through concession model to assure availability DC charging outside Kigali via partnerships or subsidies AC charging for long stay operated by private locations DC charging (urban and en-route) operated by competing CPOs MSP business case actively supported by government to assure access	Incentives for market actors to drive deployment and sales. Integrate with other (retail) earning models where kWh sales are not sufficient. Support for commercial buildings. MSP role by government or via concession to assure access, until private actor steps in. concessions with a guaranteed role for a certain period.
Public charging	DC charging not present AC charging: limited use of public locations, mainly in secondary cities without a strong business case.	PPP models for AC charging outside Kigali, operations by selected commercial actors.

Table 28: EVCI for passenger vehicles - Intermediate scenario

6.5.2 Institutional roadmap for Two Wheelers – moto-taxis

Goal Scenario

The goal scenario assumes that it is not feasible anymore to have non-interoperable charging or swapping stations for each moto-taxi provider, as this limits a scalable business case and overburdens the scarce public space and transformer capacity. Standards and interoperability will come into play; the MSP role emerges; because of the maturity of swapping station operators it is expected that MSPs will emerge without government pushes needed; most likely, the current swapping station operators will try to distinguish their swapping station services by becoming a MSP;

	Charging use case	*Financial drivers*
Kigali	Concession models for swapping stations to manage locations and capacity. Concessions for coordinated expansion; prevent parallel operators with own bikes, batteries, etc; interoperability and shared resources within the market; Competing concession owners of swapping stations, who can lose prime locations after a concession period. Standardization and interoperability as a pre-condition	Swapping station model is a service model, independent from vehicle business model. The concession has a concession period; swapping stations can lose prime locations to better service providers; Additional business models for MSPs providing access to charging services.
Outside Kigali (urban)	This follows the Kigali model	
Outside Kigali (rural)	Grid access will increase, but the amount of competing actors will be limited, reducing competition. Government may need to assure access and availability through concessions	Off-grid solutions will provide additional value, leading to integrated business cases, combining generation and charging.

Table 29: EVCI for two wheelers - goal scenario

Intermediate Scenario

In an intermediate scenario, focus should be on scaling and standardization of the existing network in Kigali. Outside Kigali, a concession model could support further expansion before private investors step in.

EVCI for two wheelers – intermediate scenario

	Charging use case	*Financial drivers*
Kigali	Moto-taxis already have a positive business case. Growing the network (with government support) is the main driver. Simplifying and standardization of grid access.	KWh sales and battery lease through current stations, Private investments for expansion
Outside Kigali (urban)	Concession model/availability-based for swapping stations outside of Kigali, thereby ensuring growth and availability	Controlled business case through concessions
Outside Kigali (rural)	Best effort by local actors	

Table 30: EVCI for two wheelers – intermediate scenario

6.5.3 Institutional roadmap for HDV - Buses

Goal Scenario

The goal scenario for buses is focusing at both the city of Kigali and other major urban centers with a dedicated urban bus transport network. It is expected that multiple bus operators will operate parts of the network, where charging happens mainly overnight. Charging depots can be an integrated part of the concession, but is expected to be provided in conjunction with the local government to assure access

to locations and access to connections. Also, to assure relative independence of a specific operator.

In the rest of Rwanda, buses connecting cities through BRT, it is essential that adequate charging facilities are available at the end-points, and where needed en-route for top-up charging.

EVCI for HDV - Buses – goal scenario

	Charging use case	*Financial drivers*
City of Kigali and other major urban centers	Concession or joint venture with City of Kigali for charging city buses. Integrate bus operation and charging.	License-based, service driven, bus depots.
Intercity/BRT	Concession models for operating charging depots and hubs. Availability is key for top-up charging Interoperability is in place to assure charging for multiple bus lines. Booking system may be an option for time-critical lines, to assure charging availability.	Availability is key, therefore a concession model or joint venture to assure availability. Income from additional services, either MSP or other retail services in these transport hubs.

Table 31: EVCI for HDV - Buses – goal scenario

Intermediate Scenario

Availability of charging infrastructure is essential for e-buses to perform their task, yet in an intermediate scenario the business case is not yet attractive. Therefore explicit support is needed, organizing market initiatives with strong government support to assure a predictable business case. Concession models with a clear end date assure the liability of charging infrastructure is essential for e-buses to perform their task, yet in an intermediate scenario the business case is not yet attractive. Therefore, explicit support is needed with a strong flexibility to move to a more mature governance once the scale of operations assures a business case for commercial actors.

EVCI for HDV - Buses – intermediate scenario

	Charging use case	*Financial drivers*
City of Kigali and other major urban centers	Assure charging facilities with every bus operator concession. Business cases and operational efficiencies need to be developed, a strong government involvement is neediness cases.	Concession driven or joint venture, with a specific end date.
Intercity/BRT	Integrated approach for e-bus operation and charging for each new line. Depots and hubs are not yet business case positive, need strong support.	Concession model or public contract to ensure availability.

Table 32: EVCI for HDV - Buses – intermediate scenario

6.5.4 Institutional roadmap for Mini-buses

Goal

Charing mini-buses will take place around taxi-parks and overnight depots. These will develop as privately run charging facilities, competing on additional services.

For prime locations, local government will provide concessions to assure optimal operations against favorable conditions.
For rural areas, guaranteed charging facilities will come from private operations of charging hubs under a concession model to assure availability and a fair price, independent from the minibus operators.

EVCI for HDV - Minibus– goal scenario

	Charging use case	***Financial drivers***
City of Kigali and other major urban centers	Customer driven license model for charging hubs at parking areas, or overnight depot charging. Concession model for prime locations, independent from bus operators.	License model from charging fees, as well as MSP services, operated by private actors.
Rural areas	Every hub will be operated by a public actor to assure access for different commercial transport actors.	Concessions or availability-based contract for rural charging hubs assure availability and an independent operation. Combination with off-grid PV and storage facilities provide additional options.

Table 33: EVCI for HDV - Minibus– goal scenario

Intermediate

As mini-buses are mostly operated by smaller enterprises, electrification relies on guaranteed availability of charging hubs and depots for overnight charging.

EVCI for HDV - Minibus– intermediate scenario

	Charging use case	*Financial drivers*
City of Kigali and other major urban centers	Concession model for the development of a basic charging network that supports initial growth; (Availability-based in secondary cities or rural areas where expansion is not feasible); MSP needed to manage demand; alignment of bus routes and charging facilities is crucial;	Alignment of bus routes and charging facilities is essential for private mini-bus operators to electrify, through concession or public contract. Customer-driven license model; for mini-bus charging depots of transport service providers on their own premises;
Rural areas	Every hub will be operated by a public actor to assure access for different commercial transport actors	Concessions or availability-based contract for rural charging hubs assure availability and an independent operation. Combination with off-grid PV and storage facilities provide additional options.

Table 34: EVCI for HDV - Minibus– intermediate scenario

6.6 Private Investment Roles

While the public sector is crucial in establishing a foundational regulatory framework for the deployment of the e-mobility ecosystem, the involvement of private stakeholders is key to creating an effective model. Public authorities need to develop basic regulations to ensure safety, interoperability, and fair pricing within the e-mobility infrastructure. However, private companies bring innovation, capital, and operational expertise, which are essential for the rapid and efficient deployment of electric vehicle charging infrastructure. Collaboration between the public and private sectors will ensure the development of a robust, accessible, and sustainable e-mobility network in Rwanda.

The roles of private stakeholders are vital in the e-mobility ecosystem, providing various services that ensure its functionality and growth.

CPOs manage the physical charging stations, ensuring they are operational and accessible. eMSPs facilitate services like charging station access, payment solutions, and user interface applications, creating a seamless user experience. These roles, often run by private entities in mature markets, are essential for developing robust charging station networks and battery swapping infrastructures. Additionally, transport companies offering electric transportation services and businesses selling electric vehicles contribute significantly to the ecosystem, promoting the adoption and sustainable operation of e-mobility solutions.

The collaboration between public regulatory efforts and private sector innovation and investment is crucial for developing a robust, accessible, and sustainable e-mobility network in Rwanda. This synergy will help accelerate the transition to electric mobility, reducing the country's carbon footprint and enhancing transportation efficiency.

To encourage investment in charging station networks, businesses must understand the critical elements involved in such an endeavor. Key considerations include conducting a thorough market assessment to identify demand and competition, selecting optimal sites for the stations, and planning the network to ensure coverage and accessibility.

Additionally, developing a detailed implementation plan is essential for smooth execution. Financial planning involves estimating capital expenditure (CAPEX) for setting up the infrastructure and operating expenditure (OPEX) for ongoing maintenance and operation. Identifying potential revenue streams, such as charging fees (direct income), or advertising, and attracting customers to nearby businesses (indirect income).

The following table outlines these essential topics for business investment models:

Investment model for an EV charging network
1. Market Assessment ▪ **Demand Assessment:** Evaluate the current and projected demand for EV charging stations in the target region. ▪ **Competition Analysis:** Identify existing competitors, their market share, and service offerings. ▪ **Regulatory Environment:** Understand government policies, subsidies, and incentives for EV infrastructure. ▪ **Technology Trends:** Stay updated on advancements in charging technology and infrastructure.
2. Site Selection and Network Planning ▪ **Location Analysis:** Choose strategic locations based on traffic patterns, proximity to amenities, and grid capacity. ▪ **Types of Charging Stations:** Decide on a mix of slow, fast, and ultra-fast chargers. ▪ **Scalability:** Plan for future expansion based on demand growth.
3. Implementation Plan ▪ **Objectives & Assumptions:** - Number of charging stations to be installed each year. - Average cost per charging station. - Expected number of users per station per day. - Average charging fee per session. ▪ **Expected growth rate in EV adoption and charging demand.** ▪ **Project Timeline:** Develop a timeline for the rollout of charging stations, including milestones and deadlines. ▪ **Resource Allocation:** Plan the allocation of resources, including financial, human, and technical resources. ▪ **Monitoring and Evaluation:** Establish metrics and KPIs to monitor performance and make adjustments as needed.
4. Capital Expenditure (CAPEX) ▪ **Charging Infrastructure Investment:** Estimate costs for purchasing and installing charging stations, software, and hardware for managing the network. ▪ **Other infrastructure investment:** grid connection fees, transformer upgrade and site preparation ▪ **Land Acquisition/Leasing:** Account for costs related to land or leasing spaces for charging stations.
5. Operating Expenditure (OPEX) ▪ **Maintenance and Repairs:** Budget for routine maintenance, equipment repairs, and replacements. ▪ **Energy Costs:** Estimate costs for electricity, including peak and off-peak rates. ▪ **Staffing Costs:** Include salaries for staff involved in operations, maintenance, and customer service.

▪ **Software and Connectivity:** Ongoing costs for network management software, connectivity, and cybersecurity.
6. Revenue Streams
▪ **Charging Fees:** Revenue from customers using the charging services, which could be time-based, energy-based, or subscription-based. ▪ **Partnerships and Advertising:** Income from partnerships with retailers, advertisers, and other stakeholders. ▪ **Government Incentives:** Include potential revenue from government subsidies and incentives for providing EV infrastructure. ▪ **Indirect Income:** Advertisement, or attracting customers to nearby businesses such as cafes, restaurants, shopping centers, supermarkets, hotels, and more

Table 35: Investment model for an EV charging network

6.7 Assessment and Recommendations on Policy Regulation

6.7.1 Key areas of regulation

The following information outlines the key areas for regulation necessary to establish a well-organized environment for the promotion and deployment of the EVCI ecosystem network.

Area of regulation	*Regulation*
Electric Vehicle Charging and Licensing of CPOs	In Rwanda, EV charging is treated as a service to the transport sector, aligning with the most common global practice and thus exempt from specific electricity regulations. CPOs establish EV charging infrastructures without licenses from RURA but must obtain business permits from the Rwanda Revenue Authority (RRA) and apply for industrial tariffs. There is no explicit national policy for EV charging that states that EV charging is treated as a service to the transport sector though current practices follow this approach. To enhance efficiency and clarity, developing explicit policies on this regard would be beneficial. Additionally, streamlining the tariff application process with the RRA is highly recommended. These measures will support the growth of the EV market by simplifying operations for CPOs and providing market clarity on system operations.
Private and Semi-Private Charging Deployment	For many developing countries including Rwanda, providing grants or subsidies to cover private charger installation costs, may create an unnecessary financial burden to the government. Perhaps an alternative option is to provide income tax deductions and other tax benefits related to the installation of domestic EV chargers. Concerning the EV charging tariff policy, extending the industrial tariff for private charging may require additional metering and transaction costs to both EV owners and distribution utilities. An alternative option is to increase the range of the lower tariff block of the increasing block tariff (IBT) scheme for residential customers to cover EV charging. In addition, new regulations could be introduced for multistorey residential buildings and office buildings to allocate parking spaces and install chargers for EVs.
Standards to locate EVCI in commercial buildings	Public policies worldwide are increasingly promoting the implementation of EVCI in commercial spaces, especially those with parking facilities. This presents a significant opportunity for integrating EVCI in new buildings, during major renovations, and within existing commercial structures through regulatory frameworks. **Erreur ! Source du renvoi introuvable.** provides a reference to the g uidelines adopted in Europe, which could serve as a general

Area of regulation	*Regulation*
	framework for Rwanda to deeply regulate the implementation of EVCI.
Public and Semi-Private Charging Infrastructure	Public institutions worldwide play a crucial role in the deployment of public EVCI to promote and sustain market growth and investments. To enhance this effort, Rwanda can align its methodology and practice by adopting one of the common contractual models for the deployment of charging infrastructure, referenced in this report. These models range from passive to active roles, enabling public institutions to accelerate the deployment of EVCI effectively. In addition to adopting a contractual role in the deployment of infrastructure, these measures could be accompanied by regulatory actions. For instance, establishing requirements for a certain number of EVCI installations along major roads would enable EV drivers to travel throughout the country conveniently. This approach could be integrated into one of the contractual models or implemented as a standalone regulatory measure. **EVCI Along the Road** Regulations around the world are promoting standards to install EVCI along roads at regular intervals. This approach aims to ensure the availability and accessibility of EV charging stations for travelers. The specific conditions and parameters for these installations will vary based on the specific case, road conditions, and vehicle flow. The **Erreur ! Source du renvoi introuvable.** provides a reference to the g uidelines adopted in Europe, which could serve as a general framework for Rwanda to deeply regulate the implementation of EVCI.
Goals on Electrification of Transport	Areas of the world, as the EU has banned the sale of fossil fuel vehicles by 2035, with various regions also mandating electrification and providing fiscal incentives. Rwanda aims for 20% bus electrification by 2030 and currently offers incentives for electric buses. Private players like BasiGo have entered the market, leveraging these incentives. To align with international best practices, Rwanda could gradually adopt policies to phase out fossil fuel vehicles, setting different goals for each vehicle category. This approach would allow for a strategic and manageable transition, leveraging existing incentives to attract private sector investments in public transport electrification and aligning with global trends towards sustainable transportation.
EVCI for Two Wheelers	Based on international best practices, subsidies for motorcycle charging infrastructure, such as battery swapping stations, are not recommended as they tend to be ineffective and create unnecessary financial burdens on the government. Current incentives should suffice to support private investments in battery swapping stations, ensuring an open market approach. The private sector in Rwanda has successfully deployed and implemented battery swap stations without the need for public sector intervention. Therefore, it is important for the government to maintain an open market guideline to foster private model investments. Regarding the standardization of batteries for electric vehicle fleets, mandating battery standardization across providers is not advisable. Such mandates can stifle competition and hinder technological innovation, limiting the potential for advancements and improvements in battery technology. It is more beneficial to allow providers to innovate and develop their own battery solutions, fostering a competitive market environment.
Tariff management assessment	Internationally, various pricing approaches for EV charging exist, often with prices set freely by the market. Business-to-consumer pricing schemes in Europe include flat rate pricing, transaction-based pricing, and time-based pricing. Some developing countries introduce lower and upper charging price caps to regulate the market. In Rwanda, the government sets the electricity tariff for CPOs equivalent to the industrial tariff, with an additional benefit from the

Area of regulation	*Regulation*
	reduced off-peak tariff. CPOs often set their own charging tariffs, competitive with petrol or diesel prices, considering their investment and operational costs. While the charging tariff may be freely determined by the CPOs, RURA may initially set a price cap to ensure fair pricing during the early stages of market development when competition is limited. During the transition phase, the price cap could be referenced to fuel prices (gasoline or diesel). As charging volumes increase, the correlation with fuel prices can be removed. The price cap should then be based on retail electricity prices, the maximum allowable profit margin, and the capital and operational costs of the CPOs. For convenience and clarity for both RURA and EV owners, price caps should align with transaction-based pricing, expressed in terms of RWF per kWh. This aligns with familiar business-to-consumer pricing models and supports clear and understandable billing practices.
Regulatory framework that governs electricity supply for EV charging	The current regulatory framework governing electricity supply for EV charging in Rwanda is detailed in the "Regulations Governing Electrical Installations (2023)," which outlines the procedures and requirements for electrical installations, including conditions for inspection, monitoring, and compliance. Electrical installations are categorized based on voltage levels and specific types of installation work, with associated fees. Recommendations for improving this framework include maintaining the robustness of the current regulations to accommodate grid connections for EV charging stations. In the medium to long term, RURA should consider issuing specific regulations for the grid interconnection of EV charging stations. In Germany, for example, charging stations with capacities over 12 kVA require approval from the network operator.
Technical Standards And Type of Connector	To standardize hardware requirements for public and semi-public charging infrastructure, Rwanda should adopt a hybrid approach that mandates the use of both Type 2 and GBT connectors as national standards. Given the diversity of EVs entering the country, particularly from Europe and China, the Type 2 connector—widely used in Europe for both AC and DC charging—will ensure alignment with international standards, ease the integration of global EV brands, and provide compatibility with most European models. This will position Rwanda as a regional leader in EV interoperability. On the other hand, the GBT connector should also be adopted to accommodate the growing influx of Chinese-manufactured EVs, as China is one of the largest EV producers and GBT is a predominant standard in Rwanda. This dual-standard approach will ensure inclusivity across major EV markets and provide a seamless experience for EV owners. It is recommended that RURA and the Rwanda Standards Board review and align the technical standards for EVCI.
Grid Stability and EV Charging	It is recommended to adopt controlled smart charging and Vehicle-to-Grid (V2G) schemes to enhance grid stability and achieve both technical and financial objectives. For uncontrolled charging, EV owners charge their vehicles based on their personal preferences, which necessitates the installation of new infrastructure and an increase in the number of charging stations. This approach is not economically optimal and does not flatten load profiles. Conversely, controlled smart charging can balance load power, implement V2G schemes, regulate voltage and frequency (V/f), and reduce losses. Financially, it can lower energy costs and increase profits for charging providers. By controlling EV charging through linking EVs to distribution networks (DNs) and considering EVs as mobile energy storage devices (ESDs), load fluctuations can be minimized. This linkage can be achieved by developing software-

Area of regulation	*Regulation*
	based methods to direct drivers to optimal charging times and locations. Adopting these strategies will ensure a more efficient, stable, and financially viable charging network, aligning with future global climate agreements and technological advancements.

Case Study Box No. 10 EVCI in new buildings, during major renovations, and within existing commercial structures

EVCI in new buildings, during major renovations, and within existing commercial structures

As a reference, the European Energy Performance Buildings Directive64, establishes key directives:

Mandatory Installations: New commercial buildings and major renovations must include EV charging stations.

Transition Plans: Existing commercial buildings are required to integrate EV charging infrastructure, particularly those with ample parking facilities.

Implementation Deadlines: Specific timelines and deadlines are set for the installation of EV charging stations in new and existing buildings to ensure compliance and facilitate the transition to electric vehicles.

Specifically and as an international best practice, the European Energy Performance Buildings Directive mandates the following:

Commercial buildings with more than 10 parking spots must install at least one charging point, with 1 in 5 parking spots equipped with a connection to a charging station.

Major renovations affecting more than 25% of a building's exterior trigger the installation of at least one charging point, provided the cost does not exceed 7% of the renovation budget.

From 2025, commercial buildings with over 20 parking spots must have at least one charging point installed.[65]

Case Study Box No. 11 EVCI Along the Road

EVCI Along the Road

For instance, in the case of Europe, The Alternative Fuels Infrastructure Regulation (AFIR) serves as a comprehensive framework aimed at standardizing and accelerating the development of EV charging infrastructure across the European Union. It offers valuable insights and directives that can be used as best practices for implementing EVCI[2], as for instance:

- The growth of recharging infrastructure for cars and vans (Light-Duty Vehicles) must match the increase in vehicle registrations. Specifically, for every battery-electric vehicle registered in a Member State, publicly accessible recharging infrastructure must provide a power output of 1.3 kW.

- Additionally, for the case of Light-Duty Vehicles, starting in 2025, fast recharging stations with a minimum power output of 150 kW need to be installed every 60 kilometers along the Trans-European Transport (TEN-T) network.

[64] https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=OJ:L_202401275&pk_keyword=Energy&pk_content=Directive
[65] https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX%3A52023DC0076

- For heavy-duty vehicles, recharging stations with a minimum power output of 350 kW must be installed every 60 kilometers along the TEN-T core network and every 100 kilometers on the larger TEN-T comprehensive network, starting from 2025. Additionally, it is required to establish charging hubs with a total power capacity of at least 400 kW, which must include at least one charging point with an individual capacity of 150 kW. These stations must also be placed at secure parking areas for overnight recharging and in urban nodes for delivery vehicles, with complete network coverage to be achieved by 2030.

6.7.2 Regulatory Approaches and Recommendations

E-mobility is a nascent sector in Rwanda. Some private sector players however have responded to the e-mobility policies that were introduced by the government and established their own e-mobility services. To further kick-start the development of the e-mobility sector, the study reviewed international policy and regulatory practices, assessed the gap in policies and regulations in Rwanda, and recommended measures that could be considered by the government to increase the deployment of EV charging infrastructures by the private sector.[66]

6.7.2.1 Licensing of CPOs

International Practice

In some Australian States, EV charging is considered as a retail sale of electricity and subjects it to regulations on electricity sales. In countries where EV charging is considered as retail sale of electricity, CPOs are required to obtain a license from the electricity sector regulatory authority.

Most of the countries that advanced EV deployment, on the other hand, consider EV charging as a service to the transport sector, which is therefore not governed by electricity regulations. For countries that consider EV charging as a transport service, EV charging is not subject to licensing but to general business permits, like any other commercial activity.

Current Practice in Rwanda

CPOs in Rwanda are not required to get a license from RURA. On the other hand, they are required to obtain business permits from Rwanda Revenue Authority (RRA).

Options and Recommended Measures for Rwanda

The current practice in Rwanda is already aligned with the international best practice. Business permits would be sufficient for CPOs to operate. CPOs on the other hand could be subject to inspections by the Rwanda Inspectorate, Competition and Consumer Protection Authority (RICA) to ensure compliance with relevant regulations and laws.

6.7.2.2 Vehicle Charging

International Practice

In countries where EV charging is considered as retail sale of electricity, CPOs are required to obtain a license from the electricity sector regulatory authority. On the

[66] On *Annex No. 35 Policy Summary Table: International Best Practice, Current Situation in Rwanda, Recommendations*, it is possible to find the contrast of the regulation measures with best practices and current situation in Rwanda.

other hand, for countries that consider EV charging as a transport service, EV charging is not subject to licensing but to general business permits, like any other commercial activity.

Current Practice in Rwanda

There is no explicit policy stipulating the approach to electric vehicle charging in the country, though the existing practice is that electric vehicle charging is considered as a service to the transport sector.

CPOs have established EV charging infrastructures without obtaining licenses from RURA. On the other hand, CPOs have to apply to RURA to avail the industrial tariff stipulated in the Strategic Paper on Electric Mobility Adaptation.

Options and Recommended Measures for Rwanda

The current practice is aligned with internal best practice. Industrial tariff application for CPOs could however be further streamlined to shorten approval period.

6.7.2.3 Regulations and Incentives for Private Charging

International Practice

In many European countries, EV owners charge their vehicles mainly at home and work relying mostly on Type 1 and Type 2 AC chargers. In many European cities, governments provide grants for the installation cost of an EV charger at domestic properties (individual houses, apartments, and condominiums) and workplaces.

In addition, new regulations were introduced in many European countries regarding the issue of lack of allocated parking spaces and chargers for EVs in apartments, multi-story buildings and workplaces. These regulations require a certain percentage of the total parking space to be dedicated for EVs. For existing apartments or condominiums, the cost of EV charger installation (including the cost of upgrading the grid connection) is being distributed according to the general rules of the associations. Each EV owner, however, pays for its own charging costs.

Current Practice in Rwanda

There are no incentives related to the installation of EV chargers in domestic properties in Rwanda.

The current fiscal incentives for EV purchase such as zero VAT, exemption of import and excise duties and exemption of withholding tax however covers charging equipment and indirectly benefits private charging of electric vehicles.

The current policy of setting a cap to EV charging tariff equal to industrial tariff to CPOs renders private charging more expensive than public charging since residential electricity tariff is more expensive than industrial tariff.

Options and Recommended Measures for Rwanda

For many developing countries including Rwanda, providing grants or subsidies to cover private charger installation costs, may create an unnecessary financial burden to the government. Perhaps an alternative option is to provide income tax deductions and other tax benefits related to the purchase and installation of domestic EV chargers.

Concerning the EV charging tariff policy, extending the industrial tariff for private charging may require additional metering and transaction costs to both EV owners

and distribution utilities. An alternative option is to increase the range of the lower tariff block of the increasing block tariff (IBT) scheme for residential customers to cover EV charging.

In addition, new regulations could be introduced for multistorey residential buildings and office buildings to allocate parking spaces and install chargers for EVs.

6.7.2.4 Regulations and Incentives for Public Charging

International Practice

In US and some European countries, governments rely on the private sector to construct EV charging infrastructures. Governments introduced various regulations defining how systems are deployed, managed, maintained and used.

Most common regulations in Europe and the US are the mandatory requirement for commercial centers and parking areas to install EV chargers.

In several European countries, municipalities issued competitive tenders for the installation and operation of public charging stations.

Governments also provide incentives to attract stakeholders and investments needed to develop the charging systems.

Current Practice in Rwanda

There are no regulations in Rwanda governing the installation of EV charging stations for public use. EV charging infrastructures in the country are being developed by electric mobility service providers to support their growing electric mobility services.

On the other hand, the fiscal and non-fiscal incentives as well as administrative measures stipulated in the Strategic Paper on Electric Mobility Adaptation are important measures to kick start the e-mobility market and stimulate investments in EV charging infrastructures.

Options and Recommended Measures for Rwanda

While the existing policy mechanisms are sufficient to kick start the development of the electric mobility market and promote investments for EV charging infrastructures, these are not sufficient to grow the market and investments beyond the current level. Rwanda can align its regulations with international best practices.

- Introduce regulations requiring commercial centers to allocate a percentage of their parking areas for EVs and to invest on both AC charges and fast DC chargers.
- Require existing petrol stations to provide fast DC charging stations.
- Issue a competitive tender for providing both AC and fast DC chargers in public areas.

6.7.2.5 Regulations and Incentives for Motorcycles

International Practice

Policies are often aimed at electric motorcycles (as opposed to electric scooters and electric bicycles) with engines above 500 W and with maximum speeds higher than 35 km per hour which require registration as a motorcycle. Electric motorcycles are categorized into 2 user categories: private (individual transport) and commercial (rent and ride hailing, and goods transport). Private users normally charge their

motorcycles at home and destinations (opportunity charging) while commercial users charge at home, destinations and swapping stations. The trip length for private users is around 10 km per day while those for commercial users are higher than 50 km per day. Battery swapping is an ideal option for commercial users.

- Taipei, China introduced huge subsidies to electric motorcycles and swapping stations (50% of construction cost and free publicly accessible land). In addition, the government also provides non-financial incentives (exclusive parking spaces, preferential parking fees and prohibition of 2-stroke motorcycles in certain areas). Despite this, the market share of electric motorcycle sales did not go beyond 15% of the total motorcycle sales.
- In People's Republic of China (PRC), local governments banned gasoline-powered motorcycles thus driving up motorcycle sales.

Key lessons learned in developing countries are the following: i) without government intervention, market uptake was mainly on e-scooters that do not replace motorcycles but bicycles and public transport, ii) financial subsides must be significantly high in order to persuade customers to prefer electric motorcycles, and iii) regulations limiting the usage of gasoline-powered motorcycles result in rapid uptake of electric motorcycles.

- Standardization of batteries were introduced in Taipei, China and People's Republic of China, but these countries later withdrew this policy since it achieved little success. Battery standardization is problematic due to the dynamics of market forces. The battery is a core element of an electric motorcycle and a main competitive distinction and cost parameter. Standardization reduces competition between electric motorcycles/battery manufacturers which diminishes the innovation speed and results in less price decreases. Standardization can hamper rather than promote the uptake of electric motorcycles.
- Charging at home or at work could be done just by plugging into a wall socket without a need for a dedicated charging infrastructure. Battery swapping stations facilitate swapping of standardized batteries, though it is also feasible to operate with various battery types and sizes, either from the same swapping site or at different locations. Swapping sites often carry from 10-30 batteries and are densely distributed (having a swap point every 4-6 km).

Current Practice in Rwanda

The private sector has responded positively to the current policies and has introduced electric motorcycle financing targeting commercial users and has established battery swapping stations to support the electric mobility ecosystem.

The current financial and fiscal incentives and other measures being planned by the government support both commercial and private usage of electric motorcycles.

There are no regulations setting the target of increasing the share of electricity motorcycles in the overall motorcycle fleet in short, medium or long terms.

Options and Recommended Measures for Rwanda

Rwanda could align with international best practice by gradually phasing out gasoline-powered motorcycles in the market. The policy could focus first on commercial users while private users could be considered later.

Subsidies on motorcycle charging infrastructures (battery swapping stations) may not be necessary as the international best practice shows that they are not very

effective and will only create an unnecessary financial burden to the government. The current incentives may be sufficient to support private investments in battery swapping stations.

An electric mobility provider can have a standardized battery for its fleet but a standardization of batteries among electric mobility providers is not a recommended policy since it can hamper competition and technological innovation.

6.7.2.6 Regulations and Incentives for Public Transport

International Practice

Several countries across the world have announced national bans on sale of internal combustion engine (ICE) vehicles. Europe is leading the ICE phase-out with several countries banning the sales of fossil-fuel based vehicles in the upcoming decade (Norway from 2025; Iceland, Netherlands, Ireland, Sweden from 2030; UK, Denmark from 2035; France, Spain from 2040).

There are several other national/local governments that are mandating electrification of fleets/ public transport in the upcoming decade in Europe, US and Asia (e.g., California has mandated 90% fleet electrification for ride-hailing players by 2030; Shenzen has already transitioned to 100% electric buses.

Most European and Asian countries provide fiscal incentives, subsidies and loans to support electric mobility in the public transport.

Current Practice in Rwanda

There is no target on banning fossil fuel-based vehicles in the country. The country has a target of 20% bus electrification by 2030.

The fiscal and non-fiscal incentives introduced in the Strategic Paper on Electric Mobility Adaptation also applies to investments in electric buses.

Options and Recommended Measures for Rwanda

Rwanda’s bus electrification target is aligned with international best practice.

A private player (BasiGo) has entered the market and offered competitive leasing rates (through pay-as-you-drive model) by taking advantage of current incentives provided under the Strategic Paper. This indicates that the current incentives are sufficient to attract private sector investments for the public transport fleet's electrification.

6.7.2.7 Electricity Tariff Management Assessment

International Practice

Various pricing approaches exist internationally.

- In most cases, prices are set freely by the market and there are no regulations about charging prices.
- Business-to-consumer pricing schemes in Europe are the following: i) flat rate pricing (e.g. €15 per month), ii) transaction-based pricing (e.g. €0.30 per kWh), and iii) time-based pricing (e.g. €1.50 per hour of charging). EV owners are more familiar with transaction-based pricing since it is like conventional business-to-consumer electricity billing. CPOs on the other hand may prefer the time-based pricing since they are interested in having as many EVs as possible charging at their stations.

- Some developing countries introduce lower and upper charging price caps.

Current Practice in Rwanda

As mentioned earlier, the government sets the electricity tariff for CPOs equivalent to the industrial tariff (large industry) and that CPOs can also benefit from the reduced off-peak tariff. The current tariff for large industries is RWF 95 per kWh and the off-peak tariff amounts to RWF 94 per kWh. As a comparison, commercial tariff amounts to RWF 179 per kWh.

CPOs set their own charging tariffs. Often, these tariffs, after considering their investment and O&M costs, are set at a rate competitive to the equivalent petrol or diesel price. In the case of BASIGO, the company estimated that the current equivalent price of diesel buses is RWF 1,500 per km. BASIGO offers a charging price of RWF 600 per km under its pay-as-you-drive financing model. KABISA's charging tariff is estimated to be around 30% higher than the industrial tariff to cover its capital and operational costs.

Options and Recommended Measures for Rwanda

While the charging tariff may be freely determined by the CPOs, in the short term when competition is limited, RURA may set initially a price cap.

In the transition, the price cap may be set with reference to fuel prices (gasoline or diesel prices). Once charging volumes increase significantly, the relationship between charging prices and fuel prices could be removed. Subsequently, the price cap should be determined based on retail electricity prices plus the maximum allowable profit margin plus the capital and operational costs of the CPOs.

It will be more convenient for both RURA and EV owners to set the price caps aligned with transaction-based pricing which is expressed in terms of RWF per kWh.

6.7.2.8 Regulatory framework that governs electricity supply for EV charging

International Practice

Grid connection requirements and fees for charging stations are the same for other consumption points. In some Latin American countries, grid connection fees depend on the costs of providing the new connection and not differentiated for EV charging.

- In Germany, charging stations with a capacity of more than 12 kilovolt amperes (kVA) require approval from the network operator. Charging devices with outputs of less than 12 kVA shall be registered with the distribution network operator.
- If a single EV charging point is connected to the low voltage grid, the connection is treated like any other connection in the low voltage grid. If the installation of a public charging station is done via direct connection to the medium voltage grid, the same regulation applies as with other devices connected to the medium voltage grid.
- Also, to accelerate the installation of charging stations at common locations (supermarkets, hotels, restaurants, etc), the German Federal Ministry of Transport and Digital Infrastructure issues calls for tenders for the construction and operation of charging sites and charging points.
- Germany and India, on the other, hand issued specific regulations for EV charging stations.

Current Practice in Rwanda

The current regulations for electrical connections in Rwanda are specified in the Regulations Governing Electrical Installations (2023). This regulation stipulates the procedures and requirements for electrical installations and conditions, inspection, monitoring and compliance.

Electrical installations are classified for various voltage levels and for specific installation works. Fees are also based according to these classifications.

Options and Recommended Measures for Rwanda

Existing regulations for electrical installations are robust and are sufficient to accommodate grid connection of EV charging stations.

RURA and Rwanda Standards Board may however consider reviewing the technical standards specified in this regulation to align with international technical standards for EV charging stations.

In the medium to long-term, RURA may consider issuing specific regulations for grid inter-connection of EV charging stations.

6.7.2.9 Technical Standards

International Practice

The main international organizations responsible for providing EV standards include the Inter-national Electrotechnical Commission (IEC), Institute of Electrical and Electronics Engineer (IEEE), International Organization of Standards (ISO), and Society of Automotive Engineers depending on the on-going trends and technologies.

In some export-oriented countries, EV standards were directly taken from these international standards.

In most countries, national standards were issued by their respective standards boards or institutes but referenced to international standards.

Current Practice in Rwanda

The Rwanda Standards Board (RSB) is the main authority responsible for issuing technical standards and undertaking quality testing, metrology and certification.

Current standards in Rwanda are benchmarked with international standards including those adopted in African countries.

No standards specific to EVs and charging infrastructures were issued Rwanda.

Options and Recommended Measures for Rwanda

The current practice in issuing standards in Rwanda is aligned with international best practice. There are however no standards for EVs and charging infrastructures issued in the country.

Rwanda can adopt international standards for EVs and charging infrastructures. The list of international standards applicable are given in the Appendix.

Issue	*International Best Practice*	*Current Practice in Rwanda*	*Recommendations*	*Responsible Agency*
Licensing of CPOs	For EV charging as service to the transport sector, there is no need to obtain license from energy regulator.	CPOs need business permit.	Current practice is aligned to international best practice.	Rwanda Revenue Authority for business permits and Rwanda Development registers the business
Regulation of Charging Infrastructure	EV charging as retail sale of electricity EV charging as service to the transport sector	EV charging as service to the transport sector. Not regulated by RURA.	Current practice is aligned to international best practice. Application process for industrial tariff could be streamlined.	RURA to streamline tariff application
Incentives and Regulation for Private Charging	Governments provide subsidies for EV charging installation at domestic properties and workplaces	Rwanda only provides fiscal and non-fiscal incentives and indirectly benefit private charging Policy on setting the industrial electricity tariff and access to off-peak industrial tariff for CPOs render private charging more expensive than public charging.	Grants and subsidies may create an unnecessary financial burden to the government. Income tax deductions and other tax benefits may be an alternative option. Adjustments to current residential electricity tariff may be an alternative solution (increasing the range of the lower bracket of the inverted tariff block).	Rwanda Revenue Authority can explore options for tax benefits for private charging. RURA may review and assess whether adjustments for residential tariff be an option.
	Regulations to install EV charging infrastructures in domestic properties (apartments, multistorey buildings) and workplaces	No regulations related to installation of EV charging infrastructures in domestic properties and workplaces.	Local governments could introduce regulations related to mandatory installation of EV charging infrastructures in large apartments, multistorey buildings and workplaces.	City of Kigali and local administration in general to introduce mandatory installation of EV charging infrastructures in large apartments, multistorey buildings and workplaces.
Incentives and Regulations for Public Charging	Mandatory installation of EV charging infrastructures in commercial/business centers and parking areas Concession through competitive bidding for public charging infrastructure provision	No regulations on public charging infrastructures	Local governments could introduce mandatory installation of EV charging infrastructures in commercial/business centers and parking areas including petrol stations. Establish concession arrangement for public charging infrastructures	City of Kigali and local administration can issue regulations mandating installation of EV charging infrastructures in commercial centers and parking areas including petrol stations. City of Kigali and local administration can introduce concession arrangements for public charging infrastructures.
	Provision of incentives such as direct funding, grants and subsidies, and other tax benefits	Introduced fiscal and non-fiscal incentives for electric mobility	Direct funding, grants and subsidies may create an unnecessary financial burden to	Rwanda Revenue Authority can explore additional tax benefits for

			the government. Additional tax benefits can be explored.	public charging infrastructure providers.
Incentives and Regulations for Electric Motorcycles	Introduce targets to electrify motorcycle fleet and banning of fossil fuel-based motorcycles	No policy on electrification of motorcycle fleet	The government can introduce targets for the electrification of the motorcycle fleet.	Rwanda Revenue Authority can explore additional tax benefits for battery swapping operators.
	Provision of subsidies and grants to battery swapping station operators.	Introduced fiscal and non-fiscal incentives for electric mobility	Direct funding, grants and subsidies may create an unnecessary financial burden to the government. Additional tax benefits can be explored.	Rwanda Revenue Authority can explore additional tax benefits for battery swapping operators.
Incentives and Regulations for Public Transport Fleet	National bans on sale of internal combustion engine (ICE) vehicles Mandating electrification of fleets/ public transport	No bans on sale of internal combustion engine (ICE) vehicles Introduced target of 20% share of electric buses by 2030	Electrification target by 2030 is aligned with international best practice. Targets beyond 2030 could be introduced.	MININFRA (Transport) could introduce targets beyond 2030.
	Provision of fiscal incentives, subsidies and loans to support electric mobility in the public transport	Introduced fiscal and non-fiscal incentives for electric mobility	Direct funding, grants and subsidies may create an unnecessary financial burden to the government. Additional tax benefits can be explored.	Rwanda Revenue Authority can explore additional tax benefits for CPOs providing services to public transport.
Tariff Management	Charging prices are set freely by the market and there are no regulations about charging prices. Business-to-consumer pricing schemes in Europe are the following: i) flat rate pricing (e.g. €15 per month), ii) transaction-based pricing (e.g. €0.30 per kWh), and iii) time-based pricing (e.g. €1.50 per hour of charging). Some countries introduce lower and upper charging price caps.	The government sets the electricity tariff for CPOs equivalent to the industrial tariff (large industry) and that CPOs can also benefit from the reduced off-peak tariff. CPOs set their own charging tariffs. Often, these tariffs, after considering their investment and O&M costs, are set at a rate competitive to the equivalent petrol or diesel price.	While the charging tariff may be freely determined by the CPOs, in the short term when competition is limited, a price cap may initially be set. Price caps can be aligned with transaction-based pricing which is expressed in terms of RWF per kWh	RURA may set initially a price cap
Regulatory Framework Governing Electricity Supply for EV Charging	Grid connection requirements and fees for charging stations are the same for other consumption points. If a single EV charging point is connected to the low voltage grid/ medium voltage grid, the	The current regulations for electrical connections in Rwanda are specified in the Regulations Governing Electrical Installations (2023). Electrical installations are classified for various voltage	Existing regulations for electrical installations are robust and are sufficient to accommodate grid connection of EV charging stations.	RURA may review interconnection standards specific to grid interconnection of EV charging stations.

	connection is treated like any other connection in the low voltage grid/ medium voltage grid.	levels and for specific installation works. Fees are also based according to these classifications.		
Technical Standards	In some export-oriented countries, EV standards were directly taken from these international standards (IEC, IEEE, ISO). In most countries, national standards were issued by their respective standards boards or institutes but referenced to international standards.	Current standards in Rwanda are benchmarked with international standards including those adopted in African countries. No standards specific to EVs and charging infrastructures were issued Rwanda.	The current practice in issuing standards in Rwanda is aligned with international best practice. Rwanda must adopt international standards for EVs in general including those for charging infrastructures.	The Rwanda Standards Board (RSB), the main authority responsible for issuing technical standards and undertaking quality testing, metrology and certifications, must review and issue EV standards.

Table 36: Summary of Policy and Regulation Assessment and Recommended Measures

6.7.3 Multi-Stakeholder Governance for EV Charging

The Implementation Models presented earlier requires a multi-stakeholder governance for EV charging. The EV charging ecosystem comprises of multiple components and processes which covers from the provision of land and supply of electricity for EV charging, specification and installation of EV charging equipment, operations and maintenance of EV charging facilities, and charging services to EV owners.

While MININFRA is the main government body responsible for electric mobility policy, there are several government agencies that play an important role in EV charging. Roles of government agencies can be classified into policy-making, regulatory functions and implementing functions.

Multi-Stakeholder Governance for EV Charging

The following table outlines potential regulatory areas for the deployment and management of EVCI in Rwanda. It identifies the most appropriate entity likely to be responsible for each regulation topic and provides an overview of the specific regulatory issues to be addressed.

Transport Sector	*Energy Sector*	*Other Sectors*
POLICY		
MININFRA: Transport EV Mobility Policy EV mobility targets (private transport, public transport, 2 and 3 wheelers) EV Charging Infrastructure Policy Public charging, semi-public, private	MININFRA: Energy Electrification of Transport Policy EV mobility targets (private transport, public transport, 2 and 3 wheelers) Renewable Energy Policy for EV Charging	
REGULATION		
RURA: Transport Licensing of CPOs	RURA: Energy Tariff policy for EV charging (for public and private charging) EV charging infrastructure grid interconnection regulations	Rwanda Development Board Business registration and licenses Monitoring of investments
		Rwanda Standards Board EV charging infrastructure standards
		Rwanda Housing Authority parking spaces to be equipped with charging infrastructure in private and commercial buildings under Building Control Regulations
		REMA Urban air quality regulations (fuel quality

Transport Sector	*Energy Sector*	*Other Sectors*
		and vehicle emissions regulations) Greenhouse gas emissions targets
		RICA Licensing of EV transport facilities Enforcement of EV infrastructure and vehicle standards and regulations
IMPLEMENTATION		
City of Kigali and Secondary Cities Concession agreement with CPOs for public charging infrastructure Public land provision for EVCI Partnerships with host establishments, fleet operators and e-mobility service providers, and enforcement of national policies related to semi-public and private charging infrastructures.	EUCL Grid connection of EV charging infrastructure Distribution network upgrades and management	
RITCO and other bus fleet operators Procurement of their own charging infrastructures through direct ownership or leasing.		
Commercial, Institutional and Residential Establishments Procurement of their own semi-public charging infrastructures		
e-mobility Service Providers Procurement of their own public charging infrastructures/battery swapping stations supporting their e-mobility services.		

Table 37: Policy Areas of regulation and institutions

6.8 Public Awareness in E-Mobility Transition

Public awareness plays a critical role in the successful transition to e-mobility. Awareness initiatives can help demystify electric vehicles (EVs), highlight their benefits, and address common misconceptions, thereby encouraging more people to adopt this sustainable mode of transportation. As important as it is to create regulations to promote EVCI and support the efforts of private stakeholders, public awareness is equally crucial. Consumers need to be informed and motivated to choose e-mobility options that align with sustainability goals. This comprehensive approach ensures a robust and well-supported transition to an electric mobility ecosystem.

The government and private stakeholders can implement various strategies to enhance public awareness about e-mobility and its charging infrastructure:

- **Nationwide Campaigns:** Launch comprehensive campaigns to educate the public on the benefits of EVs, including cost savings, environmental impact, and

technological advancements. Utilize various media channels, such as social media, television, radio, and print, to disseminate information.

- **Digital Resources:** Develop user-friendly websites and mobile apps that provide detailed information on EV models, charging station locations, and maintenance services.
- **Educational Integration.** Collaborate with schools, universities, and community organizations to incorporate e-mobility topics into educational programs.
- **Develop electric mobility vocational training program**. With industry partnerships, develop/encourage/provide funding to institutes to develop training programs for skills development in electric vehicle technology.
- **Introduce electric mobility at the elementary and secondary levels of education** in partnership with cities, municipalities, schools and universities. This would help facilitate awareness and encourage students to pursue careers in electric mobility. Initiate programs such as science fairs and tech festivals where students can get hands-on experience in projects related to electric vehicles. This would also help create awareness among parents about the advantages of electric vehicles.
- **Introduce advanced-degree and certificate programs**. Collaborate with universities and the electric vehicle industry to develop bachelor's, master's, and certificate programs dedicated to electric vehicle technology.
- **Introduce electric vehicle driving schools**. Educating drivers about electric mobility is a great opportunity to spread awareness. Integrating driving instruction with electric mobility ensures that drivers are skilled enough to operate an electric vehicle. Driving schools could cover issues such as driving range, charging time, characteristics of acceleration and regenerative breaking, and to be more conscious of pedestrians, who may not hear the silent powertrain of electric vehicles.
- **Public Engagement Events:** Organize public demonstrations and test drive events in collaboration with private stakeholders to give people firsthand experience with EVs.
- **Highlight Financial Incentives:** Emphasize the country's efforts to promote e-mobility and the benefits of adopting such practices and inform the financial and non financial incentives adopted by the country.

These measures, combined with supportive regulations and private sector efforts, will help create a well-informed public ready to embrace e-mobility solutions.

Case Study Box No. 12 Cost Savings: The case of Ampersand

Cost Savings: The case of Ampersand

One specific example that underscores the importance of public awareness is the cost savings associated with renting a charged battery from Ampersand. Drivers can save approximately 35% compared to traditional fuel and oil changes. Each Ampersand battery delivers around 350,000 km of range over its lifetime, and offers 50-110 km per swap depending on usage, roads traveled, and weight carried. Publicizing such tangible benefits can significantly motivate potential users to switch to electric vehicles.

By implementing these measures, the government can create a well-informed public that is more likely to embrace e-mobility, thereby accelerating the transition to a more sustainable transportation system.

6.8.1 Data Management, Registration, Reporting, and GIS Integration

National Data Sharing Platform.

Create a National Data Sharing Platform to mandate the sharing of both dynamic and static data by market operators within the EVCI. This centralized platform will standardize data sharing, improving interoperability and supporting smart mobility solutions. By streamlining the data exchange, this platform will enhance interoperability, efficiency, improve service quality, and increase transparency for businesses. MININFRA should lead the procurement and implementation of the National Data Sharing Platform.

A data management policy should be developed to ensure that stakeholders have clear guidelines on the information they need to share and make accessible to the public. MININFRA should lead the development of policy guidelines for the National Data Sharing Platform.

The key types of data to be shared include at least the following data on public EVCI:

Static Data:

Charging Station Locations: Precise geographical data for all public charging stations.

Operating Hours: Details about when charging stations are open for public use.

Pricing Models: Information on the pricing schemes used, such as flat rate, time-based, or consumption-based pricing.

Service Providers: Details on Charge Point Operators (CPOs) and Electric Mobility Service Providers (EMSPs) managing the infrastructure.

Charging Capacity: The capacity (kW) of each charging point, distinguishing between AC and DC charging.

Dynamic Data:

Real-Time Availability of Charging Points: Whether specific charging points are currently in use or available.

Charging Status: Information on the charging process (e.g., charging in progress, time remaining).

Reservation Status: Data on whether a charging point has been reserved by a user and when it will be available.

Real-Time Pricing: Updates on current pricing, particularly in cases where rates fluctuate based on demand or time of day.

Payment Methods: Accepted payment options (e.g., mobile payments, contactless cards).

Energy Mix/Source: The type of energy being used (renewable vs. non-renewable sources) for charging sessions.

Estimated Waiting Times: Expected wait times at busy charging stations during peak usage hours.

Regulatory and Compliance Data:

Compliance with Standards: Information on whether the charging points meet local and international standards (e.g., safety and interoperability standards).

Licenses and Certifications: Data related to the licensing and certification status of CPOs and EMSPs.

Environmental Reporting: Data on the carbon intensity or renewable energy percentage used during charging sessions, supporting sustainability tracking and reporting.

ID Registration Organization (IDRO):

Establish an ID registration entity, office (IDRO) or assign this responsibility to an existing office, which should be responsible for issuing and managing identification codes (IDs) to clearly identify and register CPOs and EMSPs. This will establish a standardized referencing system. The use of unique IDs facilitates seamless cross-border access to EV charging stations. IDRO carries out the registration of IDs of COPs and eMSP for EVs, and the registration of ID codes of EVCI.

GIS Integration:

Leverage Geographic Information Systems (GIS) in planning. With the National Data Sharing Platform as a primary data source, this will contribute to efficient monitoring and analysis of EVCI. This will provide valuable insights and a comprehensive overview to guide policy adjustments and infrastructure expansion. Although the Master Plan has developed an interactive tool displaying existing EVCI, it is crucial to keep this information up to date, including all relevant characteristics of public charging stations, to ensure accuracy and reliability.

The activities related to Data Management, Registration, Reporting, and GIS Integration for EVCI should ideally be overseen by a single entity to ensure centralized coordination and consistency in data handling. This approach will streamline processes, avoid duplication, and ensure that all relevant data is managed cohesively. This initiative is most likely to be led by MININFRA, RURA, or RTDA, given their respective roles in infrastructure, regulatory oversight, and transport development.

6.9 Final Recommendations

6.9.1 Develop a roadmap

Develop a roadmap with specific transport electrification projects per transport, to assure a consolidated approach and prevent fragmentation, using the findings from this masterplan. See also below figure.

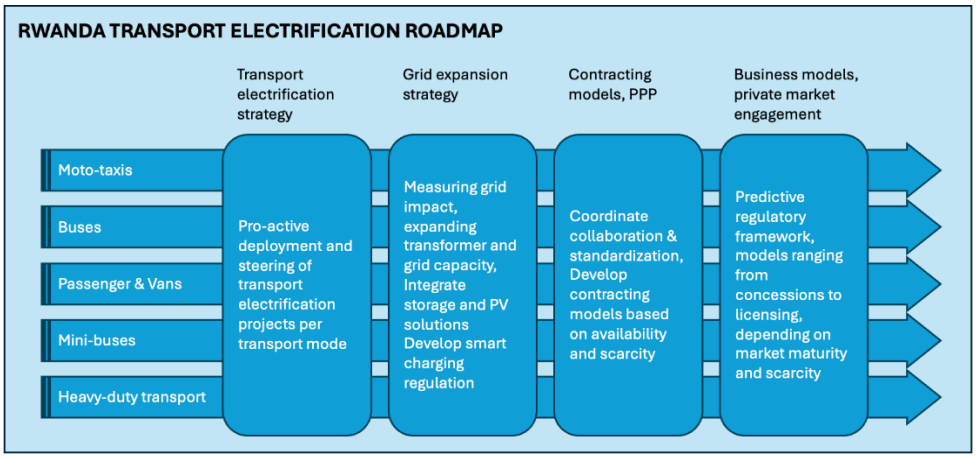


Figure 40: Rwanda Transport Electrification Roadmap

6.9.2 National Policy for EV Charging

Develop a comprehensive national policy on EV charging to provide clear guidelines for service provision and establish minimum requirements for stakeholders. This will ensure a consistent and reliable EV charging network across Rwanda.

The policy should be led by MININFRA as responsible for infrastructure policy and development throughout the country with RURA as a co-leader to define responsibilities for overseeing EV charging services, including registration, institutional roles, and integration with the broader energy grid. It should also address electricity supply, infrastructure scalability, and the future expansion of EVCI. The key provision of the policy should include:

System Principles:

Establish guidelines that ensure user accessibility, transparency, and fair transactions across all EV charging stations. The goal is to make the EV charging ecosystem easy to navigate and beneficial for all users.

Goals:

Define clear goals for the development of electric mobility and charging infrastructure, including:

Electrification Goals by Transport Mode: Set specific targets for electrification across various transport modes, such as private vehicles, public transport, and freight.

EVCI Penetration Goals: Define targets for the number and distribution of charging stations to ensure adequate coverage.

Phasing Out Gasoline Vehicles: Create a roadmap to gradually phase out gasoline-powered motorcycles, passenger vehicles, and heavy-duty vehicles, starting with the most accessible segment (motorcycles) in line with international best practices.

Electric Vehicle Charging as a Service:

Clarify that EV charging services are officially considered part of the transport sector. While this is already happening in practice, it should be explicitly stated in policy. When recognized as a transport service, CPOs typically do not require a license from the electricity sector to operate. It is essential for stakeholders to have clear guidance on which licenses or registrations they will need to comply with.

Interoperability Standards - Communication Protocols:

Ensure compatibility across different charging stations and EVs by mandating the use of the Open Charge Point Protocol (OCPP) for communication between charging

stations and Charge Point Management Systems (CPMS). OCPP is an open-source protocol that secures data transmission and enhances system interoperability.

Mandate the use of the Open Charge Point Interface (OCPI) for communication between charging stations and service providers. OCPI enables EV drivers to easily find and use charging stations, enhancing the overall user experience.

Implement ISO 15118-2, which supports Plug & Charge (PnC) and V2G (Vehicle-to-Grid) functionalities. This standard facilitates seamless authentication, authorization, and payment at the charging station, without user interaction, while also enabling bi-directional energy transfer to optimize grid stability.

Adopt IEC 61851 to establish clear guidelines for the safety and communication aspects of electric vehicle charging. This standard covers the conductive charging system for EVs, ensuring safe and reliable charging by defining the interaction between the vehicle and the charging station. It supports various charging modes and provides detailed safety protocols, facilitating a secure, standardized framework for Rwanda's EV infrastructure development.

Hardware Layer – Connector – Passenger Vehicles:

Connector Standards Standardize hardware requirements for public and semi-public charging infrastructure, adopting a hybrid approach mandating the use of Type 2 and GBT connectors as national standards. Given the diversity of electric vehicles entering Rwanda, particularly those with European and Chinese origins:

- Type 2 Connector: This connector is widely used in Europe and is considered a global standard for AC and DC charging. Making Type 2 connectors mandatory in Rwanda's public EV charging infrastructure will ensure alignment with international standards, facilitate easier integration for global EV brands, and provide compatibility with most European models. This decision would also position Rwanda as a potential regional leader in EV interoperability and cross-border compatibility with world and European vehicle markets.
- GBT Connector: As Rwanda has seen an influx of Chinese-manufactured EVs, the GBT connector standard should also be adopted to accommodate the needs of this growing segment. China is one of the largest producers of EVs, and GBT is one of the predominant standards in Rwanda. Supporting GBT connectors will help accommodate the existing and future fleet of Chinese EVs in the country.

This dual-standard approach will ensure inclusivity for the major EV markets, providing a seamless experience for EV owners.

For public recharging infrastructure equipped with at least two charging stations or two charging points, both Type 2 and GBT connectors should be installed. In cases where a charging station has only one charging point and one connector, one of these standards should be installed based on the owner's/market's preference.

The recommendation for standard connectors is focused on public EVCI for passenger vehicles. For light-duty, heavy-duty vehicles, and electric buses usually rely on depots with closed charging systems operated by private fleets. However, public charging infrastructure for heavy-duty vehicles will be necessary in the long term.

To prepare for this, it is recommended to:

1. Initiate pilot projects, such as Megawatt Charging Systems (MCS), in logistics hubs or highways to explore public charging for heavy-duty vehicles.
2. Collaborate with private fleet operators to align infrastructure needs and form public-private partnerships.

3. Ensure ongoing monitoring of global standards for heavy-duty vehicles to future-proof infrastructure as the market evolves.

While a standard connector for heavy-duty vehicles in public infrastructure may not be immediately feasible, pilot projects and collaboration will help Rwanda adapt as the market for heavy-duty EVs matures.

6.9.3 Optional Consideration for a Third Standard

While adopting Type 2 and GBT as minimum connector standards covers the majority of EVs entering Rwanda, it may be prudent to allow for an optional third standard based on market preferences in cases where both standards are already installed at a charging station. This could provide flexibility for any emerging technologies or EV manufacturers using different connector types. The decision to expand beyond two connectors should be market-driven and implemented only if demand from other vehicle manufacturers increases.

6.9.4 Motorcycle Charging Infrastructure

Maintain an open market approach and avoid mandating battery standardization to foster competition and innovation within the e-mobility sector. The battery swap model, successfully adopted by the private sector, has proven to function effectively. However, the government should encourage voluntary convergence of standards by facilitating industry forums that bring together manufacturers and operators to discuss and align on common practices. Offer incentives to companies that adopt widely compatible standards, promoting interoperability across the two wheeler market without imposing regulatory mandates.

6.9.5 Measures to Implement and Develop the EVCI Network

Adopt Contractual Frameworks for EVCI Deployment:

Implement the contractual frameworks outlined in the EVCI Master Plan to deploy public charging stations nationwide and at city or provincial levels, as described on section 6.2 of the EVCI Master Plan. The contractual models will vary depending on the mode of transport and the specific needs of each city. These frameworks will enable cities and provinces to take coordinated, consistent actions, ensuring the effective integration of EVCI into urban and regional transport systems.

A contractual model with significant private sector involvement—such as concessions, availability-based contracts, and licensed contracts—will allow private companies to take on core responsibilities, including installation, operation, demand management, and financing efforts. This approach reduces the financial burden on the public sector. However, it requires public sector involvement in strategy development and oversight of the contractual models to ensure alignment with broader public policy goals.

- The study recommended to identify/establish a body responsible for promoting public-private partnership (PPP) in the provision of EV public charging infrastructures/battery swapping stations. The convenient candidates are the City of Kigali and Secondary Cities in their geographic boundaries, supported by the Government. Under the PPP model, these government agencies have to organize competitive public tenders for the installation and operation of public charging infrastructures through a concession arrangement.

- While the e-mobility service providers will be responsible for the procurement of EV charging infrastructures under the service provider model, where a public-private partnership is not needed, the City of Kigali and Secondary Cities could ensure that their development plan are consistent and comply with the EV Charging Infrastructure Master Plan.
- In addition, City of Kigali and Secondary Cities, could also identify and designate public lands that could be used for public charging.
- On the other hand, EUCL will be responsible for grid connection of EV charging infrastructures and distribution network upgrades and management.

Integrate EVCI into New and Existing Buildings:

Implement regulatory frameworks that mandate, in a phased and transitional manner, the integration of EVCI into all new buildings, major renovations, and existing commercial structures. For example, require the allocation of at least one charging station for parking facilities with a certain number of parking spaces. It is essential that this implementation be done in stages to allow for a smooth transition and adaptation to the new requirements.

- Under the consumer-driven model, residential, commercial and institutional establishments would be required by Building Control Regulations to provide EV charging infrastructures in their parking spaces. Similarly, RITCO and other fleet operators would be required by the transport electrification policy to meet their EV targets. Though these entities will be responsible in the procurement of their EV charging infrastructure, the City of Kigali and Secondary Cities could play an enforcement role ensuring that the regulated entities comply with the regulations.

Standardize EVCI Placement Along Roads:

Establish clear standards for placing EVCI at regular intervals along major roads and highways, ensuring accessibility and convenience for EV users. These intervals should be strategically determined (e.g., one charging point every certain kilometers) to create a reliable and widespread charging network, supporting long-distance travel and reducing range anxiety for EV drivers. The placement of EVCI along roads can also be addressed through the public contractual models previously mentioned.

Charging Tariffs and Pricing Models:

Develop a phased approach to EV charging tariffs, allowing local governments and RURA to set initial price caps during the early stages of market development. Reference the price cap to fuel prices during the transition phase, then base it on retail electricity prices, profit margins, and CPO costs as charging volumes increase. As competition grows, price caps can be relaxed. Take into account different modalities in which the market could set tariffs.

- Flat Rate Pricing: A set fee for charging regardless of electricity consumed.
- Time-Based Pricing: Charges based on the duration of connection.
- Consumption-Based Pricing: Fees based on the actual kWh consumed.

Establish regular review periods to adjust price caps in line with market conditions, such as inflation, changes in electricity costs, and broader market developments. These reviews will ensure that tariffs remain fair and sustainable while promoting healthy market competition.

Ensure fair and transparent pricing in the EV charging market by implementing price regulations, with RICA overseeing enforcement. Establish clear pricing guidelines, conduct regular reviews, and educate consumers about their rights and pricing structures to promote trust and equity in the market.

Controlled Smart Charging and V2G:

Adopt controlled smart charging and Vehicle-to-Grid (V2G) schemes to enhance grid stability while meeting technical and financial objectives. Controlled smart charging effectively balances load power, regulates voltage and frequency, reduces energy losses, lowers costs, and increases profitability for charging providers. This approach ensures an efficient, stable, and financially sustainable charging network.

To support this, initiate pilot projects in selected regions to test the benefits of both smart charging and V2G technologies. Collaborate with international experts to ensure these systems are implemented using global best practices and integrate seamlessly into Rwanda's energy grid. Additionally, monitor the results of the pilots to assess scalability and long-term viability, while exploring the potential for future expansion into other regions as EV adoption grows.

EVCI Waste Management:

Integrate strategies aligned with the National Circular Economy Action Plan and existing waste disposal initiatives. Key steps include developing specific regulations for EVCI waste disposal, enforcing Extended Producer Responsibility (EPR) policies, establishing designated collection centers, and investing in advanced recycling facilities. Work closely with the private sector and recycling companies to develop economically viable waste management systems. Set up public-private partnerships for recycling facilities, with the government providing seed funding or tax breaks.

Collaborate closely with the private sector and recycling companies to develop economically viable and sustainable waste management systems. Establish public-private partnerships for recycling facilities, with the government offering seed funding, tax breaks, or other incentives to encourage private investment. Additionally, ensure these partnerships prioritize long-term operational sustainability by leveraging innovation, market-driven approaches, and global best practices in recycling. This will help reduce the environmental footprint of EVCI waste while ensuring economic viability.

Workforce Development for Sustainable Mobility: Training in EV and EVCI Maintenance

Develop specialized training programs and educational curricula in collaboration with universities and technical schools to ensure that the maintenance of EVs and EVCI aligns with the latest advancements in sustainable mobility and energy transition technologies. This initiative will provide the workforce with the critical skills needed to support the expansion of the EV industry, enhancing the reliability, efficiency, and safety of EVCI systems. Additionally, fostering a skilled workforce will ensure that Rwanda can meet the growing demands of the e-mobility sector, contributing to long-term sustainability and economic development.

Education Public Awareness and Education:

According with section 6.7, it is recommended that MININFRA (Transport), in collaboration with industry stakeholders, develop and implement various strategies to enhance public awareness about e-mobility and its supporting charging

infrastructure. These strategies should include nationwide campaigns, public engagement initiatives, and the creation of digital resources that provide information on EV technologies, charging station locations, maintenance services, and relevant policies and regulations.

Additionally, collaboration with local governments, industry stakeholders, communities, and educational institutions is suggested to integrate e-mobility topics into educational programs. This effort should focus on:

- Developing vocational training programs for e-mobility.
- Introducing e-mobility topics into elementary and secondary education curricula.
- Creating advanced degree and certificate programs related to e-mobility.
- Establishing driving schools for electric vehicles.

By fostering a better understanding of e-mobility across all educational levels and engaging the public, this initiative will support the widespread adoption of electric vehicles and the successful implementation of the required infrastructure.

Energy considerations to Develop EVCI grid interconnection regulations

To ensure the successful integration of EVCI with the national grid, it is crucial to establish comprehensive grid interconnection regulations. These regulations will support the seamless connection of charging stations to the energy grid while promoting efficiency, safety, and sustainability.

Connection Standards: Define clear technical standards for connecting EV charging stations to the grid. This should include specifications on voltage, current, and safety requirements to ensure that charging stations operate efficiently and safely without disrupting the grid.

Metering and Billing: Develop protocols for accurate metering and billing of electricity used for EV charging. This will help establish a transparent system that ensures both users and operators are billed fairly for the energy consumed.

Renewable Energy Integration: Promote or mandate the use of renewable energy sources for EV charging to align with national sustainability goals. Incentivizing the use of solar, wind, or other renewable energy sources will reduce the carbon footprint of EV charging. Furthermore, promoting the installation of modular PV power stations at LV network locations to charge EVs will reduce reliance on both Transmission and Distribution Networks. PV systems will supply power to EV chargers, reducing the load on transformers. Thus, this serves as a complementary measure to address the current transformer capacity issues.

Permitting and Approval Processes: Streamline the procedures for obtaining necessary permits and approvals for installing and operating EV charging stations. Simplified processes will accelerate the deployment of EVCI and reduce bureaucratic delays.

Grid Stability and Capacity

The increased demand from EVCI installations requires careful planning to ensure grid stability and capacity. REG should implement several measures to accommodate this growing demand:

- **Grid Interconnection Requirements**: Ensure that clear and well-defined grid interconnection requirements are in place to support EVCI deployment. This will

allow for a smooth and efficient connection of charging stations to the national grid.

- **Grid Planning/Distribution Network Upgrades**: Regularly assess and upgrade the distribution network to support the higher electricity demand driven by EVCI. Network upgrades will ensure that power is delivered reliably and efficiently to charging stations across the country. Effective grid and energy planning is crucial for expanding charging infrastructure, particularly in upgrading transformer capacity and integrating renewable energy sources. Successful planning requires a holistic approach, involving collaboration between energy, transport, and urban planning experts, with tools to facilitate data sharing. Resource allocation must be carefully balanced, ensuring grid operators manage their limited resources effectively while meeting diverse goals like eMobility. Embracing data-driven, in-house planning capabilities and advanced software tools is essential for agility and efficiency in planning future energy systems, enabling continuous improvement and transformation of grid operations. MININFRA and REG should ensure that the grid planning is equipped with the skills, tooling, and data sources to deliver high-quality plans.
- **Infrastructure Development**: Proactively develop new infrastructure to accommodate the increasing energy demands from widespread EVCI installations. This includes planning for additional substations, transformers, and other grid components necessary to handle EV charging loads.
- **Grid Monitoring**: Implement monitoring systems, such as advanced metering and transformer and substation sensors enable REG to have insight into energy usage patterns and grid load. Besides, data coming from such systems can support in the development of energy forecasts due to EVCI. This will allow for timely interventions and infrastructure expansions to maintain grid stability. MINIFRA and REG should prioritize the implementation of such systems.

Transformer Funding

- **Assessment of Regulatory Framework:** REG and RURA should assess and redesign their current regulatory framework to incentivize distribution grid operators to invest in resources critical for expanding charging infrastructure. As transformer funding is seen as a bottleneck, the regulatory framework should encourage grid operators to strategically allocate capital for grid reinforcements that support the growing demand for electric vehicle charging. A performance-based regulatory approach, targeting specific eMobility-related goals, can be introduced to ensure the expansion aligns with national energy and transportation priorities. A renewed regulatory framework should be aligned with grid planning procedures mentioned in recommendation 12.
- **Dedicated Transformer Fund:** MINIFRA could establish a dedicated transformer fund to support grid reinforcements required for the expansion of charging infrastructure. This fund would provide targeted financial resources to help alleviate the funding bottleneck for transformers and other grid upgrades, enabling faster deployment of eMobility infrastructure.
- On the other hand, promoting installation on modular PV power stations at LV network locations to charge EV will reduce the reliance on both Transmission Networks and Distribution Networks. PV will supply the EV chargers and reduce the absorbed power from transformers, thus this is a complementary measure to approach transformer capacity issues.

Connector Standards Additional Measures

Regional Collaboration for Harmonized Standards:

Currently, there is no standardized approach to EV connectors in East Africa. East Africa countries can take the lead in initiating a regional dialogue aimed at creating a unified EV charging standard across the East African Community (EAC). This would simplify cross-border EV travel, facilitate trade, and encourage regional economic integration. Over time, harmonizing EV standards across the continent could help to stimulate the adoption of electric mobility in Africa and reduce costs for infrastructure development.

Create a working group to coordinate with neighbour countries (Tanzania, Uganda, Kenya) to harmonize policies, regulations, and standards. This will ensure the interoperability of cross-border electric trucks, buses, and passenger vehicles, strengthen connectivity, and establish initial corridors for EVs along key logistics and transport routes.

Monitoring Future Technological Advancements regarding connector Standards:

While there are recommendations on hardware connector approaches for various modes of transport, Rwanda should continually monitor global technological advancements and trends in EV charging, as these may differ across transport types. As the EV ecosystem evolves, opportunities may emerge to phase out outdated standards in favor of more efficient or widely accepted alternatives.

The Rwandan government should periodically assess the adoption and effectiveness of these standards. If one of the standards proves to be significantly more adaptable or gains more widespread adoption. Rwanda may consider transitioning to a single standard to simplify the infrastructure and reduce costs in the long term. Engagement with stakeholders and participation in forums, especially within the East African Community (EAC), will be critical for aligning with regional and global trends.

Track and coordination measures for the EVCI deployment

While MININFRA (Transport) functions as the main body facilitating cooperation among EVCI stakeholders, it is recommended to establish a Working Committee for EV Charging Infrastructure to ensure effective coordination among government agencies during the implementation of the EVCI Master Plan.

The Committee should include all relevant government agencies responsible for policy (MININFRA Transport, MININFRA Energy), regulation (RURA, RDB, RSB, RHA, REMA, RICA), and implementation (City of Kigali and secondary cities, RITCO and other bus fleet operators, commercial, institutional, and residential establishments, and e-mobility service providers).

Key Functions of the Committee:

- **Coordination of Government Agencies:** Identify and address coordination issues among government agencies regarding EVCI. This will ensure clarity in roles, accountability, and effective organization within the government concerning EVCI responsibilities. The committee should meet regularly and produce progress reports to ensure timely implementation and coordination.
- **Assigning Responsibilities:** Allocate responsibilities among committee members for specific actions related to EVCI, according to the EVCI Master Plan suggestion, ensuring that all tasks are assigned and clearly understood.

- **Strategic Recommendations:** Propose strategies to accelerate the implementation of the EVCI Master Plan and to overcome any delays or barriers in the process.
- **Monitoring Progress:** Continuously monitor the implementation of the EVCI Master Plan and track progress against established milestones.
- **Addressing Policy Issues:** Identify any policy gaps or coordination issues that could delay the rollout of EV charging infrastructure and propose solutions to mitigate these challenges.
- **Private Sector Engagement:** Occasionally invite private stakeholders for targeted discussions, ensuring that public sector needs are addressed and that private operators are involved in the EVCI ecosystem. This also ensures continuous alignment and improves decision-making related to EVCI deployment.
- **Communication Channels:** Serve as an open and transparent communication channel with private stakeholders to ensure that the needs of the EVCI ecosystem are addressed, and actions are coordinated effectively.
- **Representation:** It is recommended that each entity involved in the committee designates a dedicated representative to participate in meetings and take responsibility for their instituional/agency's role in the EVCI Master Plan implementation.
- **Meeting Schedule and Legal Authority:** Set a clear and consistent meeting schedule (e.g., quarterly), ensuring that the Committee has the legal authority to enforce its recommendations. The Committee should have the power to issue binding guidance to other agencies or stakeholders, ensuring compliance and progress.

Authority and Institutional Advise:

According to the list of recommendations provided, the following are the suggested institutions responsible for their deployment and oversight.

- The **Transport Department of MININFRA** is expected to issue the EV charging infrastructure and battery swapping National Policy for EV Charging, as recommended in this study. This policy will include guidelines and standards, developed in collaboration with the **Rwanda Standards Board**, and prescribe implementation models for Rwanda.
- **MININFRA** is also responsible for leading the procurement and implementation of the National Data Sharing Platform. Data management activities, including registration, reporting, and GIS integration for EVCI, should be overseen by a single entity to ensure centralized coordination. This is an activity ideally to be shared with **Rwanda Transport Development Agency - RTDA,** or **Rwanda Development Board - RDB.**
- **MININFRA**, in collaboration with the **Rwanda Standards Board RSB** and international and The Ministry of Forening Affairs, should facilitate dialogues and work with neighboring countries to harmonize policies, regulations, and standards for EVCI. Both MININFRA and the Rwanda Standards Board are also well-positioned to advance technological developments related to connector standards.

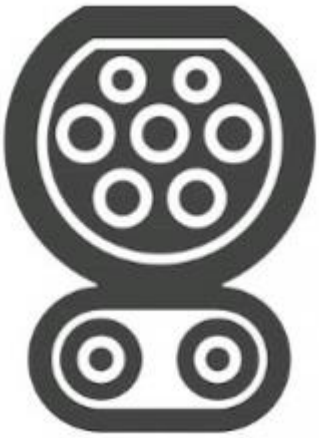
- **MININFRA** should lead the EVCI Master Plan Committee and engage all relevant institutions and cities involved in deploying EVCI measures.
- Additionally, **MININFRA** should lead workforce development guidelines for sustainable mobility and oversee public awareness and education campaigns. This could be also be supported by the Minsitry of Education.
- **The Energy Department of MININFRA** is expected to issue an electrification policy for the transport sector that includes long-term e-mobility targets, including for private, public, and motorcycle fleets. This EVCI policy should be developed jointly by MININFRA's Transport and Energy departments and in collaboration with EUCL and EDCL.
- **MININFRA** and **REG** along with **EUCL** and **EDCL** should jointly work to lead pilot projects for smart charging and Vehicle-to-Grid (V2G) technologies. Additionally, they should work in issuing grid interconnection regulations, implementing grid stability and capacity measures, and developing the transformer regulatory framework.
- The **Rwanda Development Board - RDB** is currently responsible for business registration of new investments (which include investment in charging infrastructures) as well as monitoring of investment projects to ensure that incentives are directed to projects. RDB is suggested to support on the ID registration for eMSP and CPOs.
- On the other hand, **RURA (Transport)** is also responsible for the issuance of licenses for public transport services. It needs to be clarified if charge point operators (CPOs), as operators of public infrastructures, require an additional license from RURA (Transport).
- **RURA (Energy)** is responsible for Regulations Governing Electrical Installations. RURA may also consider updating the regulations to accommodate specific requirements for EV charging infrastructures.
- **RURA (Energy)** is also currently responsible for setting tariffs for CPOs. As specified in this study, RURA may consider setting electricity price caps for EV usage. Electricity tariffs for private charging could also be defined by RURA.
- The **Rwanda Standards Board (RSB)**, the main authority responsible for issuing technical standards and undertaking quality testing, metrology and certifications, must review and issue EV standards. RSB is expected to issue technical standards regarding software and hardware standards related to EV charging infrastructure according to the recommendation no. 2. RSB is also responsible for creating the approach to foster voluntary convergence of standards regarding Battery Swap Stations.
- Building Control Regulations are currently under the **Rwanda Housing Authority - RHA**. It is expected that RHA to issue an amendment of the building regulations to cover provisions requiring parking spaces to be equipped with charging infrastructure in private, commercial and institutional buildings.
- The **Rwanda Environment Management Authority (REMA)** is responsible for protecting the urban environment and mitigating greenhouse gas emission. It is expected that REMA will issue additional regulations and incentive mechanisms improving to urban air quality focusing on transport fuel quality and vehicle emission standards as well as transport sector targets related to greenhouse gas emissions.

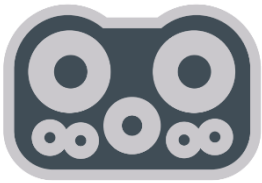
- Furthermore, **REMA** should integrate strategies aligned with the National Circular Economy Action Plan and existing waste disposal initiatives, as stated in recommendation No. 8.
- Licensing of transport facilities and premises as well as enforcement of technical standards and regulations are mandated to **Rwanda Inspectorate, Competition** and **Consumer Protection Authority (RICA)**. It is envisaged that RICA will cover the licensing of EV infrastructures and service providers and enforce compliance of EV infrastructures and vehicle technologies. Also RICA should be in charge of overseeing fair pricing within the EVCI market, implementing and enforcing pricing regulations to protect consumers and promote equitable access to charging services.
- **The City of Kigali and secondary cities** are expected to implement the contractual framework for EVCI deployment, focusing on public-private partnerships or other contractual models, with government support.
- **MININFRA**, supported by **the City of Kigali and secondary cities**, is expected to lead the development of regulatory frameworks that mandate, in a phased and transitional manner, the integration of EVCI into all new buildings, major renovations, existing commercial structures, and EVCI deployment along roads.

7 ANNEXES

Annex No. 1 Table with Type of Connectors

Type of Connector	*Explanation*
AC Connectors	
 AC - J1772 (Type 1)	The J1772 connector, initially created by Yazaki and commonly referred to as Type 1, has found extensive usage in North America and Asia. Specifically designed for AC charging, it accommodates power ratings ranging from 6.6 kW to 19.2 kW. Renowned for its compatibility with a vast majority of electric vehicles in the United States and Asia, the J1772 connector continues to be a preferred option for both residential and public charging infrastructure.
 AC - Mennekes (Type 2)	The Mennekes connector, also known as Type 2, stands as a prominent European standard plug extensively utilized throughout Europe, particularly for AC charging applications. With power delivery capacities ranging from 12 kW to 22 kW for private charging and reaching up to 43 kW for public charging, the Type 2 connector offers versatility and reliability. Noteworthy for its ergonomic design and integrated automatic locking mechanism, this connector guarantees secure and efficient charging operations for electric vehicle owners across the continent.
 AC - GB/T Connector	The GB/T connector, developed by the Guobiao Standardization Commission, stands as the primary AC charging plug within China, representing the country's dominant standard for electric vehicle charging. Supporting power ratings of up to 27.7 kW, this connector accommodates the expanding fleet of electric vehicles within the world's largest automotive market. Despite its visual resemblance to the Type 2 connector, the internal wiring of the GB/T connector sets it apart, rendering it incompatible with European standards.
DC Connectors	

Type of Connector	*Explanation*
 CCS - Type 1	The Combined Charging System (CCS), Type 1, offers a solution for rapid DC charging. In this approach, the lower pins directly facilitate charging, while the upper part's communication pin and ground conductor serve as reference points for protection systems. These connectors support power levels of up to 350 kW and can handle voltages ranging from 200 V to 1 kV at up to 350 A. One significant advantage of the CCS connector is its capability to perform both AC and DC charging simultaneously. These connectors comply with various standards, including IEC 62196–1, IEC 62196–2, and IEC 62196–3. ChargeIN oversees the adoption and standardization of Combo-type connectors.
 CCS - Type 2	The CCS Type 2, a fast DC charging system commonly used in Europe, is regulated by ChargeIn and boasts a power capacity of up to 350 kW. Its widespread adoption can be attributed to the convenience of having a single socket. However, it's important to note that CCS connectors are incompatible with CHAdeMO and GB/T charging stations due to variations in communication protocols. This disparity often requires the use of specialized adapters, which may be challenging to obtain
 CHAdeMo	The CHAdeMO connector, developed by the Japan EV Association, emerges as a prominent DC charging plug predominantly utilized by Japanese automakers. Renowned for its rapid charging capabilities, CHAdeMO connectors can deliver power up to 400 kW, facilitating swift charging experiences and enjoying widespread adoption across Asia. However, despite its initial dominance, the CHAdeMO standard encounters stiff competition from CCS connectors in regions like Europe and North America. This competition has resulted in a gradual decline in the prominence of the CHAdeMO standard as other charging standards gain traction in these markets.

Type of Connector	*Explanation*
 GBT	The DC GB/T connector, developed as the standard for electric vehicle charging in China, is known for its robust functionality and compatibility with a wide range of EVs in the region. This connector supports high power outputs of up to 250 kW, facilitating rapid charging experiences. Designed to handle voltages up to 1000V and currents up to 400A, it enables quick and efficient charging. The GB/T connector uses the CAN protocol for communication between the vehicle and the charging station, ensuring smart charging capabilities and real-time data exchange. Despite its strong adoption in China, the GB/T standard faces competition from other global standards like CCS and CHAdeMO in international markets.
 ChaoJi	ChaoJi, part of the Japanese CHAdeMO line, is anticipated to launch in 2024 with a market power of 500kW. Supported by China and Japan—a significant collaboration merging two historically distinct markets—ChaoJi is expected to replace GB/T DC chargers in China (as ChaoJi-1) and CHAdeMO chargers in Japan (as ChaoJi-2, also known as CHAdeMO 3.0). ChaoJi-1 is expected to follow the GB/T protocol and will primarily be deployed in mainland China, while ChaoJi-2 may expand to multiple global markets. With this new standard, other countries might choose to align with Japan's lead (now in sync with China) or adopt standards closer to those in Europe and North America. ChaoJi-2 is also known as CHAdeMo 3.0.
 Tesla Connector (NACS)	Initially exclusive to Tesla vehicles, Tesla's proprietary charging connector, known as the North American Charging Standard (NACS), represented a hallmark feature of Tesla's charging infrastructure. With remarkable power delivery capabilities of up to 1 MW, NACS connectors enable rapid charging, a key factor in Tesla's extensive charging network's efficiency. However, in a strategic shift towards standardization and interoperability, Tesla has embraced CCS2 connectors for its vehicles in Europe. This move aims to facilitate seamless compatibility with non-Tesla charging stations across the continent, ensuring convenience for Tesla owners and promoting broader adoption of electric vehicles.

Annex No. 2 Existing charging stations in Rwanda (Attached)

Annex No. 3 Current EVCI Landscape and Transport Infrastructure (Attached)

Annex No. 4 Planned EVCI Developments in Kigali (Attached)

Annex No. 5 Existing battery swapping stations (Attached)

Annex No. 6 Eastern Province Planned & Existing EVCI (Attached)

Annex No. 7 Northern Province Planned & Existing EVCI (Attached)

Annex No. 8 Southern Province Planned & Existing EVCI (Attached)

Annex No. 9 Western Province Planned & Existing EVCI (Attached)

Annex No. 10 Distribution map of potential charging station (Attached)

Annex No. 11 Potential EVCI for Kigali (Attached)

Annex No. 12 Potential EVCI for Nyarugenge (Attached)

Annex No. 13 Potential EVCI for Gasabo (Attached)

Annex No. 14 Potential EVCI for Kicukiro (Attached)

Annex No. 15 Potential EVCI for Southern Province (Attached)

Annex No. 16 Potential EVCI for Eastern Province (Attached)

Annex No. 17 Potential EVCI for Northern Province (Attached)

Annex No. 18 Potential EVCI for Western Province (Attached)

Annex No. 19 Table of detail EVCI combined (Existing, Planned and Potential) (Attached)

Annex No. 20 Case 1 (North-East Rwanda:RE) - Modular pV Power Station: Project Description

30.8 KWp Model EV -PV CH- R N, -1.221767, 30.411320

Report	
Project Name	EV -PV CH- R N
Project Description	EV -Charging- station In Car port
Project Address	-1.221767, 30.411320

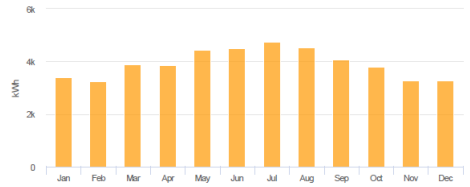
System Metrics	
Design	EV - N
Module DC Nameplate	30.8 kW
Inverter AC Nameplate	26.0 kW Load Ratio: 1.18
Annual Production	47.11 MWh
Performance Ratio	83.7%
kWh/kWp	1,529.5
Weather Dataset	TMY, 10km Grid, Meteonorm 7 (meteonorm)
Simulator Version	4a26ee4d61-bb8775eb77-328eb74c35-73d82488fd

Project Location



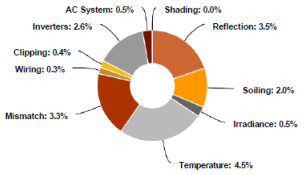
Annex No. 21 Case 1 (North-East Rwanda:RE) - Modular pV Power Station : Monthly Production and Sources of System Loss

Monthly Production



Month	GHI (kWh/m²)	POA (kWh/m²)	Shaded (kWh/m²)	Nameplate (kWh)	Grid (kWh)
January	157.3	132.9	132.9	3,832.9	3,408.2
February	141.3	127.0	127.0	3,678.9	3,256.4
March	159.4	152.3	152.3	4,435.4	3,905.6
April	146.7	150.5	150.5	4,396.5	3,872.7
May	158.3	172.7	172.7	5,061.2	4,448.9
June	155.5	174.1	174.1	5,112.6	4,506.7
July	163.5	182.5	182.5	5,363.4	4,730.5
August	166.5	175.7	175.7	5,152.3	4,548.6
September	158.7	157.5	157.5	4,592.9	4,063.7
October	160.5	147.9	147.9	4,293.0	3,816.6
November	147.5	127.0	127.0	3,664.6	3,273.0
December	152.9	127.4	127.4	3,667.2	3,277.5

Sources of System Loss



Annex No. 22 Case 1 (North-East Rwanda: RE) - Modular pV Power Station: Annual production and condition set

Annual Production

	Description	Output	% Delta
Irradiance (kWh/m²)	Annual Global Horizontal Irradiance	1,868.0	
	POA Irradiance	1,827.6	-2.2%
	Shaded Irradiance	1,827.6	0.0%
	Irradiance after Re ection	1,763.9	-3.5%
	Irradiance after Soiling	1,728.7	-2.0%
	Total Collector Irradiance	1,728.7	0.0%
Energy (kWh)	Nameplate	53,251.0	
	Output at Irradiance Levels	52,972.2	-0.5%
	Output at Cell Temperature Derate	50,609.4	-4.5%
	Output After Mismatch	48,931.8	-3.3%
	Optimal DC Output	48,771.8	-0.3%
	Constrained DC Output	48,597.2	-0.4%
	Inverter Output	47,345.0	-2.6%
	Energy to Grid	47,108.3	-0.5%
Temperature Metrics			
	Avg. Operating Ambient Temp		23.0 °C
	Avg. Operating Cell Temp		32.9 °C
Simulation Metrics			
	Operating Hours		4490
	Solved Hours		4490

Condition Set

Description	Condition Set 1											
Weather Dataset	TMY, 10km Grid, Meteonorm 7 (meteonorm)											
Solar Angle Location	Meteo Lat/Lng											
Transposition Model	Perez Model											
Temperature Model	Sandia Model											
Temperature Model Parameters	Rack Type	a	b	Temperature Delta								
	Fixed Tilt	-3.56	-0.075	3°C								
	Flush Mount	-2.81	-0.0455	0°C								
	East-West	-3.56	-0.075	3°C								
	Carport	-3.56	-0.075	3°C								
Soiling (%)	J	F	M	A	M	J	J	A	S	O	N	D
	2	2	2	2	2	2	2	2	2	2	2	2
Irradiation Variance	5%											
Cell Temperature Spread	4° C											
Module Binning Range	-2.5% to 2.5%											
AC System Derate	0.50%											
Module Characterizations	Module	Uploaded By	Characterization									
	JKM560N-72HL4-BDV (Jinko Solar)	HelioScope	Spec Sheet Characterization, PAN									
Component Characterizations	Device	Uploaded By	Characterization									
	SG12KTL (Sungrow)	HelioScope	Default Characterization									

Annex No. 23 Case 1 (North-East Rwanda:RE) - Modular pV Power Station: Components, Wiring Zones, Field Segments

Components

Component	Name	Count
Inverters	SG12KTL (Sungrow)	2 (26.0 kW)
Strings	10 AWG (Copper)	4 (72.8 m)
Module	Jinko Solar, JKM560N-72HL4-BDV (560W)	55 (30.8 kW)

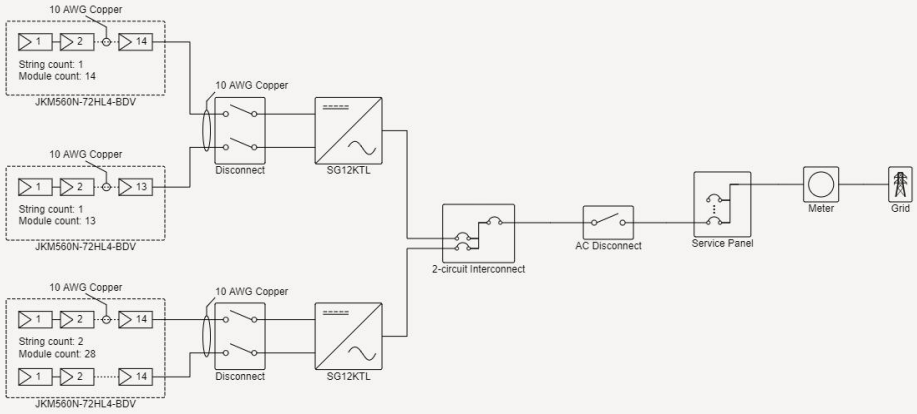
Wiring Zones

Description	Combiner Poles	String Size	Stringing Strategy
Wiring Zone	-	9-19	Along Racking

Field Segments

Description	Racking	Orientation	Tilt	Azimuth	Intrarow Spacing	Frame Size	Frames	Modules	Power
Field Segment 1	Carport	Landscape (Horizontal)	20°	0°	0.1 m	1x1	55	55	30.8 kW

Annex No. 24 Case 1 (North-East Rwanda:RE) - Modular pV Power Station: Detailed Layout



Module Specifications	
55x Jinko Solar JKM560N-72HL4-BDV	
STC Rating	560 W
Vmp	41.95 V
Imp	13.35 A
Voc	50.67 V
Isc	14.13 A

Inverter Specifications	
2x Sungrow SG12KTL	
Max AC Power Rating	13 kW
Max Input Voltage	1,000 V
Min Input Voltage	250 V

Wire Schedule		
Tier	Wire	Length
String	4x 10 AWG	73m

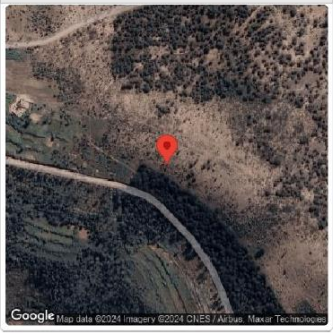
Annex No. 25 Case 2 (South-West Rwanda) - pV Power Station: Project Description

30.8 MWp Model. EV -PV CH- R S

Report	
Project Name	EV -PV CH- R S
Project Address	-2.691538, 29.602896

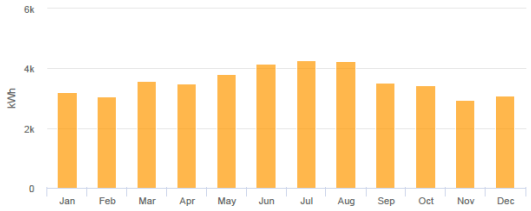
System Metrics	
Design	Design 1
Module DC Nameplate	30.8 kW
Inverter AC Nameplate	26.0 kW Load Ratio: 1.18
Annual Production	42.68 MWh
Performance Ratio	83.9%
kWh/kWp	1,385.7
Weather Dataset	TMY, 10km Grid, Meteonorm 7 (meteonorm)
Simulator Version	4a26ee4d61-bb8775eb77-328eb74c35-73d82488fd

Project Location



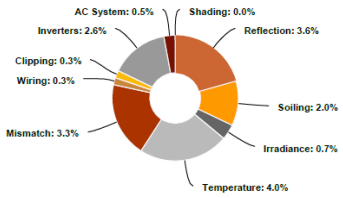
Annex No. 26 Case 2 (South-West Rwanda) - Modular pV Power Station: Monthly Production and Sources of System Loss

Monthly Production



Month	GHI (kWh/m²)	POA (kWh/m²)	Shaded (kWh/m²)	Nameplate (kWh)	Grid (kWh)
January	144.3	124.7	124.7	3,596.2	3,195.6
February	129.5	118.5	118.5	3,433.4	3,046.0
March	142.8	139.0	139.0	4,042.8	3,562.6
April	131.3	135.1	135.1	3,941.3	3,468.6
May	135.5	145.9	145.9	4,266.7	3,798.8
June	141.8	158.5	158.5	4,646.6	4,149.0
July	147.3	164.2	164.2	4,815.8	4,276.4
August	154.8	164.2	164.2	4,813.7	4,234.2
September	137.1	136.4	136.4	3,971.6	3,524.3
October	142.4	133.0	133.0	3,853.2	3,436.1
November	129.6	113.3	113.3	3,270.6	2,917.5
December	141.1	119.2	119.2	3,433.3	3,069.3

Sources of System Loss



Annex No. 27 Case 2 (South-West Rwanda) - Modular pV Power Station: Annual Production and Condition set

Annual Production

	Description	Output	% Delta
Irradiance (kWh/m²)	Annual Global Horizontal Irradiance	1,677.6	
	POA Irradiance	1,651.8	-1.5%
	Shaded Irradiance	1,651.8	0.0%
	Irradiance after Re ection	1,592.8	-3.6%
	Irradiance after Soiling	1,561.0	-2.0%
	Total Collector Irradiance	1.561.0	0.0%
Energy (kWh)	Nameplate	48,085.1	
	Output at Irradiance Levels	47,756.7	-0.7%
	Output at Cell Temperature Derate	45,853.1	-4.0%
	Output After Mismatch	44,317.9	-3.3%
	Optimal DC Output	44,178.7	-0.3%
	Constrained DC Output	44,030.3	-0.3%
	Inverter Output	42,892.9	-2.6%
	Energy to Grid	42.678.5	-0.5%
Temperature Metrics			
	Avg. Operating Ambient Temp		22.4 °C
	Avg. Operating Cell Temp		31.1 °C
Simulation Metrics			
	Operating Hours		4510
	Solved Hours		4510

Condition Set

Description	Condition Set 1											
Weather Dataset	TMY, 10km Grid, Meteonorm 7 (meteonorm)											
Solar Angle Location	Meteo Lat/Lng											
Transposition Model	Perez Model											
Temperature Model	Sandia Model											
Temperature Model Parameters	Rack Type	a	b	Temperature Delta								
	Fixed Tilt	-3.56	-0.075	3°C								
	Flush Mount	-2.81	-0.0455	0°C								
	East-West	-3.56	-0.075	3°C								
	Carport	-3.56	-0.075	3°C								
Soiling (%)	J	F	M	A	M	J	J	A	S	O	N	D
	2	2	2	2	2	2	2	2	2	2	2	2
Irradiation Variance	5%											
Cell Temperature Spread	4° C											
Module Binning Range	-2.5% to 2.5%											
AC System Derate	0.50%											
Module Characterizations	Module	Uploaded By	Characterization									
	JKM560N-72HL4-BDV (Jinko Solar)	HelioScope	Spec Sheet Characterization, PAN									
Component Characterizations	Device	Uploaded By	Characterization									
	SG12KTL (Sungrow)	HelioScope	Default Characterization									

Annex No. 28 Case 2 (South-West Rwanda) - Modular pV Power Station: Components, Wiring Zones, Field Segments

Components

Component	Name	Count
Inverters	SG12KTL (Sungrow)	2 (26.0 kW)
Strings	10 AWG (Copper)	4 (76.6 m)
Module	Jinko Solar, JKM560N-72HL4-BDV (560W)	55 (30.8 kW)

Wiring Zones

Description	Combiner Poles	String Size	Stringing Strategy
Wiring Zone	-	9-18	Along Racking

Field Segments

Description	Racking	Orientation	Tilt	Azimuth	Intrarow Spacing	Frame Size	Frames	Modules	Power
Field Segment 1	Carport	Landscape (Horizontal)	20°	0°	0.6 m	1x1	55	55	30.8 kW

Annex No. 29 Case 3: PV Power Station, Center Rwanda - pV Power Station: Project Description

30.8 KWp Model EV -PV CH- R M, -1.973694, 30.016510

Report	
Project Name	EV -PV CH- R M
Project Address	-1.973694, 30.016510

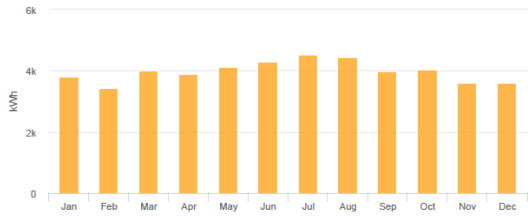
System Metrics	
Design	Design 1
Module DC Nameplate	30.8 kW
Inverter AC Nameplate	26.0 kW Load Ratio: 1.18
Annual Production	47.76 MWh
Performance Ratio	83.6%
kWh/kWp	1,550.5
Weather Dataset	TMY, 10km Grid, Meteonorm 7 (meteonorm)
Simulator Version	4a26ee4d61-bb8775eb77-328eb74c35-73d82488fd

Project Location



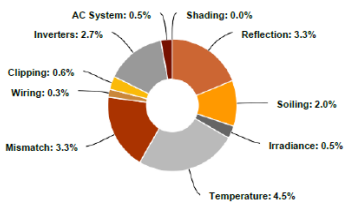
Annex No. 30 Case 3: PV Power Station, Center Rwanda- pV Power Station: Monthly Production and Sources of System Loss

Monthly Production



Month	GHI (kWh/m²)	POA (kWh/m²)	Shaded (kWh/m²)	Nameplate (kWh)	Grid (kWh)
January	158.4	147.9	147.9	4,290.0	3,802.6
February	139.5	133.9	133.9	3,892.2	3,434.2
March	157.9	157.2	157.2	4,590.8	3,998.4
April	147.4	151.4	151.4	4,428.4	3,887.2
May	152.1	161.3	161.3	4,721.4	4,134.1
June	154.2	165.8	165.8	4,861.5	4,293.9
July	162.4	174.7	174.7	5,117.1	4,535.7
August	165.6	172.7	172.7	5,062.3	4,449.1
September	155.0	156.0	156.0	4,564.4	3,994.7
October	160.4	155.8	155.8	4,534.7	4,025.9
November	145.5	138.5	138.5	4,015.5	3,595.6
December	152.2	140.5	140.5	4,072.2	3,603.7

Sources of System Loss



Annex No. 31 Case 3: PV Power Station, Center Rwanda- pV Power Station: Annual Production and Condition set

Annual Production

	Description	Output	% Delta
Irradiance (kWh/m²)	Annual Global Horizontal Irradiance	1,850.5	
	POA Irradiance	1,855.8	0.3%
	Shaded Irradiance	1,855.8	0.0%
	Irradiance after Re ection	1,793.7	-3.3%
	Irradiance after Soiling	1,757.9	-2.0%
	Total Collector Irradiance	1.757.9	0.0%
Energy (kWh)	Nameplate	54,150.6	
	Output at Irradiance Levels	53,856.4	-0.5%
	Output at Cell Temperature Derate	51,455.9	-4.5%
	Output After Mismatch	49,759.8	-3.3%
	Optimal DC Output	49,585.8	-0.3%
	Constrained DC Output	49,302.2	-0.6%
	Inverter Output	47,995.3	-2.7%
	Energy to Grid	47.755.3	-0.5%
Temperature Metrics			
	Avg. Operating Ambient Temp		22.3 °C
	Avg. Operating Cell Temp		32.3 °C
Simulation Metrics			
	Operating Hours		4489
	Solved Hours		4489

Condition Set

Description	Condition Set 1											
Weather Dataset	TMY, 10km Grid, Meteonorm 7 (meteonorm)											
Solar Angle Location	Meteo Lat/Lng											
Transposition Model	Perez Model											
Temperature Model	Sandia Model											
Temperature Model Parameters	Rack Type	a	b	Temperature Delta								
	Fixed Tilt	-3.56	-0.075	3°C								
	Flush Mount	-2.81	-0.0455	0°C								
	East-West	-3.56	-0.075	3°C								
	Carport	-3.56	-0.075	3°C								
Soiling (%)	J	F	M	A	M	J	J	A	S	O	N	D
	2	2	2	2	2	2	2	2	2	2	2	2
Irradiation Variance	5%											
Cell Temperature Spread	4° C											
Module Binning Range	-2.5% to 2.5%											
AC System Derate	0.50%											
Module Characterizations	Module	Uploaded By	Characterization									
	JKM560N-72HL4-BDV (Jinko Solar)	HelioScope	Spec Sheet Characterization, PAN									
Component Characterizations	Device	Uploaded By	Characterization									
	SG12KTL (Sungrow)	HelioScope	Default Characterization									

Annex No. 32 Case 3: PV Power Station, Center Rwanda - pV Power Station: Components, Wiring Zones, Field Segments

Components

Component	Name	Count
Inverters	SG12KTL (Sungrow)	2 (26.0 kW)
Strings	10 AWG (Copper)	4 (78.1 m)
Module	Jinko Solar, JKM560N-72HL4-BDV (560W)	55 (30.8 kW)

Wiring Zones

Description	Combiner Poles	String Size	Stringing Strategy
Wiring Zone	-	9-18	Along Racking

Field Segments

Description	Racking	Orientation	Tilt	Azimuth	Intrarow Spacing	Frame Size	Frames	Modules	Power
Field Segment 1	Carport	Landscape (Horizontal)	10°	0°	0.1 m	1x1	55	55	30.8 kW

Annex No. 33 Equipment and Installation Cost Reference

Equipment and Installation Cost, Developed Countries

Cost Element	*Study*	*Level 2*	*DCFC*			
			50 kW	150 kW	350 kW	800 kW
Equipment cost	CCT (2019)	$3,127	$28,401	$75,000	$140,000	
	NREL (2020)	$3,500	$38,000	$90,000		
	MI (2020)	$2,500 - $4,900	$20,000 - $35,800	$75,600 - $100,000	$128,000 - $150,000	
	EDF & GNA (2021)			$136,540		$481,299
Installation cost	ICCT (2019)	$2,837 - $4,148	$17,692 - $45,506	$18,577 - $47,781	$25,654 - $65,984	
	NREL (2020)	$2,500	$20,000	$60,000		
	RMI (2020)	$7,000	$62,700	$75,500	$138,200	
	EDF & GNA (2021)			$35,000		$175,000

Equipment and Installation Cost, India

Cost Element	Study	Level 1 AC	Level 2 AC	DC	CCS2
			7 kW	10 kW	25 kW
Equipment Cost	WB/ESMAP 2022	$42	$480	$2400	$8400
Civil Work	WB/ESMAP 2022	5.0%	10.0%		
Installation and Commissioning	WB/ESMAP 2022	3.0%	15.0%		
Software Cost	WB/ESMAP 2022	2.0%	5.0%		

Equipment and Installation Cost, Turkey

Cost Element	Study	Level 1 AC	Level 2 AC	DC
			22 kW	50 kW
Equipment Cost				
Home charger	WB 2023		$1650	
Public charger	WB 2023		$ 2200	$24299
Commissioning and service fee (installation fee)				
Home charger	WB 2023		$110	
Public charger	WB 2023		$132	$220
Basic infrastructural requirements (cabling)				
Home charger	WB 2023		$1100-$1320	
Public charger	WB 2023		$1100-$1320	$5500-$6600

Annex No. 34 Public Stakeholders Challenges, Interests, Resources available and Mandate.

STAKEHOLDER	STAKEHOLDER'S INTEREST with respect to deployment of EVCI	PERCEIVED PROBLEMS with respect to deployment of EVCI	RESOURCES AVAILABLE with respect to addressing the problem	MANDATE with respect to the governance of e-mobility and EVCI deployment
MININFRA - Transport	Overseeing the evolution of the electric vehicles charging infrastructures Expansion of the fast-charging network is mentioned to be of high priority	An environmental assessment of all considered charging locations is not a part of this master plan project	The grid expansion plan, based on EV uptake and charging infrastructure expansion scenarios, is an important outcome of this project Transport models are developed within the ministry. Data and models are accessible to the team	To plan, develop, manage, and maintain an efficient and integrated national transport infrastructure network, including roads, bridges, airports, railways, and water transport, to facilitate economic development, social equity, environmental improvement, and regional integration Provide modern infrastructures for sustainable economic growth and socio-economic development including overseeing the evolution of the Electric Vehicle charging infrastructure Master Plan.
MININFRA – Energy	Planning for the grid expansion, Electrification of transport at national parks should be in scope Planning for the grid expansion, based on EV uptake and charging infrastructure expansion scenarios, is an important outcome of this project	The need to upgrade transformer capacity The agricultural sector is expected to have limited electrification progress Off-grid solutions should be considered, with the following considerations: most locations can likely be connected to the grid The scarcity of land is an important factor to consider during the roll-out of charging infrastructure.	Different electricity tariff models currently exist	Standards and interoperability should support the rapid growth of the charging ecosystem, Electrification of transport at national parks should be in scope. Expansion of the fast-charging network is mentioned to be of high priority The master plan should include a set of business models to give an overview of the variety of options.
RURA	Regulate public utilities. RURA has registers of vehicles	Lack of regulation is blocking investments into different sectors,	Construction and safety regulations are in place, Close collaboration with MININFRA	Questions related to energy tariffs can be addressed to the department within RURA that is responsible for that.

				RURA has a role in the current tax exemptions for vehicles, land use, and permitting. Regulation of the public utilities, among those, Renewable and non-renewable energy, industrial gases, pipelines and storage facilities; Transport of persons and goods; Other public utilities RURA and the City of Kigali are responsible for approving permits for operating a charging point on a specific parcel.
EUCL	EUCL wants the master plan to be aligned with the reality on the ground, actionable, and focused. EUCL wants the master plan to be aligned with the reality on the ground, actionable, and focused The master plan should provide input to EUCL's next Least-Cost-Development-Plan.	Off-grid installations that were installed in the past have not always proven to be reliable. The image of off-grid installations is bad due to these experiences Dedicated transformers were funded under the "Doing Business" initiative. This initiative was focused on increasing economic activities in Rwanda and making Rwanda an attractive location for businesses/This stopped.	Upgrades of equipment in currently developed parts of the grid are within this scope of responsibilities. A plan/map exists for where to consider off-grid and where to connect. The Rwanda Electricity Sector Strengthening Plan closed last year. 70 million USD was allocated to improving energy access, 20 million on improving the distribution grid (mainly in Kigali), and 5 million on capability building and feasibility studies. Smart meter infrastructure is in place only for larger customers. The current electricity tariffs are heavily subsidized. REG moving into the eMobility space as CPO is not officially part of the current strategic plan but is something that should not be excluded from the options. Installation of home-installed PV is regulated	Responsible for operations and maintenance Providing energy utility services in the Country through operations and maintenance of existing generation plants, transmission and distribution network and retail of electricity to end-users EDCL is responsible for extending the grid. Reducing distribution losses is a high priority REG has 'take or pay' PPAs with electricity generators

			Smart meter infrastructure in place for the larger customers Dedicated transformers now must be paid by private actors themselves and EUCL provide with a payment plan Current electricity tariffs are heavily subsidized. A solar park of 8 MW is currently feeding electricity to the grid without causing challenges	
City of Kigali	Green mobility is currently within the 'Kigali Vision' and has high priority. The City of Kigali has a zero-fleet policy The City of Kigali supports the financing of vehicles as well. Accessibility of semi-public and public charging is a topic of interest for the City of Kigali The City of Kigali is interested in participating in a workshop about eMobility ecosystems and charging infrastructure.	Land scarcity to install the EV charging stations in the City	The City of Kigali has an 'Inspection Unit' that is responsible for monitoring land use according to plans and agreements. The City of Kigali responded positively to incorporating charging infrastructure requirements in the permit requirements for new buildings, such as done in Europe via the EPBD The Urban Mobility Development Department is responsible for eMobility and is expanding Green mobility is currently within the 'Kigali Vision' and has high priority Kigali Master plan with a specific component related to the "City on the move" some necessary details are written. Charging infrastructure implementation plans and the eMobility Masterplan in line with the City of Kigali's Masterplan with a broader scope. The Serena Hotel was mentioned as an example of a recently developed building with charging facilities.	Kigali is to become a city of green transport The City of Kigali would like to be an advocate of eMobility. Its responsibilities include ensuring the implementation of national policies and developing and implementing action plans for the transportation of people and goods within its boundaries. Development of a transit-oriented city with more sustainable transport modes such as mass public transport and non-motorized transport. Promote adoption of green technologies and cleaner transport modes in freight and logistics industrial sector. The City of Kigali supports companies in acquiring scarce land: by raising the importance of providing land for charging locations or by imposing to provide land against a fixed fee. The City of Kigali legally has the mandate to do this.

Annex No. 35 Policy Summary Table: International Best Practice, Current Situation in Rwanda, Recommendations

Policy Summary				
Issue	**International Best Practice**	**Current Practice in Rwanda**	**Recommendations**	**Responsible Agency**
Regulation of Charging Infrastructure	EV charging as retail sale of electricity EV charging as service to the transport sector	EV charging as service to the transport sector. Not regulated by RURA.	Current practice is aligned to international best practice. Application process for industrial tariff could be streamlined.	RURA to streamline tariff application
Licensing of CPOs	For EV charging as service to the transport sector, there is no need to obtain license from energy regulator.	CPOs need business permit.	Current practice is aligned to international best practice.	Rwanda Revenue Authority for business permits and Rwanda Development registers the business
Incentives and Regulation for Private Charging	Governments provide subsidies for EV charging installation at domestic properties and workplaces	Rwanda only provides fiscal and non-fiscal incentives and indirectly benefit private charging Policy on setting the industrial electricity tariff and access to off-peak industrial tariff for CPOs render private charging more expensive than public charging.	Grants and subsidies may create an unnecessary financial burden to the government. Income tax deductions and other tax benefits may be an alternative option. Adjustments to current residential electricity tariff may be an alternative solution (increasing the range of the lower bracket of the inverted tariff block).	Rwanda Revenue Authority can explore options for tax benefits for private charging. RURA may review and assess whether adjustments for residential tariff be an option.
	Regulations to install EV charging infrastructures in domestic properties (apartments, multistorey buildings) and workplaces	No regulations related to installation of EV charging infrastructures in domestic properties and workplaces.	Local governments could introduce regulations related to mandatory installation of EV charging infrastructures in large apartments, multistorey buildings and workplaces.	City of Kigali and local administration in general to introduce mandatory installation of EV charging infrastructures in large apartments, multistorey buildings and workplaces.
Incentives and Regulations for Public Charging	Mandatory installation of EV charging infrastructures in commercial/business centers and parking areas Concession through competitive bidding for public charging infrastructure provision	No regulations on public charging infrastructures	Local governments could introduce mandatory installation of EV charging infrastructures in commercial/business centers and parking areas including petrol stations. Establish concession arrangement for public charging infrastructures	City of Kigali and local administration can issue regulations mandating installation of EV charging infrastructures in commercial centers and parking areas including petrol stations. City of Kigali and local administration can introduce concession arrangements for public charging infrastructures.

	Provision of incentives such as direct funding, grants and subsidies, and other tax benefits	Introduced fiscal and non-fiscal incentives for electric mobility	Direct funding, grants and subsidies may create an unnecessary financial burden to the government. Additional tax benefits can be explored.	Rwanda Revenue Authority can explore additional tax benefits for public charging infrastructure providers.
Incentives and Regulations for Electric Motorcycles	Introduce targets to electrify motorcycle fleet and banning of fossil fuel-based motorcycles	No policy on electrification of motorcycle fleet	The government can introduce targets for the electrification of the motorcycle fleet.	Rwanda Revenue Authority can explore additional tax benefits for battery swapping operators.
	Provision of subsidies and grants to battery swapping station operators.	Introduced fiscal and non-fiscal incentives for electric mobility	Direct funding, grants and subsidies may create an unnecessary financial burden to the government. Additional tax benefits can be explored.	Rwanda Revenue Authority can explore additional tax benefits for battery swapping operators.
Incentives and Regulations for Public Transport Fleet	National bans on sale of internal combustion engine (ICE) vehicles Mandating electrification of fleets/ public transport	No bans on sale of internal combustion engine (ICE) vehicles Introduced target of 20% share of electric buses by 2030	Electrification target by 2030 is aligned with international best practice. Targets beyond 2030 could be introduced.	MININFRA (Transport) could introduce targets beyond 2030.
	Provision of fiscal incentives, subsidies and loans to support electric mobility in the public transport	Introduced fiscal and non-fiscal incentives for electric mobility	Direct funding, grants and subsidies may create an unnecessary financial burden to the government. Additional tax benefits can be explored.	Rwanda Revenue Authority can explore additional tax benefits for CPOs providing services to public transport.
Tariff Management	Charging prices are set freely by the market and there are no regulations about charging prices. Business-to-consumer pricing schemes in Europe are the following: i) flat rate pricing (e.g. €15 per month), ii) transaction-based pricing (e.g. €0.30 per kWh), and iii) time-based pricing (e.g. €1.50 per hour of charging). Some countries introduce lower and upper charging price caps.	The government sets the electricity tariff for CPOs equivalent to the industrial tariff (large industry) and that CPOs can also benefit from the reduced off-peak tariff. CPOs set their own charging tariffs. Often, these tariffs, after considering their investment and O&M costs, are set at a rate competitive to the equivalent petrol or diesel price.	While the charging tariff may be freely determined by the CPOs, in the short term when competition is limited, a price cap may initially be set. Price caps can be aligned with transaction-based pricing which is expressed in terms of RWF per kWh	RURA may set initially a price cap
Regulatory Framework Governing Electricity Supply for EV Charging	Grid connection requirements and fees for charging stations are the same for other consumption points.	The current regulations for electrical connections in Rwanda are specified in the Regulations Governing Electrical Installations (2023).	Existing regulations for electrical installations are robust and are sufficient to accommodate grid connection of EV charging stations.	RURA may review interconnection standards specific to grid interconnection of EV charging stations.

	If a single EV charging point is connected to the low voltage grid/ medium voltage grid, the connection is treated like any other connection in the low voltage grid/ medium voltage grid.	Electrical installations are classified for various voltage levels and for specific installation works. Fees are also based according to these classifications.		
Technical Standards	In some export-oriented countries, EV standards were directly taken from these international standards (IEC, IEEE, ISO). In most countries, national standards were issued by their respective standards boards or institutes but referenced to international standards.	Current standards in Rwanda are benchmarked with international standards including those adopted in African countries. No standards specific to EVs and charging infrastructures were issued Rwanda.	The current practice in issuing standards in Rwanda is aligned with international best practice. Rwanda must adopt international standards for EVs in general including those for charging infrastructures.	The Rwanda Standards Board (RSB), the main authority responsible for issuing technical standards and undertaking quality testing, metrology and certifications, must review and issue EV standards.

Annex No. 36 Team Composition and roles

Name	*Position*
Roland Ferwerda	Team Leader/Senior Expert
Tonny Bosch	International Transportation Expert
Clay Bonishuli	Local Transportation Expert
Jost Uwase	Socio-Economist Expert
Jan van Dinther	EV Charging Technology Expert
Walid El-Khattam	Energy Grid Analyst
Romeo Pacudan	Financial and Regulatory Expert
Ernestine Mukamboneye	GIS Expert

Alejandra Adeikalam (Freepik)	Graphic design credits